Prescription Writing Template

Dr Nathan Do
Vision Clinic, Sydney NSW 2000
Prescriber Number: 987654

Ms Patient Name
123 Red Street, Creek NSW 2308
Medicare Number: 4123 56789 1
DOB: 01/01/1960

PBS Script/Private Script

Rx: Rx: Sodium hyaluronate 0.2% **preservative-free** eye drops
Instil 1 drop both eyes qid

- Include clear instructions, including which eye to instill the drops in
- Don't use gutte (**drops**), qds (**qid**). OD (**once a day/daily**)
- Note when to **shake well before use** → active ingredients that aren't fully dissolved
    - **Suspensions**: liquid formulations where the active drug is in solid form but dispersed in a liquid.
    - **Emulsions:** some eye drops that have oil and water-based components, shaking may be required to re-mix the ingredients.
- Topical **corticosteroids** like prednisolone acetate (often in suspension form).
- **Chloramphenicol** or other antibiotic suspensions for eye infections.

- Prednisolone acetate 1% + phenylephrine 0.12% (Prednefrin Forte) - suspension
- Dexamethasone (Maxidex) 0.1% - suspension
- Fluorometholone acetate (Flarex) 0.1% - suspension → no longer
- Fluorometholone (FML Liquifilm) 0.1% - suspension
- Levocabastine 0.05% suspension (Livostin, Zyrtec) - suspension
- Chloramphenicol (generally in suspension form)

- **Note**: when to review to ensure compliance & when compounded

Quantity: 1 x 10 mL
Repeats: 5

- Acute conditions (no repeats), chronic conditions (small number of repeats, or none such as steroids, PBS has a max)

Authority: Streamlined code 1234

- **To include:** Authority Approval Number or Streamlined Authority Code (e.g. unpreserved preparation for dry eye disease) – 4-5 numbers long. No authority approval or code is required if you are prescribing non-PBS (private).

Signed: Chloe Optometrist
Dated: 06 April 2025

Medicine name pairs with 'mixed-case lettering' applied

- ERYthromycin
- aziTHROMYCIN
- cLARITHROMYcin
- ciPROFLOXAcin
- ciclosPORIN
- cyclosERINE
- valAciclovir
- valGANciclovir
- xalaTAN
- xalaCOM
- dexMETHAOne
- dexMEDETOMIDine

Example One: Acute Bacterial Conjunctivitis
Dr Isabelle Jalbert
Optometry Clinic, UNSW NSW 2052
Prescriber no: 2020

Ms Red Eye
3 Rock Street, Simple VIC 3000
Medicare no: 402491039
DOB 12/12/1929

Private script
Rx framycetin 0.5%
Instil 1 drop in both eyes four times daily

Quantity 1
Repeats 0

Signed: Isabelle Jalbert
Dated: 10 March 2025

Example Two: Uveitis
Dr Isabelle Jalbert
Optometry Clinic, UNSW NSW 2052
Prescriber no: 2020

Ms Red Eye
3 Rock Street, Simple VIC 3000
Medicare no: 402491039
DOB 12/12/1929

PBS script
Rx prednisone acetate 1% + phenylephrine 0.12% eye drops
Shake well before use
Instill 1 drop hourly

Quantity 1
Repeats 0

Signed: Isabelle Jalbert
Dated: 10 March 2025

Appendix B

Board-approved list of topical schedule 2, 3 and 4 medicines that optometrists with a scheduled medicines endorsement are qualified to administer, obtain, possess, prescribe, supply or use for the purposes of the practice of optometry[5]

Schedule 2 Pharmacy Medicine

Anti-infectives
Dibromopropamidine
Propamidine

Decongestants/ anti-allergics
Antazoline
Azelastine
Ketotifen
Levocabastine
Lodoxamide
Naphazoline
Pheniramine
Sodium Cromoglycate

Miotics, mydriatics and cycloplegics
Phenylephrine ≤2.5%

Schedule 3 Pharmacist Only Medicine

Anti-infectives
Chloramphenicol

Schedule 4 Prescription Only Medicine

Anti-infectives
Aciclovir
Azithromycin
Bacitracin
Cephazolin
Ciprofloxacin
Framycetin
Ganciclovir
Gentamicin
Gramicidin
Neomycin
Ofloxacin
Polymyxin
Tetracycline
Tobramycin

Anti-inflammatories
Cyclosporin
Dexamethasone
Diclofenac
Fluorometholone
Flurbiprofen
Hydrocortisone
Ketorolac
Lifitegrast
Loteprednol
Prednisolone

Decongestants/ anti-allergics
Olopatadine

Anti-glaucomas
Apraclonidine
Betaxolol
Bimatoprost
Brimonidine
Brinzolamide
Dorzolamide
Latanoprost
Pilocarpine
Tafluprost
Timolol
Travoprost

Miotics, mydriatics and cycloplegics
Atropine
Cyclopentolate
Homatropine
Pilocarpine
Phenylephrine
Tropicamide

Local anaesthetics
Amethocaine
Lignocaine
Oxybuprocaine
Proxymetacaine

- Prescriptions are necessary for Schedule 4 medicines → recommended for Schedule 2 or 3
- Panadol (commercial name) → Paracetamol (active ingredient)
    - You can include the brand name, but the active ingredient must appear first. EXCEPTION: medicines have 4+ active ingredients.
- Do NOT prescribe repeats for acute conditions e.g. presumed bacterial conjunctivitis (topical chloramphenicol eye drops)
- A e.g. case you may prescribe repeats → topical lubricants for management of chronic dry eye disease OR anti-glaucoma drug for open angle glaucoma (stable and non progressive)
- AVOID repeats for corticosteroid eye drops due to significant side effect profile
- 30 drops are in 1 mL → 5 mL has 150 drops
    - Px using drops twice a day in both eyes will be using 30 x 2 eyes x twice a day (120 drops a month)

PBS

Browse by Optometrical Items

Optometrist PBS prescribing

PREPARATIONS WHICH MAY BE PRESCRIBED BY AUTHORISED OPTOMETRISTS

From 1 January 2008, optometrists accredited to prescribe under State or Territory legislation can apply for approval as PBS prescribers (*authorised optometrists*). Information for optometrists on becoming a PBS prescriber is available from Services Australia.

The medications listed for prescribing by authorised optometrists are identified by 'OP' in the PBS Schedule. Optometrists must not write PBS prescriptions for any other medicines listed on the PBS.

The Pharmaceutical Benefits Advisory Committee (PBAC) is responsible for making recommendations to the Minister of Health regarding preparations for prescribing by authorised optometrists.

Some products are included in more than one section of the Schedule, and for more than one prescriber type. For a prescription to be eligible for subsidy, prescribers must ensure that they prescribe under the PBS only those medicines, and in accordance with the restrictions, listed for their practitioner type.

Optometrist PBS prescriptions are identifiable by colour, and include the words 'PBS/RPBS optometrist'. Prescriptions must include the optometrist's PBS prescriber number. The same optometrist prescription form is used to prescribe unrestricted, restricted or authority items. Only one item is allowed per form. Optometrist PBS prescriptions may include repeats.

Regulation 24 does not apply for optometrist prescribing. An optometrist cannot direct that original and repeat supplies of pharmaceutical benefits be supplied at the one time.

Authority prescriptions: Authority prescriptions for authority required items, or for increased quantities or repeats, require prior approval from Services Australia or the Department of Veterans' Affairs (DVA) for each prescription. (Refer to Prescribing Medicines — Information for PBS prescribers and Supplying Medicines — What Pharmacists Need to Know, for more information on authority prescriptions.) DVA approval for non-Schedule items is not available for optometrist prescribing.

RPBS: Optometrists approved as PBS prescribers may write prescriptions for supply under the RPBS. The medicines available for prescribing by authorised optometrists under the RPBS are the same as those available under the PBS. There are no optometrist listings in the Repatriation Schedule for prescribing for veterans only. There is no provision for optometrist prescribers to request approval to prescribe items that are not included in the PBS Schedule (non-Schedule items).

State and Territory requirements: Optometrists may prescribe medications as private prescriptions according to their State/Territory prescribing accreditation. The medicines which can be prescribed differ between States and Territories. It is the optometrist's responsibility to ensure adherence to State/Territory law for all prescriptions (PBS and private) and additionally to all PBS requirements for PBS/RPBS prescriptions.

LEGEND Medical Practitioner Midwife Nurse Practitioner Optometrist Supply Only 60-day prescriptions

Prescriber code	Item code	Name, manner of administration and form & strength	Max qty packs	Max qty units	No. of repeats
	5501M	ACICLOVIR aciclovir 3% eye ointment, 4.5 g	1	1	0
	14425F	BETAXOLOL betaxolol 0.5% eye drops, 5 mL	2	2	5
	5544T	BETAXOLOL betaxolol 0.5% eye drops, 5 mL	1	1	5
	14315K	BIMATOPROST bimatoprost 0.03% eye drops, 3 mL	2	2	5
	5551E	BIMATOPROST bimatoprost 0.03% eye drops, 3 mL	1	1	5
	10053D	BIMATOPROST bimatoprost 0.03% eye drops, 30 x 0.4 mL ampoules	1	1	5
	14422C	BIMATOPROST bimatoprost 0.03% eye drops, 30 x 0.4 mL ampoules	2	2	5
	14317M	BIMATOPROST + TIMOLOL bimatoprost 0.03% + timolol 0.5% eye drops, 3 mL	2	2	5
	5558M	BIMATOPROST + TIMOLOL bimatoprost 0.03% + timolol 0.5% eye drops, 3 mL	1	1	5
	10108B	BIMATOPROST + TIMOLOL bimatoprost 0.03% + timolol 0.5% eye drops, 30 x 0.4 mL ampoules	1	1	5
	14331H	BIMATOPROST + TIMOLOL bimatoprost 0.03% + timolol 0.5% eye drops, 30 x 0.4 mL ampoules	2	2	5
	14496Y	BRIMONIDINE brimonidine tartrate 0.15% eye drops, 5 mL	2	2	5
	5563T	BRIMONIDINE brimonidine tartrate 0.15% eye drops, 5 mL	1	1	5
	5534G	BRIMONIDINE brimonidine tartrate 0.2% eye drops, 5 mL	1	1	5
	14497B	BRIMONIDINE brimonidine tartrate 0.2% eye drops, 5 mL	2	2	5
	14491Q	BRIMONIDINE + TIMOLOL brimonidine tartrate 0.2% + timolol 0.5% eye drops, 5 mL	2	2	5
	5535H	BRIMONIDINE + TIMOLOL brimonidine tartrate 0.2% + timolol 0.5% eye drops, 5 mL	1	1	5
	5540N	BRINZOLAMIDE brinzolamide 1% eye drops, 5 mL	1	1	5
	14321R	BRINZOLAMIDE brinzolamide 1% eye drops, 5 mL	2	2	5
	10547D	BRINZOLAMIDE + BRIMONIDINE brinzolamide 1% + brimonidine tartrate 0.2% eye drops, 5 mL	1	1	5
	14423D	BRINZOLAMIDE + BRIMONIDINE brinzolamide 1% + brimonidine tartrate 0.2% eye drops, 5 mL	2	2	5
	14495X	BRINZOLAMIDE + TIMOLOL brinzolamide 1% + timolol 0.5% eye drops, 5 mL	2	2	5
	5562R	BRINZOLAMIDE + TIMOLOL brinzolamide 1% + timolol 0.5% eye drops, 5 mL	1	1	5
	14420Y	CARBOMER-980 carbomer-980 0.2% eye drops, 30 x 600 mg ampoules	6	6	5
	5504Q	CARBOMER-980 carbomer-980 0.2% eye drops, 30 x 600 mg ampoules	3	3	5
	14385D	CARBOMER-980 carbomer-980 0.2% eye gel, 10 g	2	2	5
	5503P	CARBOMER-980 carbomer-980 0.2% eye gel, 10 g	1	1	5
	11853W	CARMELLOSE SODIUM carmellose sodium 0.5% eye drops, 10 mL	1	1	5
	14319P	CARMELLOSE SODIUM carmellose sodium 0.5% eye drops, 10 mL	2	2	5
	11112W	CHLORAMPHENICOL chloramphenicol 0.5% eye drops, 10 mL	1	1	2
	13264E	CICLOSPORIN ciclosporin 0.09% eye drops, 60 x 0.25 mL ampoules	1	1	5
	12663L	CICLOSPORIN ciclosporin 0.1% eye drops, 30 x 0.3 mL ampoules	1	1	5
	5564W	CIPROFLOXACIN ciprofloxacin 0.3% eye drops, 5 mL	2	2	0
	5565X	DEXAMETHASONE dexamethasone 0.1% eye drops, 5 mL	1	1	0
	14521G	DEXTRAN-70 + HYPROMELLOSE dextran-70 0.1% + hypromellose 0.3% eye drops, 15 mL	2	2	5
	5526M	DEXTRAN-70 + HYPROMELLOSE dextran-70 0.1% + hypromellose 0.3% eye drops, 15 mL	1	1	5
	14524K	DORZOLAMIDE dorzolamide 2% eye drops, 5 mL	2	2	5
	5541P	DORZOLAMIDE dorzolamide 2% eye drops, 5 mL	1	1	5
	14366E	DORZOLAMIDE + TIMOLOL dorzolamide 2% + timolol 0.5% eye drops, 5 mL	2	2	5
	5542Q	DORZOLAMIDE + TIMOLOL dorzolamide 2% + timolol 0.5% eye drops, 5 mL	1	1	5
	5557L	FRAMYCETIN SULFATE framycetin sulfate 0.5% eye/ear drops, 8 mL	1	1	2
	14364L	HYALURONATE SODIUM hyaluronate sodium 0.1% eye drops, 10 mL	2	2	5
	2184Y	HYALURONATE SODIUM hyaluronate sodium 0.1% eye drops, 10 mL	1	1	5
	14494W	HYALURONATE SODIUM hyaluronate sodium 0.2% eye drops, 10 mL	2	2	5
	2171G	HYALURONATE SODIUM hyaluronate sodium 0.2% eye drops, 10 mL	1	1	5
	5516H	HYDROCORTISONE ACETATE hydrocortisone acetate 1% eye ointment, 5 g	1	1	0
	11849P	HYPROMELLOSE hypromellose 0.3% w/v eye drops, 10 mL	1	1	5

14492R	HYPROMELLOSE hypromellose 0.3% w/v eye drops, 10 mL	2	2	5
14320Q	HYPROMELLOSE hypromellose 0.5% eye drops, 15 mL	2	2	5
5517J	HYPROMELLOSE hypromellose 0.5% eye drops, 15 mL	1	1	5
11634H	HYPROMELLOSE hypromellose 0.3% w/w eye drops, 10 mL	1	1	5
14318N	HYPROMELLOSE hypromellose 0.3% w/w eye drops, 10 mL	2	2	5
5519L	HYPROMELLOSE + CARBOMER-980 hypromellose 0.3% + carbomer-980 0.2% eye gel, 10 g	1	1	5
14453Q	LATANOPROST latanoprost 0.005% eye drops, 2.5 mL	2	2	5
5552F	LATANOPROST latanoprost 0.005% eye drops, 2.5 mL	1	1	5
14350G	LATANOPROST + TIMOLOL latanoprost 0.005% + timolol 0.5% eye drops, 2.5 mL	2	2	5
5553G	LATANOPROST + TIMOLOL latanoprost 0.005% + timolol 0.5% eye drops, 2.5 mL	1	1	5
12612T	LIQUID PARAFFIN + GLYCEROL + TYLOXAPOL + POLOXAMER-188 + TROMETAMOL HYDROCHLORIDE + TROMETAMOL + CETALKONIUM CHLORIDE liquid paraffin + glycerol + tyloxapol + poloxamer-188 + trometamol hydrochloride + trometamol + cetalkonium chloride eye drops, 10 mL	1	1	5
14352J	LIQUID PARAFFIN + GLYCEROL + TYLOXAPOL + POLOXAMER-188 + TROMETAMOL HYDROCHLORIDE + TROMETAMOL + CETALKONIUM CHLORIDE liquid paraffin + glycerol + tyloxapol + poloxamer-188 + trometamol hydrochloride + trometamol + cetalkonium chloride eye drops, 10 mL	2	2	5
5567B	OFLOXACIN ofloxacin 0.3% eye drops, 5 mL	2	2	0
14353K	PARAFFIN paraffin 1 g/g eye ointment, 3.5 g	4	4	5
5523Q	PARAFFIN paraffin 1 g/g eye ointment, 3.5 g	2	2	5
14493T	PARAFFIN paraffin 1 g/g eye ointment, 2 x 3.5 g	2	2	5
5522P	PARAFFIN paraffin 1 g/g eye ointment, 2 x 3.5 g	1	1	5
11439C	PERFLUOROHEXYLOCTANE perfluorohexyloctane 100% eye drops, 3 mL	1	1	5
14424E	PERFLUOROHEXYLOCTANE perfluorohexyloctane 100% eye drops, 3 mL	2	2	5
14355M	PILOCARPINE pilocarpine hydrochloride 1% eye drops, 15 mL	2	2	5
5536J	PILOCARPINE pilocarpine hydrochloride 1% eye drops, 15 mL	1	1	5
14523J	PILOCARPINE pilocarpine hydrochloride 2% eye drops, 15 mL	2	2	5
5537K	PILOCARPINE pilocarpine hydrochloride 2% eye drops, 15 mL	1	1	5
14550T	PILOCARPINE pilocarpine hydrochloride 4% eye drops, 15 mL	2	2	5
5538L	PILOCARPINE pilocarpine hydrochloride 4% eye drops, 15 mL	1	1	5
13113E	POLYETHYLENE GLYCOL-400 + PROPYLENE GLYCOL polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 30 x 0.8 mL unit doses	2	2	5
14520F	POLYETHYLENE GLYCOL-400 + PROPYLENE GLYCOL polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 30 x 0.8 mL unit doses	4	4	5
14421B	POLYETHYLENE GLYCOL-400 + PROPYLENE GLYCOL polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 15 mL	2	2	5
5524R	POLYETHYLENE GLYCOL-400 + PROPYLENE GLYCOL polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 15 mL	1	1	5
5568C	PREDNISOLONE ACETATE + PHENYLEPHRINE prednisolone acetate 1% + phenylephrine hydrochloride 0.12% eye drops, 10 mL	1	1	0
14768G	PROPYLENE GLYCOL propylene glycol 0.6% eye drops, 10 mL	1	1	5
14426G	SOY LECITHIN + TOCOPHEROL + VITAMIN A soy lecithin 1% + tocopherol 0.002% + vitamin A palmitate 0.025% spray, 100 actuations	4	4	5
5545W	SOY LECITHIN + TOCOPHEROL + VITAMIN A soy lecithin 1% + tocopherol 0.002% + vitamin A palmitate 0.025% spray, 100 actuations	2	2	5
14634F	TIMOLOL timolol 0.5% eye drops, 2.5 mL	1	1	5
5550D	TIMOLOL timolol 0.5% eye drops, 2.5 mL	1	1	5
14649B	TIMOLOL timolol 0.5% eye drops, 5 mL	1	1	5
5548B	TIMOLOL timolol 0.5% eye drops, 5 mL	1	1	5
5569D	TOBRAMYCIN tobramycin 0.3% eye drops, 5 mL	1	1	2
5570E	TOBRAMYCIN tobramycin 0.3% eye ointment, 3.5 g	1	1	0
14549R	TRAVOPROST travoprost 0.004% eye drops, 2.5 mL	2	2	5
5554H	TRAVOPROST travoprost 0.004% eye drops, 2.5 mL	1	1	5
14316L	TRAVOPROST + TIMOLOL travoprost 0.004% + timolol 0.5% eye drops, 2.5 mL	2	2	5
5555J	TRAVOPROST + TIMOLOL travoprost 0.004% + timolol 0.5% eye drops, 2.5 mL	1	1	5

Scheduled medicines list for optometrists

Effective 10 December 2021

The Optometry Board of Australia (OBA) approved list of topical schedule 2, 3 and 4 medicines that optometrists with a scheduled medicines endorsement are qualified to administer, obtain, possess, prescribe, supply or use for the purposes of the practice of optometry. This list can also be found on the OBA's website: https://www.optometryboard.gov.au/Policies-Codes-Guidelines.aspx

Medicines below that are not listed on the PBS can still be prescribed, but must be written as a private prescription. Private prescriptions do not have restriction on maximum quantity of repeats, e.g., Patanol (olopatadine).

Optometry Australia keeps an updated list of Medicines shortages and supply updates on our website: https://www.optometry.org.au/practice-professional-support/patient-practice-management/prescribing-medicines-pbs/medicines-shortages-and-supply-updates/

Schedule 2 pharmacy medicine

Anti-infectives	Decongestants/anti-allergics	Miotics, mydriatics and cycloplegics
Dibromopropamidine	Antazoline	Phenylephrine ≤ 2.5%
Propamidine	Azelastine	
	Ketotifen	
	Levocabastine	
	Lodoxamide	
	Naphazoline	
	Pheniramine	
	Sodium cromoglycate	

Schedule 3 pharmacist-only medicine

Anti-infectives
Chloramphenicol

Schedule 4 prescription-only medicine

Anti-infectives	Anti-inflammatories	Decongestants/ anti-allergics	Anti-glaucomas	Miotics, mydriatics and cycloplegics	Local anaesthetics
Aciclovir	Ciclosporin	Olopatadine	Apraclonidine	Atropine	Amethocaine
Azithromycin	Dexamethasone		Betaxolol	Cyclopentolate	Lignocaine
Bacitracin	Diclofenac		Bimatoprost	Homatropine	Oxybuprocaine
Cephazolin	Fluorometholone		Brimonidine	Pilocarpine	Proxymetacaine
Ciprofloxacin	Flurbiprofen		Brinzolamide	Phenylephrine	
Framycetin	Hydrocortisone		Dorzolamide	Tropicamide	
Ganciclovir	Ketorolac		Latanoprost		
Gentamicin	Lifitegrast		Pilocarpine		
Gramicidin	Loteprednol		Tafluprost		
Neomycin	Prednisolone		Timolol		
Ofloxacin			Travoprost		
Polymyxin					
Tetracycline					
Tobramycin					

Opthalmic compounding pharmacists

Optometry Australia keeps an updated list of ophthalmic compounding pharmacists on our website: https://www.optometry.org.au/practice-professional-support/patient-practice-management/prescribing-medicines-pbs/.
Please keep us updated if you know of a certified ophthalmic compounding pharmacist that is not on our list by emailing national@optometry.org.au.

Note: Do you prescribe low dose atropine for myopia control?

If you are prescribing low-concentration atropine, please clearly label the prescription with **'To Be Compounded'** to avoid confusion in the dispensing process. There have been incidents where prescriptions for low-concentration atropine have been incorrectly processed by pharmacists as Atropt (atropine sulphate 1%). Low-concentration atropine eye drops must be prepared by a compounding pharmacy that is equipped to compound eye drops.

Eikance 0.01% (atropine sulfate) eye drops are available through prescription. This new product is subject to additional monitoring under the Black Triangle Scheme. Healthcare professionals are asked to report any suspect adverse events at www.tga.gov.au/reporting-problems.

To report an adverse event or problem with any medication: https://www.tga.gov.au/report-adverse-event-or-problem-health-professionals.

PBS list of medicines prescribed by optometrists

Revised October 2024

60-day dispensing

From 1 September 2024, optometrists will be able to offer patients a 60-day prescription for certain medications listed on the PBS, provided they are assessed to be stable on their current treatment and meet other eligibility criteria specific to each medicine. These include topical dry eye drops and glaucoma medications. The 60-day dispensing policy will give twice the amount of their medication on a single prescription. Optometrists should use their clinical judgement to decide whether patients receiving PBS medicines are eligible for, or require, 60-day prescriptions.

The 60-day dispensing policy may reduce both time spent in consultations to renew prescriptions for stable patients and visits to the pharmacy, as well as savings for the patient's annual medication costs.

Optometrists should use PBS's online search function at https://www.pbs.gov.au/pbs/search to search for specific eligibility criteria for each medication as well as other details such as the maximum number of packs and number of repeats. For each eye drop, look for the '60D' and 'OP' logo.

Anti-glaucoma preparations

Active ingredients	Product	Restrictions	Maximum quantity#		Repeats
			30-day	60-day	
Betaxolol 0.5% eye drops, 5 mL	Betoptic, BetoQuin	**Restricted:** Condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription.	1	2	5
Bimatoprost 0.03% eye drops, 3 mL	Bimatoprost-WGR, Bimprozt, Bimtop, Lumigan		1	2	5
* Bimatoprost 0.03% eye drops, 30 x 0.4 mL unit doses	Lumigan PF		1	2	5
Bimatoprost 0.03% + timolol 0.5% eye drops, 3 mL	Ganfort 0.3/5		1	2	5
* Bimatoprost 0.03% + timolol 0.5% eye drops, 30 x 0.4 mL unit doses	Ganfort PF 0.3/5		1	2	5
Brimonidine tartrate 0.15% eye drops, 5 mL	Alphagan P 1.5		1	2	5
Brimonidine tartrate 0.2% eye drops, 5 mL	Alphagan, Enidin		1	2	5
Brimonidine tartrate 0.2% + timolol 0.5% eye drops, 5 mL	Combigan		1	2	5
Brinzolamide 1% eye drops, 5 mL	Azopt, BrinzoQuin		1	2	5
Brinzolamide 1% + brimonidine tartrate 0.2% eye drops, 5 mL	Simbrinza 1%/0.2%		1	2	5
Brinzolamide 1% + timolol 0.5% eye drops, 5 mL	Azarga		1	2	5
Dorzolamide 2% eye drops, 5 mL	Trusamide, Trusopt		1	2	5
Dorzolamide 2% + timolol 0.5% eye drops, 5 mL	Cosdor, Cosopt, Vizo-PF Dorzolatim		1	2	5
Latanoprost 0.005% eye drops, 2.5 mL	APO-Latanoprost, Latanoprost-WGR, Latanoprost Sandoz, Xalaprost, Xalatan		1	2	5
Latanoprost 0.005% + timolol 0.5% eye drops, 2.5 mL	APO-Latanoprost/Timolol 0.05/5, Xalacom, Xalamol 50/5		1	2	5
Pilocarpine hydrochloride 1% eye drops, 15 mL	Isopto Carpine		1	2	5
Pilocarpine hydrochloride 2% eye drops, 15 mL	Isopto Carpine		1	2	5
Pilocarpine hydrochloride 4% eye drops, 15 mL	Isopto Carpine		1	2	5
Timolol 0.5% eye drops, 2.5 mL	Timoptol XE, Timoptol-LA		1	-	5
Timolol 0.5% eye drops, 5 mL	Timoptol		1	-	5
Travoprost 0.004% eye drops, 2.5 mL	Travatan		1	2	5
Travoprost 0.004% + timolol 0.5% eye drops, 2.5 mL	DuoTrav		1	2	5

* = unit doses

Note: To satisfy PBS criteria for combination anti-glaucoma preparations, patient must have been inadequately controlled with monotherapy and must have open angle glaucoma or ocular hypertension.

https://www.optometry.org.au/wp-content/uploads/States/WA/ECOWA/PBS-List-Oct-2022.pdf

Anti-viral eye preparations				
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**	**Repeats**
Aciclovir 3% eye ointment, 4.5 g	ViruPOS, XOROX	**Restricted:** herpes simplex keratitis.	1	0

Antibiotics				
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**	**Repeats**
Chloramphenicol 0.5% eye drops, 10 mL	Chlorsig	**Restricted:** for treatment of a patient identifying as Aboriginal or Torres Strait Islander.	1	2
† Ciprofloxacin 0.3% eye drops, 5 mL	CiloQuin, Ciloxan	**Authority required:** bacterial keratitis.	2	0
Framycetin sulfate 0.5% eye/ear drops, 8 mL	Soframycin		1	2
† Ofloxacin 0.3% eye drops, 5 mL	Ocuflox	**Authority required:** bacterial keratitis.	2	0
Tobramycin 0.3% eye drops, 5 mL	Tobrex	**Restricted:** suspected Pseudomonal eye infection, perioperative use in ophthalmic surgery.	1	2
Tobramycin 0.3% eye ointment, 3.5 g	Tobrex		1	0

† = Note: must be in consultation with an ophthalmologist if prescribed under PBS scheme.

Anti-inflammatory agents				
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**	**Repeats**
Dexamethasone 0.1% eye drops, 5 mL	Maxidex		1	0
Hydrocortisone acetate 1% eye ointment, 5 g	Hycor		1	0
Prednisolone acetate 1% + phenylephrine hydrochloride 0.12% eye drops, 10 mL	Prednefrin Forte	**Restricted:** uveitis.	1	0

Topical ocular lubricant ointments					
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**		**Repeats**
			30-day	**60-day**	
Paraffin 1 g/g eye ointment, 2 x 3.5 g	Poly Visc, Refresh Night Time	**Restricted:** Condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription.	1	2	5
Paraffin 1 g/g eye ointment, 3.5 g	Poly Visc		2	4	5

Tear supplements					
Active ingredients	**Product**	**Restrictions**	**Maximum quantity***		**Repeats**
			30-day	**60-day**	
Carbomer-980 0.2% eye gel, 10 g	Optifresh Eye Gel, PAA, Viscotears	**Restricted:** severe dry eye syndrome, including Sjögren's syndrome. **Restricted:** Condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription.	1	2	5
* Carmellose sodium 0.5% eye drops, 15 mL	Refresh Tears Plus		1	-	5
* Carmellose sodium 1% eye drops, 15mL	Refresh Liquigel		1	-	5
* Carmellose sodium 0.5% + glycerol 0.9% eye drops, 15 mL	Optive		1	-	3
Dextran-70 0.1% + hypromellose 0.3% eye drops, 15 mL	Poly-Tears, Tears Naturale		1	2	5
Hypromellose 0.3% w/w eye drops, 10 mL	Genteal, In A Wink Moisturising, Revive Tears		1	2	5
Hypromellose 0.5% eye drops, 15 mL	Methopt		1	2	5
* Hypromellose 0.3% + carbomer-980 0.2% eye gel, 10 g	HPMC PAA, Genteal Gel		1	-	5
Polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 15 mL	Optix, Systane		1	2	5

* = not available on 60-day prescription

PBS LIST OF MEDICINES
PRESCRIBED BY OPTOMETRISTS

New Streamline Authority Code 15559

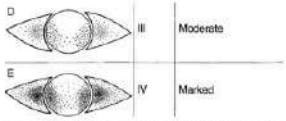
A new streamlined authority code for unit-dose tear supplements (15559) has been introduced for specifically 60-day prescriptions. The 6172 and 4105 codes remain unchanged for one-month prescriptions. Optometrists still maintain discretion over what quantity of medicine is prescribed, meaning a one-month prescription can still be supplied where clinically appropriate.

For tear supplements, click on the red font 'authority required (streamlined)' associated with each medication on PBS's search function at https://www.pbs.gov.au/pbs/search to find the authority streamline code 15559.

Unpreserved tear supplements					
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**		**Repeats**
			30-day	**60-day**	
* Carbomer-980 0.2% eye drops, 30 x 600 mg ampoules	Viscotears Gel PF	**Restricted:** severe dry eye syndrome in patients sensitive to preservatives in multi-dose eye drops. **Authority required (STREAMLINED):** Optometrists have 3 authority codes that are streamlined. You do not need to contact PBS to obtain an authority number. Write the pre-approved code on the prescription. **4105** - Hylo-Fresh & Hylo-Forte **6172** - all other unit-dose tear supplements **15559** - **(new from 1 Sept 2024)** unit dose tear supplements (60-day prescription). The condition must be stable for the prescriber to consider the listed maximum quantity of this medicine suitable for this patient AND patient must be sensitive to preservatives in multi-dose eye drops	3	6	5
Carmellose sodium 0.5% eye drops, 10 mL	Evolve Carmellose		1	2	5
* Carmellose sodium 0.5% eye drops, 30 x 0.4 mL ampoules	Cellufresh		3	6	5
* Carmellose sodium 1% eye drops, 30 x 0.4 mL ampoules	Celluvisc		3	6	5
Hyaluronate sodium 0.1% eye drops, 10 mL	Hylo-FreshS		1	2	5
Hyaluronate sodium 0.2% eye drops, 10 mL	Hylo-Forte		1	2	5
Hypromellose 0.3% w/v eye drops, 10 mL	Evolve Hypromellose		1	2	5
Liquid paraffin + glycerol + tyloxapol + poloxamer-188 + trometamol hydrochloride + trometamol + cetalkonium chloride eye drops, 10 mL	Cationorm		1	2	5
Perfluorohexyloctane 100% eye drops, 3 mL	Novatears		1	2	5
* Polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 30 x 0.8 mL unit doses	Systane		2	4	5
Soy lecithin 1% + tocopherol 0.002% + vitamin A palmitate 0.025% eye spray, 100 actuations	Tearsagain		2	4	5

* = unit doses

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Immunosuppressant therapy				
Active ingredients	**Product**	**Restrictions**	**Maximum quantity**	**Repeats**
Ciclosporin 0.09% eye drops, 60 x 0.25 mL ampoules	Cequa	Treatment of chronic severe dry eye disease with keratitis. The treatment must be in combination with a preservative-free artificial tear substitute. **Clinical criteria – patient must have:** ■ a corneal fluorescein staining (CFS) grade of 4 at treatment initiation (using the modified Oxford scale or equivalent – below);  ■ an ocular surface disease index (OSDI) score of at least 23 ■ and a condition not be adequately controlled by monotherapy with a preservative-free artificial tear substitute. The treatment must be the sole PBS-subsidised therapy for this condition. Applications for authorisation under this restriction may be made in real time using the Online PBS Authorities system (see www.servicesaustralia.gov.au/HPOS) or by telephone by contacting Services Australia on 1800 888 333. **More information, including criteria for continuing treatment beyond 180 days:** Cequa: https://www.pbs.gov.au/medicine/item/13284E Ikervis: https://www.pbs.gov.au/medicine/item/12663L	1	5
Ciclosporin 0.1% eye drops, 30 × 0.3 mL ampoules	Ikervis		1	5

■ = unit doses

Safe prescribing

From 1 February 2021, Department of Health regulations require the inclusion of active ingredients on all Pharmaceutical Benefits Scheme (PBS) and Repatriation PBS (RPBS) prescriptions, except for:

- handwritten prescriptions;
- paper based medication charts in the residential aged care sector
- medicinal items with 4 or more active ingredients.

Prescribers may continue to include a product/brand name on prescriptions. Where a brand name is included on prescriptions, the active ingredient must appear first. Consumers' ability to identify a medicine's active ingredient is critical for medicines safety.

In NSW only: From 1 November 2022, the patient's date of birth, in addition to the name and address, must be added to all handwritten and computer generated S4 medication prescriptions.

Closing the Gap (CTG) PBS Co-payment Program

Since 1 July 2021, after some advocacy work from Optometry Australia, optometrists are able to register eligible Aboriginal and Torres Strait Islander patients for PBS co-payment via HPOS: https://www.pbs.gov.au/info/publication/factsheets/closing-the-gap-pbs-co-payment-measure

Those who are registered but do not have a concession card will pay $7.50 for the majority of PBS general medicines. If registered people have a concession card, they will receive the majority of PBS general medicines for free.

How to check which patients are eligible: https://www.servicesaustralia.gov.au/which-patients-are-eligible-for-closing-gap-pbs-co-payment?context=23016

How to register patients: https://www.servicesaustralia.gov.au/how-to-register-patients-for-closing-gap-pbs-co-payment?context=23016

How to write a CTG prescription: https://www.servicesaustralia.gov.au/how-to-write-closing-gap-pbs-co-payment-prescription?context=23016

emiership fees. Please do not share or distribute.

8. In-office intense pulsed light therapy

Product name	Manufacturer	Product type	Comments	Supporting studies	Distributor
Tearstim IRPL	**ESW Vision**	IPL	→ Used in over 50 countries for the genuine treatment of Dry Eye Disease since 2014 → Optimal footprint, designed for all clinic & practice types (Travel suitcase available as option for multiple locations) → Patented Regulated Flash Technology (IRPL) dedicated to Dry Eye relief since 2014 (No spike inconsistency, each pulse of light is uniform and consistent from the beginning of the pulse to the end, more efficient and effective than unregulated generic IPL) → 85% satisfaction rate, proven efficacy on mild, medium and severe dry eyes (ESW-V Survey 2021) → Streamlined protocol with touch screen features allows any trained staff to perform the treatment efficiently and safely → Optimal surface coverage and flash count with high efficiency and efficacy → No ongoing maintenance or service cost → Patented cooling system with NO alteration of the flash wavelength delivered to the patient's skin (No water is used, pulse light does not pass through water)	→ Albietz JM, Schmid KL. Intense pulsed light treatment and meibomian gland expression for moderate to advanced meibomian gland dysfunction. Clin Exp Optom. 2018;101(1):23-33. DOI:10.1111/cxo.12541 → Xue AL, Wang MTM, Ormonde SE, Craig JP. Randomised double-masked placebo-controlled trial of the cumulative treatment efficacy profile of intense pulsed light therapy for meibomian gland dysfunction. Ocul Surf. 2020;18(2):286-297. DOI: 10.1016/j.jtos.2020.01.003 → Vigo L , Taroni L, Bernabei F, et al. Ocular surface workup in patients with meibomian gland dysfunction treated with intense regulated pulsed light. Diagnostics. 2019;9(4):147. DOI: 10.3390/diagnostics9040147 → Gedar Totuk ÖM, Kabadayı K, Özkapı C, Aykan Ü. Efficacy of intense pulsed light treatment for moderate to severe acute blepharitis or blepharoconjunctivitis A retrospective case series. Turk J Ophthalmol. 2021;51(2):89-94. DOI: 10.4274/tjo.galenos.2020.28924 → Craig JP, Chen Y-H, & Turnbull PRK. Prospective trial of intense pulsed light for the treatment of meibomian gland dysfunction. Invest Ophthalmol Vis Sci. 2015;56(3):1965-1970. DOI: 10.1167/iovs.14-15764 → Karaca EE, Evren Kemer Ö, Özek D. Intense regulated pulse light for the meibomian gland dysfunction Eur J Ophthalmol. 2020;30(2):289-292.DOI: 10.1177/1120672118817687 → Jiang X Lv H, Song H, et al. Research article evaluation of the safety and effectiveness of intense pulsed light in the treatment of meibomian gland dysfunction. J Ophthalmol. 2016;2016:1910694. DOI: 10.1155/2016/1910694 → Egri S, Van Hollebecke I, Guindolet D, et al. Efficacy of intense pulsed light therapy in the treatment of meibomian gland dysfunction-related severe dry eye. J Fr Ophtalmol. 2021;44(2):169-175. DOI: 10.1016/j.jfo.2020.04.061 → Monaco G, Casalino G. Superficial keratectomy followed by intense pulsed light for Salzmann's nodular degeneration and coexisting meibomian gland dysfunction. Eur J Ophthalmol. 2022;32(1):NP27-NP30. DOI: 10.1177/1120672120964691 → Miotto-Montesinos G, Rojas-Alvarez E, Torres-Piedra DM. Preliminary results of treatment with Intense Regulated Pulsed Light in Dry Eye Syndrome. Rev. Mex. Oftalmol. 2018;92(6E):253-259. DOI:10.24875/RMOE.M18000039 → S Guilloto Caballero, García Madrona JL, Colmenero Reina E. Effect of pulsed laser light in patients with dry eye syndrome. Arch Soc Esp Oftalmol. 2017;92(11):509-515. DOI: 10.1016/j.oftal.2016.12.018 A full list of supporting studies is available at www.eyetek.com.au	**Eyetek** 1300 618 336
Thermaeye PLUS	**MDS Medical Technologies**	IPL	→ One hand piece, 3 filters. More treatments to treat all skin types. → 520nm, 650nm and 800mm wavelength in combination with Fractional Square Pulsed technology. Plus water cooled delivery system providing epidermal cooling	→ Verges C, et al. Prospective evaluation of a new intense pulsed light, Thermaeye Plus, in the treatment of dry eye disease due to meibomian gland dysfunction. J Optom. 2021;14(2):103-113. DOI:10.1016/j.optom.2020.08.009 → Fuentes Paez G, et al. Intense pulsed light: Results in chronic dry eye syndrome after LASIK. Arch Soc Esp Oftalmol. 2020;95(5):226-230. DOI:10.1016/j.oftal.2020.02.007	**Optimed** 1300 657 720

1. Diagnostics

Brand & model	Diagnostic tests & functionality	Device / Mounting	Interface to software	Other comments	Further info
CSO Antares+	→ Corneal topography → Dynamic tear break up time analysis (NITBUT), time lapse play back for explanation and education of DED/MGD to patients. (NIF-BUT, NIAvg-BUT graph) → Comparison module (before/after) → Advanced OSDI Questionnaire integrated → Limbal and conjunctival hyperemia → IR meibography (scaling) with comparison → Tear meniscus height → Contact lenses fitting → Keratoconus screening → Videokeratoscope → Pupillography → Blue light, flourescein cornea staining diagnostic	Table mounted unit	Data are fully dicom compatible	→ Full patient history with 2022 CSO Phoenix 4.0 Software for follow-ups → Advanced Contact Lens, Ortho-K Auto-Fit Module based on topography → Auto focus, fast acquisition topographic → Wavefront Aberrometric maps → Advanced AI Keratoconous screening automation → Customisable maps that suit specific user applications → Intuitive modern interface and assessment report for operator and patient education → Patient historical assessments for follow up and progression monitoring	**Eyetek** 1300 618 336
DEA	→ DEQ-5 Questionaire → Non-invasive TBUT → NaFl TBUT → Tear meniscus height → Interferometry → Meibography → NaFl staining grading → Redness grading → Eyelid margin grading → Anterior imaging	**3 Options:** → Mounted to existing slit-lamp → Mounted to slit-lamp style holder → Handheld	Can be integrated to Sunix and Optomate	→ Auto grading → Customisable testing protocols → Customisable reporting	**OphthalmoPro** 1300 578 288
ESW Vision Tearcheck	→ Unique usability → Touchscreen display → Fully automatic → Wireless use → Wireless data transfer & print → Patient education **9 Exams:** → TFSE® (Tear Film Surface Evaluation) → NITBUT → Meibography IR → Ocular redness → Tear meniscus → OSDI → OSIE® (Ocular Surface Inflammatory Evaluation) → Incomplete blinks → Demodex		ESW Vision's Tearcheck is an educational and diagnostic tool to screen dry eye symptoms. Tearcheck's touch-based functionality enhances the experience for practitioners, trained staff and patients. It is designed to teach patients about MGD while increasing practice efficiency		**Eyetek** 1300 618 336
Huvitz HRT-1A	→ Non-invasive TBUT → Imaging of meibomian glands				**Opticare** 1800 251 852
Huvitz HRK-9000A	→ TFBUT → Meibography	External unit	NA	NA	**Optimed** 1300 657 720
Huvitz HTR-1A	→ Tear film break up time → Meibography	External unit	NA		**Optimed** 1300 657 720
Luneau VX120+ Dry Eye	→ Non-invasive TBUT → Imaging of meibomian glands				**Opticare** 1800 251 852
MeCheck	→ 3D Meibography → 12-level Meibography → Five-zone Evaluation of Glands → OSDI-6 Test → Demodex screening	**3 Options:** → Mounted to existing slit-lamp → Mounted to floorstand → Handheld	Can be integrated to Sunix and Optomate	→ Full automated analysis → Stand-alone device (can be operated without a PC)	**OphthalmoPro** 1300 578 288

Brand & model	Diagnostic tests & functionality	Device / Mounting	Interface to software	Other comments	Further info
Medmont Meridia™ Professional	→ Corneal topography (limbus-to-limbus) → Anterior eye imaging and video (white, infrared and cobalt light) → Meibography → Fluorescein imaging and video → Dry eye reports → Tear meniscus height → Non-invasive tear breakup time → Corneal and scleral contact lens simulation → Pupil measurement → Horizontal visible iris diameter **Grading (Efron, BHVI and Meiboscale used as reference)** → Blepharitis → Conjunctival/bulbar redness → Conjunctival staining → Corneal staining → Meibomian gland dysfunction → Limbal redness → Papillary conjunctivitis → Palpebral redness → Palpebral roughness	External unit	Integration with Sunix and Optomate Medmont DV2000 Imaging Software integrates with various slit lamp cameras and imports images and videos DICOM compatible		**Medmont International** (03) 9259 0800
OCULUS Keratograph 5M	→ (Infrared) Non-invasive tear breakup time → (Infrared) Tear meniscus height → (Infrared) Video of incomplete blinks → (Infrared) Meibography → (Visible) Auto redness grading – limbal & bulbar → (Visible) Tear film dynamic → (Visible) Image of lid margin → (Visible) Lipid layer thickness (qualitative only) → (Visible) Flourescein imaging (video & still) → JENVIS pro Dry Eye Patient Report → Corneal topography	External unit	Links to Sunix and Optomate	Ability to compare reports over time (A vs B)	**Designs For Vision** 1800 225 307
Rexxam RET-700 Dry Eye Option	→ Meibomian gland observation → Dry eye observation mode	External unit	NA	NA	**Optimed** 1300 657 720
Topcon CA-800	→ Corneal placido topography (Infrared) → Keratoconus screening → Non-invasive tear breakup time (Infrared) → Tear meniscus height (Infrared & visible) → Blinks analysis (Infrared & visible) → Table of blink rate & blink completion → Meibography (Infrared) → Fluorescein imaging → Corneal aberrations between blinks	Table mounted unit	Simple links to Sunix & Optomate Links to Eyespace Links to Topcon iMap		**Device Technologies** 1800 429 551
Topcon Myah	→ Corneal placido topography (Infrared) → Keratoconus screening → Non-invasive tear breakup time (Infrared) → Tear meniscus height (Infrared & Visible) → Video of incomplete blinks (Infrared & Visible) → Table of blink rate & blink completion → Meibography (Infrared) → Fluorescein imaging → Corneal aberrations between blinks → Capable of axial length & myopia management	Table mounted unit	Simple links to Sunix & Optomate Links to Eyespace Links to Topcon iMap Pro		**Device Technologies** 1800 429 551
Topcon SL-D701 (FAG)& DC4 camera	→ Full colour imaging (Visible) → 'Blue free' imaging (Visible) → Fluorescein imaging (Visible) → Meibography (Infrared)	Table mounted unit	Simple links to Sunix & Optomate		**Device Technologies** 1800 429 551
TearView	→ Dynamic real time visualisation of tear film over cornea and sclera → Visualises active meibomian glands	Attaches to turret of slit lamp	USB connection to tablet with proprietary software	Does not interfere with tear film Tear film observed under natural blinking Can be used with contact lens inserted	**Beyond 700** +61 420 410 862
Trukera ScoutPro (successor of TearLab Osmolarity System)	→ Real-time and precise osmolarity measurement of human tears to aid in the diagnosis of dry eye disease → Analyse the effects of tear osmolarity on residual refractive outcomes in cataract surgery	External, portable unit ScoutPro Pen sitting in the charging base		→ Measurement by individual test cards → Microfluidic Lab on a chip technology → Specimen collection and analysis in one → Results displayed on LCD screen	**IQ Medical** sales@iqmedical.com.au 08 8357 8022

2. Oral essential fatty acid supplements

Brand name	Active ingredients	Shelf life	Available sizes + delivery	Therapeutic indications	Contraindications	Further information	Supplier
Dry Eye Complete	**Per serving (2 capsules):** 1,133mg Omega-3 (triglyceride, Alaskan wild caught) 1,000IU D3 60mg Maqui Berry 10mg Lutein 2mg Zeaxanthin 66mg Omega-7	See each individual package	60 capsules, 2 capsules per day	Moderate dry eye disease	Consult with GP before starting	Unique anti-inflammatory and antioxidant formula	**OphthalmoPro** 1300 578 288
DRYEYE Forte	**Each soft gel capsule contains:** Concentrated Omega-3 triglycerides-fish 667mg Equiv. Eicosapentaenoic acid (EPA) 240mg Equiv. Docosahexaenoic acid (DHA) 160mg Borago officinalis seed oil fixed 455mg Equiv. gamma-Linolenic acid (GLA) 100mg Colecalciferol 5mg Equiv. Vitamin D3 200IU d-alpha Tocopherol 27mg Equiv. Vitamin E 40IU	3 years	60 soft gel capsules Treatment typically starts with 3 capsules per day for 6 weeks and then 1-2 capsules per day thereafter	Targeting the root causes of dry eye and working from the inside out, DRYeye FORTE increases tear production, improves the quality of the tear film, and reduces inflammation. Systemically relieves dry eyes. Ideal for dry eye syndrome associated with contact lens intolerance, air-conditioning, air travel, environmental factors, symptoms of menopause, LASIK and other eye surgery patients	Consult with GP before starting	Features pharmaceutical grade re-esterified omega 3 fish oil. Treatment typically starts with 3 capsules per day for 6 weeks, then 1–2 capsules per day thereafter	**MD EyeCare** 1300 95 2001
Lacritec	Flax seed (linseed) oil 334mg Borage oil (Borago officinalis) 434mg Concentrated Omega-3 re-esterified triglycerides (fish oil) 332mg	3 years from manufacture	60 oral capsules 200 oral capsules economy pack 3 per day for 6 weeks, then 2 per day ongoing	Lacritec is an innovative product that works from the inside out to provide continuous dry eye relief. Lacritec delivers a proprietary blend of omega fatty acids sourced from flax seed, fish oil and borage oil. This Australian developed, patented formulation works to support a healthy tear film and reduce inflammation. Lacritec dry eye relief capsules are ideal for: ➔ Ongoing dry eye syndrome ➔ Contact lens wearers ➔ Air travel and air conditioning ➔ Eye strain ➔ Eye procedures including laser eye surgery	Check with GP before starting	A follow up of 209 Lacritec patients showed significant symptom improvement and a halving of their eye drop usage after 8 weeks (data on file)	**Stiltec Pty Ltd** 1800 622 883

3. Ocular lubricants

(includes drops, gels, sprays and ointments; with preservative and preservative-free)

Brand name	Active ingredient	Shelf life	Available sizes	Dry eye subtype	Therapeutic indications	Contraindications	Preservative	PBS listing	Supplier
Drops									
Blink® Intensive Tears	Polyethylene glycol 400 (peg 400) Hyaluronic acid	Use within 45 days of first opening	15mL bottle	Aqueous deficient	Mild-to-moderate dry eye. Irritated, uncomfortable eyes. May be used with contact lenses.		Yes - Ocupure	No	**Bausch & Lomb (Australia) Pty Ltd** 1800 251 150
Blink® Intensive Gel Tears	Polyethylene glycol 400 (peg 400) Hyaluronic acid	Use within 45 days of first opening	10mL bottle	Aqueous deficient	Moderate-to-severe dry eye. Longer lasting relief.		Yes - Ocupure	No	**Bausch & Lomb (Australia) Pty Ltd** 1800 251 150
Blink® Intensive Tears Preservative Free	Polyethylene glycol 400 (peg 400) Hyaluronic acid	Discard immediately after use	20 x 0.4mL vials	Aqueous deficient	Mild-to-moderate dry eye. Irritated, uncomfortable eyes. May be used with contact lenses.		No	No	**Bausch & Lomb (Australia) Pty Ltd** 1800 251 150
Cationorm®	Mineral oils, glycerol, tyloxapol, poloxamer 188, tris hydrochroloride, tromethamine, cetalkonium chloride	10mL: 3/12 after opening Vial: single use only	30 x 0.4mL Single-unit dose (SUD) 10mL Preservative Free Multidose bottle (PFMD)	Evaporative & aqueous deficient and mixed	Cationorm is a hydrating and lubricating emulsion which protects and heals the ocular surface and relieves the discomfort and irritation due to dry eye. Cationorm is the only positively-charged AT in Australia and is designed to optimise contact time and spreadability on the negatively-charged ocular surface. Suitable for use with all types of contact lenses. Dosage: ONE drop in affected eye(s) up to 4 times daily as required.	Hypersensitivity to any ingredients of the product	No	Yes (10mL PFMD only). Yes: Authority Required (streamlined) – 6172. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**CSL Seqirus** Good Optical (03) 9645 1066

3. Ocular lubricants (includes drops, gels, sprays and ointments; with preservative and preservative-free)

Brand name	Active ingredient	Shelf life	Available sizes	Dry eye subtype	Therapeutic indications	Contraindications	Preservative	PBS listing	Supplier
Drops									
GenTeal® Drops	Hypromellose	3/12 after opening	10mL bottle	Aqueous deficient	Lubricant ophthalmic solution for people who experience sensation of dryness, fatigue or discomfort of the eyes due to environmental irritants.	Allergy to any ingredients of the product	Yes - GenAqua®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
Hylo Forte	Hyaluronic acid	6/12 after opening	10mL Comod pump bottle	Aqueous deficient	Treatment of severe or persistent dry eye where there is an aqueous tear deficiency.		No	Yes: Authority Required (streamlined) – 4105. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**AFT** Good Optical (03) 9645 1066
Hylo Fresh	Hyaluronic acid	6/12 after opening	10mL Comod pump bottle	Aqueous deficient	Treatment of mild to moderate dry eye where there is an aqueous tear deficiency.		No	Yes: Authority Required (streamlined) – 4105. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**AFT** Good Optical (03) 9645 1066
In a Wink® Moisturising Drops	Hypromellose	3/12 after opening	10mL bottle	Aqueous deficient	Lubricant ophthalmic solution for people who experience sensation of dryness, fatigue or discomfort of the eyes due to environmental irritants.	Allergy to any ingredients of the product	Yes - GenAqua®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
Liquifilm Tears	Polyvinyl alcohol 14mg	1/12 after opening	15mL	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use)	Allergy to any ingredients of the product	Benzalkonium chloride	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**AbbVie** 1800 252 224
Nova Tears	Perfluorohexyloctane	8/52 after opening	3mL bottle	Evaporative dry eye	Treatment of evaporative dry eye where there is a lipid deficiency.	Some contact lenses	No	Yes: Authority Required (streamlined) – 6172. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**AFT** Good Optical (03) 9645 1066
Nova Tears + Omega-3	Perfluorohexyloctane and omega-3 fatty acids	8/52 after opening	3mL bottle	Evaporative dry eye	Treatment of evaporative dry eye where there is a lipid deficiency. Concentrated high quality Omega-3 helps to provide lubrication and protection	N/A	No	No	**AFT** Good Optical (03) 9645 1066
Oculocin Propo	Active ingredients: propolis, aloe vera, chamomilla Active agents: phenols, aloins & emodins	2/12 after opening	10 single vials (5mL) or 10mL multidose	Evaporative & aqueous deficient	A 'natural' eye-drop for reducing bacteria and inflammation in dry eye. Has analgesic effects. Safe for long term use; no antibiotics, no stinging, no known side effects.	Known hypersensitivity to any of the ingredients in the formulation	No	No	**CLCA** 1800 125 023
Optimel® Manuka+® Dry Eye Drops	Leptospermum (spp honey)	Use within 2 months of first opening	10mL	Evaporative & aqueous deficient	Containing 16.5% manuka honey, it helps to stabilise the surface health of the eye and may provide relief from dry eye symptoms such as sore, irritated eyes and eyelids.	Do not use if there is a history of hypersensitivity or sensitive to honey, food gums, benzoate preservatives	Yes - Benzoic acid	No	**Designs For Vision** 1800 225 307
Optive Advanced	5mg carmellose sodium 10mg glycerin 5mg polysorbate 80	3/12 after opening	15mL bottle	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	Yes - Purite®	No	**AbbVie** 1800 252 224
Optive Advanced Nonpreserved	5mg carmellose sodium 10mg glycerin 5mg polysorbate 80	3/12 after opening	30 x 0.4mL vials	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g. medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	No	No	**AbbVie** 1800 252 224
Optive Fusion	5mg carmellose sodium 9mg glycerin 1mg sodium hyaluronate	3/12 after opening	10mL bottle	Aqueous deficient	Intended to lubricate the surface of the eye(s) to relieve the feeling of dryness, burning, irritation, stinging, foreign body sensation and/or discomfort caused by dry eye disease or environmental factors such as dust, glare, pollen and air pollution.	Allergy to any ingredients of the product	Yes - Purite®	No	**AbbVie** 1800 252 224
Optive Fusion Preservative Free	5mg carmellose sodium 10mg glycerin 1mg sodium hyaluronate	Single use vial	30 x 0.4mL vials	Aqueous deficient	Intended to lubricate the surface of the eye(s) to relieve the feeling of dryness, burning, irritation, stinging, foreign body sensation and/or discomfort caused by dry eye disease or environmental factors such as dust, glare, pollen and air pollution.	Allergy to any ingredients of the product	No	No	**AbbVie** 1800 252 224
Optive Gel Drops	10mg carmellose sodium 9mg glycerin	3/12 after opening	10mL bottle	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	Yes - Purite®	No	**AbbVie** 1800 252 224

3. Ocular lubricants (includes drops, gels, sprays and ointments; with preservative and preservative-free)

Brand name	Active ingredient	Shelf life	Available sizes	Dry eye subtype	Therapeutic indications	Contraindications	Preservative	PBS listing	Supplier
Drops									
Optive Lubricant	Carmellose Sodium 5mg Glycerin 9mg	3/12 after opening	15mL	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	Purite®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**AbbVie** 1800 252 224
Optive Sensitive	Carmellose sodium 5mg Glycerin 9mg	Single use vial	30 x 0.4mL vials	Aqueous deficient	A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	No	No	**AbbVie** 1800 252 224
Poly-Tears®	Dextran-70, hypromellose	6/12 after opening	15mL bottle	Aqueous deficient	Specially formulated to provide temporary relief of burning, irritation and discomfort due to dryness of the eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
Systane® BALANCE	Propylene glycol (PG)	6/12 after opening	10mL bottle	Evaporative	For the temporary relief of burning and irritation due to dryness of the eye. Contains mineral oil. For evaporative dry eye and dry eye symptoms caused by meibomian gland dysfunction.	Allergy to any ingredients of the product	Yes - PolyQuad®	No	**Alcon*** 1800 224 153
Systane® COMPLETE Preservative Free	Propylene glycol (PG)	3/12 after opening	10mL bottle	Evaporative & aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. Contains mineral oil (nano-lipid technology), suitable for aqueous deficient, evaporative and mixed dry eye.	Allergy to any ingredients of the product	No	No	**Alcon*** 1800 224 153
Systane® COMPLETE	Propylene glycol (PG)	6/12 after opening	10mL bottle	Evaporative & aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	No	**Alcon*** 1800 224 153
Systane® GEL DROPS	Propylene glycol (PG), polyethylene glycol 400 (PEG 400)	3/12 after opening	10mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due to dryness of the eye. Droppable gel formulation. For severe dry eye[5].	Allergy to any ingredients of the product	Yes - PolyQuad®	No	**Alcon*** 1800 224 153
Systane® HYDRATION Presevative Free	Sodium hyaluronate, propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	3/12 after opening	10mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For mild dry eye symptoms[5] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	No	No	**Alcon*** 1800 224 153
Systane® HYDRATION	Sodium hyaluronate, propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	3/12 after opening	10mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For mild dry eye symptoms[5] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	No	**Alcon*** 1800 224 153
Systane® HYDRATION UD	Sodium hyaluronate, propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	Single use (vials)	30 x 0.7mL vials	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For mild dry eye symptoms[5] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	No	No	**Alcon*** 1800 224 153
Systane® Lubricant Eye Drops	Propylene glycol (PG), polyethylene glycol 400 (PEG 400)	6/12 after opening	15mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due to dryness of the eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
Systane® Lubricant UD	Propylene glycol (PG), polyethylene glycol 400 (PEG 400)	Single use (vials)	30 x 0.8mL vials	Aqueous deficient	For the temporary relief of burning and irritation due to dryness of the eye.	Allergy to any ingredients of the product	No	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome). Authority Required (streamlined) for preservative-free (6172), patient must be sensitive to preservatives in multi-dose eye drops.	**Alcon*** 1800 224 153
Systane® ULTRA Preservative Free	Propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	3/12 after opening	10mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For mild dry eye symptoms[5] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	No	No	**Alcon*** 1800 224 153
Systane® ULTRA	Propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	6/12 after opening	10mL bottle	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For mild dry eye symptoms[5] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	No	**Alcon*** 1800 224 153
Systane® ULTRA UD	Propylene glycol (PG) & polyethylene glycol 400 (PEG 400)	Single use (vials)	25 x 0.5mL vials	Aqueous deficient	For the temporary relief of burning and irritation due of dryness of the eye. For moderate to severe dry eye symptoms[3] due to aqueous deficient dry eye.	Allergy to any ingredients of the product	No	No	**Alcon*** 1800 224 153
Tears Naturale®	Dextran-70, hypromellose	6/12 after opening	15mL bottle	Aqueous deficient	Specially formulated to provide temporary relief of burning, irritation and discomfort due to dryness of the eye.	Allergy to any ingredients of the product	Yes - PolyQuad®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome)	**Alcon*** 1800 224 153
Xailin Fresh	Carmellose sodium 0.5%	Unopened - 24 months from date of manufacture. 28 days after opening	30 x 0.4mL vials	Aqueous deficient	Treatment of mild to moderate dry eye where there is tear insufficiency/aqueous deficiency, evaporative and MGD.	In case of infections and to patients with known hypersensitivity to any of the ingredients in the formulation	No	No	**Optimed** 1300 657 720
Xailin Hydrate	Hypromellose 0.3%	Unopened - 24 months from date of manufacture. 28 days after opening	10mL bottle	Aqueous deficient	Treatment of mild to moderate dry eye where there is tear insufficiency/aqueous deficiency, evaporative and MGD.	Known hypersensitivity to any of the ingredients in the formulation	No	No	**Optimed** 1300 657 720

3. Ocular lubricants (includes drops, gels, sprays and ointments; with preservative and preservative-free)

Brand name	Active ingredient	Shelf life	Available sizes	Dry eye subtype		Therapeutic indications	Contraindications	Preservative	PBS listing	Supplier
Gel										
Celluvisc	Carmellose sodium 10mg	Single use vial	30 x 0.4mL vials	Aqueous deficient		A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome, ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	No	Yes: Authority Required (streamlined) – 6172. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**AbbVie** 1800 252 224
GenTeal®Gel	Hypromellose & carbomer-980	3/12 after opening	10g tube	Aqueous deficient		Lubricant eye gel for people who experience sensation of dryness, fatigue or discomfort of the eyes due to environmental irritants.	Allergy to any ingredients of the product	Yes - GenAqua®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
HPMC PAA Gel	Hypromellose & carbomer-980	3/12 after opening	10g tube	Aqueous deficient		Lubricant eye gel for people who experience sensation of dryness, fatigue or discomfort of the eyes.	Allergy to any ingredients of the product	Yes - GenAqua®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**Alcon*** 1800 224 153
Optimel® Manuka+® Forte Eye Gel	Leptospermum (spp honey)	Indefinite	10g	Evaporative & aqueous deficient		For more difficult cases, containing 98% Manuka honey, it helps to stabilise the surface health of the eye and may provide relief from dry eye symptoms such as sore, irritated eyes and eyelids.	Do not use if there is a history of hypersensitivity or sensitive to honey, food gums, benzoate preservatives without seeking medical advice	No	No	**Designs For Vision** 1800 225 307
Refresh Liquigel®	Carmellose Sodium 10mg	3/12 after opening	15mL	Aqueous deficient		A sterile substance used to provide supplemental lubrication/hydration to the natural eye to treat dry, tired and/or strained eyes resulting from dry eye syndrome (a disorder resulting in dry, gritty, burning symptoms), ageing/hormone changes (menopause) or environmental factors (e.g, medications, pollution, computer use, air conditioning and contact lens use).	Allergy to any ingredients of the product	Purite®	Yes: Restricted Benefit (severe dry eye syndrome, including Sjögren's syndrome).	**AbbVie** 1800 252 224
Viscotears Eye Gel	Carbomer 2mg/g	Unopened: Use by date Opened: 6 weeks after opening	10g tube	Aqueous deficient		Viscotears Eye Gel is used as substitute for tear film in the treatment of dry eye conditions and for unstable tear film.	Patients with known hypersensitivity to carbomer (polycid) or to any of the excipients. Do not use with concurrent therapy of neurotrophic keratitis with Oxervate (cenegermin)	Yes	Yes	**Bausch & Lomb** 1800 251 150
Viscotears Gel PF	0.2% Carbomer	Unopened: Use by date	30 x 0.6g single-dose units	Aqueous deficient		Viscotears Gel PF has a moistening and wetting effect of the eyes. It improves the moistening of the ocular surface in the presence of dry eye sensation, burning or tiredness of the eyes, caused, for instance, by watching television for a long time or extensive computer work, air from heating systems or air conditioning.	Do not use this product if you are sensitive to any of the ingredients	No	Yes: Authority Required (streamlined) – 6172. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops.	**Bausch & Lomb** 1800 251 150
Xailin Gel	Carbomer 0.2%	Unopened - 24 months from date of manufacture. 28 days after opening	10g tube	Aqueous deficient		Treatment of mild to severe dry eye where there is tear insufficiency/aqueous deficiency, evaporative and MGD.	Known hypersensitivity to any of the ingredients in the formulation	No	No	**Optimed** 1300 657 720
Ointment										
Poly Visc®Ointment	Paraffin	4/52 after opening	3.5g, 2 x 3.5g tube	Evaporative dry eye		Specially formulated eye ointment designed to provide soothing relief from the symptoms associated with dry, irritated eyes, particularly while you sleep.	Allergy to any ingredients of the product	No	Yes	**Alcon*** 1800 224 153
VitAPOS	Retinol palmitate	6/12 after opening	5g tube	Aqueous deficient		Dry eye overnight		No	Yes	**AFT** Good Optical (03) 9645 1066
Xailin Night	White soft paraffin w/w 57.3%	Unopened - 24 months from date of manufacture. 60 days after opening	6g tube	Aqueous deficient		Treatment of moderate to severe dry eye where there is tear insufficiency/aqueous deficiency, evaporative and MGD.	Known hypersensitivity to any of the ingredients in the formulation	No	No	**Optimed** 1300 657 720
Spray										
Tears Again Spray	Soy lecithin	6/12 after opening	10mL spray bottle	Evaporative dry eye		Provides soothing relief to dry, tired, irritated eyes. Applied onto closed eyelids, TearsAgain eye spray hydrates dry eyes and reduces moisture evaporation. Suitable for use with contact lenses and eye makeup and gentle on sensitive eyes.		Yes - phenoxyethanol	Yes: Authority Required (streamlined) – 6172. For severe dry eye syndrome, patient must be sensitive to preservatives in multi-dose eye drops	**Good Optical** (03) 9645 1066

4. Lid hygiene

(includes warm compresses, wipes, cleansers and products containing tea tree oil)

Product name	Manufacturer	Product type	Comments	Distributor
Aelo Eyelid & Eyelash Cleanser	**Aelo**	Lid cleanser	Perfect for maintaining daily eyelid and eyelash hygiene, these cleansers are ideal for anyone who wears false lashes, uses eye cosmetics, works in dusty environments, or just enjoys the sensation of clean eyes. With a formulation of carefully selected ingredients, they are gentle enough for sensitive eyes and the delicate skin surrounding them.	**Aelo** hello@aelo.com.au
Aelo Self-Heating Eye Mask	**Aelo**	Moist heat warm compress	Designed to hydrate and warm the delicate eye area, these masks reach 40°C for 30 minutes and are great for styes. Each box contains 10 individually packed self-heating eye masks, perfect for travel and planes. They are disposable for hygienic purposes.	**Aelo** hello@aelo.com.au
Allergy 30's Wipes	**Cynacon OcuSoft**	Lid cleanser	Seasonal allergic conjunctivitis.	**Good Optical** (03) 9645 1066
Avenova®	**Novabay**	Hypochlorous acid spray	0.01% hypochlorous acid spray solution that kills a broad spectrum of bacteria and successfully helps relieve chronic eye conditions like MGD, blepharitis and dry eye, without stinging or irritation. No bleach, and no impurities.	**Designs For Vision** 1800 225 307
Blephadex Eyelid Applicators	**Blephadex**	Eyelid applicator	In-practice manual debridement treatment whilst simultaneously depositing Blephadex solution.	**Optimed** 1300 657 720
Blephadex Eyelid Foam	**Blephadex**	Foam bottle	Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex.	**Optimed** 1300 657 720
Blephadex Eyelid Wipes	**Blephadex**	Wipes	Tea tree oil, coconut, aloe barbadensis formulation in foam designed for cleaning lashes, eyelids and treatment of demodex.	**Optimed** 1300 657 720
Blephadex Pro with Manuka Honey	**Blephadex**	Wipes	Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex. With added benefits of manuka honey (UMF 10+).	**Optimed** 1300 657 720
Blephadex Warming Eyelid Wipes	**Blephadex**	Wipes	Warming wipe is activated with water droplet. Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex.	**Optimed** 1300 657 720
BlephEx	**BlephEx**	In-practice mechanical debridement	In-practice procedure to remove bacterial biofilm, scurf, debris from lashes and lid margin by means of microblepharoexfoliation (MBE).	**Optimed** 1300 657 720
Blephex Lid Cleanser	**BlephEx**	Lid cleanser	Suitable for moistening Blephex Procedural Tips. Contains Manuka honey and Dead Sea salt to fight inflammation on the biofilm and offers anti-demodex properties.	**Optimed** 1300 657 720
Bruder Hygienic Eyelid Solution	**Bruder**	Spray	Pure 0.02% hypochlorous acid solution in amber bottle (preservative free).	**Optimed** 1300 657 720
Bruder Hygienic Eyelid Wipes	**Bruder**	Wipes	No Rinse, leave on formula. Gentle enough for daily use. Low residue, non-soapy formula. No stinging or burning.	**Optimed** 1300 657 720
Bruder Moist Heat Eye Compress	**Bruder**	Compress	Available in double eye and single eye. Patented Medibead technology to deliver uniform moist heat.	**Optimed** 1300 657 720
D.E.R.M®	**Eyeeco**	Moist heat therapy	D.E.R.M®, Dry Eye Relief Mask is an entry level solution for immediate dry eye relief. The mask contains silica beads, which naturally attract moisture from the air and release it when microwaved.	**Designs For Vision** 1800 225 307
Dry Eye Compress	**Ocusci**	Moist heat therapy Cold therapy	➔123% improvement in DE Symptoms - 5 days of use ➔Machine-washable cover with Silver+ antimicrobial ➔Just 25 seconds in the microwave ➔Patented Hydroblock coating - delivers 10 minutes of moist, therapeutic heat ➔Custom logo options	**OphthalmoPro** 1300 578 288
Dry Eye Compress (USB)	**Ocusci**	USB powered warm compress	➔Delivers therapeutic heat at your convenience ➔USB powered, plug into any PC or portable battery ➔3 temperature settings and 3 timer settings ➔Automatic shut off	**OphthalmoPro** 1300 578 288
Eyecloud®	**Eyeeco**	Moist heat therapy Cold therapy	A controlled moist heat therapy system designed to provide 600, 20 minute treatments. The expanded mask coverage and pressure control design evenly distributes therapy to the eyelids and surrounding skin. Moist heat therapy: ➔Stimulates tear production ➔Promotes blood circulation ➔Returns oil along eyelids to healthy state. Cold treatments reduce: ➔Eyelid and undereye puffiness ➔Inflammation and itchiness associated with allergies.	**Designs For Vision** 1800 225 307
Eye Doctor Essential	**Body Doctor**	Warm compress	Microwaveable. Brandable for practices.	**Good Optical** (03) 9645 1066
Eye Doctor Stye Mask	**Body Doctor**	Warm compress for one eye	Great one eye mask for styes.	**Good Optical** (03) 9645 1066

4. Lid hygiene (includes warm compresses, wipes, cleansers and products containing tea tree oil)

Product name	Manufacturer	Product type	Comments	Distributor
Eye Doctor Mask Premium	**Body Doctor**	Warm/cool compress	Use hot or cold. Can heat in conventional oven. Brandable for practices.	**Good Optical** (03) 9645 1066
Eye Doctor Click n Go Mask	**Body Doctor**	Travel compress	Great for planes and camping. Up to 100 uses.	**Good Optical** (03) 9645 1066
Eyegiene InstaWarmth Mask Compress	**Eyegiene**	Compress	Disposable wafers utilise unique warming technology when removed from packet and slipped into the ultra-soft eye mask.	**Optimed** 1300 657 720
Eyeleve by Bruder Contact Lens Compress	**Bruder**	Compress	Patented Medibead technology to deliver uniform moist heat. Comfort stitch to reduce the pressure off the eyelid.	**Optimed** 1300 657 720
Eyelid & Lash Foaming Cleanser EVERYDAY	**Helbron Laboratories**	Cleanser	Designed for use in a general eyelid and eyelash hygiene regime. A perfect option for wearers of false lashes and makeup, as well as those who work in dusty and dirty environments - or those who just love the feeling and look of clean bright eyes.	**Ocul Brands Pty Ltd** (03) 9593 6501
Eyelid & Lash Foaming Cleanser FORTE	**Helbron Laboratories**	Cleanser	A higher strength formulation designed for the control of pre existing eyelid and eyelash conditions such as blepharitis, demodex, dry eye, sore and itchy eyes. With boosted tea tree oil to provide additional antibacterial help, this formulation is a front line tool for control of many common eyelid and eyelash conditions.	**Ocul Brands Pty Ltd** (03) 9593 6501
Eye Warmer	**Silinfra**	Compress	USB powered heating pad for the eyes.	**Optimed** 1300 657 720
Hypochlor Gel	**Cynacon OcuSoft**	Hypochlor Gel	Great results with Rosacea.	**Good Optical** (03) 9645 1066
Hypochlor Spray	**Cynacon OcuSoft**	Hypochlor Spray	Great results with Rosacea.	**Good Optical** (03) 9645 1066
I Lid 'n Lash®	**I-Med Pharma**	Eyelid Wipes	Clean and remove ocular debris and makeup from around the eyes. Used daily, I-LID 'N LASH® is effective in maintaining lid and lash health for patients with dry eyes or sensitive skin. Available in standard and 5% tea tree for demodex management.	**Designs For Vision** 1800 225 307
Nulids	**Nulids**	At home 'mi-croblepharoex-foliation' device	Works well with Zocular Gel.	**Good Optical** (03) 9645 1066
Optimel® Manuka+ Eyelid Cream	**Melcare**	Eyelid Cream	Helps restore skin's protective acid mantle. Dry eye conditions such as blepharitis and meibomian gland disease (MGD) are frequently present with various dermatological conditions, including seborrheic dermatitis, rosacea, and eczema. Nearly half of people suffering from MGD have had seborrheic dermatitis at some time. Skin pH in atopic dermatitis is often high and the resulting cascade of changes contribute to the development of atopic dermatitis leading to dryness and flaky skin. Optimel® Manuka+® Eyelid Cream is a low pH emollient cream containing Leptospermum spp honey and Jojoba oil to help restore skin's natural structure. The humectant properties of honey also help maintain a lasting skin softness.	**Designs For Vision** 1800 225 307
Opti-Soothe Moist Heat Mask	**Scope**	Moist heat mask	The Opti-Soothe Moist Heat Mask is reusable and designed to give you effective relief from dry eye and other related eye conditions.	**Good Optical** (03) 9645 1066
Opti-Soothe Preservative Free Eyelid Wipes	**Scope**	Eyelid wipe	Contains a unique and gentle formulation of tea tree oil, hyaluronic acid, chamomile, aloe vera. Preservative-free nature makes it ideal for daily eyelid hygiene. No need to rinse off formulation after use.	**Good Optical** (03) 9645 1066
Original Foam 50ml	**Cynacon OcuSoft**	Lid cleanser	General maintenance and cleanser.	**Good Optical** (03) 9645 1066
Original 30's Wipes	**Cynacon OcuSoft**	Lid cleanser	Demodex Infestation	**Good Optical** (03) 9645 1066
Oust Foam 50ml	**Cynacon OcuSoft**	Lid cleanser	Demodex Infestation	**Good Optical** (03) 9645 1066
Platinum Plus Foam 50ml	**Cynacon OcuSoft**	Lid cleanser	Anterior blepharitis	**Good Optical** (03) 9645 1066
Plus 30's Wipes	**Cynacon OcuSoft**	Lid cleanser	Anterior blepharitis	**Good Optical** (03) 9645 1066
Plus Foam 50ml	**Cynacon OcuSoft**	Lid cleanser	Anterior blepharitis	**Good Optical** (03) 9645 1066
Systane® LID WIPES	**Alcon***	Daily eyelid cleanser	Non-medicated lid wipe for daily eyelid hygiene.	**Alcon*** 1800 224 153
Tranquileyes XL®	**Eyeeco**	Moist heat therapy	Tested to sustain optimum temperature and humidity for 20 minute treatments. Moist heat therapy: ➔Stimulates tear production ➔Promotes blood circulation ➔Returns oil along eyelids to healthy state. Cold treatments reduce: ➔Eyelid and undereye puffiness ➔Inflammation and itchiness associated with allergies.	**Designs For Vision** 1800 225 307

4. Lid hygiene (includes warm compresses, wipes, cleansers and products containing tea tree oil)

Product name	Manufacturer	Product type	Comments	Distributor
TearRestore Thermal Mask	**TearRestore Inc**	Warm compress	Improve compliance, patient satisfaction and meibomian gland expression with the first open-eye warm compress. The TearRestore Thermal Mask delivers consistent heat, the ability to see, and natural meibomian gland expression (blinking) throughout compress. Reusable, self-heating, & durable heat packs. Each heat pack produces 60-90 treatments.	**Good Optical** (03) 9645 1066
Tranquileyes Tea Tree Eyelid & Facial Cleanser™	**Eyeeco**	Foaming eyelid and facial cleanser	Infused with chamomile and shea butter, these facial cleansers gently remove all dirt and makeup, leaving the skin feeling smooth, silky, and refreshed. Combined with tea tree oil to help manage demodex and related ocular surface diseases. Available in Gentle (1%) or Advanced (2%) tea tree formulations.	**Designs For Vision** 1800 225 307
ZocuFoam	**Okra LLC**	Foaming cleanser	Penetrating foam with okra micelles, bid or qd. Each bottles lasts 3 months.	**Good Optical** (03) 9645 1066
ZocuKit for ZEST	**Okra LLC**	In-office procedure	Therapeutic deep lid cleansing (debridement) to immediately and dramatically improve ocular surface and periorbital inflammation.	**Good Optical** (03) 9645 1066
ZocuShield	**Okra LLC**	Okra gel	Leave-on gel for moderate-severe cases. Only one drop of gel needed per eye and each ZocuShield bottle lasts for 3 months.	**Good Optical** (03) 9645 1066
ZocuWipe	**Okra LLC**	Towelette	Leave-on formulation effective at just once a day.	**Good Optical** (03) 9645 1066

5. Tear conservation

(punctal occlusion/moisture chamber goggles)

Product name	Manufacturer	Product type	Comments	Distributor
Dissolvable VisiPlug Lacrimal Plug	**Lacrimedics**	1x sterile pair of synthetic plugs per box	Available in 0.4mm and 0.5mm	**Optimed** 1300 657 720
Duraplug	**EagleVision**	Long term temporary plug	60 to 180 days 0.2–0.4mm sizes	**Device Technologies** 1800 429 551
Eyeseals®	**Eyeeco**	Hydrating sleep mask	Eyeseals creates a moisture rich environment for the eyes, eyelids, and surrounding skin to help your eyes feel bright, refreshed, and hydrated when you awake. Soft and flexible unitary design includes adjustable headband. Protective ridge prevents chaffing from CPAP masks with additional material between eyebrows. Recommended for sensitive skin. Clear mask recommended for CPAP users.	**Designs For Vision** 1800 225 307
Eyeseals® 4.0	**Eyeeco**	Hydrating sleep mask	Eyeseals® 4.0 creates a moisture rich environment for the eyes, eyelid, and surrounding skin such that your eyes and surrounding skin are hydrated when you awake. Soft and flexible unitary design includes adjustable headband. Ideal for air travel. Clear mask recommended for CPAP users.	**Designs For Vision** 1800 225 307
Form Fit® Intracanicular Plugs	**Oasis**	Punctal plugs	Form Fit is truly a one size plug solution. Once inserted into the canaliculus, Form Fit hydrates over a 10 minute period. As it hydrates, Form Fit increases in size until it completely fills the vertical canalicular cavity.	**Designs For Vision** 1800 225 307
Soft Plug® Extended Duration	**Oasis**	Punctal plugs	Extended duration absorbable plugs are designed to provide patient comfort for up to 3 months. This long lasting plug is effective and economical for transient dry eye symptoms following surgery. Extended duration plugs are 2mm in length.	**Designs For Vision** 1800 225 307
Soft Plug® – Silicone Punctum Plugs	**Oasis**	Punctal plugs	The distinctive low profile dome head combined with a softer more flexible silicone maximises patient comfort. The pointed nose design simplifies insertion and the large anchor with wide shelf firmly secures plug making dislocation less likely.	**Designs For Vision** 1800 225 307
Onyix® and Quartz® Silicone Sleep Mask	**Eyeeco**	Hydrating sleep mask	Create a moisture rich environment for the eyes, eyelid, and surrounding skin during sleep. The eyes and surrounding skin will feel refreshed and hydrated each morning. Suitable for use with CPAP machines.	**Designs For Vision** 1800 225 307
SuperEagle	**EagleVision**	Preloaded permanent punctal plug	Simplified sizing Sml (0.4 to 0.6mm), med (0.6 to 0.8mm), lge (0.8mm+)	**Device Technologies** 1800 429 551
Superflex	**EagleVision**	Preloaded permanent punctal plug	0.5 to 1.3mm sizes	**Device Technologies** 1800 429 551
Ziena Eyewear	**7Eye**	Moisture chamber goggles	Prescription Optifog (antifog) lenses available, including single vision, multifocal to wholesale customers. Transitions, polarised and tinted Available. These moisture chamber glasses can also be used for motorcycle or sports eyewear.	**SBR Optics** (03) 5231 3405

6. Neurostimulation devices

Product name	Model	Functions	Other comments	Usage guidelines	Distributor
Olympic Ophthalmics	iTear 100	External stimulation of nasociliary nerve Simulates lacrimal gland, meibomian glands, and mucous secretion	→ 5 levels of stimulation → Programmable via phone	Indicated for temporary use (up to 30 days) to increase acute tear production during vibratory stimulation of the external nasal nerve in adults	**Beyond 700** +61420410862

7. Electric field therapy

Product name	Model	Functions	Other comments	Usage guidelines	Distributor
Resono Ophthalmic	REXON-EYE	Non invasive device which applies low-power, high frequency electric fields, capable of stimulating the metabolism and natural regeneration of cells and reduces local inflammation (https://www.resono.it/qmr-technology/)	→ No skin type restrictions / no pain or discomfort → No ongoing cost or consumables → No labour or time to dispensing treatment → Contact-less treatment → No heavy cleaning required	Indicated for evaporative and aqueous deficient dry eye syndrome	**Eyetek** 1800 618 336

8. In-office intense pulsed light therapy

for MGD and physical heating and expression of the meibomian glands (including device-assisted therapies)

Product name	Manufacturer	Product type	Comments	Supporting studies	Distributor
EYE-LIGHT	**Espansione SPA**	IPL and LLLT	IPL and Low Light Level Therapy	32 publications for DED & MGD, as well as additional evidence for Chalazion, Sjögren's syndrome, Demodex and Refractive surgery Full Bibliography can be found at: https://espansionegroup.it/bibliography/	**OphthalmoPro** 1300 578 288
InMode	**Envision**	IPL and RF	→ Envision is the latest in-office ocular care platform for the treatment of meibomian gland dysfunction (MGD) and the symptoms of dry eye disease (DED) → Deploys multiple programmable modalities which work synergistically to deliver precision ocular treatments → Small-size Intense Pulsed Light (IPL) and bipolar radiofrequency (RF) applicators with precise depth control allowing for highly efficacious procedures in the small, more delicate subdermal layers of the periorbital region → Lumecca-I is an effective IPL treatment helps to decrease the inflammation, particularly reducing the population of Demodex mites, often associated with blepharitis, rosacea as well as redness, swelling, and discomfort. Lumecca-I was developed with an advanced xenon flash lamp producing 40% of total pulse energy in the 500-600 nm range. IPL energy penetrates the skin and the selective photo-thermolysis will provide a non-invasive, effective treatment for managing gland function, also minimising inflammation → Forma-I (bipolar radiofrequency) addresses the symptoms of dry eye disease (DED) caused by meibomian gland dysfunction (MGD). Intended for use in the periorbital area and upper and lower lids and relieves inflammation of meibomian glands and eye irritation → Morpheus8 can affect vascular lesions on the face through Vasoconstriction: The heat generated by the RF energy can cause blood vessels to constrict (vasoconstriction), which may temporarily reduce the appearance of vascular lesions like rosacea → The Morpheus8 Prime fractional radio-frequency microneedling technology allows for droopy eyelids to be tightened with a minimally invasive fractional technology	Sean P, Cohen A, Parsa K Jaccoma E, Burrell K, Carruthers J. Transcutaneous radiofrequency-mediated meibomian gland expression is an effective treatment for dry eye: A prospective cohort trial. The Open Ophthalmology Journal. 2023;17. DOI: 10.2174/18743641-v17-e230328-2022-63	**InMode ANZ** +61 02 7209 5678
Meibomask	**Espansione SPA**	LLLT	Low Light Level Therapy	32 publications for DED & MGD, as well as additional evidence for Chalazion, Sjögren's syndrome, Demodex and Refractive surgery Full bibliography can be found at: https://espansionegroup.it/bibliography/	**OphthalmoPro** 1300 578 288
MGrx	**Ocusci**	3 in 1 thermal treatment device	→ 3 treatment modes – Thermal debridement, thermal massage and thermal expression → Reusable instruments → No consumable costs	McMurren BJ, Kling MA, Fasciani A, Nymark-McMahon H. MGrx - A novel multi-modal thermal device for treating moderate to severe meibomian gland dysfunction and dry eye. The Open Ophthalmology Journal. Oct 2023;17. DOI: 10.2174/18743641-v17-231005-2023-13	**OphthalmoPro** 1300 578 288

8. In-office intense pulsed light therapy

Product name	Manufacturer	Product type	Comments	Supporting studies	Distributor
Optilight	**Lumenis**	IPL	→ Optimal Pulse Technology (OPT) (delivers repeatable, controlled flashes each time with a nospike consistency) → OPT handpiece (pen-like tip designed to treat the delicate facial contours of each individual patient) → SapphireCool tip (provides continuous contact cooling to maximise comfort for patients) → Able to treat darker skin types (Fitzpatrick type V) → Proven settings and protocols based on Lumenis' clinically validated treatment protocols for safe and effective treatment → Variable spot size (interchangeable lightguides/OPT hand piece) with ability to easily apply treatment directly to upper & lower lids → Lumenis developed & first brought to market IPL as a technology in 1993. → OptiLight is a 7th generation system specifically designed for eyecare professionals → Streamlined DE programme for patient safety & clinical efficiencies → Fastrack results with shorter duration between treatments possible – maximum patient comfort & enhanced protection of epidermis	→ Stapleton F, et al. TFOS DEWS II Epidemiology Report. The Ocular Surface. 2017;15(3):334-365. DOI: 1016/j.jtos.2017.05.003. → Lemp MA, et al. Distribution of aqueous-deficient and evaporative dry eye in a clinic-based patient cohort: a retrospective study. Cornea. 2012 May;31(5):472-8. DOI: 10.1097/ICO.0b013e318225415a. → Liu R, et al. Analysis of cytokine levels in tears and clinical correlations after intense pulsed light treating meibomian gland dysfunction Am J Ophthalmol. 2017 Nov;183:81-90. DOI: 10.1016/j.ajo.2017.08.021. → Yin Y, et al. Changes in the meibomian gland after exposure to intense pulsed light in meibomian gland dysfunction (MGD) patients. Curr Eye Res. 2018 Mar;43(3):308-313. DOI: 10.1080/02713683.2017.1406525. → Kassir R, et al. Intense pulsed light for the treatment of rosacea and telangiectasias. J Cosmet Laser Ther. 2011 Oct;13(5):216-22. DOI: 10.3109/14764172.2011.613480. → Papageorgiou P, et al. Treatment of rosacea with intense pulsed light: significant improvement and long-lasting results. Br J Dermatol. 2008 Sep;159(3):628-32. DOI: 10.1111/j.1365-2133.2008.08702.x. → Prieto VG, et al. Effects of intense pulsed light on sundamaged human skin, routine, and ultrastructural analysis. Lasers Surg Med. 2002;30(2):82-5. DOI: 10.1002/lsm.10042. → Rong B, et.al. Intense pulsed light applied directly on eyelids combined with meibomian gland expression to treat meibomian gland dysfunction. Photomed Laser Surg. 2018 Jun;36(6):326-332. DOI: 10.1089/pho.2017.4402. → Dell SJ, et al. Prospective evaluation of intense pulsed light and meibomian gland expression efficacy on relieving signs and symptoms of dry eye disease due to meibomian gland dysfunction. Clin Ophthalmol. 2017 May 2;11:817-827. DOI: 10.2147/OPTH.S130706. → Toyos R, Desai NR, Toyos M, Dell SJ. Intense pulsed light improves signs and symptoms of dry eye disease due to meibomian gland dysfunction: A randomized controlled study. PLoS One. 2022 Jun 23;17(6):e0270268. DOI: 10.1371/journal.	**Lumenis ANZ** 1800 586 364
Tear Science	**LipiFlow**	Vectored Thermal Pulse Therapy	Automated heating and expression of Meibomian Glands	Clinically proven MGD treatment in over 20 peer review articles	**Johnson & Johnson Surgical Vision** 1800 266 111

9. Prescription drugs to manage dry eye disease

Brand name	Active ingredient	Shelf life	Available sizes + delivery	Therapeutic indications	Contraindications	TGA status (as at Sept 2024)	PBS listing	Further info	Supplier
Cequa®	Ciclosporin 0.09%	18 months	60 vial pack single use ampoules	To increase tear production in patients with moderate to severe keratoconjunctivitis sicca (dry eye) where prior use of artificial tears has not been sufficient.	Hypersensitivity to the active substance or to any of the excipients. Active or suspected ocular or peri-ocular infection. Ocular or peri-ocular malignancies or premalignant conditions.	Registered	Yes Authority Required (Online/Phone application)	Please review Cequa Product Information before prescribing. For further HCP information and resources please visit www.cequa.com.au Further PBS information available at https://www.pbs.gov.au/medicine/item/13284E	**Sun Pharma ANZ** 1800 726 229
Ikervis®	Ciclosporin 0.1%	3 years	30 x 0.3mL single use ampoules	Treatment of severe keratitis in adult patients with dry eye disease which has not improved despite treatment with tear substitutes.	Hypersensitivity to the active substance or to any of the excipients. Active or suspected ocular or peri-ocular infection. Ocular or peri-ocular malignancies or premalignant conditions.	Registered	Yes Authority Required PBS code 12663L	Administered once daily. Formulated with a cationic vehicle. Please review Ikervis® Product Information before prescribing.	**CSL Seqirus** 1800 642 865

10. Steroids

Brand name	Active ingredient	Shelf life	Available sizes + delivery	Therapeutic indications	Contraindications	PBS listing	Supplier
FML	Fluorometholone alcohol 0.1%	Discard unused contents 4 weeks after opening	5mL drops	A steroid responsive inflammation of the palebral and bulbar conjunctiva, cornea and anterior segment of the globe.	→Acute superficial (or epithelial) Herpes simplex keratitis (dendritic keratitis). →Fungal diseases of ocular structures. →Vaccinia, varicella, mycobacterial infection of the eye and most other viral diseases of the cornea and conjunctiva. →Tuberculosis of the eye. →Hypersensitivity to the constituents of this medication.	No	**AbbVie Mascot NSW 2020** 1800 224 153
Hycor	Hydrocortisone acetate 1%	Discard any of the ointment still left after 4 weeks from first using it	5g ointment	Inflammation of the eyelid or eyelid margin.	→Active eyelid infection. →Hypersensitivity to the constituents of this medication.	Yes	**Aspen**
Loteprednol	Loteprednol etabonate 0.5%	Single use vials	20 single use drop vials	A topical anti-inflammatory eye drop for inflammation in dry eye disease.	→Acute superficial (or epithelial) Herpes simplex keratitis (dendritic keratitis). →Fungal diseases of ocular structures. →Vaccinia, varicella, mycobacterial infection of the eye and most other viral diseases of the cornea and conjunctiva. →Tuberculosis of the eye. →Hypersensitivity to the constituents of this medication.	No	*Compounded*
Maxidex	Dexamethasone 1mg/mL (preservative: benzalkonium chloride 0.1 mg/mL)	Discard container 4 weeks after opening	5mL drops	Dexamethasone is indicated for use in the treatment of steroid responsive inflammatory conditions of the palpebral and bulbar conjunctiva, cornea and anterior segment of the eye.	→Epithelial herpes simplex (dendritic keratitis). →Vaccinia, varicella, and most other viral diseases of the cornea and conjunctiva. →Tuberculosis of the eye. →Fungal disease of ocular structures or untreated parasitic eye infections, mycobacterial ocular infections or untreated bacterial eye infections. →Hypersensitivity to the constituents of this medication.	Yes	**Novartis**
Prednefrin Forte	Prednisolone acetate 1% + phenylephrine 0.12% suspension	Discard unused contents 4 weeks after opening	10mL drops	Prednefrin Forte is indicated for use in the treatment of steroid responsive inflammatory conditions of the palpebral and bulbar conjunctiva, cornea and anterior segment of the eye.	→Epithelial herpes simplex (dendritic keratitis). →Vaccinia, varicella, and most other viral diseases of the cornea and conjunctiva. →Tuberculosis of the eye. →Fungal disease of ocular structures or untreated parasitic eye infections, mycobacterial ocular infections or untreated bacterial eye infections. →Hypersensitivity to the constituents of this medication.	Yes Restricted benefit PBS code 5568C	**AbbVie Mascot NSW 2020** 1800 224 153
Prednisolone Minims	Prednisolone phosphate 0.5%	Single use vials	20 single use drop vials	A topical anti-inflammatory glucosteroid eye drop for inflammation in dry eye disease.	→Acute superficial (or epithelial) herpes simplex keratitis (dendritic keratitis). →Fungal diseases of ocular structures. →Vaccinia, varicella, mycobacterial infection of the eye and most other viral diseases of the cornea and conjunctiva. →Tuberculosis of the eye. →Hypersensitivity to the constituents of this medication.	No	**Bausch & Lomb** 1800 251 150

2.2. Antibacterials

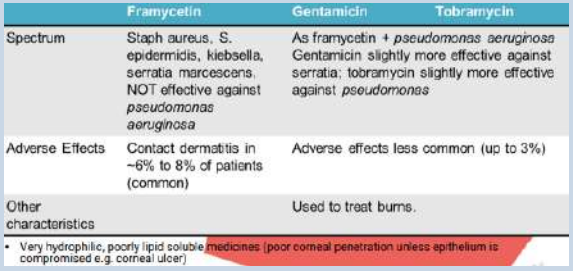
Protein Synthesis Inhibitors – act on bacterial ribosomes

Mode of Action	Spectrum	Indications	Dosage	Contraindications + Precau	Side Effects
30S SUBUNIT					
TETRACYCLINES (Tetracycline, Minocycline, Doxycycline) 1. TETRACYCLINE - not commercially available → half life (12-22 hrs) a. Compound - tetracycline HCl 1% eye ointment 2. MINOCYCLINE → half life (11-23 hrs) a. tab, 50 mg, 60, Akamin, Minomycin, **3. DOXYCYCLINE - main choice** → half life (6-1`1 hrs) ***a. Doxycycline (Doxsig) 50 mg tablet, 25 - GP*** b. doxycycline 0.025%/0.1% eye drops c. Tab 50mg (Doxylin, Frakas), 100mg (Doxylin), 100mg scored (Doxsig); Capsules 50mg (Doryx), 100mg (Doryx)					
Anti bacterial + inflam. > Bacteriostatic > Prevents sequential attachment of tRNA & stops protein biosynthesis from occurring > Interferes with t-RNA anticodon reading of m-RNA codon > Reversibly binds to 30S subunit Inhibits production of: > **bacterial lipase** - lipases releases irritating fatty acids > MMP (**matrix metalloproteinase**) - MMP degrades collagen in ocular tissues	> Broad spectrum - high rates of **resistance** > Effective against **rarer organisms** (e.g. Mycoplasma, Chlamydia, Lyme disease) > Eyelid colonisers (S Epidermidis and S Aureus) Counseling: Irregular gut absorption → take w food **(2) (3)** - well absorbed orally **(1) (2)** - likely to GI upset **(3)** - well tolerated + good safety profile + longer H-L	**> Meibomian Gland Dysfunction (MGD) (chronic)** **> persistent epithelial defects** **> non infectious corneal conditions** - neurotrophic ulcers and RCE > **Acne rosacea** > phlyctenular conjunctivitis, > **Chlamydia** - conjunctivitis > prophylaxis against malaria > trachoma SHORT TERM (≤21 days) use for serious infections > Short courses (7-10 days) SAFE during breastfeeding	DOXYCYCLINE: MIC = 200mg/day - antimicrobial effect <200mg - anti-inflam. effect. *Take in the morning. > Adult: 100 mg 1-2x/day > **MGD**: 50mg 2x/day 1-2 wks, then 25mg 1x/day 1-6 months **> Recurrent Corneal Erosion (RCE)**: 50mg 2x/day + topical steroid 1 month > **Corneal ulcers with stromal thinning:** 100mg 2x/day > Acne: 50mg 1x/daily for 6 wks > Rosacea: 50mg 1x/daily > Chlamydia: 100mg 2x/day 1-3 wks > Gonococcal infections: 100mg 2x/day 5-7 days > Malaria prophylaxis: 100mg daily - start 2 days before and continue 4 wks after endemic area * Take with food or milk to reduce stomach upset	**CONTRA**: tetracyclines allergy PRECAUTIONS: > Systemic lupus erythematosus > tx w oral retinoid (↑ risk of benign **intracranial HTN** → papilledema) > renal and hepatic disease > Not in children < 8 y.o. Due to affinity to calcium in bones + teeth > Renal impairment – doxycycline and minocycline can be used with dose adjustment > Absorption ↓ if admin. w/ iron tonics, antacids, sodium bicarbonate > Potentiate effects of anticoagulant drug warfarin → uncontrolled bleeding > Stop any osteoporosis tx w/ strontium ranelate > AVOID during pregnancy (unless 1st 18 wks) + <8y/o	**Common (>1%):** nausea, vomitting, diarrhoea, epigastric burning; tooth discolouration, enamel dysplasia, photosensitivity **Infrequent (0.1-1%):** rash, stomatitis, fungal overgrowth **Rare (<0.1%):** photo onycholysis and nail discolouration, esophageal ulcers, hepatitis, fatty liver degeneration, benign intracranial HTN, allergic reactions, toxic epidermal necrolysis, worsening of SLE, **(2)** high incidence of vestibular + CNS adverse effects GI upset: (2)/(1) > (3) 1-2x day **Photosensitivity**: (3) > (2)

AMINOGLYCOSIDES (Tobramycin, Gentamicin, Framycetin, Neomycin) → S4

1. **FRAMYCETIN (PBS)**
   - ***a. Framycetin sulfate (Soframycin) 0.5% eye drops, 8 mL***
2. **TOBRAMYCIN (PBS restricted)** → fortified (higher concentrations, 1.3% to 1.5% preparations can be compounded for Bacterial Keratitis, in combination with fortified cephalosporin or vancomycin)
   - ***a. Tobramycin (Tobrex) 0.3% eye drops, 5 mL***
   - ***b. Tobramycin (Tobrex) 0.3% eye ointment, 3.5 g***
3. **GENTAMICIN** → not commercially available; needs compounding (eye drops between 0.9% to 1.4% can be compounded)
4. **NEOMYCIN** → not commercially available; needs compounding

- **Tobrex (Alcon) ophthalmic eye drops 0.3%, 5 mL bottle → PBS Restricted**
- **Tobrex (Alcon) ophthalmic ointment 0.3%, 3.5g tube → PBS Restricted**



	Framycetin	Gentamicin	Tobramycin
Spectrum	Staph aureus, S. epidermidis, klebsella, serratia marcescens. NOT effective against *pseudomonas aeruginosa*	As framycetin + *pseudomonas aeruginosa* Gentamicin slightly more effective against serratia; tobramycin slightly more effective against *pseudomonas*	
Adverse Effects	Contact dermatitis in ~6% to 8% of patients (common)	Adverse effects less common (up to 3%)	
Other characteristics		Used to treat burns.	

- Very hydrophilic, poorly lipid soluble medicines (poor corneal penetration unless epithelium is compromised e.g. corneal ulcer)

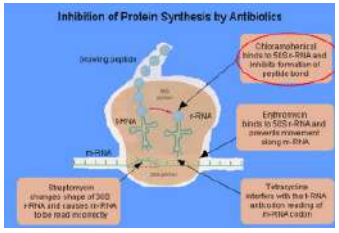
• **Soframycin (Sanofi-Aventis) eye drops 0.5%, 8 mL bottle → on PBS**					
Antibacterial > Protein synthesis inhibitor > Rapidly bactericidal (concentration dependent) → because main site of action is the membrane-associated bacterial ribosome > Cell wall disruption > Binds to 30S → interrupt protein translation > Postantibiotic effect > Very hydrophilic - poor lipid solubility and corneal Penetration → doesn't matter when treating bacterial keratitis since the epithelium is compromised by a corneal ulcer	> Mainly **gram-ve** (e.g. P aeruginosa → beside Framycetin), reasonable gram +ve > Highly effective against rapidly multiplying microbes **> Synergistic effect w β-lactam antibiotics → administer separately or else inactivated ([penicillins, cephalosporins] monobactams) (2) (3)** anti-pseudomonal activity + well tolerated. **>** Inactivated due to opening of B-lactam ring and reaction with amino group of aminoglycoside to form an inactive amide **>** Cell-wall injury induced by B-lactam compounds allows increased uptake of aminoglycoside in bacteria due to accessibility of bacterial cell membrane **>** Access on PBS for multi dose tobramycin (**restricted to suspected pseudomonas eye infections** or perioperative use in ophthalmic surgery)	> **Bacterial eye infections - conjunctivitis,** keratitis (fortified preparations?) **> Prophylaxis after ocular surgery (if chloramphenicol unsuitable)** > Ocular surface dmg/burns (gentamicin and tobramycin used systemically) > Topical preparations SAFE in pregnancy & breastfeeding	FRAMYCETIN + TOBRAMYCIN: 1 drop every 2–4 hrs for 2 days;= if there is improvement, 1 drop 4x/day for 5 days (Framycetin 3x) > Tobramycin ointment: adjunct to drops at night OR 3x/day GENTAMICIN: 1 drop/hr (including overnight) > **Corneal ulcers:** fortified prep ~1.3-1.5%, hrly w/ loading dose (every 10 mins), combined w/ fortified cefazol 5% (g+ and g- cover) --- • **Minimum**: 4x/day (3 for Framycetin) • Every 30 min - 1hr for severe infections • Tobramycin (2x/day), but every 3 hrs when indicated • Fortified Gentamicin or Tobramycin at concentrations of 1.3% or 1.5% (for corneal ulcers) ○ Normally administered with fortified B-lactam antibiotic (to provide full coverage) ○ E.g, topical cephalosporin (cephazolin 5%), NOT administered at the same time	**CONTRAIND**: kidney disease, pre-existing hearing loss (test CNVIII), neuromuscular disease (e.g. myasthenia gravis) **PRECAU**: allergic reaction to an ocular aminoglycoside; cross-allergenicity can occur > Used topically → can be inactivated by purulent exudates (rinse with sterile saline prior to instillation) > Framycetin is **NOT** effective against P Aeruginosa → Gentamicin and Tobramycin are • Tobramycin more effective at P Aeruginosa, Gentamicin more effective against Serratia	**Common:** ocular irritation, SPK **Rare**: delayed wound healing, retinal toxicity, hypersensitivity **(1)** more likely contact Dermatitis **Adv**: ocular irritation and superficial punctate keratitis and rarely delayed wound healing, retinal toxicity (in presence of leakage due to corneoscleral wound) and hypersensitivity reactions Systemically → irreversible ototoxicity or hearing loss, frequent but reversible nephrotoxicity, rare but serious neuromuscular blockade Contact dermatitis more common with framycetin
50S SUBUNIT					
MACROLIDES (Erythromycin, Clarithromycin, Azithromycin and Roxithromycin) > No ophthalmic preparations commercially available - need compounding **1. ERYTHROMYCIN** ***a. Compound erythromycin 0.5% ointment (or eyedrops)*** 2. CLARITHROMYCIN **3. AZITHROMYCIN** ***a. Azithromycin (Zithromax) 500 mg tablet - only GP can prescribe (main choice)*** *b. Compound azithromycin 1% to 1.5% eye drops (or ointment)* 4. ROXITHROMYCIN					
Antibacterial **Immunomodulatory** **Anti-inflammatory** > Bacteriostatic - can be	> Very effective against **g +ve** esp **Chlamydia** - some g-ve > ↓ biofilm production in mucoid strains of Pseudo. aeruginosa + inhibit virulence	**> Treat + prevent bacteria infections** **> Chlamydial infections** **> MGD - TOPICAL** > Good 2nd choice if px has	AZITHROMYCIN ORAL: > 500 mg 1x on day 1, then 1x/day for 2 days or 250 mg 1x/day for 4 days. Higher dose can be given for >3 days	**CONTRAIND**: allergy to macrolides, hearing disturbances/loss, blood disorders **PRECAUTION**: Myasthenia gravis	**Common**: N, V, diarrhoea, abdominal pain/cramps, candida infections (2) **Infrequent**: rash, HA **Rare**: hypersensitivity, fixed

Bactericidal under right conc > Binds to 50S r-RNA and prevents movement along m-RNA ➡ suppresses cytokines + chemokines and ⬇ MMP > Prevent translocation and inhibits protein synthesis **Effective, low toxicity - ALTERNATIVE to tetracyclines**	Factors **(3)** > Oral: Long H-L, several days > Topical: long H-L, effective against g+ve lid flora, strong penetration to the eyelids **(2) (3)** better tolerated + shorter courses than **(1)**	Penicillin/cephalosporin allergy > Trachoma > Pneumonia > Streptococcal pharyngitis/ Tonsillitis > Traveller's diarrhoea Topical **(1)** - public health measure; prophylaxis of ophthalmia neonatorum	**> Chlamydial/gonococcal:** 1g as a single dose. **> Trachoma + Chlamydia trachomatis conj:** 20 mg/kg (max 1 g), single dose. TOPICAL: **> MGD:** 1 drop 2x/day for 2-3 days then 1 drop 1x/day for 1 month	> Cross-resistance between diff. Macrolides > SAFE in pregnancy [except **(2)**] and breastfeeding (may cause loose bowels in infants)	Stevens-Johnson syndrome, psychiatric disturbances, ototoxicity (e.g. tinnitus, dizziness, hearing loss), pancreatitis, prolonged QT interval, blood dyscrasias > Oral **(1)** - higher incidence of GI adverse effects > **(1) (2)** More drug interactions

CHLORAMPHENICOL (S3)

- ***Chloramphenicol (Chlorsig) 0.5% eye drops, 10 mL (PBS Restricted)***
  - Termolabile ➡ loses effectiveness with heat; keep refrigerated
- ***Chloramphenicol (Minims Chloramphenicol) 0.5%, 0.5 mL (single use), 20***
  - Thermolabile
- ***Chloramphenicol (Chlorsig) 1% eye ointment, 4g***
  - Keep at room temp

- **Chlorig (Aspen) ointment 1%, 4g tube**
- **Cholfen (Alphapharm) ointment 1%, 5g tube**
- **Chlorsig (Aspen) eye drops 0.5%, 10 mL bottle ➡ PBS RESTRICTED (Aboriginal + Torres Strait)**
- **Cholfen (Alphapharm), Ocuzo (AFT Pharmaceuticals) eye drops 0.5%, 10 mL bottle (Generic Brands Available)**
- **Minims Chloramphenicol (B&L) eye drops, 0.5%, 0.5mL (single use), 20 per pack ➡ non-preserved**

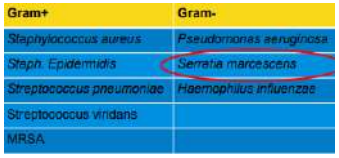
Antibacterial > Binds to 50S r-RNA and inhibits formation of peptide bond (**inhibits peptidyl translocation)** > Prevents activity of peptidyl transferase.  	> Broad spectrum - most g+ve & g-ve > Bacteriostatic > Very lipophilic ➡ good penetration to AC topically, poor skin penetration > most gram+ and gram-, including: Haemophilus influenza, Neisseria Rickettsia, Chlamydia spirochetes and Mycoplasma > **NOT** effective against Pseudomonas Aeruginosa (gram negative bacterium) > PBS restricted ➡ Aboriginal Torres Strait Islander	**> Bacterial eye infections** **> Blepharitis** **> Bacterial conjunctivitis** **> Other superficial ocular infection** **> Prophylaxis after ocular surgery/corneal abrasion/ superficial trauma** > Superficial ocular infection > SAFE in pregnancy & breastfeeding • Ointment is useful in a paediatric setting • Eyedrops: refrigerate until opened "thermolabile" • Ointment: room temperature is okay	**> Bacterial conj:** **Initial**: 1 drop every two hours (if severe) for 1-2 days **Drops**: 4x daily **Ointment:** 3x daily (or at night adjunct to eye drops) ➡ up to 5 days **Minims** is indicated in event of frequent installation to limit toxicity associated with preserved eye drops **> Blepharitis:** treat inflammation NOT infection massage 1-2x/day ointment on lid margin - 1wk (up to 3wks if needed) **> Prophylaxis:** 1 drop/ointment 4x/day until healed (~3-5 days)	**CONTRAINDICATIONS**: hx of blood dyscrasia, bone marrow problems (e.g. aplastic anaemia) > **NOT** for bacterial keratitis or against pseudo. Aeruginosa (active corneal infection) > Evidence for corneal abrasion lacking - limit use in higher risk of infection; infectious/vegetative matter > potential **competitive inhibition** with other protein synthesis inhibiting drugs like **clindamycin** and **macrolides**	**Common**: stinging/burning after instillation **Infrequent** : unpleasant taste **Rare**: hypersensitivity reactions (e.g. anaphylaxis, dermatitis) • Cases of **aplastic anemia** or other blood dyscrasias **Adv**: unpleasant taste (infrequent), allergic reactions (rare)

Fusidic Acid

- Available commercially oversea
- Require compounding and a prescription from an ophthal or a GP - Not on OBA list
  - ***Compounded as 1% eye ointment (by an ophthal)***

Prevents turnover of elongation factor G from ribosome Primarily g+ve - Staph, Strep and Corynebacterium Bacteriostatic - resistance is frequent	**Indications**: bacterial conjunctivitis.

2.3. Antibacterials

Nucleic Acid Synthesis Inhibitors					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
FLUOROQUINOLONES (Ciprofloxacin, Ofloxacin) → XXXfloxacin (e.g. levofloxacin, moxifloxacin) **1. CIPROFLOXACIN** *a. Ciprofloxacin (CiloQuin or Ciloxan) 0.3% eye drops, 5 mL - PBS restricted (Authority Required)* *b. DEEP CORNEAL ULCERS: Oral ciprofloxacin (Cifran) 500 mg tablet - GP* **2. OFLOXACIN** *a. Ofloxacin (Ocuflox) 0.3% eye drops, 5 mL - PBS restricted (Authority Required)*					
Antibacterial > inhibit bacterial DNA synthesis by blocking DNA gyrase (topoisomerase IV) > Impair DNA replication, transcription and repair > Bactericidal and broad spectrum > Enters bacterial cells easily via porins **(2)** better corneal penetration w/ low dosage > ↑ rate of corneal perforation	> Gram–ve: bacteria, DNA gyrase - target (e.g. Pseudomonas, Serratia) > Gram +ve: topoisomerase IV-target (e.g. Staphylococcus, Streptococcus species) > Broad spectrum > Emerging resistance - US and India > PBS Restricted → Bacterial Keratitis  	> Alternatives - ineffective or contra **> Bacterial eye infections: keratitis, corneal ulcers, severe conjunctivitis** Alt. tx bacterial keratitis: > dual fortified antibiotics (e.g. fortified aminoglycoside + fortified cephalosporin). > SAFE in pregnancy & breastfeeding - low dose for minimal systemic absorption	CIPROFLOXACIN **> Bacterial keratitis:** Day 1 - 1 drop every 15 mins first 6 hrs, then every 30 mins. Day 2 - 1 drop every hr. Day 3 - 1 drop every 4 hrs until healed (at least 14 days) **> Severe bacterial conj:** 1 drop every 2 hrs while awake for 2 days; if improved, qid while awake for 5 days. **> Deep Corneal ulcer:** topical + oral 2x daily OFLOXACIN **> Bacterial keratitis:** Day 1-2: 1 drop every 30 min while awake for 2 days, and at 4 & 6 hrs after going bed Days 3-7: 1 drop hourly for 5 days Day 8 until healed: 1 drop qid (< 21 days) **> Severe bacterial conj:** 1 drop every 4 hrs for 2 days; if improved, qid up to 8 days.	**CONTRA** h/o hypersensitivity w/quinolone, tendon damage **PRECAU**: > Not tested in < 1y/o > Systemic use: tendon problem, photosensitivity > Use with systemic NSAIDS ↑ CNS toxicity > Evidence suggest topical fluoroquinolones **upregulate MMP** → ↑ corneal perforation risk + CNS effects (HA, dizziness, convulsions) → due to GABA receptor (inhibitory neurotransmitter) Ciprofloxacin commonly precipitates on corneal ulcers – can slow epithelial healing (but reversible once tx is stopped) Ofloxacin has better corneal penetration but associated with increased rate of **corneal perforation**	**Common**: mild transient ocular irritation (e.g. burning, stinging) **Infrequent/rare: white corneal precipitates**, keratitis (photophobia, corneal infiltrates and staining), allergic reactions (very rare hypersensitivity, anaphylaxis, Stevens-Johnson syndrome) **Ocular side effects (oral):** uveitis, retinal detachment, diplopia (rare) **(1)** common transient white ppt on corneal ulcers - slow epithelial healing; reversible + unpleasant taste **(2)** occasional ppt on corneal ulcers Systemic fluoroquinolones → diplopia in rare cases → due to "tendinitis" of the EOM
Sulfonamides (topical) • no commercially available ophthalmic preparations					
> Inhibit enzymes which metabolise folate → blocks folate synthesis → bacteria dies > Folate found in out diet > Bacteriostatic			> Commonly associated with **Stevens-Johnson syndrome** (one of the possible forms of **erythema multiforme)** - acute, non-progressive inflammatory disorder of the skin and mucous membranes → can be fatal > Allergies to sulfonamides very common - widespread resistance, use declined > May cause **erythema multiforme** → an acute, self-limited, non progressive inflammatory disorder of the skin and mucous membranes		

Cell Wall Synthesis Inhibitors - no commercially available topical ophthalmic preparations					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
BETA LACTAM ANTIBIOTICS (penicillins, cephalosporins, carbapenems and monobactams)					

- No commercially available topical ophthalmic preparations in Australia

1. PENICILLINS
    a. **AMOXICILLIN**
        i. Amoxicillin tablets 1000 mg scored (Maxamox)
        ii. Combined preparation: Amoxycillin tablets 500mg, clavulanic acid 125mg (Augmentin Duo) ➔ effective - broader spectrum
        iii. Amoxycillin capsules 250mg or 500 mg (Alphamox, Amiloyn, Amoxil, Cilamox) ➔ also available as an oral liquid or injection powder
    b. CEPHALOSPORINS
        i. Cefazolin on OBA
        ii. Cephalothin 5% eye drops (not on OBA list) ➔ compounding by ophthal
    c. Penicillin G - 0.3% eye drops ➔ compounding by ophthal (rare)
2. CARBAPENEMS
3. MONOBACTAMS

(1) interferes w bacterial cell wall peptidoglycan synthesis by binding to penicillin binding proteins ➔ cell death **(2)** interferes with linking of peptidoglycan layers > Bactericidal	Resistance common - **β-lactamase inhibitors** (e.g. clavulanic acid) given in combo with β-lactam antibiotics **Amoxicillin/Penicillin (oral):** narrow/mod/broad range ➔ NOTE: amoxicillin + clavulanic acid is broad range **Cefazolin (oral):** G+ve **Penicillin G:** G+ve Amoxicillin & Cefazolin: SAFE in pregnancy/breastfeeding	AMOXICILLIN > **hordeolum** **> preseptal cellulitis,** **> dacryocystitis** > other g+ve CEFAZOLIN: **> Gonococcal infection,** **> Staphylococcal/ streptococcal infections ➔ px with penicillin allergy,** > Surgical prophylaxis > UTI	CEFAZOLIN Loading dose of one drop (fortified) every 10 mins for 1-2 hrs, then every hr w combo drop (fortified gentamicin 1.5%) hourly on the half hour	CONTRA: **Amoxicillin** - h/o allergy to penicillin Cefazolin - Hypersensitivity to cephalosporins/penicillin. PRECAU: < 1 month old, warfarin, aminoglycosides PENICILLIN ALLERGY: cross-reactivity between cephalosporin and penicillins	Penicillins generally well tolerated **Common**: diarrhoea, N, pain and inflammation at injection site superinfection. Allergy **Infrequent**: Vomiting

Vancomycin (NOT OBA - not commercially available)

- Can be compounded as 2.5% eye drops by **ophthal**

> inhibit bacterial cell wall synthesis by preventing formation of peptidoglycan polymers > Glycopeptide antibiotic > Bactericidal > Primarily g+ve	> Serious infections by susceptible organisms **> resistance to penicillins** **> severe bacterial keratitis** > endophthalmitis	PRECAUTIONS: h/o blood disorders, hearing/renal impairment, allergy to teicoplanin, inflammatory GI conditions, toxicity in elderly	Oral - indigestion, nausea, vomiting, diarrhoea and chills. **Common**: local pain (may be severe), thrombophlebitis **Rare**: interstitial nephritis, serious skin reactions, eg linear IgA bullous disease, Clostridioides difficile-associated disease, chemical peritonitis (intra-peritoneal use)

Bacitracin (OBA - not commercially available, needs compounding)

- Compound as 500u/mL ophthalmic solution

Polypeptide antibiotic, Inhibits cell wall synthesis. Primarily g+ve

Penicillin G (needs compounded by opthal, e.g. 0.3% eye drops or 500u/ml solution)

Beta-lactam antibiotic, primarily gram+ve

Polymyxins and gramicidin (OBA - not commercially available, needs compounding)

Disrupt cell wall/membrane. Primarily g+ve

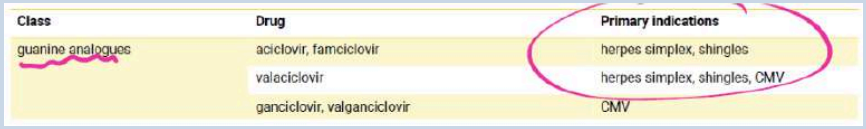
2.4. Antivirals and Anti-infectives

Anti-amoebal, antifungals and antiseptics

Mode of Action	Spectrum	Indications	Dosage	Contraindications + Precau	Side Effects
PROPAMIDINE/DIBROMOPROPAMIDINE (other antibacterial) • ***Brolene (Sanofi-Aventis): 0.1% eye drops (propamidine), 10mL bottle OR 0.15% ointment (dibromopropamidine), 5g tube***					
Antiseptic > Prevents replication & induces encystment of Acan	> Not cell type selective > Amoebistatic > Very weakly amoebicidal - disrupt trophozoite membrane & cyst walls → amoeba death.	**> Acanthamoeba keratitis (refer to ophthal)** **> mild acute conjunctivitis** SAFE to use in pregnancy and breastfeeding	**> Acanthamoeba keratitis:** hourly drops for several months (combined w cationic antiseptic) **> Mild acute conjunctivitis:** 3-4x/day	**Common**: stinging and burning after instillation **Infrequent**: corneal epithelial microcysts (prolonged use), conjunctival hyperaemia, punctate corneal erosions, local allergic reactions (BAK preservative in drops) **Significant toxicity when used hourly over a long term	
Cationic antiseptics - need compounding • ***Chlorhexidine gluconate 0.02% eye drops*** • ***Polyhexamethylene biguanide ophthalmic suspension 0.02%***					
Antiseptic and anti-amoeba		**> Acanthamoeba keratitis -** used in combo, w propamidine		Similar to propamidine	
Povidone-Iodine - no ophthalmic preparations available in AU • Different types of preparation under trade name Betadine e.g. pre-surgical skin scrubs, mouth gargles, ointments					
Antiseptic > Disrupts cell membrane		**> Adenovirus (not clinically proven - off label use)** **> Single use in-office "rinse" 5% conc - prior to cataract surgery as prophylaxis against secondary infections**		CONTRA: Iodine allergy SE: Stinging, burning, conjunctival hyperaemia, punctate corneal erosions	
Topical Interferon alfa-2b - not widely available • Compounded as formulated injections - 1 million IU/mL IFN alfa-2b					
> Triggers immune system, activates antiviral proteins → inhibit viral replication once virus enters > Protein (cytokine) activates intracellular pathway - upregulates host genes affecting antiviral responses > Enhances phagocytosis by macrophages and augments cytotoxicity of lymphocytes > Interferon and a nucleoside antiviral may have synergistic interaction			**Topically:** **> Ocular surface squamous neoplasia** **> HSV epithelial keratitis** **> Systemic injections e.g. cancer** Use in combo with antivirals to manage HSV keratitis		
Antifungals • *Natamycin 5% + Voriconazole 1%*					

- Fungi eukaryotes containing a distinct nucleus with the cell's DNA
- Can be uni or multicellular
- **Yeast**: round, unicellular & slightly larger than bacteria
- **Molds**: multicellular & consists of long branching filaments (hyphae) that intertwine & branch
- **Antifungals** consists of 4 major classes:
    - **Polyenes**: bind the ergosterol in the fungal cell membrane, increasing cell permeability
        - 0.15% to 0.5% **Amphotericin B** (against candida, also effective against aspergillus, penicillium, cryptococcus, fusarium [less effective])
        - 5% Natamycin (in US as a suspension; effective against filamentous fungi e.g. Fusarium, aspergillus)
        - Often combined with Azoles (below)
    - **Azoles**: impairs biosynthesis of ergosterol in fungal cell membrane & increases cell permeability through the fungal cytochrome P450 enzymes → fungistatic
        - Voriconazole 1%

        - Itraconazole 1% ointment
        - Miconazole 1% eye drops
    - **Pyrimidines**: block thymidine synthesis in fungal organisms which impairs DNA & protein synthesis ➡ fungistatic
    - **Echinocandins**: inhibit glucan synthesis, causing abnormal fungal cell wall
- Antifungals are not listed on OBA ➡ not commercially available as ophthalmic preparations
- Modern treatment of fungal keratitis would involve a combination of polyene & azole
- Fungal infections of the eye required **prolonged treatment**

Guanine Analogues					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
<u>GUANINE ANALOGUES (ORAL)</u> **1. ACICLOVIR** ***a. Aciclovir (Aciclovir APOTEX) 200 mg tablet, 90*** ***b. Aciclovir, 800 mg, tablet, 35*** **2. FAMCICLOVIR** a. Famciclovir (Famvir) 125 mg tablet, 40 b. Famciclovir 250 mg tablets (**Famvir, Ezovir, Favic**, Famlo), 500 mg tablets (Famvir, Favic, Ezovir, Elovax) **3. VALACYCLOVIR** a. Valaciclovir (Vaclovir) 500 mg tablet, 30 ⇒ Valtex, Zelitrex, Valacor, Shilova b. Valaciclovir tablets 1g (Valtex) **4. GANCICLOVIR** a. Ganciclovir (Cymevene) 500 mg injection, 5 vials  					
> Suppress viral DNA replication ➡ inhibit viral DNA polymerase + DNA synthesis. > Polymerase inhibitors - act on nucleic acid synthesis Counseling: > Drink lots of fluids > No machinery - meds may make you dizzy	**(2)** & **(3)**: better oral bioavailability + longer H-L compared to **(1)** **(1)** poor GI absorption **(3)** is a prodrug of **(1)**	> Tx/prevent **herpes simplex infections ➡** (1) (2 - preferred) (3 - preferred) Reduced by 50% HSV keratitis > Tx/prevent **CMV ➡** (3) (4 - preferred) PREGNANCY (1) safe (2) (3) - limited data (4) - contraindicated BREASTFEEDING (1) (3) Safe (2) - limited data (4) - contraindicated	<u>ACICLOVIR</u> **> Herpes simplex: Infection** - 400mg 5x day until resolution (~5-10 days) **Prophylaxis** - 400mg 2x daily for 6 months (studies found to reduce recurrence by 50% HSV keratitis) > **Herpes zoster:** 800mg 5x day until resolution (~7-10 days) <u>FAMCICLOVIR</u> **> Initial infection:** 250 mg 3x day for 5–10 days. **> Recurrent infection:** 125 mg 2x daily for 5 days **> Prevention:** 250 mg 2x daily. <u>VALACICLOVIR</u> **> Genital herpes:** Initial - 500 mg 2x daily for 5–10 days. **Recurrent:** 500 mg 2x daily for 3 days. **> Initial/recurrent infection,** HIV +ve: oral 1 g 2x for 5–14 days. **> Shingles:** 1 g 3x daily for 7 days. > Prevention of CMV: 2 g 4x daily for 90 days. <u>GANCICLOVIR</u> - **CMV retinitis** **> Induction**, IV 5 mg/kg every 12rs for 14–21 days. **> Maintenance,** IV 6 mg/kg once daily for 5 days/wk; or 10 mg/kg once daily 3x/wk; or 5 mg/kg once daily every day.	PRECAUTIONS: > Neurological abnormalities: ⬆ risk of reversible encephalopathic changes (e.g. seizures, hallucinations) > Renal impairment ⬆ risk of nephrotoxicity + neurological adverse effects > Elderly ⬆ risk of neurological adverse effects **Common**: nausea, vomiting, diarrhoea, HA,
<u>ACICLOVIR (OCULAR)</u> • ***Aciclovir (ViruPOS or Xorox) 3% eye ointment, 4.5 g tube***					
> **Prodrug** - converted by phosphorylation of herpes thymidine kinase ➡ allows selective inhibition of viral DNA polymerase		> Selectively taken up by HSV > Rare resistance	**> HSV, VZV** **> epithelial keratitis**	Apply 1 cm ointment into lower conjunctival sac 5x daily;	**Common**: transient mild stinging after instillation

> 30x more potent against virus enzyme than host enzyme ➡ limited host toxicity > SAFE - pregnancy and breastfeeding > Poor gastric absorption	> Good penetration - high conc. In aq. Humour **> NOT effective against adenovirus**	**> some CMVs** HSV - oral and topical best HZV - oral best	continue for 3 days after epithelium is healed Keep spare tube at home	**Infrequent**: SPK, allergic reactions Avoid direct eye contact	
GANCICLOVIR (OCULAR) - Needs compounding, not commercially available ➡ refer to OPHTHAL ● ***Idoxuridine 0.1% eye drops; 0.5% ointment*** ● ***Trifluorothymidine (1% eye drops) can be compounded for HSV keratitis*** ● ***Cidofovir (Vistideis) - CMV injectable Rx can be compounded as 2.5mg/mL eye drops*** ● Available In Europe as Virgan for HSV keratitis - **0.15% eye gel**					
> Phosphorylated to ganciclovir monophosphate by a CMV viral kinase	**CMV retinitis** - immunocompromised px or after transplant **Induction**: IV 5 mg/kg every 12 hrs for 14–21 days. **Maintenance**: IV 6 mg/kg once daily 5 days/wk; or 10 mg/kg once daily 3x/wk; or 5 mg/kg once daily/day.	CONTRAINDICATION: allergy to ganciclovir, tx w/ imipenem - increases risk of seizures AVOID in pregnancy & breastfeeding **Common**: anaemia, fever, N, V, anorexia, dyspepsia, diarrhoea, abdominal pain, raised liver enzymes, HA, confusion, hallucinations, seizures,, sweating, rash, itch **Infrequent**: chest pain, chills, mouth ulceration, cough, dry mouth, drowsiness, arthralgia			

2.5. Corticosteroids and Immunomodulatory Medicines

Topical Corticosteroids					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
CORTICOSTEROIDS (Dexamethasone, Prednisolone, Fluorometholone, Hydrocortisone) ➡ REMEMBER TO SHAKE SUSPENSIONS BEFORE USE In order of topical efficacy: **1. Combo drug - prednisolone acetate 1% + phenylephrine hydrochloride 0.12% (Prednefrin Forte) eye drops, 10 mL ➡ suspension** **2. Dexamethasone (Maxidex) 0.1% eye drops, 5 mL ➡ suspension** **3. Fluorometholone acetate (Flarex) 0.1% eye drops, 5 mL ➡ suspension (no longer on PBS)** **4. Fluorometholone (FML Liquifilm) 0.1% eye drops, 5 mL ➡ suspension (no longer on PBS)** **5. Prednisolone sodium phosphate (Prednisolone) 0.5%, 0.5 mL (single use), 20, Minims ➡ refrigerate** ● **Hydrocortisone acetate (Hycor) 1% eye ointment, 5 g**					
Anti-inflammatory + immunosuppressive ➡ regulate gene expression > Intracellular receptor mediated inhibition of inflam. cascade - fibroblast and keratocyte activity. > Upregulate lipocortins (phospholipase A2 inhibitory proteins) ➡ prevents AA from producing cyclo + lipoxygenases **Acetate**: very lipid soluble = enhanced corneal penetration **Salts**: more water soluble = limited corneal penetration Good penetration ≠ good efficacy SAFE in pregnancy & breastfeeding	PREDNISOLONE: *MOST effective anti-inflam. > Low potency & penetrating drug. > **Acetate** (high potency/ penetration/IOP rise) > **Sodium phosphate** (mid potency/IOP rise, low penetration)	**> Acetate: Severe/deep inflam. Of anterior eye - penetrate intact epithelium** > acute anterior uveitis, episcleritis, resistant ocular allergy, post-op inflam > Prednefrin Forte (mild): vasoconstrictive effects	**Usual**: 1 drop 2–6x daily (10-14 days) **Severe inflammation:** 1 drop every hr then gradually ⬇ freq. **Maintenance**: 4x/day	CONTRA: > Active ocular infection /epithelial disease (esp. HSV epithelial/fungal keratitis) PRECAUTIONS: > h/o of HSV keratitis > Glaucoma, ocular HTN, narrow angles > h/o steroid response > Current tx with topical NSAIDs (⬆ risk of delayed healing) > Children: ⬆ susceptibility to ⬆ IOP & cataract. ⬆ risk of adrenal suppression w intensive or long-term use > ⬇ resistance to microbes by dampening immune system - in cornea & AC ➡ **may mask evidence of progression of infection/allow**	**Common**: ocular HTN (reversible) ocular infection, delayed corneal healing, rebound inflam. **Rare**: Cataracts: posterior subcapsular with long-term (>1 year; partially reversible), adrenal suppression (intensive or long-term use) **Steroid Induced IOP Rise** ● Involves a and B receptors of trabecular meshwork cells leading to overproduction of ECM (clogs TM) ● Alteration of actin cross-linking (⬆ rigidity) ● Inhibition of phagocytosis of TM cells ● ⬆ endothelial blood vessel tight junctions ➡ reduces passage into TM
	DEXAMETHASONE: high potency/IOP rise, lower penetration than prednisolone acetate	**Allergic and inflammatory** conditions of anterior eye incl. **Post-op inflam.**	Same as Prednisolone		
	FLUOROMETHOLONE: Forms not interchangeable > **acetate** (more potent, mid penetration) > **normal** (poor penetration) > mid IOP rise for both	**Inflammation of anterior eye** **> Acetate:** severe inflams on ocular; lower penetration **> Normal:** mild conditions	Same as Prednisolone ****ideal when you want to minimise corticosteroid side effects****		
	HYDROCORTISONE OINTMENT:	**> Eyelid inflammation**	**> Eyelid inflam:** 2–4x daily.		

	> unpreserved > Mid penetration, low potency/IOP rise > ⬇ potential adverse effects of inflam. once infection is under control.	**> Adjunct to corticosteroid drops (reduce freq of overnight dosing)**	**> Adjunct to steroid drops:** apply eye at night.	**reactivation**	**Counseling**: > DO NOT undertreat - review often > Rebound inflam ➡ taper > Care in CL px > Side effects more common w long term use > high prevalence of subconjunctival haem in long term steroids to treat chronic uveitis
Loteprednol ● Not commercially available in AU ➡ needs compounding ○ Used in USA and Canada ● ***Lotemax 0.5% suspension***					
> Lipophilic, good penetration to AC > Induction of lipocortins > Activity consistent but less than other potent corticosteroids		> Corticosteroid analogue (similar) of prednisolone > Ester base: Rapidly inactivated after therapeutic effect ➡ minimises significant side effects ➡ lower IOP raise than others		**TGA**: post-op. Inflam (cataract surgery), CL papillary conjunctivitis **Off label:** DED, allergic conjunctivitis, acute anterior uveitis Dosage: 4x daily	
Immunosuppressants					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
CICLOSPORIN - Black Triangle Scheme 1. Ciclosporin (Cequa) 0.09% eye drops, 60 x 0.25 mL ampoules 2. Ciclosporin (Ikervis) 0.1% eye drops, 30 x 0.3 mL ampoules 3. Available commercially as a topical ophthalmic preparation in the USA under brand Restasis - accessed by some px under PBS Special Access Scheme					
Calcineurin inhibitor: > binds to cyclophilin within T-cell lymphocytes > blocks action of calcineurin in activated T cells > Prevents production of IL-2 and other cytokines ➡ normally amplify immune response > Suppresses immune system's response to antigens; ⬆ tear production	> Onset may be delayed (1 month) > Modest evidence of effectiveness at 3 months >Long term safety/efficacy lacking ➡ monitor closely and review 6 monthly	Usually - orally to prevent rejection in transplant px **Moderate-severe dry eye -** lubricants insufficient	Cequa: 1 drop 2x daily Ikervis: 1 drop 1x daily at bedtime	PRECAUTIONS: peri ocular /ocular infections, uncontrolled HTN, malignancy – may worsen > Low topical absorption - pregnancy & breastfeeding (consult)	Common: transient stinging/ burning on instillation, conjunctival hyperaemia
LIFITEGRAST - Black Triangle Scheme ● Lifitegrast (Xiidra), 5%, 0.2 mL (single use), 60, eye drop ● New - approved for use in AU in 2019 ➡ withdrawn in AU marketplace in 2022. ● Can potentially still be compounded					
> Lymphocyte function associated antigen-1 (LFA-1) antagonist ➡ prevents	> Suppress inflam. response associated with DED. > Improvement may be seen	**Moderate-severe DED -** lubricants insufficient	1 drop 2x daily Store single use in o.g foil pouch until used > active	> Safety in <18y/o, pregnant and lactating women is not established	**Very common (5-25%):** ⬇ VA instillation site irritation, metallic taste

signalling to ICAM-1 and T-cell from adhering to the ocular surface. > inhibits release of cytokines and other inflam mediators.	within 14 days of tx; assess after 2–3 months. regular review for ongoing tx		ingredient degrade when exposed to light		**Common**: blurred vision, conjunctival hyperemia, HA, ⬆ lacrimation, sinusitis, eye discharge/discomfort/pruritus/irritation

3.1. NSAIDs

NSAIDs (Non-Steroidal Anti-Inflammatory Drugs)					
Mode of Action	**Spectrum**	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
TOPICAL NSAIDS (Ketorolac, Diclofenac and Flurbiprofen) 1. KETOROLAC a. **Ketorolac (Acular) 0.5% eye drop, 5mL. Protect from light.** 2. DICLOFENAC - not commercially available a. Compound in 0.1% eye drop formulation 3. FLURBIPROFEN - not commercially available 4. NEPAFENAC - Ilevro 0.3% eyedrops, 3ml. Suspension. **NOT ON OBA** a. Very lipophilic - really good corneal penetration. Prodrug and a suspension ➡ recommended in the setting of cataract surgery					
Analgesics, antipyretics and anti-inflammatory effects **> Stronger anti-inflam in acetic acids (1) (2)** **> Inhibit COX-1 and COX-2** by preventing prostaglandin (PT) synthesis and PT-mediated inflammation. > COX-1 inhibition = impaired gastric cytoprotection and antiplatelet effects. > COX-2 inhibition = anti inflam. and analgesic actions	COX 1 and 2 inhibition = ⬇ glomerular filtration rate & renal blood flow - ⬇ risk of peptic ulceration > Prevent ⬆ vasodilation, vascular permeability & oedema with inflam **(1)** - longer H-L **(1) (2)** are non selective > Synergistic activity w topical steroids post-op > Good corneal penetration to AC (& vitreous?)	**> Inflam. eye conditions.** **> Seasonal allergic conjunctivitis (short term), GPC, inflamed pterygium,** **> Surgical settings e.g. post op** **> ocular pain (post refractive surgery, abrasion, FB).** **> Prevent/tx CMO** (1) mild-mod postop pain, SAC (2) **Rheumatoid arthritis**, Osteoarthritis, Local pain and inflam. in soft tissues (topical)	**4x a day** **Counseling**: 60% respond to one NSAID but not the other > no response within 10 days ➡ try another NSAiD > DO NOT use more than one at a time > Pregnancy ➡ seek advice, ⬆ risk of miscarriage > SAFE in breastfeeding - non selective and ibuprofen preferred	CONTRA: H/o aspirin/NSAID induced hypersensitivity PRECAUTIONS: > concurrent use w/ ocular corticosteroids; delay healing. > may ⬆/⬇ effect of PT analogues on IOP; monitor. > existing ocular surface diseases, i.e. concurrent fluoroquinolones/steroid use **Topical NSAIDs may mask symptoms of ocular infections/delay healing**	**Common**: stinging on instillation, ocular irritation, delayed healing. **Infrequent**: local allergic reaction (skin irritation, erythema, itching, rash). **Rare**: dyspnoea, exacerbation of asthma, systemic allergic reactions.

Nonselective NSAIDs (oral)

Drug	Half-life (hours)	Oral dose/day	Routes	Comments	Brand name
Aspirin[1]	0.25 (2-19)[2]	3 or 4	Oral	Nonselective; analgesic & antiplatelet agent (inhibits platelet COX for life of platelet unlike other nonselective NSAIDs)	Aspro Clear, Disprin, Solprin, etc.
Diclofenac[1]	1-2	2 or 3	Oral, rectal, topical	Nonselective; risk of cardiovascular events appears highest	Voltaren, Difenac, Clonac, Fenac, Vidofen, Cambia
Ibuprofen[1]	2-2.5	3 or 4 (2[3])	Oral, topical	Nonselective	Advil, Bugesic, Nurofen, Brufen, Ibuprofen, Rafen, FenPaed
Indomethacin	4.5-8	2-4	Oral, rectal	Nonselective	Arthrexin, Indocid
Ketorolac	4-8	4	IM, oral	Nonselective	Toradol
Mefenamic acid[1]	3-4	3	Oral	Nonselective	Ponstan, Fenin
Naproxen[1]	12-15	2 (1)[3]	Oral	Nonselective	Inza, Naproxen, Naprosyn, Proxen SR, Naprogesic, Anaprox, Crysanal
Piroxicam	30-50	1	Oral	Nonselective; severe skin reactions may be more common	Mobilis, Feldene, Piroxicam

Classes of NSAIDs

Chemical class	Examples	Analgesic	Antipyretic	Anti-inflammatory
Salicylic acids	Aspirin	+	+	+
Propionic acids	Ibuprofen Flurbiprofen	+	+	+
Acetic acids	Indometacin Diclofenac Ketorolac *Nepafenac*	+	+	++
Oxicams	Piroxicam	+	+	++
Pyrazoles	Phenylbutazone	+/-	+	++
Fenamates	Mefenamic acid	+	+	+/-
Para-Aminophenols	Paracetamol	+	+	-

Nonselective NSAIDs (COX-1 and COX-2 inhibitors
Aspirin, diclofenac, ibuprofen, indomethacin, ketoprofen, ketorolac, mefenamic acid, naproxen, piroxicam

Selective NSAIDs (COX-2 inhibitors)
Celecoxib, etoricoxib, meloxicam, parecoxib

Pink: OTC

3.2. Anti-Allergic Medicines

Anti-allergic				
Mode of Action	**Indications**	**Dosage**	**Contraindications + Precau**	**Side Effects**
ANTIHISTAMINES (Oral) ● **Sedating** ○ E.g. Polaramine, Allersoothe, Nausicalm ● Less sedating ○ E.g. cetirizine or Zyrtec, loratadine or Claratyne and fexofenadine or Telfast and Fexo.				
> Sedating (older) - more lipid soluble, penetrate blood brain barrier effectively. Anticholinergic activity: alpha-blocking or anti-serotonin activity ➡ CNS effects (depress/stimulate) **> Less sedating (newer)** - less lipid soluble, no CNS effects. Preferred.. > Binds to H1 receptors and stabilises it in an inactive form	**Sedating**: > Allergic conditions (rhinitis, urticaria) > Itch, N and V e.g. motion sickness **Less sedating:** > Allergic rhinitis > chronic urticaria. **Dosage: 1-2x a day (long H-L)** SAFE in pregnancy and breastfeeding (short term only in sedating) > Response also varies per person	PRECAUTIONS **Sedating**: GI and bladder outlet obstructions, AAC crisis may be rarely precipitated. Avoid in elderly and <2y/o **Less sedating:** ↑ risk of sedation & anticholinergic effects in elderly (preferred to older sedating); monitor carefully ➡ more likely with cetirizine or Zyrtec & levocetirizine **Counseling**: “may make you sleepy” > Don’t drive or operate machinery > Avoid alcohol/meds	SE - Sedating **Common**: sedation, psychomotor impairment, dizziness, confusion, HA, blurred vision, mydriasis, dry eyes/mouth, constipation,, urinary retention Infrequent : nausea, vomiting, diarrhoea, hypotension **Rare**: leucopenia, agranulocytosis, haemolytic anaemia, allergic reactions, arrhythmias, dyskinesia, hallucinations, elevated liver enzymes Psychomotor impairment: impaired alertness, cognition, learning, memory and performance. May continue day after a dose and can occur in the absence of sedation Paradoxical stimulation: CNS stimulation (e.g. excitation, hallucinations, seizures) rarely, especially in children **SE - Less Sedating:** **Common**: drowsiness, fatigue, headache, nausea, dry mouth **Infrequent**: elevated liver enzymes, weight gain **Rare** : rash, hypersensitivity (eg anaphylaxis, bronchospasm)	
ANTIHISTAMINES (Topical) - Antazoline, Levocabastine, Pheniramine **1. ANTAZOLINE - OTC** **2. LEVOCABASTINE - pharmacy** ***a. Levocabastine (Livostin) 0.05%, 4 mL, (suspension)*** i. Age 6+yrs b. Zyrtec Levocabastine 0.05% suspension, 4ml bottle **3. PHENIRAMINE - OTC** 4. Eye drops on AMH are combined with vasoconstrictor - naphazoline a. Not recommended - benefit doubtful and rebound hyperaemia leads to overuse				
2nd gen H1 antagonists - target histamine receptors in conjunctival BV and ocular mucous membranes	**Allergic conjunctivitis** – seasonal and perennial conjunctivitis.	2x daily, can be increased to 3-4x SAFE in pregnancy & breastfeeding.	Associated with drowsiness (same as above) ➡ minimal CNS secondary adverse effects	**Common**: stinging on instillation, mild eye irritation, headache.
MAST CELL STABILISERS (Cromoglycate, Lodoxamide) **1. CROMOGLYCATE** **a. ibbCromoglycate (Cromo-Fresh), 2%, 10mL, eye drop** i. Preservative free - good for dry eyes **2. LODOXAMIDE** **a. Lodoxamide (Lomide) 0.1%, 10mL eye drop**				
> Inhibits mast cell degranulation & release of inflam. mediators from mast cells (often remain in tissue for some time) > prevent Ca+ influx across mass cell membrane	**Allergic conjunctivitis (SAC, PAC)** **VKC (vernal keratoconjunctivitis)** SAFE in pregnancy & breastfeeding.	**Dosage: 1 drop 4-6x daily.** Counseling: Takes at least 1 wk for mast cell stabilisers to work > Max effect at 3-6 weeks > Start tx 1 months before onset of hayfever		NO contraindications/precautions Transient stinging on installation.

		> Use in combo with oral/topical antihistamines (or steroids if severe)	

ANTIHISTAMINE + MAST CELL STABILISERS (Azelastine, Ketotifen, Olopatadine)

1. **AZELASTINE**
   a. **Azelastine (Eyezep) 0.05%, 6 mL, eye drop**
      i. Age 3y/o+
2. **KETOTIFEN**
   a. **Ketotifen (Zaditen) 0.025%, 5 mL, eye drop**
   b. **Ketotifen (Zaditen) 0.025%, 0.4 mL (single use)**
      i. Age 4y/o+
3. **OLOPATADINE**
   a. **Olopatadine (Paladopt) 0.1%, 5 mL, eye drop**
      i. Age 4y/o+
      ii. Patanol

Selective H1 antagonist + inhibition of mast cell degranulation. SAFE in pregnancy & breastfeeding.	**Allergic conjunctivitis, SAC, PAC, VKC** **Azelastine**: 1 drop 2x daily, max 4x day **Ketotifen** & Olopatadine: 1 drop 2x daily > Ketotifen - narrow therapeutic range; higher dose may worsen allergies > Olopatadine may have better short term efficacy than ketotifen	**CNS effects**: sedation and muscarinic receptor effects e.g. dry eye and dry mouth **Azelastine** - bitter taste, rarely causes allergy. **Olopatadine** - hyperaemia, keratitis and dry eye.

3.3. Mydriatics and Cycloplegics

Mydriatics and Cycloplegics (atropine, cyclopentolate, homatropine, pilocarpine, phenylephrine and tropicamide)

Mode of Action	Indications	Contrain/Precautions	Side Effects	Practice Pts/Counselling
CYCLOPLEGICS (Topical Anticholinergics) - Atropine, cyclopentolate, homatropine*, tropicamide				
> Reversibly blocks ACh receptors on iris sphincter > Causes BOTH mydriasis and cycloplegia SAFE in pregnancy & breastfeeding.	**1. Mydriasis (dilation):** > DFE exam > Uveitis: prevent/break posterior synechiae, prevent constriction, ↓ leakage of cell and flares from iris vessel into AC **2. Cycloplegia (accom. paralysis):** > pain relief (uveitis, corneal abrasion, severe keratitis), > chemical occlusion (amblyopia), > prevent myopia progression > ciliary block glaucoma (rare).	CONTRAINDICATIONS: > significant head injury > narrow angles, > h/o CAG; can ppt an AAC crisis. > lenticular subluxation - risk of anterior lens displacement during cycloplegia PRECAUTIONS: > darkly pigmented irides: resistant to cycloplegia → higher doses? > diabetes: inadequate mydriasis; consider adjunct phenylephrine **> extreme caution** in neonates, preterm infants and children w spastic paralysis or brain damage	**Common**: intolerance to bright light (glare), stinging on installation (esp w cyclopent. 1%), blurred vision (esp near), transient IOP elevation. **Infrequent**: contact allergic blepharitis, persistent ocular irritation, insomnia. **Rare**: systemic toxicity (dose related), e.g. dryness of skin and mouth, fever, facial flushing. "As blind as a bat, dry as a bone, red as a beetroot and mad as a hatter."	> Punctal occlusion esp. in children and neonates to minimise systemic adverse effects. > Phenylephrine used as an adjunct. > Wear sunnies > Impaired ability to judge distance - caution driving Return immediately if eye becomes painful, red or if vision deteriorates **If ocular pain ↑ with light/accom:** > Test by shining a light in the contralateral eye and if there is increased pain in the affected eye.
PHENYLEPHRINE • Phenylephrine 2.5% eye drop (Minims Phenylephrine), 0.5mL x 20 • Phenylephrine 10% eye drop (Minims Phenylephrine), 0.5mL x 20				
> Relatively selective ∝-1 agonist – stimulates pupil dilator muscle. > Does not affect accom-, MYDRIASIS	**> Diagnostic - mydriasis, conjunctival blanching** **> Relief of mild ocular congestion** **> Difficult to dilate px**	**PRECAUTIONS**: > AAC attack in those at risk > BP elevation with 10% drops w/ repeated doses; contra - recent	**Common**: rebound miosis, hyperaemia, stinging on installation. **Infrequent**: liberation of iris	**1 drop of 2.5%** > Adjunct if mydriasis difficult > Diagnostic vasoconstriction; episcleral vessels should blanch after 5

ONLY > Used ADJUNCT to cycloplegics i.e. tropicamide	Maximal mydriasis occurs after 60-90 mins and duration of action is 5-7 hours. SAFE in breastfeeding.	myocardial infarction, unstable angina, insulin dependent diabetes. > HTN reaction in those treated with MAOI > Elderly + children - higher risk of systemic adverse effects: do NOT use 10% > Avoid in **pregnancy** particularly 10% drops and in the 1st trimester	pigment, allergic contact dermatitis. **Rare**: systemic effects – most common w 10%; HTN, tachycardia, tremor, anxiety, sweating	mins **10%:** > 1 drop for rapid maximal mydriasis in adults > 1 drop up to 4x daily as an adjunct to ACh in uveitis
Indications		**Dosage**		**Counselling**
ATROPINE • ***Atropine (Eikance) 0.01%, 0.3 mL (single use), 30, eye drop*** • ***Atropine sulfate monohydrate (Atropt) 1% eye drops, 15 mL*** • ***Atropine (Minims Atropine) 1%, 0.5 mL (single use), 20, Minims***				
> Cycloplegic refraction: 1 drop, 1hr before exam (adult) or 1 drop once daily for 1-3 days (child >1y/o) **> Uveitis:** 1 drop (1%) 1-2x daily; max 4x daily **> Progressive myopia:** 1 drop (0.01% or 0.05%) diluted atropine in both eyes 1x daily, at night for up to 5 years **> Penalisation therapy in amblyopia:** 1 drop (1%) in unaffected eye 1x daily each Sat and Sun for up to 2 years			"your eyes may remain dilated for several days after stopping the drop" > Safe in keratoplasty and keratoconus px > HIGH systemic adverse effects **Atropine overdose: accidental poisoning. 10mL volume if ingested letha**	
HOMATROPINE • Discontinued but can be compounded - homatropine hydrobromide 2% ophthalmic solution.				
> Uveitis: 1 drop of 2%, 2-4x daily for management of severe pain, up to every 3-4 hours.				Some systemic side effects
CYCLOPENTOLATE • ***Cyclopentolate 0.5%, 0.5 mL (single use), 20, Minims eye drop,*** • ***Cyclopentolate (Cyclogyl) 1%, 15 mL, Cyclogyl eye drop,*** • ***Cyclopentolate 1%, 0.5 mL (single use), 20 Minims eye drop,***				
> Diagnostic use (mydriasis, cycloplegia) **> Uveitis:** 1 drop 0.5% (1% for deeply pigmented eyes) every 6-8 hrs for 3 months **> Cycloplegic refraction:** Adult + child - 1 drop of 0.5% or 1% 30 mins before, (wait 30 mins) repeat in 5–15 minutes if necessary **<1y/o: use 0.5%**				> Medium systemic adverse effects **> Care in px with down syndrome - can exacerbate syndrome**
TROPICAMIDE • Tropicamide, 0.5%, 0.5 mL (single use), 20, Minims eye drop • Tropicamide (Mydriacyl) eye drop, 0.5%, 15 mL, eye drop • Tropicamide, 1%, 0.5 mL (single use), 20, Minims eye drop • Tropicamide (Mydriacyl), 1%, 15 mL, Mydriacyl eye drop				
> Diagnostic use (mydriasis, cycloplegia) NOT for management of pain/AC reactions in uveitis		**> Mydriasis for DFE:** 1 drop 0.5% 20 mins before exam (wait 30 minutes for darkly pigmented eyes). Consider using 1% for darker eyes. **> Cycloplegic refraction:** 1 drop 1% repeat after 5 mins. May need additional drop after 20-30 mins if not examined yet.		> Rare systemic adverse effects **> Care in px with down syndrome - can exacerbate syndrome**

Medicine / use	Peak effect[1]	Duration[1]	Systemic adverse effects
Atropine 1%			
mydriasis	30-40 minutes	7-10 days	+++
cycloplegia	1-3 hours	7-14 days	
Cyclopentolate 0.5-1%			
mydriasis	30-60 minutes	1 day	++
cycloplegia	25-75 minutes	6-24 hours	
Tropicamide 0.5-1%			
mydriasis	20-40 minutes	6 hours	Rare
incomplete cycloplegia	30-40 minutes	2-6 hours	
Homatropine 2% (NOT COMMERCIALLY AVAILABLE)			
Mydriasis	20-30 minutes	6 hours – 4 days	+
Cycloplegia	30-90 minutes	10 hours – 2 days	





Table – Comparison of ocular anticholinergics

Drug/use	Peak effect[1]	Duration[1]	Comments
Atropine 1%			
mydriasis	30–40 minutes	7–10 days	• rarely used for diagnostic indications as it can affect near vision for at least a week after use
cycloplegia	1–3 hours	7–14 days	• may be considered for refraction when maximal cycloplegia is required
Cyclopentolate 0.5–1%			
mydriasis	30–60 minutes	1 day	• preferred for cycloplegic refraction • may be considered for mydriasis if other agents fail
cycloplegia	25–75 minutes	6–24 hours	
Tropicamide 0.5–1%			
mydriasis	20–40 minutes	6 hours	• preferred for mydriasis (may be combined with phenylephrine if additional dilation is required)
incomplete cycloplegia	30–40 minutes	2–6 hours	• may be considered for cycloplegic refraction, although its effects are less reliable

[1] when given as a single dose

3.4.-3.5. Pain Management

Pain Management – Analgesics				
Mode of Action	**Indications**	**Precautions**	**Contraindications**	**Side Effects**
LOCAL (TOPICAL) ANAESTHETICS: Office use only **1. Tetracaine (also amethocaine):** a. Minims Amethocaine 0.5% eyedrops, 20 x 0.5mL (single use) b. Minims Amethocaine 1.0% eyedrops, 20x 0.5mL (single use) **2. Oxybuprocaine (also benoxinate):** a. Minims Oxybuprocaine 0.4% eyedrops, 20x 0.5mL (single use) **3. Proxymetacaine (also proparacaine):** a. Proxymetacine (**Alcaine**) 0.5% eyedrops, 15mL bottle **4. Lidocaine (previously lignocaine):** a. Combo drug: Minims Lidocaine 4% + fluorescein 0.25%, 20x 0.5mL (single use)				
> Block nerve conduction (reversibly) SAFE in pregnancy and breastfeeding.	**> Short term (≤20mins) ocular surface anaesthesia** (in-office to control pain, FB removal, minor surgery, cataract extraction, pterygium surgery) **> ↑ corneal permeability and intraocular bioavailability of other topical drugs** **> ↓ initial stinging of other topical drugs** (e.g. DFE).		> Preservative-free drops for corneal scrapings - affect culture. > Prolonged use can impair corneal healing and mask infections > NEVER PRESCRIBE for home use. **Common**: stinging on installation, punctate epithelial corneal damage. **Rare**: allergy.	
ASPIRIN/ACETYLSALICYLIC ACIDS (Oral) • tab, 300 mg (dispersible), 96, Solprin • tab, 320 mg, 20, Aspro • tab, 100 mg, 112, Astrixa				
Non-selective NSAID **Analgesic, antipyretic, anti-inflam and antiplatelet effects.** > Non-opioid > Unscheduled > Available in multiple formats (coated, suppository). > Antiplatelet - prevent platelets from sticking together, ↓ body's	**Mild to moderate pain, fever, migraines etc.** **Analgesic, anti-inflam/pyretic:** 300-900 mg every 4-6 hrs; **Antiplatelet**: 100mg per day > Take with/after food > Remove tablets from packaging just before use	> Stop aspirin at least 7 days prior to surgery (i.e. cataract) - highlight use in referral > Avoid pregnancy - increase risk of miscarriage + foetal development impairments. > Avoid breastfeeding - risk of Reye's syndrome → Ibuprofen preferred > Avoid in children <16y/o	**CONTRAINDICATIONS**: > Allergic reactions: bronchospasm, angioedema. > Allergic reactions to aspirin/other NSAID > Px with ↑risk of severe bleeding > Avoid drug interactions which may cause – heart failure, uncontrolled HTN, asthma, gout, elderly, Reye's syndrome	**Common**: N, dyspepsia, V, GI ulceration/bleeding, asymptomatic blood less, increasing bleeding time, headaches, dizziness. **Infrequent**: skin reaction, iron deficiency anaemia, renal Impairment. **Rare**: major haemorrhages, blood

ability to form blood clots				dyscrasias.
PARACETAMOL (acetaminophen)				
Analgesic, antipyretic only > Non-opioid > less GI effect, > unscheduled SAFE in pregnancy, breastfeeding and children	**Mild to moderate pain, fever** Preferred to NSAIDs ➡ fewer side-effects. > Can combine w/ NSAIDs > Pain relief ~30mins after admin	> 0.5-1g every 4-6 hours, max dose 4g daily (8 tablets) > 1330 mg controlled release (2x 665mg tablets) every 6-8 hours, maximum dose 3.99 g daily (6 tablets) CONTRAINDICATIONS: > chronic liver disease - ⬆ risk of liver damage following therapeutic dose or overdose of paracetamol; evidence is lacking. > Hepatotoxicity follows paracetamol overdose - px at risk: malnourished, severe renal/hepatic/ cardiac disease, dehydrated or chronic alcohol intake		**Common**: increased Aminotransferases. **Rare**: hypersensitivity reactions, acute hepatitis
NSAIDs/Non opioid analgesics ● **Indications**: Pain due to inflammatory arthropathies (joint diseases), inflammation and tissue injury. ● Pregnancy ➡ seek advice, potential ⬆ risk of miscarriage ● SAFE in breastfeeding - non selective and ibuprofen preferred				
IBUPROFEN ● tab, 200 mg, Advil, Nurofen, Rafen ● tab, 400 mg, 12, 24, Nurofen Double Strength				
Nonselective - analgesic, antipyretic, anti-inflammatory Well tolerated in children and infants (no Reye's syndrome).	**> Pain associated with inflammation (e.g. period pain)** **> Postoperative pain** **> HA, Fever** **> Rheumatoid arthritis, JIA, Osteoarthritis**	> 200-400 mg 3-4x per day (analgesia); > 400-600mg 4x per day (anti-inflammatory); maximum 2.4g per day > Controlled release tablet: 600mg every twice daily > Take with a glass of water - ideally with food > Can reduce antiplatelet activity of low-dose aspirin ➡ choose alt. NSAID if regular use of both is needed.		
DICLOFENAC ● tab, 25 mg, 30, Difenac ● tab, 50 mg, 20, Voltaren Rapid				
Nonselective - analgesic, antipyretic, anti-inflammatory	**> Pain, esp due to inflammation (e.g. period pain),** **> Rheumatoid arthritis, Osteoarthritis**	> 25-50mg, 2-3x per day; maximum 200mg per day > Swallow whole - do not crush or chew > Withdrawn from UK OTC market - reports of cardiovascular adverse effects > Check aminotransferase conc. Periodically. > Infrequent: atherothrombotic events – avoid use if heart failure, angina, peripheral arterial disease or CVD ➡ sudden atherosclerotic plaque disruption (rupture or erosion), and thrombus (blood clot) formation.		
INDOMETACIN ● cap, 25 mg, 50, Arthrexin, Indocid				
Nonselective - analgesic, antipyretic, anti-inflammatory effects.	**> Pain, esp due to inflammation (e.g. period pain),** **> Rheumatoid arthritis, Osteoarthritis.** **> Acute gout** **> Ankylosing spondylitis.**	> 25-50mg, 2-4x per day > Take capsules with or shortly after food. CONTRAINDICATION: Proctitis - inflammation of the lining of the rectum. SIDE EFFECTS: Vertigo is common - affects alertness and coordination. > Avoid driving or operating equipment until you know how you react.		
NAPROXEN ● tab, 250 mg, 50, Inza ● tab, 500 mg, 50, Inza				

Nonselective - analgesic, antipyretic, anti-inflammatory effects.	**> Pain, esp due to inflammation (e.g. period pain),** **> Rheumatoid arthritis, JIA, Osteoarthritis** **> Ankylosing spondylitis.**	> 250-500mg, 2x daily; controlled release product 750-1000 mg once daily; maximum 1250 mg daily > Take with or shortly after food. > Controlled release tablets: swallow whole ➙ Can take at night to relieve nocturnal/morning symptoms. SIDE EFFECTS: Pseudoporphyria - stop if it occurs > Can reduce antiplatelet activity of low-dose aspirin ➙ choose alternative NSAID if regular use of both meds is needed.

OPIOID ANALGESICS

- **MOA**: target opioid receptors in CNS and GIT; produce analgesia respiratory depression, sedation and constipation
  - No ceiling effect (there is one with peripheral acting agents).
- **Indications: severe, acute and/or chronic pain.**
- Most are S8, some are S4
- **Adverse effects:** death through respiratory depression, addiction, constipation, nausea and vomiting, sedation, cognitive impairment
- Pregnancy - can cause respiratory depression in newborn + withdrawal effects on mother
- Breastfeeding - use with caution, avoid codeine.
- Children: initiated/recommended by specialists.
  - < 12 months more susceptible to respiratory depression

CODEINE + NSAIDs - fixed dose combinations:

- Codeine 8mg + aspirin 300mg tablets (Aspalgin)
- Codeine 12.8mg + ibuprofen 200mg tablets (Brufen Plus, Nurofen Plus, Ibuprofen/Codeine, Ibudeine)
- Codeine 8mg + paracetamol 500mg (Panamax)
  - No inflam effects
- Codeine 30mg + paracetamol 500mg (Panadeine Forte, Codalgin Forte, Codapane Forte, Comfarol Forte, Prodeine Forte)
  - No inflam effects

Analgesic (low dose) + antipyretic + anti-inflam. (some preparation). **S4**	**Mild to moderate pain.** DOSAGE: > Codeine + aspirin: 1-2 tablets every 4hrs; max 8 tablets/day > Codeine + ibuprofen: 1-2 tablets every 4 hrs; max 6 tablets/ day > Codeine + paracetamol: 1-2 tablets every 4-6 hrs; max 8 tablets daily	> Narcotic (possible dependence) > Liver toxicity. > Potential for misuse leading to dependence and overdose ➙ Can result in toxicity from the non-opioid analgesic SIDE EFFECTS: GI upset, Dizziness/drowsiness (avoid driving),

TOPICAL/ORAL CORTICOSTEROIDS

- Analgesics and anti-inflam properties
- **Indications: SEVERE inflam. Conditions; significant side effect**

Act higher up in cyclooxygenase pathway

  - More effective anti-inflam. And analgesic properties than NSAIDs
- **Dosage**: 40mg/day

ANTIDEPRESSANTS AND ANTIEPILEPTICS (neuropathic pain)

- **Tricyclic antidepressants (TCAs)**
  - Amitriptyline (Endept, Entrip) 10-25 mg at night, titrate up to maximum 150mg
  - Duloxetine (painful diabetic neuropathy) (Andepra, Cymbalta, Depreta, Dytrex, Tixol) 60mg once daily
    - Effective in painful diabetic neuropathy
  - Evidence limited + conflicting
- **Antiepileptics**
  - Gabapentin (Gapentin, Gabapentin, Neurontin, Nupentin, Gabacor) 100- 300 mg at night, increase dose graduate according to response (up to 1.8- 3.6g daily in 3 doses usual)
  - Pregabalin (Lypralin, Lyrica, Lyzalon) 75 mg at night, increase to 150mg daily in 1-2 doses (maximum 300 mg twice daily)
  - Carbamazepine (Tegretol) 50-100 mg twice daily increase gradually to 400-800 mg daily in 2-4 doses. Up to 1.2 g daily.

> Complaints of symptoms out of proportion to clinical signs > Injury is to the nerves themselves and not a tissue injury.	Determine whether peripheral or central signalling - in office trial of topical anaesthetic to reduce pain signalling: • Pain disappears w/ topical anaesthetic use = **Peripherally** present. • Pain remains present = **centrally** present.	**> Post-herpetic neuralgia, Neuropathic dry eye, Post refractive surgery** > Consider when all other treatments have failed > Combined with non-therapeutic tx

	○ Allodynia and hyperalgesia are common features of neuropathic pain.	> If they fail to respond to one - may respond to another in same class > Combining 2x meds from diff classes may be more effective. > Referral to pain specialist and/or ophthal if required.

Comparison of ocular local anaesthetics

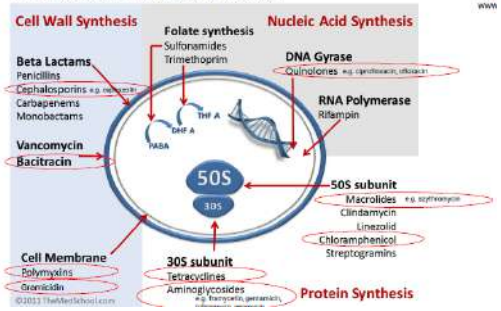
Drug	Type	Onset of action	Duration	Duration of initial sting	Punctate epithelial damage
Tetracaine	ester	20 seconds	20 minutes	30 seconds	+++
Oxybuprocaine	ester	20 seconds	10-20 minutes	10 seconds	++
Proxymetacaine	ester	20 seconds	10-20 minutes	10 seconds	++
Lidocaine	amide	2-5 minutes	20-30 minutes	30 seconds	+

Drugs	Acute pain (nociceptive)	Neuropathic pain	Chronic pain (nociceptive with / without neuropathic)
Paracetamol	Effective*	Less effective	Effective
NSAIDs	Effective†	Less effective	Effective
Antidepressants (e.g.low dose TCAs, duloxetine)	Rarely used	May be effective; first and second line	May be effective if neuropathic component
Antiepileptics (e.g. gabapentin, pregabalin, carbamazepine)			
Opioids	Effective	May be effective; third line	May be effective **but** limited or no benefit long-term (>90 days) for non-cancer pain

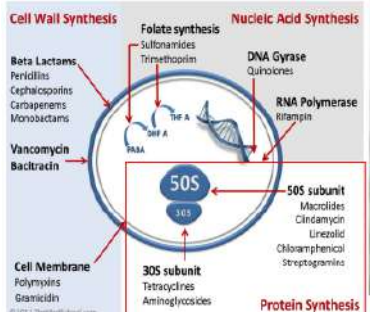
Drug and form	Potency[1]	Penetration[1]	IOP rise	Brand Name
Prednisolone acetate suspension, 0.5%	high	+++ go thru intact epi	++++	Prednefrin Forte
Dexamethasone suspension, 0.1%	high	++ for ocular surface	++++	Maxidex
Fluorometholone acetate suspension, 0.1%	high	++	+++	Flarex
Prednisolone sodium phosphate solution, 0.5%	mid	+	+++	Minims Prednisolone
Fluorometholone suspension, 0.1%	mid	-	+++	FML
Hydrocortisone acetate ointment 1%	low	++	++	Hycor

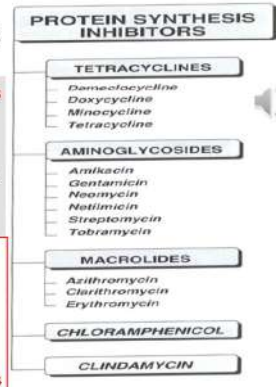
Class	Action	Antimicrobial effect
Fluoroquinolones	DNA gyrase	Broad spectrum (G-ve>G+ve)
Aminoglycosides	Protein synthesis	G-ve
β Lactams (penicillins & cephalosporins)	Cell wall structure	G+ve
Macrolides	Protein synthesis	G+ve/Chlamydia
Tetracyclines	Protein synthesis	G+ve
Chloramphenicol	Protein synthesis	G+ve
Fusidic acid	Protein synthesis	G+ve
Sulphonamides	Metabolism	G+ve

Antibiotics: Mode of action



Protein Synthesis Inhibitors



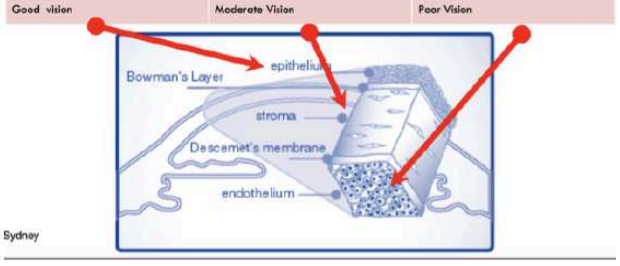


Approved Optometrist Rx Anti-inflammatory Medicines

		Potency
Dexamethasone (PBS) Prednisolone (PBS restricted) Lotepredno l * Fluorometholone Hydrocortisone (PBS)	Corticosteroid	+++ to — Potency
Diclofenac * Flurbiprofen * Ketorolac § (see note)	NSAID	
Ciclosporin (PBS authority) Lifitegrast *	Immunosuppressive	

*Not commercially available – compound Rx
§ Note: **Nepafenac** is an NSAIDs suspension eyedrop commercially available in Australia but not on OBA list

Epithelium	Stroma	Endothelium
Spots/Ulcers/Dendrites Infiltrates	Wesley Ring, Stromal infiltrate, Interstitial K.	KP Oedema
FB Sensation	Pain	Misty vision
Glare	Glare	
Epiphora		
Good vision	Moderate Vision	Poor Vision



Tissue	Histamine Receptor	Effects / Function
Vascular system		
Capillary (microcirculation)	H_1 and H_2	Dilation, increased permeability
Arteriole (smooth muscle)	H_1 and H_2	Dilation (allergic response)
Smooth muscle		
Bronchial, bronchiolar	H_1	Contraction (bronchospasm)
Gastrointestinal	H_1	Contraction
Exocrine glands		
Gastric	H_2	Gastric acid secretion (HCl)
Epidermis	H_1	Triple response (flush, flare, wheal)
Adrenal medulla	H_1	Adrenaline and noradrenaline release
Central nervous system	H_1 H_3	Motion sickness, nausea, alertness Decrease release of neurotransmitters
Intestine Blood leukocytes	H_4	Unknown

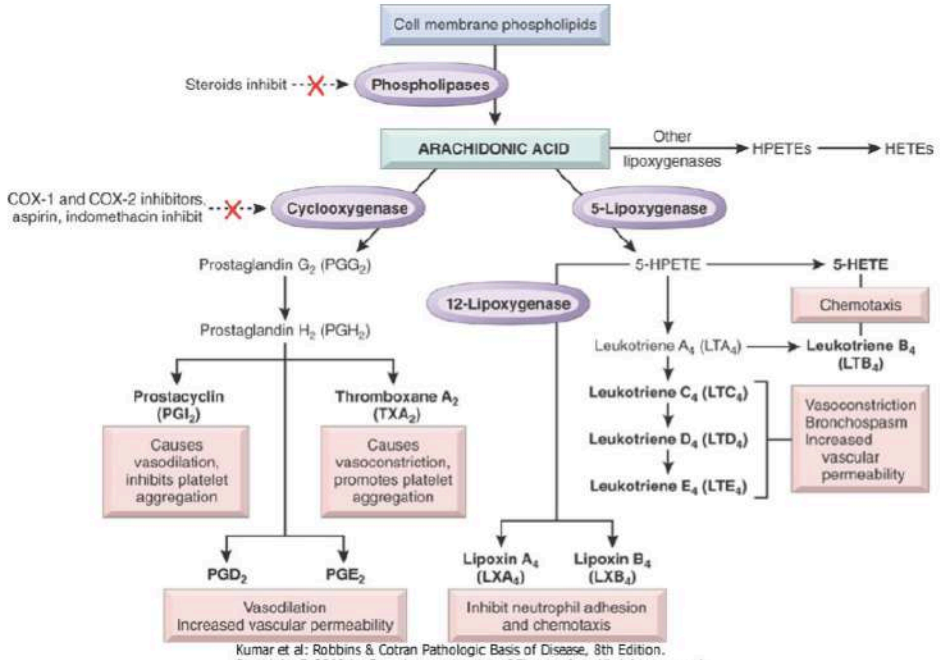
Sympathomimetics - enhances adrenergic function/receptors
- increase heart rate, blood pressure, and dilate bronchioles
- Can also affect other systems such as the central nervous system, gastrointestinal tract, and smooth muscles.

Sympatholytics - interrupts/blocks adrenergic function/receptors
- Reduce heart rate, blood pressure, and smooth muscle tone
- Can have effects on other systems like the central nervous system and gastrointestinal tract
- Mainly for CVD.

Parasympathomimetics - activate the parasympathetic nervous system by mimicking or modifying the effects of acetylcholine -> rest and digest
- Decreased heart rate (bradycardia)
- Increased secretion of saliva, sweat, and digestive juices
- Constriction of the pupils (miosis)
- Constriction of the bronchi
- Increased gastrointestinal motility

Parasympatholytics - inhibit the parasympathetic systems + acetylcholine effects -> flight or fight
- Increased heart rate (tachycardia)
- Decreased secretion of saliva, sweat, and digestive juices (dry mouth, reduced sweating)
- Dilation of the pupils (mydriasis)
- Relaxation of the bronchi
- Decreased gastrointestinal motility

E.g. atropine



Kumar et al: Robbins & Cotran Pathologic Basis of Disease, 8th Edition.
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Drug	Half-life (hours)	Oral doses/day	Routes	Comments	Brand name
Aspirin[1]	0.25 (2-19)[2]	3 or 4	Oral	Nonselective; analgesic & antiplatelet agent (inhibits platelet COX for life of platelet unlike other nonselective NSAIDs)	Aspro Clear, Disprin, Solprin, etc.
Diclofenac[1]	1-2	2 or 3	Oral, rectal, topical	Nonselective; risk of cardiovascular events appears highest	Voltaren, Difenac, Clonac, Fenac, Viclofen, Cambia
Ibuprofen[1]	2-2.5	3 or 4 ([3])	Oral, topical	Nonselective	Advil, Bugesic, Nurofen, Brufen, Ibuprofen, Rafen, FenPaed
Indomethacin	4.5-6	2-4	Oral, rectal	Nonselective	Arthrexin, Indocid
Ketoprofen	1.5-2	1[3]	Oral	Nonselective	Orudis, Oruvail
Ketorolac	4-6	4	IM, oral	Nonselective	Toradol, Ketoral
Mefenamic acid[1]	3-4	3	Oral	Nonselective	Ponstan, Femin
Naproxen / Naproxen sodium[1]	12-15	2 (1)[3]	Oral	Nonselective	Inza, Naproxen, Naprosyn, Proxen SR, Naprogesic, Anaprox, Crysanal
Piroxicam	30-50	1	Oral, topical	Nonselective; severe skin reactions may be more common	Mobilis, Feldene, Piroxicam

Role in inflammation	Mediators
Vasodilation	**Prostaglandins** Nitric oxide Histamine
Increased vascular permeability	Histamine and serotonin C3a and C5a (by liberating vasoactive amines from mast cells, other cells) Bradykinin **Leukotrienes C4, D4,** E4 PAF Substance P
Chemotaxis, leukocyte recruitment and activation	TNF, Il-1 Chemokines C3a, C5a **Leukotriene B4** (Bacterial products, e.g., N-formyl methyl peptides)
Fever	IL-1, TNF **Prostaglandins**
Pain	**Prostaglandins** Bradykinin
Tissue damage	Lysosomal enzymes of leukocytes Reactive oxygen species Nitric oxide

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Eye drugs

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General principles: topical eye drugs

Topical drugs act either within the eye or on its surface. Those with an intraocular site of action (eg drugs for glaucoma) require a balance of lipid- and water-solubility to cross into the aqueous humour.

Blinking distributes and eliminates drugs dissolved in tears. Excess drug drains over the lids onto the cheek, and through the tear duct to the nasal mucosa where it is absorbed into the systemic circulation (without first pass through the liver) and may lead to systemic adverse effects.

Topical eye medications (particularly if used frequently) may interfere with the lubricating and protective effects of the tear film and can lead to damage to the eye surface, as seen with dry eye syndrome (p 501).

Choice of form

Adverse effects due to systemic absorption are more common with eye drops than with ointments.

Ointments have a longer duration of effect than drops as drug dilution and drainage is slower. They are appropriate to use at night, in children, and when action depends on sustained drug concentrations. However, they are often difficult to self-administer, and blur vision.

Preservatives

Preservatives are required to reduce contamination in most multidose eye drops. Single-use units are preservative free but may be more difficult to use.

Preservatives can irritate and damage the corneal and conjunctival epithelium, particularly if it is already inflamed. Normally, tears quickly dilute and remove preservatives, but if tear secretion decreases, the risk of preservative toxicity increases.

Adverse effects include mild irritation (burning or stinging), symptoms of dry eye syndrome (with disruption of tear film), severe conjunctivitis and, rarely, corneal scarring. Toxicity is related to the type and concentration of preservative, the frequency and duration of use and whether the tear film is intact.

Patients with chronic ocular diseases, eg glaucomas or dry eye syndrome, may need to use several eye drops each day. Using less irritant or preservative-free eye drops reduces the risk of damage to the conjunctiva and cornea without affecting efficacy.

Benzalkonium chloride is the most frequently used preservative in ophthalmic products, but is the most irritant:

- consider drops containing benzalkonium chloride only if used infrequently or for short periods
- consider the number of drops containing this preservative being used
- stopping drops containing benzalkonium chloride may not improve irritation immediately because of its accumulation in ocular tissues.

Polyquaternium, sodium chlorite, stabilised oxychloro complex and sodium perborate are less irritant than benzalkonium chloride and may be used when drops are required more frequently. However, they may still be irritating, especially in severe dry eye syndrome.

Counselling

It is important to write the date on the bottle or tube when you first open it and to discard it 28 days later (unless told otherwise).

Before using eye medications, wash your hands then sit or lie down. Tilt your head back, look up and gently pull down your lower eyelid to form a pouch. Avoid letting the tip of the dropper or tube touch your eyes, skin or any other surface.

Use a clean tissue to mop up any excess.

Some people find it easier to use eye medications properly if they have someone help them or if they use a mirror.

Drops

Shake suspensions (cloudy liquids) gently before use.

Put 1 drop into the pouch (see above). If you think that the first drop missed your eye, put in another.

Try not to blink after putting in a drop. Close your eyes and gently press against the inner corner of your eye with your finger (over the tear duct) for 1–2 minutes. This increases the effectiveness of the eye drop and helps reduce the amount of medicine that gets into the rest of your body where it may cause side effects.

The eye pouch will be full after a single drop. If you need to use another eye drop at the same time of day, wait at least 5 minutes before using it.

If you find it difficult to tell whether a drop has gone into your eye and you don't have someone to help, try storing your eye drops in a refrigerator, so you can feel when the cold drop goes in your eye.

An administration aid may help if you have difficulty using eye drops.

Ointments

Squeeze a small amount (about 1 cm length) along the pouch (see above) then blink several times to spread the ointment. If you need to use drops at a similar time of day, use the ointment last.

Children

Hold the child's eyelids open between the index finger and thumb of one hand and put drops in with the other. If this is difficult, put a couple of drops onto the skin at the inner corner of the eye and wait for the eyes to open.

Infants and toddlers may need to be held still during administration. If you don't have someone to help you may need to swaddle the child with a sheet or lay them on the floor and gently hold their head still between your knees.

Contact lenses

Do not wear contact lenses when using eye ointments or if there is an infection.

Soft contact lenses

Wearing soft contact lenses (other than those disposed of daily) is generally not recommended while using eye drops containing preservatives (p 482). They may be removed before using an eye drop, then reinserted after at least 15 minutes.

If drops are used twice daily you can insert the lens after putting in the morning drop and remove the lens before the evening drop.

If drops are used accidentally while soft lenses are in, remove them and rinse well with an appropriate solution before reinserting.

Practice points

- do not use topical drugs in open eye injuries
- the shelf life for preserved eye products is 28 days after opening, unless otherwise stated; each hospital may have its own discard policy
- in general, there are 20 drops in each mL of an eye drop

11.1 Drugs for eye infections

Eye infections

Urgently refer any patient with a red eye who also has eye pain, photophobia or visual disturbance or wears contact lenses to an ophthalmologist or optometrist for further assessment.

Advise patient with conjunctivitis or keratitis who wears contact lenses to discontinue use until symptoms have completely resolved.

Viral conjunctivitis

Viral conjunctivitis is the most common type of infectious conjunctivitis and is mainly caused by adenoviruses. Symptoms include redness, irritation, a watery or mucoid discharge and lymphadenopathy; it is often associated with upper respiratory tract infection.

Topical ocular antivirals or antibacterials are not indicated; ocular lubricants and/or cool compresses may help to relieve symptoms.

Practice points

- use a clean cotton ball or tissue soaked with sterile water or sodium chloride 0.9% solution to remove any discharge
- highly contagious until discharge resolves; ensure hygiene measures are followed (eg avoid touching eyes, wash hands frequently, do not share towels or bedding)
- seek advice from an ophthalmologist or optometrist if symptoms are severe or do not improve within 7–10 days

Bacterial conjunctivitis

Bacterial conjunctivitis is more common in children than in adults. It typically has a rapid onset with symptoms such as redness, mucopurulent discharge and crusting of eyelids.

Infection is most commonly caused by *S. aureus*, *S. pneumoniae* or *H. influenzae*. Swabs for culture are not usually required for initial diagnosis.

When to use drug treatment

Topical ocular antibacterials may hasten recovery, prevent complications and limit the spread of infection to other people but their benefits are modest. They are indicated in young children or if symptoms are severe.

As bacterial conjunctivitis is frequently self-limiting (resolving within 7 days without treatment), a delayed approach (where antibacterials are only started if symptoms do not improve within 2–3 days) may be considered in uncomplicated cases.

Drug choice

Chloramphenicol is preferred as it has a broad spectrum of action (although it is not active against *Pseudomonas*), has good ocular penetration, is well tolerated and inexpensive.

Framycetin may be used if chloramphenicol is unsuitable but is associated with contact allergic reactions.

Other antibacterials (eg tobramycin) should be reserved for use by ophthalmologists and for serious infections not responding to other treatment (after taking sample for microbiological culture).

Practice points

- use a clean cotton ball or tissue soaked with sterile water or sodium chloride 0.9% solution to remove any discharge or crusting
- highly contagious until discharge resolves; ensure hygiene measures are followed (eg avoid touching eyes, wash hands frequently, do not share towels or bedding)
- seek advice from an ophthalmologist or optometrist if there is no improvement after 2 days treatment with antibacterials

Chlamydial conjunctivitis

The strains of *C. trachomatis* that cause inclusion conjunctivitis (and urogenital infections) differ from the strains that cause trachoma.

Inclusion conjunctivitis is an acute infection that requires treatment with oral antibacterials. In adults, it usually occurs in those with a new sexual partner. In neonates, who require urgent referral, the usual onset is 5–12 days after birth. Treat with azithromycin (p 120); in adults, doxycycline (p 136) is an alternative.

Trachoma is a chronic conjunctivitis that affects rural and remote populations of Aboriginal and Torres Strait Islander people. Azithromycin (p 120) is used for prevention and treatment; for further information, see the CDNA guidelines for *Management of trachoma*.

Gonococcal conjunctivitis

Gonococcal conjunctivitis requires urgent referral as it can result in corneal ulceration and perforation that can lead to blindness. In neonates, the usual onset is 2–5 days after birth.

Treat with a single dose of ceftriaxone (p 113) or cefotaxime. In adults, also give azithromycin (p 120) as there is often coexisting chlamydia (combination may also increase efficacy and reduce development of resistance).

Blepharitis

Blepharitis can be classified as either anterior or posterior; however, they often co-exist and symptoms frequently overlap.

Anterior blepharitis includes:

- *seborrhoeic blepharitis*: characterised by greasy scales around the eyelashes
- *staphylococcal blepharitis*: characterised by redness and swelling of the lid margin, hard scales and eyelash loss or misdirection.

Posterior blepharitis may be caused by meibomian gland dysfunction, which is characterised by blocked glands and/or changes in glandular secretion (eg frothy discharge along eyelid margin).

Treatment aims to control symptoms and prevent exacerbations; cure is usually not possible. Treat associated conditions, eg seborrhoeic dermatitis (p 360), rosacea (p 403) or dry eyes (p 501). Seek

advice from an ophthalmologist or optometrist if symptoms are severe or do not respond to treatment.

Eyelid hygiene

Recommended for all patients with blepharitis. A common regimen is twice daily while disease is active, then daily to prevent exacerbations:

- apply a warm compress (eg face cloth soaked in warm water) to closed eyes for 2–5 minutes and gently massage eyelids
- clean the eyelids using a damp cotton pad (or similar material) dipped in diluted baby shampoo (eg 1 part shampoo to 10 parts water); proprietary eyelid cleansers are also available.

Drug treatment

Evidence for efficacy of antibacterials for blepharitis is limited. However, if symptoms do not improve with an adequate trial of eyelid hygiene, consider the following:

- *anterior blepharitis*: adding chloramphenicol (p 486) eye ointment
- *posterior blepharitis*: adding an oral antibacterial (eg a tetracycline), especially if associated with rosacea.

Chalazion

A chalazion is a chronic granuloma of a meibomian gland presenting as a minimally tender focal lid mass. Topical or systemic antibacterials are usually not indicated. Most cases resolve over several weeks with conservative treatment (eg an ocular lubricant and warm or cold compresses), but incision and drainage may be required.

Stye

Acute infection (staphylococcal abscess) of a sebaceous gland associated with an eyelash follicle, causing pain, tenderness and focal swelling.

Hasten 'pointing' with hot water compresses 3 or 4 times daily, pulling out associated eyelash may aid recovery. Antibacterials are not usually required.

Styes are often secondary to blepharitis.

Keratitis due to infection

Inflammation of the cornea caused by bacteria, viruses, fungi or protozoa. Risk factors include contact lens wear, corneal trauma or abrasion, and immunosuppression. Refer promptly to an ophthalmologist as it is sight-threatening.

Bacterial keratitis

Common signs and symptoms include pain, redness, blurred vision, photophobia and tearing. Causative organisms often include *Staphylococci*, *Streptococci* and Gram-negative bacilli (eg *Pseudomonas*).

Treatment involves frequent administration of topical ocular antibacterials; either monotherapy with ciprofloxacin or ofloxacin, or combined therapy, eg with high-strength cefazolin and gentamicin (prepared by some pharmacies).

Herpes simplex epithelial keratitis

Characterised by branching ulceration of the cornea, and is caused by primary (or reactivation of latent) herpes simplex virus infection. It usually affects only one eye; bilateral infection is more common in immunosuppression. Typically there is a foreign body sensation (often mild due to associated loss of corneal sensation), photophobia and tearing.

Treat with topical aciclovir (p 486). Never use corticosteroids for herpes simplex epithelial keratitis (proven or suspected). Refer all new cases to an ophthalmologist within 24 hours, and refer recurrent cases that fail to respond to treatment within 3 days or that have corneal opacity.

Herpes zoster ophthalmicus

The ophthalmic division of the trigeminal nerve is involved in about 10–20% of herpes zoster cases and, without antivirals, half of these will develop eye complications, eg conjunctivitis, keratitis, uveitis or raised intraocular pressure.

Seek specialist advice:

- start oral aciclovir, famciclovir or valaciclovir as soon as possible (within 72 hours of rash onset) to help resolve rash and to decrease pain, corneal damage and the risk of anterior uveitis; treat for at least 7 days
- use IV aciclovir if sight is threatened
- regular use of ocular lubricants is recommended
- systemic analgesics and cool compresses may also be used as required for symptom relief
- uveitis and keratitis may be controlled with topical corticosteroids.

11.1.1 Aminoglycosides

Framycetin p 484
Gentamicin p 485
Tobramycin p 485

Indications

Bacterial eye infections

Precautions

Allergic reaction to an ocular aminoglycoside—cross-allergenicity may occur; increased risk of allergic reaction.

Pregnancy

Safe to use.

Breastfeeding

Safe to use.

Adverse effects

Ocular irritation and superficial punctate keratitis are common; other adverse effects include delayed corneal epithelial wound healing, retinal toxicity (if there is leakage through corneoscleral wound) and, rarely, hypersensitivity reactions.

Comparative information

Gentamicin and tobramycin have antipseudomonal activity and are generally well tolerated.

Framycetin is not active against *Pseudomonas* species, and is more likely to cause contact allergy than gentamicin or tobramycin; cross-allergenicity between aminoglycosides may occur.

Framycetin

For additional information see Aminoglycosides p 484
See also Bacterial conjunctivitis p 483

Indications

Bacterial conjunctivitis

Dosage

Dose frequency depends on severity of infection and is adjusted according to response. The following is a guide only:

Adult, child, 1 eye drop every 2–4 hours for 2 days; then if there is improvement, 1 drop 4 times daily for 5 days.

eye drop, 0.5%, 8 mL, *Soframycin*, PBS

Gentamicin

For additional information see Aminoglycosides p 484
See also Bacterial keratitis p 484

Indications

Bacterial keratitis (with cefazolin eye drops), see also Practice points below

Dosage

Dose frequency depends on severity of infection and is adjusted according to response. The following is a guide only:

Adult, child, initially 1 eye drop (0.9–1.4%) every hour (including overnight).

Practice points

- only use under close supervision by, or following discussion with, an ophthalmologist (after taking sample for microbiological culture)
- high-strength gentamicin and cefazolin eye drops are prepared by some pharmacies

Tobramycin

For additional information see Aminoglycosides p 484
See also Bacterial conjunctivitis p 483

Indications

Bacterial conjunctivitis (see also Practice points below)

Dosage

Dose frequency depends on severity of infection and is adjusted according to response. The following is a guide only:

Adult, child, 1 eye drop every 2–4 hours for 2 days; then if there is improvement, 1 drop 4 times daily for 5 days.

Eye ointment may be used as an adjunct to drops at night, or as a single agent 3 times daily.

Practice points

- tobramycin should be reserved for use by ophthalmologists and for serious infections not responding to treatment with other topical antibacterials (after taking sample for microbiological culture)
- higher strengths of tobramycin eye drops (0.9–1.4%) are used for treating bacterial keratitis and are prepared by some pharmacies

eye drop, 0.3%, 5 mL, *Tobrex*, PBS-R[1]
eye oint, 0.3%, 3.5 g, *Tobrex*, PBS-R[1]

[1] perioperative use; suspected pseudomonal eye infection; invasive ocular infection

11.1.2 Quinolones

See also Bacterial conjunctivitis p 483, Bacterial keratitis p 484

Also known as fluoroquinolones.

Ciprofloxacin p 485
Ofloxacin p 486

Indications

Bacterial eye infections

Precautions

History of hypersensitivity with quinolone use—avoid use if possible.

History of tendon damage with quinolone use—use with caution as there may be a risk of recurrence.

Pregnancy

Safe to use as systemic absorption is minimal.

Breastfeeding

Safe to use.

Adverse effects

Common (>1%)

mild transient ocular irritation (eg burning, stinging)

Infrequent or rare

white corneal precipitates (common with ciprofloxacin), keratitis, allergic reactions (very rare reports of severe hypersensitivity including angioedema, anaphylaxis, Stevens-Johnson syndrome)

Comparative information

Ciprofloxacin and ofloxacin have a similar spectrum of antibacterial activity.

Ciprofloxacin commonly forms a transient, white precipitate on corneal ulcers that may slow the rate of corneal epithelial healing. Precipitation of ofloxacin on corneal ulcers has been reported occasionally.

Practice points

- due to concern about emerging resistance:
  - reserve quinolones for treatment of bacterial keratitis (under close supervision by, or following discussion with, an ophthalmologist)
  - other antibacterials are preferred for empirical treatment of conjunctivitis

Ciprofloxacin

For additional information see Quinolones p 485

Indications

Bacterial keratitis or severe bacterial conjunctivitis (see also Practice points p 485 in Quinolones)

Adverse effects

Common (>1%)

unpleasant taste

Dosage

Adult, child

Bacterial keratitis

Dose frequency depends on severity of infection and is adjusted according to response. The following is a guide only:

Day 1, 1 eye drop every 15 minutes for the first 6 hours, then once every 30 minutes.

Day 2, 1 eye drop every hour.

From day 3 until healed, 1 eye drop every 4 hours.

Severe bacterial conjunctivitis
1 eye drop every 2 hours while awake for 2 days; then if there is improvement, 1 drop every 4 hours while awake for 5 days.

eye drop, 0.3%, 5 mL, *Ciloquin, Ciloxan*, PBS-A[2][1]

[1] bacterial keratitis

Ofloxacin

For additional information see Quinolones p 485

Indications
Bacterial keratitis or severe bacterial conjunctivitis (see also Practice points p 485 in Quinolones)

Dosage

Adult, child

Bacterial keratitis
Dose frequency depends on severity of infection and is adjusted according to response. The following is a guide only:
Days 1–2, 1 eye drop every 30 minutes while awake, then at 4 and 6 hours after going to bed.
Days 3–7, 1 eye drop every hour while awake.
From day 8 until healed, 1 eye drop 4 times a day.

Severe bacterial conjunctivitis
1 eye drop every 4 hours for 2 days; then if there is improvement, 1 drop 4 times daily for up to 8 days.

eye drop, 0.3%, 5 mL, *Ocuflox*, PBS-A[2][1]

[1] bacterial keratitis

11.1.3 Other antibacterials

Chloramphenicol

See also Bacterial conjunctivitis p 483

Mode of action
Inhibits bacterial protein synthesis by binding to the 50S subunit of the bacterial ribosome and preventing the activity of peptidyltransferase.

Indications
Bacterial conjunctivitis or blepharitis

Accepted
Prevention of infection after corneal abrasion, see Practice points below
Prevention of infection after ocular surgery, seek specialist advice

Precautions

Pregnancy
Safe to use.

Breastfeeding
Safe to use.

Adverse effects

Common (>1%)
stinging or burning after instillation

Infrequent (0.1–1%)
unpleasant taste

Rare (<0.1%)
hypersensitivity reactions, eg angioedema, anaphylaxis, dermatitis

Dosage

Bacterial conjunctivitis
Adult, child, 1 eye drop every 2 hours while awake for 1–2 days then, if there is improvement, 1 drop 4 times daily for up to 5 days.
Eye ointment may be used at night if drops are used during the day, or as a single agent 3 or 4 times daily.

Bacterial blepharitis
Adult, child, massage eye ointment into lid margin once or twice daily for 1 week (in some cases, up to 3 weeks' treatment may be needed).

Prevention of infection after corneal abrasion
Adult, child, 1 eye drop (or eye ointment) 4 times daily until healed (usual duration 3–5 days).

Practice points
- broad-spectrum bacteriostatic activity against Gram-positive, Gram-negative (except *Pseudomonas* spp.) and anaerobic bacteria; good ocular penetration
- chloramphenicol is available without prescription for bacterial conjunctivitis:
  - for more information, including when to refer, see the Pharmaceutical Society of Australia's guidelines for the provision of ophthalmic chloramphenicol
  - ensure patient consults a doctor or optometrist if symptoms do not improve within 48 hours
- a randomised double-blind study in children in primary care with mild acute infective conjunctivitis found no significant difference in terms of clinical cure between topical chloramphenicol and placebo
- evidence of a benefit in corneal abrasion is lacking; consider limiting use to cases with a higher risk of infection, eg abrasion caused by infectious or vegetative matter
- large population-based studies have found no association between use of chloramphenicol eye drops and aplastic anaemia

eye drop, 0.5%, 10 mL, *Chlorsig*, PBS-R[1]/RPBS
eye drop, 0.5%, 10 mL, *Choifen, Oczuza*[a]
eye drop, 0.5%, 0.5 mL (single use), 20, *Minims Chloramphenicol*
eye oint, 1%, 4 g, *Chlorsig*
eye oint, 1%, 5 g, *Choifen*

[1] Aboriginal or Torres Strait Islander patients
[a] generic brands available

11.1.4 Antivirals

Aciclovir

See also Herpes simplex epithelial keratitis p 484

Indications
Herpes simplex epithelial keratitis

Precautions

Pregnancy
Safe to use.

Breastfeeding
Safe to use.

Adverse effects

Common (>1%)
transient mild stinging after instillation

Infrequent (0.1–1%)
superficial punctate keratitis, allergic reactions

Dosage
Adult, child, apply about 1 cm of eye ointment into the lower conjunctival sac 5 times daily for 14 days, or for 3 days after corneal epithelium healed, whichever is shorter.

eye oint, 3%, 4.5 g, *ViruPOS, Xorox*, PBS-R[1]

[1] herpes simplex keratitis

11.2 Drugs for glaucoma

Glaucomas

Glaucomas are optic neuropathies characterised by optic disc changes and irreversible, progressive visual field loss. They are classified as either open-angle or angle-closure and may be primary or secondary.

Open-angle glaucoma
Although increased intraocular pressure (IOP) is not a defining characteristic of glaucoma, IOP is its main modifiable risk factor. Treatments that lower IOP reduce the rate of glaucoma progression (whether or not IOP is increased).
Initial treatment of glaucoma is generally with topical drugs or selective laser trabeculoplasty; surgery may be considered in some cases (eg advanced glaucoma).
Management of secondary glaucomas may also include removal or treatment of the underlying cause (if possible), eg stopping corticosteroids or treating inflammatory eye disease.

Before starting treatment
Establish a target IOP for each patient using factors such as baseline IOP, glaucoma severity and other risk factors. The target, which is used to assess treatment efficacy, should be reviewed regularly.

Drug choice
See also Table 11–1 Comparison of topical drugs for open-angle glaucoma p 487
Prostaglandin analogues are generally first-line due to their higher efficacy, once-daily dosing and minimal systemic adverse effects. Topical beta-blockers are also commonly used but may cause systemic adverse effects. If both prostaglandin analogues and beta-blockers are not appropriate, a topical alpha$_2$ agonist or topical carbonic anhydrase inhibitor may be used. Pilocarpine is less commonly used due to adverse effects.
When choosing topical treatment, also consider the irritant effects of preservatives (p 482) (particularly in patients with ocular surface disease, eg dry eye syndrome). If possible, use products containing less irritant or no preservatives.
When a single drug is not tolerated or doesn't seem effective, consider switching to an alternative drug. If response to a prostaglandin analogue is poor, switching to another one may be effective as drugs in this class are structurally different.
When a single drug is effective but the target IOP is still not reached, consider adding a second drug. Some drugs are available in fixed-dose combinations and although there is less dose flexibility, these are generally preferred as they are convenient, aid compliance, reduce exposure to preservatives and reduce cost to the patient.
The only oral treatment, acetazolamide, has adverse effects that limit its use to refractory glaucoma or short-term treatment pre- and post-surgery.

Practice points
- review treatment (including efficacy and adverse effects) 2–8 weeks after any change in medication and regularly thereafter
- check compliance and administration technique before changing drug treatment; as glaucoma is usually asymptomatic, compliance is a major problem (only about 30–70% of prescribed doses are taken)
- information on eye drop administration is available from *Glaucoma Australia*; see also Counselling p 482

Table 11–1 Comparison of topical drugs for open-angle glaucoma

Class	Drugs	Doses per day	Comments
prostaglandin analogues	bimatoprost, latanoprost, travoprost	1	• first line • increase aqueous humour outflow • the most effective class • may cause iris hyperpigmentation and eyelash changes
beta-blockers	betaxolol, timolol	1–2	• first line • decrease aqueous humour production • may cause systemic adverse effects, eg bradycardia, bronchospasm
alpha$_2$ agonists	apraclonidine, brimonidine	2–3	• second line • increase aqueous humour outflow and decrease its production • apraclonidine can only be used short term
carbonic anhydrase inhibitors	brinzolamide, dorzolamide	2–3	• second line • decrease aqueous humour production
cholinergic	pilocarpine	3–4	• less commonly used (usually third or fourth line) • increases aqueous humour outflow • high incidence of adverse effects, eg blurred vision, headache

- in some cases topical treatment may be started in just one eye, using the other as a control; if effective, treatment is then started in the other eye
- although patients with ocular hypertension (increased IOP without glaucoma changes) have an increased risk of glaucoma, not all will develop glaucoma; monitoring alone is often appropriate; treatment may be considered for those with additional risk factors for developing glaucoma

Angle-closure glaucoma

Up to 75% of patients with angle-closure glaucoma will have intermittent or chronic angle closure with asymptomatic progressive loss of visual field rather than an acute crisis. Initial treatment is laser iridotomy; if unsuccessful, surgical procedures are considered. Topical treatment may be used, particularly in those with persistent raised IOP despite laser or surgical procedures.

Risk factors for angle-closure glaucoma

Risk factors include Asian ethnicity, older age, female gender, family history and long sight. Use of topical or systemic drugs that cause pupil dilation may precipitate acute angle-closure crisis.

Acute angle-closure crisis

This is an ophthalmic emergency that requires urgent referral to an ophthalmologist. Treatments used to reduce IOP prior to laser iridotomy may include a topical beta-blocker, topical alpha$_2$ agonist, topical carbonic anhydrase inhibitor, pilocarpine, oral or IV acetazolamide or IV mannitol.

11.2.1 Beta-blockers

See also Glaucomas p 487
For drug interactions see Beta-blockers p 990

Betaxolol p 488
Timolol p 489

Mode of action

Reduce aqueous humour production, probably by blockade of beta receptors on the ciliary epithelium.

Indications

Glaucoma
Ocular hypertension

Precautions

Treatment with a systemic beta-blocker—reduces intraocular pressure (but less than with a topical beta-blocker). Adding a topical beta-blocker further reduces intraocular pressure (but this may be less effective than using an alternative class of topical agent). Risk of systemic adverse effects is increased.

Treatment with drugs that cause bradycardia—may further decrease heart rate; monitor cardiac function. Avoid combination with verapamil (unless under specialist supervision).

Cardiovascular

As topical beta-blockers are absorbed systemically, consider potential for cardiovascular effects (see Precautions p 267 in Beta-blockers). Topical beta-blockers are contraindicated in bradyarrhythmia, second- or third-degree atrioventricular block or uncontrolled heart failure.

Respiratory

As topical beta-blockers may precipitate bronchospasm, they are generally contraindicated in asthma. However, betaxolol (which is beta$_1$ selective) may be used with caution if required. They may be used with caution in COPD; betaxolol may be preferred.

Surgery

Theoretical increased risk of bradycardia and hypotension during surgery.

Elderly

Systemic adverse effects are more common, eg hypotension (may cause falls).

Pregnancy

May be used. Monitor fetus for possible adverse effects, eg bradycardia.

Breastfeeding

Considered safe to use. Although adverse effects are unlikely, consider monitoring the infant.

Adverse effects

The most important adverse effects are systemic.

Common (>1%)

stinging on instillation (especially betaxolol solution), bradycardia, blurred vision

Infrequent (0.1–1%)

decreased corneal sensation, hypotension, fainting, fatigue, confusion, hallucinations, bronchospasm

Comparative information

Timolol is a nonselective beta$_1$ and beta$_2$ receptor blocker whereas betaxolol selectively blocks beta$_1$ receptors. Although studies suggest betaxolol is less likely to cause bronchospasm than timolol, it should be used with caution in patients with asthma. Betaxolol may be less effective at reducing intraocular pressure and is more likely to cause local stinging compared to timolol.

Practice points

- to reduce systemic effects, advise patients to close their eyes and apply punctal occlusion after administration, see Counselling p 482

Betaxolol

For additional information see Beta-blockers p 488
For drug interactions see Beta-blockers p 990

Indications

Glaucoma
Ocular hypertension

Dosage

Adult, 1 eye drop twice daily.

Practice points

- betaxolol 0.25% suspension (which may be available through the SAS, see Drug availability in Australia p xviii) is as effective as betaxolol 0.5% solution; use of suspension reduces local stinging, which may help compliance

eye drop, 0.5% (solution), 5 mL, *Betoptic, Betoquin*, PBS

Timolol

For additional information see Beta-blockers p 488
For drug interactions see Beta-blockers p 990

Indications

Glaucoma
Ocular hypertension

Dosage

Adult

Conventional drops, 1 eye drop once or twice daily.

Gel-forming drops (Timoptol-XE®), 1 eye drop once daily.

eye drop, 0.5%, 2.5 mL, *Timoptol-XE*, PBS
eye drop, 0.5%, 5 mL, *Timoptol*, PBS

Timolol combinations

Bimatoprost with timolol p 490
Brimonidine with timolol p 491
Brinzolamide with timolol p 492
Dorzolamide with timolol p 492
Latanoprost with timolol p 490
Travoprost with timolol p 490

11.2.2 Prostaglandin analogues

See also Glaucomas p 487
For drug interactions see Prostaglandin analogues p 1078

Bimatoprost p 489
Bimatoprost with timolol p 490
Latanoprost p 490
Latanoprost with timolol p 490
Travoprost p 490
Travoprost with timolol p 490

Mode of action

Reduce intraocular pressure by increasing uveoscleral outflow of aqueous humour.

Indications

Glaucoma
Ocular hypertension

Precautions

Intraocular inflammation (iritis, uveitis)—use with caution as prostaglandin analogues may worsen inflammation.

Herpetic keratitis—avoid use if active, use with caution if history of infection as reactivation may occur.

Pseudophakia with torn posterior lens capsule, aphakia, risk factors for macular oedema—use with caution as prostaglandin analogues may further increase the risk of macular oedema.

Pregnancy

Limited human data. Avoid use if possible (due to theoretical risk of uterine contractions) or seek specialist advice.

Breastfeeding

Safe to use.

Adverse effects

Common (>1%)

iris hyperpigmentation (below), eyelash changes (below), periorbital changes (below), conjunctival hyperaemia, eye irritation (eg grittiness, itching, stinging), allergic conjunctivitis, blurred vision, punctate keratitis, blepharitis, headache

Infrequent (0.1–1%)

reversible macular oedema (including cystic oedema), eyelid and conjunctival oedema, iritis or uveitis

Rare (<0.1%)

reactivation of herpetic keratitis

Iris hyperpigmentation

Gradual (over months to years), usually irreversible, increase in iris pigmentation in treated eyes, especially those of mixed colour, eg blue/brown.

Eyelash changes

Gradual darkening, lengthening and thickening of the eyelashes; reversible when treatment stopped.

Periorbital changes

May include darkening of eyelid skin, deepening of upper eyelid sulcus, loss of periorbital fat and upper lid ptosis. Some of the effects appear to be reversible after stopping treatment.

Comparative information

All prostaglandin analogues appear to have similar efficacy.

Bimatoprost is available preservative-free or preserved with benzalkonium chloride; latanoprost with benzalkonium chloride; travoprost is preserved with polyquaternium.

Counselling

These eye drops may slowly and permanently change the colour of your eye making the iris appear darker.

Practice points

- evening dosing is preferred for optimal effect; however, administration at another time may be considered if this improves compliance
- to reduce systemic effects, advise patients to close their eyes and apply punctal occlusion after administration, see Counselling p 482
- some adverse effects (eg periorbital changes) will be more noticeable if only 1 eye is treated; in some cases, a drug from a different class may be preferred
- drugs in this class are structurally different; if response to one is poor, consider trying another
- a paradoxical increase in intraocular pressure may occur if 2 ocular prostaglandin analogues are used together; avoid combination

Bimatoprost

For additional information see Prostaglandin analogues p 489
For drug interactions see Prostaglandin analogues p 1078

Indications

Glaucoma
Ocular hypertension

Dosage

Adult, 1 eye drop once daily, preferably at night.

eye drop, 0.03%, 3 mL, *Bimprozt, Lumigan*[a], PBS
eye drop, 0.03%, 0.4 mL (single use), 30, *Lumigan PF*, PBS

[a] generic brands available

Bimatoprost combinations

Bimatoprost with timolol p 490

Bimatoprost with timolol

Go to Bimatoprost p 489 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop once daily.

eye drop, bimatoprost 0.03%, timolol 0.5%, 3 mL, *Ganfort*, PBS-R[1]

eye drop, bimatoprost 0.03%, timolol 0.5%, 0.4 mL (single use), 30, *Ganfort PF*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

Latanoprost

For additional information see Prostaglandin analogues p 489
For drug interactions see Prostaglandin analogues p 1078

Indications
Glaucoma
Ocular hypertension

Adverse effects
Common (>1%)
periorbital rash

Dosage
Adult, 1 eye drop once daily, preferably at night.

eye drop, 0.005%, 2.5 mL, *Xalaprost*, *Xalatan*[a], PBS

[a] generic brands available

Latanoprost combinations
Latanoprost with timolol p 490

Latanoprost with timolol

Go to Latanoprost p 490 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop once daily.

eye drop, latanoprost 0.005%, timolol 0.5%, 2.5 mL, *Xalacom*, *Xalamol*[a], PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS
[a] generic brands available

Travoprost

For additional information see Prostaglandin analogues p 489
For drug interactions see Prostaglandin analogues p 1078

Indications
Glaucoma
Ocular hypertension

Dosage
Adult, 1 eye drop once daily, preferably at night.

eye drop, 0.004%, 2.5 mL, *Travatan*, PBS

Travoprost combinations
Travoprost with timolol p 490

Travoprost with timolol

Go to Travoprost p 490 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop once daily.

eye drop, travoprost 0.004%, timolol 0.5%, 2.5 mL, *DuoTrav*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

11.2.3 Alpha$_2$ agonists

See also Glaucomas p 487

Apraclonidine p 491
Brimonidine p 491
Brimonidine with timolol p 491

Mode of action
Reduce intraocular pressure by suppressing production of aqueous humour and increasing uveoscleral outflow.

Indications
Glaucoma

Precautions
Severe cardiovascular disease—may worsen; use with caution.

Adverse effects
Common (>1%)
ocular allergy (below), hyperaemia, ocular irritation, dry mouth and nose, taste disturbance, conjunctival blanching, lid retraction, headache

Ocular allergy
Usually presents as a follicular conjunctivitis with symptoms such as hyperaemia, tearing and foreign body sensation. There may also be periocular dermatitis. In patients with a previous allergy to apraclonidine, the risk of allergy to brimonidine does not appear to be further increased.

Comparative information
Apraclonidine is indicated for short-term use (up to 3 months) as its effect usually declines over time. Brimonidine is effective when used long term and has a lower incidence of ocular allergy compared to apraclonidine.

Practice points
- may be used by ophthalmologists in acute angle-closure crisis or to prevent ocular hypertension following laser surgery
- to reduce systemic effects, advise patients to close their eyes and apply punctal occlusion after administration, see Counselling p 482

Apraclonidine

For additional information see Alpha$_2$ agonists p 490

Indications
Short-term (<3 months) reduction of intraocular pressure in patients on maximally tolerated treatment for glaucoma

Precautions
Pregnancy
Limited data; seek specialist advice.
Breastfeeding
No data; unlikely to be a concern.

Dosage
Adult, 1 eye drop 2 or 3 times daily.

Practice points
- apraclonidine is only indicated for short-term use as its effect diminishes over time and it has a high incidence of ocular allergy

eye drop, 0.5%, 5 mL, 10 mL, *Iopidine*

Brimonidine

For additional information see Alpha$_2$ agonists p 490

Indications
Glaucoma
Ocular hypertension

Precautions
Children
Contraindicated in children <2 years. Serious adverse effects (eg CNS depression) are more likely in children than adults, especially if <6 years or <20 kg.
Pregnancy
Limited data but considered safe to use. Avoid use close to delivery due to theoretical risk of CNS depression and apnoea in neonate.
Breastfeeding
Limited data. Avoid use due to theoretical risk of CNS depression and apnoea in infant.

Adverse effects
Common (>1%)
fatigue, drowsiness, dizziness
Infrequent (0.1–1%)
bradycardia, hypotension
Rare (<0.1%)
uveitis

Dosage
Adult, 1 eye drop twice a day (of either strength).

Counselling
This eye drop may cause drowsiness or dizziness; avoid driving or operating machinery if you are affected.

Practice points
- the 0.15% drops (preserved with Purite®) appear to lower intraocular pressure to a similar extent as the 0.2% drops (preserved with benzalkonium chloride) and may cause fewer adverse effects

eye drop, 0.15%, 5 mL, *Alphagan P 1.5*, PBS
eye drop, 0.2%, 5 mL, *Alphagan*, *Enidin*, PBS

Brimonidine combinations
Brimonidine with timolol p 491
Brinzolamide with brimonidine p 492

Brimonidine with timolol

Go to Brimonidine p 491 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop twice daily.

eye drop, brimonidine 0.2%, timolol 0.5%, 5 mL, *Combigan*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

11.2.4 Carbonic anhydrase inhibitors

See also Glaucomas p 487

Brinzolamide p 492
Brinzolamide with brimonidine p 492
Brinzolamide with timolol p 492
Dorzolamide p 492
Dorzolamide with timolol p 492

Mode of action
Inhibit carbonic anhydrase II (predominant subtype found in the eye), which reduces aqueous humour production.

Indications
Glaucoma
Ocular hypertension

Precautions
Compromised corneal endothelium—may increase risk of corneal oedema.
Allergy to sulfonamides—may increase risk of allergy to carbonic anhydrase inhibitors.
Pregnancy
Limited data. Avoid use if possible, or seek specialist advice.
Breastfeeding
Limited data but considered safe to use.

Adverse effects
Common (>1%)
ocular irritation (more common with dorzolamide), transient blurred vision (more common with brinzolamide), foreign body sensation, bitter taste
Infrequent (0.1–1%)
GI disturbance, headache, paraesthesia, dizziness, dermatitis
Rare (<0.1%)
allergic reactions, eg urticaria, angioedema

Counselling
Your eye may feel uncomfortable for a little while after you have put in the drop. If you have blurred vision, avoid driving or operating machinery until your sight improves.

Practice points
- to reduce systemic effects, advise patients to close their eyes and apply punctal occlusion after administration, see Counselling p 482

Brinzolamide

For additional information see Carbonic anhydrase inhibitors p 491

Indications
Glaucoma
Ocular hypertension

Adverse effects
Infrequent (0.1–1%)
conjunctivitis, blepharitis, keratitis

Dosage
Adult, 1 eye drop twice daily.

eye drop, 1% (suspension), 5 mL, *Azopt, BrinzoQuin*, PBS

Brinzolamide combinations
Brinzolamide with brimonidine p 492
Brinzolamide with timolol p 492

Brinzolamide with brimonidine

Go to Brinzolamide p 492 and Brimonidine p 491 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop twice daily.

eye drop, brinzolamide 1%, brimonidine 0.2% (suspension), 5 mL, *Simbrinza*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

Brinzolamide with timolol

Go to Brinzolamide p 492 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop twice daily.

eye drop, brinzolamide 1%, timolol 0.5% (suspension), 5 mL, *Azarga*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

Dorzolamide

For additional information see Carbonic anhydrase inhibitors p 491

Indications
Glaucoma
Ocular hypertension

Adverse effects
Common (>1%)
allergic conjunctivitis and blepharitis, superficial punctate keratitis

Dosage
Adult
Single agent, 1 eye drop 3 times daily.
Adjunct to beta-blocker, 1 eye drop twice daily.

eye drop, 2%, 5 mL, *Trusamide*, PBS

Dorzolamide combinations
Dorzolamide with timolol p 492

Dorzolamide with timolol

Go to Dorzolamide p 492 and Timolol p 489 for information on individual components; limited detail below.

Indications
Glaucoma or ocular hypertension, when monotherapy is inadequate

Dosage
Adult, 1 eye drop twice daily.

eye drop, dorzolamide 2%, timolol 0.5%, 5 mL, *Cosdor, Cosopt, Vizo-PF Dorzolatim*, PBS-R[1]

[1] open-angle glaucoma or ocular hypertension, see PBS

11.2.5 Other drugs for glaucoma

Acetazolamide p 492
Mannitol p 493
Pilocarpine p 494

Acetazolamide

Systemic carbonic anhydrase inhibitor
See also Glaucomas p 487, Epilepsy p 735
For drug interactions see Acetazolamide p 964

Mode of action
Inhibits carbonic anhydrase, resulting in:
- reduced aqueous humour production, thereby reducing intraocular pressure
- carbon dioxide accumulation in the brain, thereby reducing neuronal activity
- mild diuresis and metabolic acidosis (which increases respiration rate and improves oxygenation at high altitude).

Indications
Acute reduction of intraocular pressure, eg perioperatively, acute angle-closure crisis
Chronic open-angle glaucoma where other treatments have failed
Adjunctive treatment of refractory epilepsy, including menstrual-related epilepsy
Accepted
Prevention of acute mountain sickness in those at moderate or high risk, see also Practice points below

Precautions
Adrenal or respiratory failure or metabolic acidosis—contraindicated (increased risk of profound acidosis).
Hyponatraemia or hypokalaemia—contraindicated as acetazolamide may worsen electrolyte disturbances.
Allergy to sulfonamides—may increase risk of allergy to acetazolamide; manufacturer contraindicates use.
Renal
Contraindicated when CrCl <10 mL/minute (increased risk of profound acidosis); reduce dose if CrCl 10–50 mL/minute.
Acetazolamide increases risk of recurrence of urolithiasis.
Hepatic
Contraindicated in hepatic impairment or cirrhosis due to the risk of hepatic encephalopathy.
Elderly
Increased risk of adverse effects, eg metabolic acidosis.
Pregnancy
Contact one of the pregnancy drug information centres (p 1142).
Breastfeeding
Contact one of the pregnancy drug information centres (p 1142).

Adverse effects
Common (>1%)
paraesthesia (of face, hands, feet), fatigue, drowsiness, depression, decreased libido, bitter or metallic taste, nausea, vomiting, abdominal cramps, diarrhoea, polyuria, renal stones, metabolic acidosis, electrolyte disturbances (hypokalaemia, hyponatraemia), rash
Rare (<0.1%)
ocular effects (eg transient myopia, angle-closure glaucoma), severe skin reactions (eg Stevens-Johnson syndrome), aplastic anaemia, thrombocytopenia, agranulocytosis or neutropenia, anaphylaxis

Dosage
Acute reduction of intraocular pressure
Adult, oral/IV, initially 250–500 mg, followed by 125–250 mg every 4–6 hours as required. Usual maximum 1 g daily.
Chronic open-angle glaucoma
Adult, oral, initially 125 mg twice daily, increase to a maximum of 250 mg 4 times daily.
Epilepsy
Seek specialist advice. The following dose has been used.
Adult, child >1 month, oral, initially 4–8 mg/kg (maximum 250 mg) daily; increase according to response up to 30 mg/kg (maximum 1 g) daily in 2–4 doses.
Prevention of acute mountain sickness
Start acetazolamide the day before ascent and continue until acclimatised (usually 2 days after reaching the highest altitude) or once final descent has begun.
Adult, oral, 125 mg twice daily.
Renal impairment
Adult, CrCl 10–50 mL/minute, halve the dose or double the dosing interval (maximum 250 mg twice daily).
Administration advice
Give IV injection over 3–5 minutes. The injection has a high pH and extravasation may lead to tissue damage.

Practice points
- if possible, check renal function, electrolytes and complete blood count before starting acetazolamide; monitor regularly if used for more than a few weeks
- when acetazolamide is used for epilepsy:
  - tolerance may develop with continuous use
  - it may be given intermittently for menstrual-related epilepsy
- consider acetazolamide for prevention of acute mountain sickness when rapid ascent is unavoidable; for further information (including risk categories), see the chapter on high-elevation travel and altitude illness in *CDC Yellow Book*
- although acetazolamide is still marketed for use as a diuretic in heart failure, other agents are preferred

tab, 250 mg (scored), 100, *Diamox*, PBS
inj, 500 mg (powder), 5, *Glaumox*

Mannitol

Hyperosmotic agent
See also Glaucomas p 487

Mode of action
Increases plasma osmolality and draws water out of body tissues, producing an osmotic diuresis that results in reduced intraocular and intracranial pressure.

Indications
Reduction of intraocular pressure unresponsive to conventional treatment including:
- acute angle-closure crisis
- before intraocular surgery (to prevent prolapse of intraocular contents)

Reduction of intracranial pressure and treatment of cerebral oedema (seek specialist advice)
Cystic fibrosis (see Mannitol p 941)

Precautions
Cardiac disease—avoid use or reduce dose as mannitol can increase extracellular volume which may precipitate heart failure. Contraindicated in severe heart failure.
Pulmonary oedema—avoid use or reduce dose as mannitol can increase extracellular volume which may exacerbate pulmonary oedema. Contraindicated in severe pulmonary oedema.
Dehydration—use with caution as mannitol-induced osmotic diuresis may worsen dehydration. Contraindicated in severe dehydration.
Renal
Mannitol is predominantly renally cleared. Renal impairment or treatment with nephrotoxic drugs may increase the risk of renal failure due to mannitol; reduce dose. Contraindicated in renal failure.

Adverse effects
fluid and electrolyte imbalance (below), injection site reactions (eg local pain, thrombophlebitis, skin necrosis due to extravasation), nausea, vomiting, chills, dizziness, hypotension, tachycardia, fever, angina-like chest pains, hypersensitivity reactions (including urticaria, anaphylaxis/anaphylactoid reactions), acute renal failure (with large doses)

Fluid and electrolyte imbalance
Fluid and/or electrolyte shifts can produce pulmonary oedema, acidosis, electrolyte loss, dry mouth, thirst, oedema, headache, blurred vision, seizures and heart failure.

Dosage

Reduction of intraocular pressure
Consult local protocols. The following dose has been used.
When used pre-operatively, give 1–1.5 hours before surgery for maximum reduction of intraocular pressure.
Adult, IV 1–2 g/kg (5–10 mL/kg of 20% solution); maximum 100 g (500 mL).
Elderly, renal impairment or other precautions (see above), IV 0.5–1 g/kg (2.5–5 mL/kg of 20% solution); maximum 50 g (250 mL).

Administration advice
Mannitol crystallises at low temperatures (especially 20% solution); if necessary, redissolve using dry heat (eg a warming cabinet) and agitation; allow to cool to body temperature before administration.
Infuse IV over 30–60 minutes via an in-line filter. Infuse into a large peripheral vein or central venous catheter; avoid extravasation.

Practice points
- monitor renal function, serum electrolytes, serum osmolality, fluid balance, BP, heart rate and respiratory rate
- produces a rapid (within 30 minutes) but temporary (6 hours) drop in intraocular pressure

inj, 10%, 1000 mL, *Osmitrol*
inj, 20%, 500 mL, *Osmitrol*

Pilocarpine

Cholinergic
See also Glaucomas p 487

Mode of action
Cholinergic effect contracts:
- iris sphincter causing miosis (pupil constriction)
- ciliary muscle, which increases outflow through the trabecular meshwork lowering intraocular pressure.

Indications
Chronic open-angle glaucoma
Acute angle-closure crisis, seek specialist advice
Reversal of mydriatic effect of phenylephrine (not recommended, see Practice points)

Precautions
Acute inflammatory disease of the anterior segment and conditions where pupillary constriction is not recommended, eg uveitis, iritis and some forms of secondary glaucoma—contraindicated.
Previous retinal detachment, pre-existing retinal damage, young patients with myopia—increases risk of retinal detachment.
Heavily pigmented eyes—pigment can reduce pilocarpine's effect; a higher concentration or more frequent doses may be required.

Pregnancy
Limited data but considered safe to use.

Breastfeeding
Limited data but considered safe to use.

Adverse effects

Common (>1%)
headache (below), blurred vision, myopia, reduced vision in low light, constriction of visual field, ocular irritation (eg burning, itching, smarting)

Infrequent (0.1–1%)
lacrimation

Rare (<0.1%)
retinal detachment, systemic effects (eg increased sweating, diarrhoea, bronchospasm, bradycardia)

Headache
Frontal headache (brow ache) due to ciliary spasm is common initially (particularly in younger people) but usually improves after a few weeks.

Long-term use
Miosis may be permanent with long-term use. Lens opacities have also been reported.

Dosage

Chronic open-angle glaucoma
Adult, initially 1 eye drop of 1% 3 or 4 times daily, then increase concentration slowly according to response and tolerability.

Reversal of mydriasis
Adult, 1 eye drop of 1%.

Counselling
These eye drops may sting and make your eyes water, which may dilute and reduce the effectiveness of subsequent eye drops. If you are using more than one type of eye drop, put pilocarpine drops in last.
These eye drops may cause blurred vision. Do not drive or operate machinery if you are affected. Take particular care in poor light.

Practice points
- to reduce systemic effects, advise patients to close their eyes and apply punctal occlusion after administration, see Counselling p 482

Reversal of mydriasis
- as phenylephrine's mydriatic effect is short, pilocarpine is not routinely used to reverse it
- pilocarpine is ineffective in reversing mydriasis due to topical anticholinergics

eye drop, 1%, 15 mL, *Isopto Carpine*, PBS
eye drop, 2%, 15 mL, *Isopto Carpine*, PBS
eye drop, 2%, 0.5 mL (single use), 20, *Minims Pilocarpine*
eye drop, 4%, 15 mL, *Isopto Carpine*, PBS

11.3 Drugs for allergic and inflammatory eye conditions

Allergic conjunctivitis

A common condition characterised by itchy eyes; other symptoms may include redness, swelling and tearing. It may occur in association with allergic rhinitis (p 412).
There are 2 main types, seasonal (hay fever) and perennial. Drug treatment is aimed at symptom control.

Before starting treatment
Identify and avoid triggers (eg pollens, animal dander, house dust mites) where possible.
Refer patients who wear contact lenses or who have eye pain, photophobia, visual disturbance or severe conjunctivitis to an eye specialist for further investigation.

Topical treatment
When selecting treatment, consider potential adverse effects, convenience (including frequency of use), cost and patient preferences.
General measures include:
- application of cold compress to eyelids as required
- irrigation with sodium chloride 0.9% solution (eg twice daily)
- use of ocular lubricants (p 502).

Antihistamines (p 496): topical antihistamines are preferred to oral antihistamines (particularly for isolated ocular symptoms) as they have a more rapid onset; however, they can sting on instillation. Antihistamines with mast cell stabilising properties (azelastine, ketotifen and olopatadine) may also prevent symptoms of allergic conjunctivitis.
Mast cell stabilisers (p 497) are not useful for acute symptoms as they can take up to 2 weeks to reach full effect. They may be useful if started 2–4 weeks before hay fever season or if used with a topical antihistamine for initial symptom control.
Ketorolac (p 497) is an NSAID occasionally used for seasonal allergic conjunctivitis when response to other treatments is inadequate and additional anti-inflammatory effects are required. However, it can cause ocular irritation.
Corticosteroids (p 498) should only be used under specialist supervision for severe symptoms unresponsive to other treatments. They have serious adverse effects that require monitoring (eg raised intraocular pressure).
Vasoconstrictors (p 495) are available in combination with antihistamines. They are not recommended as their benefit is doubtful and rebound hyperaemia can lead to overuse.

Practice points
- treatment for allergic rhinitis (p 412), eg intranasal corticosteroids, may also relieve the symptoms of allergic conjunctivitis

11.3.1 Vasoconstrictors

See also Allergic conjunctivitis p 494

Naphazoline p 495
Naphazoline with antazoline p 495
Naphazoline with pheniramine p 496
Tetryzoline p 496

Mode of action
Vasoconstrictors are alpha-adrenoreceptor agonists that constrict conjunctival blood vessels, reducing ocular redness and discomfort.

Indications
Temporary relief of eye redness due to minor irritation, see Practice points below

Precautions
Risk factors for angle-closure glaucoma (p 488)—acute angle-closure crisis may rarely be precipitated.

Pregnancy
Safe to use.

Breastfeeding
Safe to use.

Adverse effects

Common (>1%)
rebound hyperaemia, stinging on instillation, mild mydriasis, blurred vision

Comparative information
Naphazoline and tetryzoline have similar efficacy and adverse reactions.

Counselling
Seek medical attention if symptoms do not improve within 48 hours.
Do not use these drops regularly for prolonged periods as this can make eye redness worse.

Practice points
- vasoconstrictor eye drops, used widely in OTC products, are not recommended:
  - their benefit is doubtful
  - rebound hyperaemia leads to overuse
  - although advertised as being useful for relieving eye redness due to minor irritants such as dust, smoke and contact lens wear, a cool compress is as beneficial and is safer; ocular lubricants (p 502) may also relieve irritation due to dry eyes

Naphazoline

For additional information see Vasoconstrictors p 495

Indications
Temporary relief of eye redness due to minor irritation, see Practice points p 495 in Vasoconstrictors

Dosage
Adult, child >12 years, 1 eye drop every 6–12 hours as required.

eye drop, 0.01%, 10 mL, *Optrex*
eye drop, 0.012%, 15 mL, *Murine Clear eyes*
eye drop, 0.1%, 0.5 mL (single use), 10, *Murine Red eyes*
eye drop, 0.1%, 15 mL, *Albalon, Naphcon Forte, Systane Red Eyes*

Naphazoline combinations
Naphazoline with antazoline p 495
Naphazoline with pheniramine p 496

Naphazoline with antazoline

Go to Naphazoline p 495 and Antihistamines p 496 for information on individual components; limited detail below.

Indications
Mild allergic conjunctivitis

Dosage
Adult, 1 eye drop every 6–12 hours as required.

Practice points
- this combination product is not recommended as benefit is doubtful and rebound hyperaemia (caused by vasoconstrictor) leads to overuse

eye drop, naphazoline 0.05%, antazoline 0.5%, 15 mL, *Albalon-A*

Naphazoline with pheniramine

Go to Naphazoline p 495 and Antihistamines p 496 for information on individual components; limited detail below.

Indications
Mild allergic conjunctivitis

Dosage
Adult, child >12 years, 1 eye drop every 6–12 hours as required.

Practice points
- this combination product is not recommended as benefit is doubtful and rebound hyperaemia (caused by vasoconstrictor) leads to overuse

eye drop, naphazoline 0.025%, pheniramine 0.3%, 15 mL, *Naphcon-A, Visine Allergy*

Tetryzoline

For additional information see Vasoconstrictors p 495

Also known as tetrahydrozoline.

Indications
Temporary relief of eye redness due to minor irritation, see Practice points p 495 in Vasoconstrictors

Dosage
Adult, child >6 years, 1 eye drop every 6–12 hours as required.

eye drop, 0.05%, 15 mL, *Murine Sore Eyes, Visine Advanced Relief, Visine Clear*

11.3.2 Antihistamines

See also Allergic conjunctivitis p 494

Antihistamines
Antazoline (see Naphazoline with antazoline p 495)
Levocabastine p 496
Pheniramine (see Naphazoline with pheniramine p 496)

Antihistamine-mast cell stabilisers
Azelastine p 496
Ketotifen p 496
Olopatadine p 496

Mode of action
Reduce the effects of histamine by binding to the H_1 receptor and stabilising it in an inactive form. Azelastine, ketotifen and olopatadine also stabilise mast cells.

Indications
Allergic conjunctivitis

Precautions

Pregnancy
Safe to use.

Breastfeeding
May be used.

Adverse effects

Common (>1%)
stinging on instillation, mild eye irritation, headache

Comparative information
Antazoline and pheniramine are only available in combination with naphazoline (a vasoconstrictor). Although direct comparisons are limited, all the other antihistamines appear to have similar efficacy. In addition to rapid relief of symptoms (related to their antihistamine effects), azelastine, ketotifen and olopatadine may also prevent symptoms due to their mast cell stabilising properties.

Azelastine

For additional information see Antihistamines p 496

Indications
Allergic conjunctivitis

Adverse effects

Infrequent (0.1–1%)
bitter taste

Rare (<0.1%)
allergy

Dosage
Adult, child >4 years, initially 1 eye drop twice daily; if needed increase to 1 drop 4 times daily.

eye drop, 0.05%, 6 mL, *Eyezep*

Ketotifen

For additional information see Antihistamines p 496

Indications
Allergic conjunctivitis

Adverse effects

Infrequent (0.1–1%)
allergy, blurred vision (on instillation), dry eye, dry mouth, sedation

Dosage
Adult, child >3 years, 1 eye drop twice daily.

eye drop, 0.025%, 0.4 mL (single use), 20, *Zaditen*
eye drop, 0.025%, 5 mL, *Zaditen*

Levocabastine

For additional information see Antihistamines p 496

Indications
Allergic conjunctivitis

Adverse effects

Common (>1%)
drowsiness

Dosage
Adult, child >6 years, 1 eye drop twice daily, increasing to 3 or 4 times daily if necessary.

Counselling
You may feel drowsy and the effects of alcohol may be increased; do not drive or operate machinery if you are affected.

eye drop, 0.05% (suspension), 4 mL, *Livostin, Zyrtec Levocabastine*

Olopatadine

For additional information see Antihistamines p 496

Indications
Allergic conjunctivitis

Adverse effects

Common (>1%)
hyperaemia, keratitis, dry eye

Dosage
Adult, child >3 years, 1 eye drop twice daily.

eye drop, 0.1%, 5 mL, *Paladopt, Patanol*

11.3.3 Mast cell stabilisers

See also Allergic conjunctivitis p 494

Cromoglycate p 497
Lodoxamide p 497

Mode of action
Inhibit release of inflammatory mediators from mast cells.

Indications
Allergic conjunctivitis and keratoconjunctivitis

Adverse effects

Common (>1%)
transient stinging and burning on instillation

Counselling
It can take several weeks for these eye drops to reach full effect.

Practice points
- mast cell stabilisers have a delayed onset of action:
  - use for at least 2 weeks before evaluating effect
  - if possible, start treatment 2–4 weeks before the onset of hay fever season, or use with oral or topical antihistamines for initial symptom control
- some ocular antihistamines (p 496) also have mast cell stabilising properties; they are generally preferred to mast cell stabilisers as the antihistaminic effect provides rapid symptom relief

Cromoglycate

For additional information see Mast cell stabilisers p 497

Also known as cromoglicate and cromolyn.

Indications
Allergic conjunctivitis and vernal keratoconjunctivitis

Precautions

Pregnancy
Safe to use.

Breastfeeding
Safe to use.

Dosage
Adult, child, 1 eye drop 4–6 times daily.

eye drop, 2%, 10 mL, *Cromo-Fresh*

Lodoxamide

For additional information see Mast cell stabilisers p 497

Indications
Allergic conjunctivitis and vernal keratoconjunctivitis

Precautions

Pregnancy
No human data but considered safe to use.

Breastfeeding
No human data but unlikely to be a concern.

Dosage
Adult, child >4 years, 1 eye drop 4 times daily.

eye drop, 0.1%, 10 mL, *Lomide*

11.3.4 NSAIDs

Ketorolac p 497
Nepafenac p 498

Mode of action
NSAIDs inhibit cyclo-oxygenase, decreasing prostaglandin synthesis and prostaglandin-mediated inflammation.

Precautions
Aspirin- or NSAID-induced hypersensitivity reactions (eg asthma, urticaria or rhinitis)—contraindicated.
Treatment with ocular corticosteroids—may increase risk of healing problems as both ocular NSAIDs and ocular corticosteroids may delay healing.
Treatment with prostaglandin analogues—NSAIDs may increase or decrease the effect of prostaglandin analogue eye drops on intraocular pressure (IOP); consider monitoring IOP if NSAID eye drops are being used long term.
Ocular infection—due to their anti-inflammatory effects, NSAIDs may mask symptoms of infection.

Pregnancy
Short-term low-dose use should not be a concern.

Breastfeeding
Safe to use.

Adverse effects

Common (>1%)
stinging on instillation, ocular irritation, delayed healing

Infrequent (0.1–1%)
local allergic reactions

Rare (<0.1%)
dyspnoea, exacerbation of asthma, systemic allergic reactions

Corneal effects
Include keratitis, corneal thinning, epithelial breakdown and corneal ulceration. These effects appear to be rare and are often associated with risk factors such as pre-existing epithelial defects.

Practice points
- ocular NSAIDs are used as an alternative or adjuvant to ocular corticosteroids

Ketorolac

For additional information see NSAIDs p 497
See also Allergic conjunctivitis p 494

Indications
Short-term treatment of seasonal allergic conjunctivitis
Prevention and reduction of inflammation associated with cataract surgery

Dosage

Allergic conjunctivitis

Adult, child >3 years, 1 eye drop 4 times daily for up to 4 weeks.

Cataract surgery

Adult, child >3 years, 1 eye drop 4 times daily. Treatment may start 24 hours before surgery; continue for 2–4 weeks if needed.

Practice points

- may be used for prevention and treatment of cystoid macular oedema following cataract surgery

eye drop, 0.5%, 5 mL, *Acular*

Nepafenac

For additional information see NSAIDs p 497

Indications

Cataract surgery:

- to prevent or reduce pain and inflammation
- to reduce risk of postoperative macular oedema in patients with non-proliferative diabetic retinopathy

Dosage

Adult, 1 eye drop once daily. Start 24 hours before cataract surgery, give an additional drop 30–120 minutes before surgery.

After surgery, continue treatment for up to 14 days for prevention of pain and inflammation, or up to 60 days in patients with non-proliferative diabetic retinopathy.

eye drop, 0.3% (suspension), 3 mL, *Ilevro*

1

11.3.5 Corticosteroids

See also Allergic conjunctivitis p 494

Dexamethasone p 499
Fluorometholone p 499
Hydrocortisone p 499
Prednisolone p 499
Prednisolone with phenylephrine p 499

Mode of action

Corticosteroids regulate gene expression resulting in local anti-inflammatory and immunosuppressive effects.

Precautions

History of, or risk factors for, glaucoma—increased risk of ocular hypertension due to corticosteroids; monitor intraocular pressure carefully.

Treatment with ocular NSAIDs—may increase risk of healing problems as both ocular corticosteroids and ocular NSAIDs may delay healing.

Ocular infection

Corticosteroids suppress the immune system and can increase the risk and severity of infections; their anti-inflammatory effects can also mask signs of infection. Ocular corticosteroids are contraindicated in most ocular infections, particularly in herpes simplex epithelial keratitis, fungal keratitis and untreated infections. They may be used in conjunction with anti-infectives in some specific infections with close supervision by, or after discussion with, an ophthalmologist.

Children

Children are at increased risk of ocular hypertension and cataract. They may also be at increased risk of adrenal suppression with intensive or long-term use of ocular corticosteroids.

Pregnancy

Safe to use.

Breastfeeding

Safe to use.

Adverse effects

Do not use without close supervision by, or discussion with, an ophthalmologist: their serious adverse effects can threaten vision.

Common (>1%)

ocular hypertension (below), ocular infection, delayed corneal healing, rebound inflammation (after stopping treatment abruptly)

Rare (<0.1%)

posterior subcapsular cataracts (may be common with long-term use), adrenal suppression (with intensive or long-term use)

Ocular hypertension

Risk depends on the individual as well as the drug's dose, duration of treatment and anti-inflammatory efficacy. Increased intraocular pressure is usually reversible after stopping treatment.

Comparative information

Anti-inflammatory efficacy of topical ocular corticosteroids varies due to factors such as frequency of administration, the drug's potency and its ability to penetrate into the eye. Penetration into the aqueous humour is also affected by the drug's formulation and whether the corneal epithelium is inflamed or damaged.

The relative anti-inflammatory efficacy of corticosteroid eye drops (with an intact cornea), in descending order, is thought to be:

- prednisolone acetate suspension, 1%
- dexamethasone suspension, 0.1%
- fluorometholone suspension, 0.1%
- prednisolone sodium phosphate solution, 0.5%.

Hydrocortisone acetate eye ointment is usually used for eyelid skin inflammation or as an adjunct to corticosteroid eye drops (to reduce the frequency of overnight dosing).

Dexamethasone is also available as an intravitreal implant for specialist use.

Practice points

- only use ocular corticosteroids under close supervision by, or following discussion with, an ophthalmologist
- regular monitoring is required, including:
  - slit lamp examination
  - measurement of intraocular pressure (if used for >10 days)
- when stopping treatment, the dosage may need to be reduced gradually (over several days to weeks) to avoid rebound inflammation
- other ocular corticosteroids may be available via the SAS (see Drug availability in Australia p xviii) including:
  - loteprednol eye drops
  - a preservative-free formulation of triamcinolone suitable for intravitreal injection

Dexamethasone

For additional information see Corticosteroids p 498
For drug interactions see Corticosteroids p 1008

Indications

Selected allergic and inflammatory conditions of lids, conjunctiva, cornea, iris and ciliary body, including postoperative inflammation

Intravitreal implant (seek specialist advice)

Macular oedema due to:

- diabetes
- central retinal vein occlusion
- branch retinal vein occlusion

Non-infectious posterior uveitis

Dosage

Allergic and inflammatory conditions

Use the same way for adults and children.

Dose depends on the indication and severity of inflammation and is adjusted according to response.

Usual range, 1 eye drop 2–6 times daily.

Severe inflammation, initially 1 eye drop every hour then gradually reduce frequency.

eye drop, 0.1% (suspension), 5 mL, *Maxidex,* PBS
implant, 700 mcg, 1, *Ozurdex,* PBS-A[1]

1 macular oedema, non-infectious posterior segment uveitis, see PBS

Fluorometholone

For additional information see Corticosteroids p 498

Indications

Selected allergic and inflammatory conditions of lids, conjunctiva and cornea

Dosage

Adult, child >2 years

Dose depends on the indication and severity of inflammation and is adjusted according to response.

Usual range, 1 eye drop 2–4 times daily.

eye drop, 0.1% (suspension), 5 mL, *FML*

Hydrocortisone

For additional information see Corticosteroids p 498

Also known as cortisol.

Indications

Selected inflammatory conditions of lids, eg eyelid dermatitis

Adjunct to corticosteroid eye drops (to reduce frequency of overnight dosing), eg for anterior uveitis

Dosage

Adult, child

Dose depends on the indication and severity of inflammation and is adjusted according to response.

Eyelid inflammation, apply eye ointment 2–4 times daily.

Adjunct to corticosteroid eye drops, apply eye ointment at night.

eye oint, 1%, 5 g, *Siguent Hycor,* PBS

Prednisolone

For additional information see Corticosteroids p 498

Indications

Selected allergic and inflammatory conditions of conjunctiva and cornea

Dosage

Dose depends on the indication and severity of inflammation and is adjusted according to response.

Adult, child

Usual range, 1 eye drop 2–4 times daily.

Severe inflammation, initially 1 eye drop every hour then gradually reduce frequency.

eye drop, prednisolone sodium phosphate 0.5%, 0.5 mL (single use), 20, *Minims Prednisolone*

Prednisolone combinations

Prednisolone with phenylephrine p 499

Prednisolone with phenylephrine

Go to Prednisolone p 499 for more information; limited detail below.

Indications

Selected inflammatory conditions of the eye, including postoperative inflammation

Precautions

Risk factors for angle-closure glaucoma (p 488)—use with caution as phenylephrine may rarely precipitate acute angle-closure crisis.

Dosage

Adult, child

Dose depends on the indication and severity of inflammation and is adjusted according to response.

Usual range, 1 eye drop 2–6 times daily.

Severe inflammation, initially 1 eye drop every hour then gradually reduce frequency.

eye drop, prednisolone acetate 1%, phenylephrine 0.12% (suspension), 10 mL, *Prednefrin Forte,* PBS-R[1,2]

1 corneal grafts, uveitis
2 severe eye inflammation pre- or post-cataract surgery in Aboriginal or Torres Strait Islander patients

Table 11–2 Comparison of ocular lubricants
See also Dry eye syndrome p 501

Brand®	Preservative	Key ingredients	Pack size, PBS status
Drops (multidose)			
With preservative			
Blink Intensive Tears	oxychloro complex[1]	macrogol 400, sodium hyaluronate	15 mL
GenTeal, In a Wink Moisturising	sodium perborate	hypromellose	10 mL, PBS-R[2]
Liquifilm Tears, PVA Tears	benzalkonium chloride[3]	polyvinyl alcohol	15 mL
Methopt	benzalkonium chloride[3]	hypromellose	15 mL, PBS-R[2]
Murine Tears	benzalkonium chloride[3]	polyvinyl alcohol, povidone	15 mL
Optive	oxychloro complex[1]	carmellose, glycerin	15 mL, PBS-R[2]
Optive Advanced	oxychloro complex[1]	carmellose, glycerin, polysorbate 80	15 mL
Optive Fusion	oxychloro complex[1]	carmellose, glycerin, sodium hyaluronate	10 mL
Optive Gel Drops	oxychloro complex[1]	carmellose, glycerin	10 mL
Optix	oxychloro complex[1]	macrogol 400, propylene glycol	15 mL, PBS-R[2]
Poly-Tears, Tears Naturale	polyquaternium	hypromellose, dextran 70	15 mL, PBS-R[2]
Refresh Liquigel, Refresh Tears Plus	oxychloro complex[1]	carmellose	15 mL, PBS-R[2]
Revive Tears	sorbic acid	hypromellose	10 mL, PBS-R[2]
Systane	polyquaternium	hydroxypropyl guar, macrogol 400, propylene glycol	15 mL, PBS-R[2]
Systane Balance, Systane Complete	polyquaternium	hydroxypropyl guar, mineral oil, propylene glycol, sorbitol	10 mL
Systane Ultra	polyquaternium	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	10 mL
Preservative-free			
Cationorm	none	heavy mineral oil, light mineral oil, glycerin	10 mL, PBS-A[2]
Evolve Carmellose	none	carmellose	10 mL, PBS-A[2]
Evolve Hypromellose	none	hypromellose	10 mL, PBS-A[2]
Hylo-Forte, Hylo-Fresh	none	sodium hyaluronate	10 mL, PBS-A[2]
Murine Dry Eyes	none	sodium hyaluronate	10 mL
NovaTears	none	perfluorohexyloctane	3 mL, PBS-A[2]
Systane Complete Preservative-free	none	hydroxypropyl guar, mineral oil, propylene glycol, sorbitol	10 mL
Systane Ultra Preservative-free	none	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	10 mL
Drops (single use)			
Blink Intensive Tears	none	macrogol 400, sodium hyaluronate	20
Cationorm	none	heavy mineral oil, light mineral oil, glycerin	30
Cellufresh, Celluvisc	none	carmellose	30, PBS-A[90][2]
Optifresh Plus, Optifresh Tears, Refresh Plus	none	carmellose	30
Optive Advanced	none	carmellose, glycerin, polysorbate 80	30
Optive Fusion	none	carmellose, glycerin, sodium hyaluronate	30
Optive Sensitive	none	carmellose, glycerin	30
Refresh	none	polyvinyl alcohol, povidone	10, 30
Systane	none	hydroxypropyl guar, macrogol 400, propylene glycol	30, PBS-A[60][2]
Systane Hydration	none	hydroxypropyl guar, macrogol 400, propylene glycol, sodium hyaluronate	30
Systane Ultra	none	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	24
TheraTears	none	carmellose	24

1 also known as Purite®
2 see *PBS* for details
3 most irritant to the eye, see also Preservatives p 482

Brand®	Preservative	Key ingredients	Pack size, PBS status
Eye gel (multidose)			
GenTeal, HPMC PAA	sodium perborate	hypromellose, carbomer 980	10 g, PBS-R[2]
Optifresh, PAA, Viscotears	cetrimide	carbomer 980	10 g, PBS-R[2]
Systane Gel Drops	polyquaternium	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	10 mL
Eye gel (single use)			
TheraTears	none	carmellose	28
Viscotears Gel PF	none	carbomer 980	30, PBS-A[90][2]
Eye ointment			
Refresh Night Time	none	paraffin	3.5 g, PBS[2]/PBS-R[2][2]
Poly Visc	none	paraffin, lanolin	3.5 g, PBS[2]/PBS-R[2][2]
VitA-POS	none	retinol palmitate, paraffin, lanolin	5 g
Eye spray			
Murine Eye Mist	none	phospholipid liposomes	15 mL
Optrex Actimist	phenoxyethanol		10 mL
Tearsegain	phenoxyethanol		10 mL, PBS-A[2][2]

1 also known as Purite®
2 see *PBS* for details
3 most irritant to the eye, see also Preservatives p 482

11.4 Drugs for dry eyes

Ciclosporin p 502
Lifitegrast p 502
Ocular lubricants p 502

Dry eye syndrome

A common chronic condition; its prevalence increases with age and is most often due to increased evaporation of tears and/or decreased tear production. Symptoms include pain, irritation, blurred vision, itching and gritty or foreign body sensation.

Before starting treatment

Consider secondary causes, eg meibomian gland dysfunction, Sjogren's syndrome, rheumatoid arthritis, contact lenses, or adverse drug effects (eg anticholinergics, estrogens, isotretinoin).

Drug choice

Ocular lubricants

See also Table 11–2 Comparison of ocular lubricants p 500

Ocular lubricants (often called artificial tears) relieve symptoms and are usually used first line. There is no evidence that any lubricant is superior to another; patient preference after an empirical trial usually determines choice.

Consider referral to an optometrist or ophthalmologist if there is an inadequate response after 2–4 weeks' treatment.

Eye drops and gels

Drops may require frequent administration, sometimes up to hourly, depending on severity. Gels provide longer relief than eye drops and can blur vision (but less so than ointments).

Products may contain preservatives (p 482). Usually, tears quickly dilute and remove preservatives, but in dry eye syndrome (particularly when severe) reduced tear secretion increases risk for preservative toxicity. Preservative-free products are preferred for patients who require frequent lubricant use (eg >4 times daily) or who use other eye drops; this is more important than choice of lubricant for these patients.

Eye ointments

Ointments can blur vision and are most often used overnight for symptoms that interrupt sleep or occur on awakening. They may also be used throughout the day if symptoms are severe. Ointments don't contain preservatives, however, lanolin (found in some products) may cause irritation.

Eye sprays

Sprays contain phospholipid (usually soy lecithin) liposomes which are thought to stabilise the lipid layer of the tear film to reduce tear evaporation. They are sprayed onto closed eyelids (from about 10 cm away) and eyes briefly remain closed afterwards.

Other drug treatments

Corticosteroid eye drops may sometimes be used for short periods in more severe cases to reduce ocular surface inflammation. They should only be used under specialist supervision with appropriate monitoring for adverse effects.

Ciclosporin or lifitegrast eye drops are used for their anti-inflammatory effects in moderate-to-severe dry eye syndrome when ocular lubricants are insufficient. Direct comparative trials are lacking but ciclosporin appears to have a slower onset of action

Acetylcysteine may occasionally be used for its mucolytic effect in filamentary keratitis associated with severe dry eyes. It is prepared by some pharmacies as 5% or 10% eye drops.

Practice points

- treat any underlying meibomian gland dysfunction and/or blepharitis; treatment may include warm compresses and eyelid scrubs, see Blepharitis p 483

- low humidity and/or increased air movement (eg air conditioners) can increase tear evaporation and worsen symptoms; consider use of a humidifier or wrap-around glasses; avoid prolonged screen viewing which can reduce blinking and contribute to dry eyes
- other treatment options may include punctal plugs (which prevent tear drainage) or autologous serum eye drops

Ciclosporin

See also Dry eye syndrome p 501

Also known as cyclosporin.

Mode of action

Suppresses the inflammatory response associated with dry eye syndrome, resulting in increased tear production.

Indications

Moderate-to-severe dry eye syndrome when ocular lubricants are insufficient (Cequa®)

Severe keratitis (associated with dry eye syndrome) when ocular lubricants are insufficient (Ikervis®)

Precautions

Peri-ocular or ocular infection—contraindicated as ciclosporin may worsen infection.

Peri-ocular or ocular malignancy—contraindicated.

Pregnancy

Unlikely to be of concern due to low topical absorption.

Breastfeeding

Unlikely to be of concern due to low topical absorption.

Adverse effects

Common (>1%)

pain on instillation, conjunctival hyperaemia, lacrimation

Ikervis®: eye irritation, eyelid erythema

Dosage

Adult

Cequa®, 1 eye drop twice daily.

Ikervis®, 1 eye drop once daily at bedtime.

Counselling

If you need to use an ocular lubricant at the same time of day, leave a 15-minute gap between this medicine and the ocular lubricant.

Practice points

- onset of effect may be delayed (some improvement may be seen after 1 month of treatment); reassess response at least every 6 months
- other formulations of ciclosporin eye drops may be available through the SAS (see Drug availability in Australia p xviii) or prepared by some hospital pharmacies

eye drop, 900 mcg/mL, 0.25 mL (single use), 60, *Cequa*, PBS-A[1]

eye drop, 1 mg/mL, 0.3 mL (single use), 30, *Ikervis*, PBS-A[1]

[1] chronic severe dry eye disease with keratitis, see PBS

Lifitegrast

See also Dry eye syndrome p 501

This is a new drug; unreported adverse effects or interactions may occur.

Mode of action

Lymphocyte function-associated antigen-1 (LFA-1) antagonist thought to suppress the inflammatory response associated with dry eye syndrome.

Indications

Moderate-to-severe dry eye syndrome when ocular lubricants are insufficient

Precautions

Pregnancy

No human data but unlikely to be of concern due to low systemic absorption.

Breastfeeding

No human data but unlikely to be of concern due to low systemic absorption.

Adverse effects

Common (>1%)

ocular irritation (eg pain, itch), lacrimation, blurred vision, taste disturbance

Rare (<0.1%)

hypersensitivity reactions (including anaphylaxis)

Dosage

Adult, 1 eye drop twice daily.

Practice points

- improvement in some dry eye symptoms may be seen within 14 days of starting treatment; assess response after 2–3 months and, if continued, regularly review need for ongoing treatment

eye drop, 5%, 0.2 mL (single use), 60, *Xiidra*

Ocular lubricants

See also Dry eye syndrome p 501, Table 11–2 Comparison of ocular lubricants p 500

Mode of action

Provide lubrication.

Indications

Dry eye syndrome

Relief of mild ocular irritation (eg viral conjunctivitis, after dust, wind or sun exposure, contact lenses, allergic conjunctivitis)

Exposure keratitis (eg facial nerve palsy)

Neurotrophic keratitis

Precautions

Pregnancy

Safe to use.

Breastfeeding

Safe to use.

Adverse effects

Ocular irritation

The frequency of ocular irritation differs slightly with the preservative used, see Preservatives p 482.

Dosage

Use the same way for adults and children.

Eye drops, 1 drop every 1–12 hours as required.

Eye ointment, gel: use at night as required, or throughout the day if symptoms are severe.

Eye spray, 1–2 sprays on closed eyelids 3 or 4 times daily or as required.

Practice points

- encourage regular use of eye drops (eg up to hourly)
- damage to the conjunctiva and cornea caused by preservatives (p 482) can be reduced by switching to preservative-free drops

For products, see Table 11–2 Comparison of ocular lubricants p 500.

11.5 Drugs for eye examinations and procedures

Mydriasis and cycloplegia

Topical anticholinergics aid eye examination by causing mydriasis (pupil dilation) and cycloplegia (paralysis of accommodation). Phenylephrine (a sympathomimetic) may be used as an adjunct to anticholinergics for mydriasis (eg to ensure maximal dilation in difficult-to-dilate pupils). An ocular topical anaesthetic may be used beforehand in some cases.

Practice points

- dilating the pupils may reduce ability to assess head injury
- although pilocarpine (a cholinergic agent) may be used to reverse mydriasis due to phenylephrine, it is not routinely recommended; note that pilocarpine is not effective for reversal of mydriasis due to topical anticholinergics

11.5.1 Anticholinergics

See also Mydriasis and cycloplegia p 503, Anticholinergics p 767
For drug interactions see Anticholinergics p 970

Atropine p 504

Cyclopentolate p 504

Tropicamide p 505

Mode of action

Block muscarinic actions of acetylcholine resulting in relaxation of the iris sphincter and ciliary muscles, leading to pupil dilation (mydriasis) and paralysis of accommodation (cycloplegia), respectively.

Indications

Mydriasis (for examination of ocular fundus)

Cycloplegia (for diagnostic refraction)

Uveitis (for pain relief and prevention or freeing of posterior adhesions)

Precautions

Significant head injury—use short-acting agents only, with care; consult patient's neurosurgeon or intensive care specialist. Before use, note appearance and light reaction of pupils. Always make a note that pupils were dilated intentionally.

Risk factors for angle-closure glaucoma (p 488)—acute angle-closure crisis may rarely be precipitated. Use anticholinergics under specialist supervision in patients who have had treatment for angle-closure crisis.

Lenticular subluxation—small risk of anterior lens displacement.

Diabetes—adequate mydriasis may not be achieved; consider phenylephrine as an adjunct when anticholinergics are used for diagnostic purposes.

Darkly pigmented irides—compared to light irides, mydriatic and cycloplegic effects may be reduced and their onset delayed; when used for diagnostic purposes, consider using a higher concentration, instilling a second drop or adding phenylephrine.

Pregnancy

Safe to use.

Breastfeeding

Safe to use; unlikely to cause adverse effects in breastfed infants at usual doses.

Adverse effects

Systemic effects appear to be dose-related. They are particularly rare with tropicamide.

Common (>1%)

intolerance to bright light (glare), stinging on instillation (especially 1% cyclopentolate), blurred vision (especially near vision), transient intraocular pressure elevation (especially in pre-existing ocular hypertension)

Infrequent (0.1–1%)

contact allergic blepharitis (atropine), persistent ocular irritation (mucus discharge, severe watering discharge, superficial punctate keratopathy and characteristically no itch), punctal stenosis with prolonged use (years), insomnia, drowsiness (especially cyclopentolate)

Rare (<0.1%)

systemic toxicity, eg dryness of skin and mouth, fever, facial flushing, tachycardia, irritability, disorientation, ataxia, visual hallucinations, incoherent speech, delirium, psychosis, seizures, hyperactivity (children)

Comparative information

See Table 11–3 Comparison of ocular anticholinergics p 504

Counselling

Enlarged pupils cause blurred vision and sensitivity to bright light; wearing dark glasses may help. Your ability to judge distance may be impaired. Do not drive or operate machinery while vision is disturbed.

Tell your health professional if the eye becomes painful or red, or if vision deteriorates.

Parents or carers of children: following pupil dilation, your child may not be able to perform skilled tasks until their vision clears.

Table 11–3 Comparison of ocular anticholinergics

Drug/use	Peak effect[1]	Duration[1]	Comments
Atropine 1%			
mydriasis	30–40 minutes	7–10 days	• rarely used for diagnostic indications as it can affect near vision for at least a week after use • may be considered for refraction when maximal cycloplegia is required
cycloplegia	1–3 hours	7–14 days	
Cyclopentolate 0.5–1%			
mydriasis	30–60 minutes	1 day	• preferred for cycloplegic refraction • may be considered for mydriasis if other agents fail
cycloplegia	25–75 minutes	6–24 hours	
Tropicamide 0.5–1%			
mydriasis	20–40 minutes	6 hours	• preferred for mydriasis (may be combined with phenylephrine (p 505) if additional dilation is required) • may be considered for cycloplegic refraction, although its effects are less reliable
incomplete cycloplegia	30–40 minutes	2–6 hours	

[1] when given as a single dose

Atropine

For additional information see Anticholinergics p 503
For drug interactions see Anticholinergics p 970, Atropine p 970

Indications

Cycloplegic refraction (when maximal cycloplegia required), see also Practice points below
Uveitis
Penalisation of unaffected eye in treatment of amblyopia
Reduce progression of myopia

Precautions

Children

One drop of 0.5% atropine can cause systemic effects in infants. Use with particular caution, if at all, in neonates, preterm infants and children with spastic paralysis or brain damage (increased susceptibility to systemic reactions).
Also use with caution in children <3 months as prolonged cycloplegia is possibly associated with development of amblyopia.
Observe children for 30–45 minutes after instillation of first dose.

Dosage

Adult

Cycloplegic refraction, 1 eye drop (1%) 1 hour before examination. In some cases, 1 drop (1%) once daily for up to 3 days before examination may be used.
Uveitis, 1 eye drop (1%) once or twice daily (maximum 4 times daily).

Child

For children <1 year, refer to hospital protocols.
Cycloplegic refraction (child >1 year), usually 1 eye drop (1%) once daily for 1–3 days before examination.
Uveitis (child >2 years), 1 eye drop (1%) once daily. If inadequate, consider alternative treatment rather than increasing dose.
Amblyopia (child >3 years), 1 eye drop (1%) in the unaffected eye once daily each Saturday and Sunday.
Myopia (child 4–14 years), 1 eye drop (0.01%) in each eye once daily, at night.

Counselling

This medicine is long-acting; your eye(s) may remain dilated for several days after stopping the eye drops.

Practice points

- atropine is rarely used for cycloplegic refraction in school-age children or adults as it can affect near vision for at least a week after administration
- in amblyopia, penalisation with atropine appears to be as effective as occlusive patching; although most evidence for atropine is in children aged 3–12 years, it should also be considered as an option in older children
- atropine reduces myopia progression; however, long-term safety and the optimal dose, strength and duration of treatment are uncertain

eye drop, 0.01%, 0.3 mL (single use), 30, *Eikance*
eye drop, 1%, 15 mL, *Atropt*, PBS
eye drop, 1%, 0.5 mL (single use), 20, *Minims Atropine*

Cyclopentolate

For additional information see Anticholinergics p 503
For drug interactions see Anticholinergics p 970

Indications

Diagnostic use (mydriasis, cycloplegia)
Uveitis

Precautions

Children

Use with caution in neonates, preterm infants and children with spastic paralysis or brain damage (increased susceptibility to systemic reactions).
Moderate-to-severe drowsiness is common; young children (<6 years) with low BMI are at greatest risk.
Observe children for 30–45 minutes after instillation (when used for uveitis, this is only required after the first dose).

Dosage

Diagnostic use

Instil 30–60 minutes before examination.

Adult

1 eye drop of 0.5% or 1%, repeated in 5–15 minutes if necessary.

Child

1–12 years, 1 eye drop of 0.5% or 1%, repeated in 5–15 minutes if necessary.
1 month – 1 year, 1 eye drop of 0.5%, repeated in 5–15 minutes if necessary.

Uveitis

Adult, child >3 months, 1 eye drop of 0.5% (1% for deeply pigmented eyes) every 6–8 hours.

Practice points

- combination cyclopentolate-phenylephrine eye drops may be available through the SAS (see Drug availability in Australia p xviii) or prepared by some hospital pharmacies

eye drop, 0.5%, 0.5 mL (single use), 20, *Minims Cyclopentolate*
eye drop, 1%, 15 mL, *Cyclogyl*
eye drop, 1%, 0.5 mL (single use), 20, *Minims Cyclopentolate*

Tropicamide

For additional information see Anticholinergics p 503
For drug interactions see Anticholinergics p 970

Indications

Diagnostic use (mydriasis, cycloplegia)

Precautions

Children

Use with caution in neonates, preterm infants and children with spastic paralysis or brain damage (although systemic reactions with tropicamide are rare there may be an increased risk in these patients).

Dosage

For children <1 year, refer to hospital protocols.

Mydriasis for examination of fundus

Adult, child >1 year, 1 eye drop (0.5%) 15–20 minutes before examination. Consider using 1% eye drops for darkly pigmented eyes.

Cycloplegic refraction

Adult, child >1 year, 1 eye drop (1%); repeat after 5 minutes. Another drop may be needed to prolong the effect if patient is not examined within 20–30 minutes.

eye drop, 0.5%, 0.5 mL (single use), 20, *Minims Tropicamide*
eye drop, 0.5%, 15 mL, *Mydriacyl*
eye drop, 1%, 0.5 mL (single use), 20, *Minims Tropicamide*
eye drop, 1%, 15 mL, *Mydriacyl*

11.5.2 Other drugs for mydriasis

Phenylephrine

See also Mydriasis and cycloplegia p 503
For drug interactions see Phenylephrine p 1070

Mode of action

Relatively selective alpha$_1$ agonist; stimulates pupil dilator muscle. Maximal mydriasis occurs after 60–90 minutes; duration of action is 5–7 hours. Does not affect accommodation.

Indications

Diagnostic mydriasis

Accepted

Prevention or freeing of posterior adhesions in uveitis

Precautions

Risk factors for angle-closure glaucoma (p 488)—acute angle-closure crisis may rarely be precipitated.
Recent MI, unstable angina, insulin-dependent diabetes—BP elevation with the 10% drops may be significant if repeated doses given.
Treatment with MAOI—may cause a hypertensive reaction; the 10% drops carry more risk than weaker strengths.

Elderly

Increased risk of systemic adverse effects; do not use 10% drops.

Children

Increased risk of systemic adverse effects, especially hypertension and intraventricular bleeding in the first 2–4 weeks of life in preterm babies. Do not use 10% drops in children.

Pregnancy

Avoid use, particularly the 10% drops and in the first trimester.

Breastfeeding

Safe to use.

Adverse effects

Common (>1%)

rebound miosis, hyperaemia, stinging on instillation

Infrequent (0.1–1%)

liberation of iris pigment (probably has no deleterious effects), allergic contact dermatitis

Rare (<0.1%)

systemic effects (most frequently with 10% drops), eg hypertension, tachycardia, tremor, anxiety, sweating

Dosage

Mydriasis

Adult, child, 1 eye drop (2.5%) once only as adjunct to anticholinergic if mydriasis difficult.

Uveitis

Adult, 1 eye drop (10%) up to 4 times daily as adjunct to anticholinergic.

Counselling

Enlarged pupils cause blurred vision and sensitivity to bright light; wearing dark glasses may help. Your ability to judge distance may be impaired. Do not drive or operate machinery while vision is disturbed.
Tell your health professional if the eye becomes painful or red, or if vision deteriorates.

eye drop, 2.5%, 0.5 mL (single use), 20, *Minims Phenylephrine*
eye drop, 10%, 0.5 mL (single use), 20, *Minims Phenylephrine*

Phenylephrine combinations

Prednisolone with phenylephrine p 499

11.5.3 Local anaesthetics

For additional information see Local anaesthetics p 27

Lidocaine with fluorescein (see Fluorescein p 1129)
Oxybuprocaine p 506
Proxymetacaine p 506
Tetracaine p 506

Mode of action
Block nerve conduction reversibly.

Indications
Short-term ocular surface anaesthesia

Precautions
Corneal scrapings—use preservative-free drops (preservative may affect microbiological culture).

Pregnancy
Safe to use.

Breastfeeding
Safe to use.

Adverse effects

Common (>1%)
stinging on instillation, punctate epithelial damage of cornea (do not use long term because of epithelial toxicity, ie acute corneal ulceration)

Rare (<0.1%)
allergy

Comparative information
See Table 11–4 Comparison of ocular local anaesthetics p 506

Counselling
Close eyes after instillation and dab away tears without rubbing eyes.
These eye drops may sting at first.

Practice points
- topical anaesthetics mask the protective mechanism of pain
- prolonged use impairs corneal epithelial healing, prevents reflex ocular protection and masks progression of keratopathy; use for short procedures only (<20 minutes)
- never prescribe for home use
- topical anaesthetics increase corneal permeability and intraocular bioavailability of other topical drugs; they also reduce the initial stinging of other topical drugs and should be instilled first

Oxybuprocaine

For additional information see Local anaesthetics p 506

Also known as benoxinate.

Indications
Short-term ocular surface anaesthesia

Dosage
Adult, child, 1 eye drop; repeat in 1–2 minutes if necessary. Up to 6 doses may be used for foreign body removal or minor surgery; up to 10 doses for pterygium surgery.

eye drop, 0.4%, 0.5 mL (single use), 20, *Minims Oxybuprocaine*

Proxymetacaine

For additional information see Local anaesthetics p 506

Also known as proparacaine.

Indications
Short-term ocular surface anaesthesia

Dosage
Adult, child, 1 eye drop, repeated in 5 minutes if necessary. Up to 2 doses may be used for foreign body removal or minor surgery; up to 7 doses for minor corneal or conjunctival procedures.

eye drop, 0.5%, 15 mL, *Alcaine*

Tetracaine

For additional information see Local anaesthetics p 506

Also known as amethocaine.

Indications
Short-term ocular surface anaesthesia

Dosage
Adult, child, 1 eye drop, repeated in 5 minutes if necessary. Up to 3 doses may be used for foreign body removal or minor surgery; up to 5 doses for prolonged anaesthesia (eg cataract extraction).

eye drop, 0.5%, 0.5 mL (single use), 20, *Minims Amethocaine*
eye drop, 1%, 0.5 mL (single use), 20, *Minims Amethocaine*

Table 11–4 Comparison of ocular local anaesthetics

Drug	Type	Onset of action	Duration	Duration of initial sting	Punctate epithelial damage
lidocaine[1]	amide	2–5 minutes	20–30 minutes	30 seconds	+
oxybuprocaine	ester	20 seconds	10–20 minutes	10 seconds	++
proxymetacaine	ester	20 seconds	10–20 minutes	10 seconds	++
tetracaine	ester	20 seconds	20 minutes	30 seconds	+++

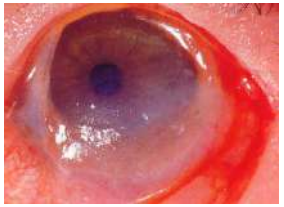
[1] not available as a single-ingredient product

Chapter 12

Gastrointestinal drugs

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Lid Diseases + Blepharitis (Week 4)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Erythema Multiforme (Week 2)   	**Definition**: • Acute, immune-mediated, self-limited condition. • Targets primarily the skin but can involve mucous membranes including ocular surfaces. Causes and Triggers: • Infections: ○ Herpes Simplex Virus (HSV) — most common overall cause, particularly in recurrent EM. ○ Mycoplasma pneumoniae. • Drugs: ○ **Sulfonamides** (important): Sulfonamide antibiotics can trigger hypersensitivity reactions, including Erythema Multiforme, Stevens-Johnson Syndrome (SJS), and Toxic Epidermal Necrolysis (TEN) ○ Other drugs: NSAIDs, antiepileptics, antibiotics like penicillins. • Other Triggers: ○ Malignancies, autoimmune conditions. Clinical Presentation: • Skin: Target lesions (“iris” or “bull’s-eye” appearance). • Distribution: Symmetric, predominantly on extremities. • Mucosal involvement (EM Major): • Oral ulcers • Conjunctivitis • Genital erosions. • Ocular involvement: • Conjunctival injection (redness) • Membranous conjunctivitis • Keratitis (corneal inflammation) • Potential for severe sequelae like symblepharon (adhesions). Systemic Symptoms: Mild fever, malaise can precede lesions. **Differential Diagnoses:** • Stevens-Johnson Syndrome (SJS) • Toxic Epidermal Necrolysis (TEN) • Urticaria • Viral exanthems • Drug-induced eruptions • Autoimmune blistering disorders (e.g., bullous pemphigoid) **Important clinical point:** SJS and TEN are on a spectrum with EM but are typically more severe, involving extensive mucocutaneous necrosis and systemic illness.	**Workup**: • Identify infectious trigger (especially HSV). • Review medication history carefully, especially recent sulfonamide or antibiotic use. • Ocular examination if there are eye symptoms. **Non Pharmacological Treatment:** • Discontinue any offending medication immediately (e.g., sulfonamide antibiotic if suspected). • Supportive care: ○ Skin and mucosal care ○ Hydration ○ Ocular lubrication with preservative-free artificial tears for ocular surface protection. **Pharmacological Treatment:** 1. Mild Cases (EM Minor): • Symptomatic management. • Antihistamines for pruritus. • Topical corticosteroids for skin if inflamed. 2. Moderate to Severe Cases (EM Major, mucosal involvement): • Systemic corticosteroids (e.g., oral prednisolone) may be used to control inflammation. • If HSV-triggered: ○ Oral antivirals such as acyclovir or valacyclovir. ○ Consider long-term antiviral prophylaxis if recurrent. 3. Ocular Management: • Preservative-free artificial tears to prevent dryness and mechanical trauma. • Topical corticosteroids (e.g., fluorometholone or prednisolone acetate) for conjunctivitis or keratitis if inflammation is significant. • Topical antibiotic prophylaxis (e.g., chloramphenicol 0.5% drops) if there is an epithelial defect at risk of infection. • Urgent referral to ophthalmology if corneal ulceration, symblepharon, or conjunctival scarring is developing.	**General Management:** • Patient Education: Counsel about trigger identification and avoidance. • Drug Allergy Documentation: If drug-induced (e.g., sulfonamide allergy), ensure documentation in medical records. • Ophthalmology follow-up: If ocular involvement occurs to prevent chronic complications. **Prognosis**: • Good in EM minor — often self-limiting in 2–4 weeks. • Recurrent EM is common in HSV-triggered cases. • Severe cases (particularly drug-induced EM major) can lead to permanent complications like: ○ Severe dry eye ○ Symblepharon ○ Vision loss if not appropriately managed.
Hordeolum ☺	• Acute, tender, erythematous lump on eyelid (infection). • **External hordeolum:** involves eyelash follicle or glands of	**Non-Pharmacological:** • Warm compresses (5–10 minutes, 3–4× daily) to promote	• Self-limiting in most cases. • Internal hordeola may evolve into



Zeis/Moll; points anteriorly.
- **Internal hordeolum:** involves meibomian gland; points inward through the conjunctiva.
- Often caused by **Staphylococcus aureus.**
- Frequently associated with blepharitis.
- **Acute** (hordeolum), chronic (chalazion)

Differentials:
- **Chalazion** – chronic, **non-tender**, non-infectious granuloma.
- **Preseptal cellulitis** – diffuse eyelid swelling without a focal lump.
- **Sebaceous gland carcinoma** – persistent, recurrent, or atypical lesion (especially elderly).

drainage.
- Gentle lid hygiene and lid scrubs (avoid massage during acute phase).
- Lash epilation for external hordeolum.
- Hot spoon bathing and stye mask.

Pharmacological:
- Topical **chloramphenicol** 0.5% qid if lesion is discharging.
- Systemic **antibiotics** (e.g., Augmentin Duo or **cephalexin**) if severe, recurrent or spreading.
  - Amoxicillin + clavulanic acid 875/125 mg (augemntin Duo) twice a day or:
  - Cephalexin 500 mg 2-4 times a day
- Treatment should last for 7 days

Surgical:
- Incision and drainage for non-resolving internal hordeolum.

chalazia.
- Monitor for **preseptal/orbital cellulitis.**
- Refrain from lid massaging during active/acute stage to limit the potential of spreading
- **Palpation** to determine the hardness of the lesion.
- Routine referral to ophthalmologist if persistent/recurrent.

Treatment	Duration	Infection	Size	Hardness
Compresses	Less than 6 wk	Yes	Small-medium	Soft
Intralesional steroids/incision	2 wk-6 mo	No/yes	Small-large	Rubbery
Incision/drainage	Longer than 6 mo	Yes	Small-large	Hard

Chalazia



Definition:
- A chalazion is a localized, chronic, sterile, lipogranulomatous inflammation caused by obstruction of a meibomian gland duct, often evolving from an unresolved internal hordeolum [OBJ].

Signs:
- Subcutaneous nodule within the tarsal plate of the eyelid.
- Typically well-defined, non-tender, rubbery to firm lesion.
- Diameter between 2 to 8 mm [OBJ].
- May cause mechanical ptosis or induce astigmatism if large enough.

Symptoms:
- Painless swelling or lump in the eyelid.
- Cosmetic concerns (cosmetically unacceptable to patient).
- Occasionally mild tenderness if secondarily infected.

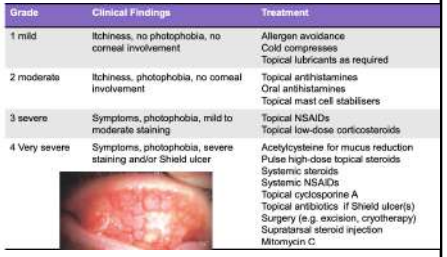
Differentials:
- Hordeolum:
  - Hordeolum is typically acute, tender, erythematous, painful — important to distinguish because chalazion is usually non-tender, chronic.
  - Hordeola are infectious; chalazia are non-infectious, inflammatory.
  - Hordeola usually present for less than 6 weeks, chalazia longer than 6 weeks [OBJ].
- Meibomian gland carcinoma (rare but important):
  - Suspect if lesion is recurrent, unilateral, non-resolving, or associated with lash loss, distortion of lid margin, or other concerning features.
  - Requires urgent referral (within 1 week)

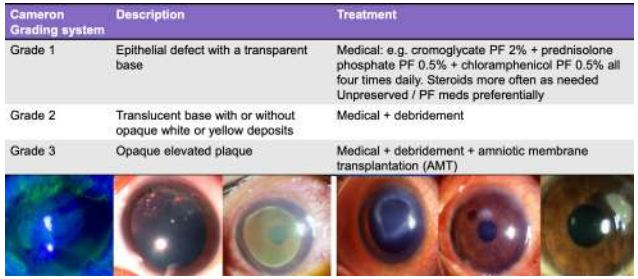
Initial / Optometric Management:
- Alleviation/Palliation:
  - Warm compresses:
    - Apply for 5–10 minutes several times a day to soften secretions and promote spontaneous drainage [OBJ].
  - Lid massage:
    - Gentle massage after warm compresses to help express gland contents.
  - Hygiene:
    - Encourage gentle lid scrubs with appropriate lid cleanser (NOT baby shampoo) .
  - Pharmacological (rarely needed for chalazion itself):
    - If secondary infection suspected (rare), consider prophylactic topical antibiotics (e.g., chloramphenicol 0.5% qid) [OBJ].

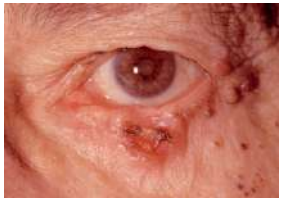
Ophthalmological Management:
- Routine Referral to Ophthalmologist if:
  - Persistent beyond 6 weeks despite conservative management.
  - Recurrent lesions.
  - Causing significant astigmatism.
  - Cosmetically unacceptable to patient [OBJ].
- Specialist Treatments:
  - Intense Pulse Light (IPL) with Meibomian Gland Expression (MGX):
    - In-office application of IPL therapy with or without radiofrequency.
    - A series of 3–4 sessions every few weeks can be used [OBJ].
    - Emerging evidence shows

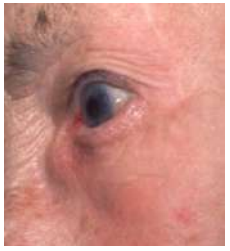
General Management Tips:
- Educate the patient about the benign nature of chalazia.
- Emphasize the importance of consistent warm compresses and lid hygiene to prevent recurrence [OBJ].
- Manage patient expectations that resolution may take several weeks to months.

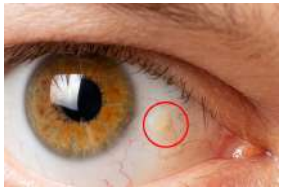
Prognosis:
- Most small chalazia resolve spontaneously with conservative treatment.
- Larger or persistent lesions typically resolve after surgical or intralesional intervention.
- Recurrences are possible, especially in patients with underlying meibomian gland dysfunction (MGD) or ocular rosacea, requiring long-term lid hygiene and management [OBJ] [OBJ].

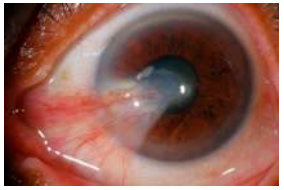
		equivalence to surgical curettage but requires higher-level confirmation [OBJ]. • Incision and Curettage: ○ Traditional gold-standard treatment for persistent chalazion. ○ Surgical drainage under local anaesthetic [OBJ]. • Intralesional Corticosteroid Injection: ○ Alternative to incision and curettage, especially useful in children or those unwilling/unable to undergo surgery. ○ Usually triamcinolone acetonide is injected into the lesion.	
Cellulitis 	• **Preseptal cellulitis:** infection anterior to orbital septum; eyelid edema, warmth, erythema, tenderness, no proptosis or ophthalmoplegia. • **Orbital cellulitis:** posterior to septum; presents with proptosis, painful/restricted eye movements, fever, possible RAPD → ocular dysfunction, **severe and sight/life-threatening** • **Common causes:** sinusitis, trauma, hordeolum, insect bite. **Differentials**: • **Allergic edema** – bilateral, non-tender, non-infectious. • **Hordeolum** – localized painful lump. • **Contact dermatitis** – itching with history of allergen exposure. • **Orbital tumors or abscesses.**	• **Preseptal**: **Oral antibiotics** (e.g. amoxicillin-clavulanic acid, **cephalexin**). ○ Oral **dicloxacillin** or **cephalosporin** (compounded by an opthal) for 10 days • **Orbital**: Emergency referral for IV antibiotics, imaging (CT), hospital admission. • Monitor visual acuity, pupil reactions, and ocular motility.	• Prompt treatment essential to prevent complications like abscess, meningitis, or vision loss.
Vernal Keratoconjunctivitis (VKC)  	• Chronic allergic eye condition, typically in young boys in hot, dry climates. • Bilateral intense itching, photophobia, ropy discharge, blepharospasm. • 23% are perennial → long-lasting • **Signs**: giant papillae on upper tarsus “cobblestone” appearance, Trantas dots, conjunctival hyperemia. • Can lead to shield ulcers and keratopathy. • Booth a **Type 1 and Type 4** hypersensitivity reaction **Differentials**: • **Atopic keratoconjunctivitis (AKC)** – older age onset, eczema. • **Giant papillary conjunctivitis (GPC)** – contact lens wearers. • **Chlamydial conjunctivitis** – follicular reaction. • **Herpetic keratitis** – dendritic lesions.	• **Grade 1 (Mild):** itchiness, no photophobia, no corneal involvement ○ Allergen avoidance, cold compresses, preservative-free lubricants. • **Grade 2 (Moderate):** itchiness, photophobia, no corneal involvement ○ Add topical **antihistamines, mast cell stabilisers** (e.g., **olopatadine**). • **Grade 3 (Severe):** symptoms, photophobia, mile to moderate staining ○ Topical **NSAIDs**, short-term topical corticosteroids. • **Grade 4 (Very severe):** symptoms, photophobia, severe staining and/or shield ulcer ○ Supratarsal steroid injections, systemic immunosuppressants (e.g., **cyclosporine**), amniotic membrane if corneal involvement. ○ Pulse high dose topical steroids ■ **PredForte** 1% every 1-2 hours while awake for 4-7 days, then 4x a day for 4-7 days	• Typically resolves post-puberty. • Recurrence and steroid dependence are concerns. • An attempt is normally made to limit steroid therapy to 1-2 weeks in pulse therapy • Systemic NSAIDs, systemic steroids and topical cyclosporine A are possible options for severe disease presentations  Grade / Clinical Findings / Treatment 1 mild — Itchiness, no photophobia, no corneal involvement — Allergen avoidance; Cold compresses; Topical lubricants as required 2 moderate — Itchiness, photophobia, no corneal involvement — Topical antihistamines; Oral antihistamines; Topical mast cell stabilisers 3 severe — Symptoms, photophobia, mild to moderate staining — Topical NSAIDs; Topical low-dose corticosteroids 4 Very severe — Symptoms, photophobia, severe staining and/or Shield ulcer — Acetylcysteine for mucus reduction; Pulse high-dose topical steroids; Systemic steroids; Systemic NSAIDs; Topical cyclosporine A; Topical antibiotics if Shield ulcer(s); Surgery (e.g. excision, cryotherapy); Supratarsal steroid injection; Mitomycin C
VKC Shield Ulcer	• Non-infectious corneal ulcer from mechanical trauma by giant papillae. • Appears superiorly as grey-white, oval ulcer with possible eosinophilic plaque. • Causes photophobia, pain, reduced vision.	• **Grade 1:** topical **corticosteroids** and **mast cell stabilisers.** • **Grade 2:** add epithelial debridement. • **Grade 3:** surgical intervention (e.g., amniotic membrane grafting).	• Requires long-term VKC control to prevent recurrence. • Visual prognosis good with timely treatment. • Could take months to

	• Thought to be caused by giant papillae abrading the corneal surface – or inflammatory mediators secreted by **eosinophils** **Differentials:** • **Bacterial ulcer –** purulent discharge, pain. • **HSV keratitis** – dendritic pattern. • **Neurotrophic ulcer** – reduced corneal sensitivity. • **Marginal keratitis** – peripheral corneal infiltration.	 **Cameron Grading system** / **Description** / **Treatment** Grade 1 – Epithelial defect with a transparent base – Medical: e.g. cromoglycate PF 2% + prednisolone phosphate PF 0.5% + chloramphenicol PF 0.5% all four times daily. Steroids more often as needed Unpreserved / PF meds preferentially Grade 2 – Translucent base with or without opaque white or yellow deposits – Medical + debridement Grade 3 – Opaque elevated plaque – Medical + debridement + amniotic membrane transplantation (AMT)	re-epithelialise • Translucent dense plaque deposited over the sterile ulcer – toxic eosinophilic granule major basic protein (cytotoxic and delays re-epithelialisation) • Debridement ➔ scraping of base and margin of ulcer using Bard Parker blade
Atopic Dermatitis (Eczema) 	• Chronic relapsing eczematous skin condition involving eyelids. • Red, dry, itchy skin, scaling, sometimes crusting. • Frequently associated with asthma, allergic rhinitis (immune disorders). **Differentials:** • **Contact dermatitis** – acute onset with clear allergen exposure. • **Seborrheic dermatitis** – greasy scale, less itchy. • **Blepharitis** – more lid-margin focused, meibomian involvement.	• **General care:** soap substitutes, cold compresses, fragrance-free emollients. Medications: • Topical **steroids** (e.g., Hycor 1% bid–qid for 5–10 days). • Weak topical corticosteroids: **hydrocortisone** 1% ophthalmic ointment, used twice daily • **2nd line treatment:** calcineurin inhibitors (pimecrolimus, tacrolimus) • Oral **antihistamines** for the itchiness (non-sedating day, sedating night). • Tacrolimus ointment (non-steroidal immunosuppressant) for chronic cases. • Refer to GP/dermatologist if unresponsive.	• Chronic but manageable. • Important to treat fully during flares and prevent triggers. • Tell the patient to avoid rubbing. • Perhaps consultation with an allergist.
Contact Blepharoconjunctivitis 	• Acute allergic response due to direct contact with sensitizing agent (Type IV hypersensitivity). • Eyelid swelling, redness, itching, tearing. • Sudden onset. • **Common triggers:** neomycin, preservatives (BAK), cosmetics. **Differentials**: • **Atopic dermatitis** – chronic and relapsing. • **Blepharitis** – lid margin inflammation, not acute. • **Infectious conjunctivitis** – purulent discharge. • **Hordeolum** – focal painful swelling.	• Avoid allergen exposure. • Cold compresses to reduce inflammation. • Removal of the irritant. • **Topical corticosteroids: Hycor** 1% ointment, FML 0.1% (but no longer on PBS). • Oral **antihistamines** for symptomatic relief. • Refer for patch testing if persistent/recurrent.	• Rapid resolution if allergen is removed. • Educate on avoiding offending products. • Hypersensitivity involves a cell-associated antigen, **cytotoxic T-lymphocytes** and cytotoxicity NOT mast-cell activation
Molluscum Contagiosum  	• Viral infection (**poxvirus**) producing small, firm, umbilicated papules. • Due to skin-to-skin contact and mildly contagious. • Can affect eyelid and shed virus into conjunctiva ➔ chronic **follicular conjunctivitis.** • More common in children, immunocompromised. • Grow over 1-2 weeks and normally asymptomatic • Pannus in long-standing cases. **Differentials**: • **Warts** – rougher surface. • **Chalazion** – deeper, non-umbilicated. • **Sebaceous carcinoma** – older age, persistent.	• **Observation**: most resolve in 6–18 months. • **If needed:** excision, curettage, cryotherapy. • Lubricants for conjunctivitis. • Refer if recurrent/chronic ocular involvement. • Cautery and expression can also be successful	• Highly contagious. • Good hygiene to prevent spread. • Self-limiting with excellent prognosis.
Phthiriasis [pediculosis ciliaris (lice)]	• Infestation of eyelashes by pubic lice (Pthirus pubis). • Intense lid itch, visible nits and adult lice.	• Mechanical epilation of affected lashes. • Treat underlying lice source – refer for STI screen.	• Excellent prognosis if treated. • Reassess in **1 week for hatchlings**.

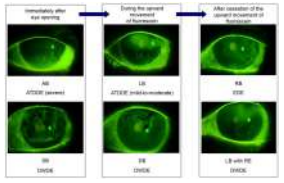
	• Blood/fecal debris at lash base. • Usually sexually transmitted, but can also be spread by contact with trowels, undergarments and bedding. **Differentials**: • **Demodex** – cylindrical sleeves, no visible lice. • **Blepharitis** – no motile organisms. • **Contact dermatitis** – no lice or debris.	• Wash bedding, clothing. Notify contacts. • Child case = **mandatory safeguarding referral.** **Pharmacological** • Smothering with bland **ointment** (e.g. Vaseline) BID for 10–14 days. • **Pediculicidal** agent (**permetrin** 1% lotion) – possible for 10 minutes with eyes closed BUT risk of ocular surface toxicity	• Symptoms of genital itchiness ➔ referring them to the GP or sexual health clinic
Stings & Insect bites	• **Localised** allergic or irritant response. • Redness, lid swelling, pain or itch. • May mimic hordeolum or cellulitis. **Differentials**: • **Preseptal cellulitis** – history of trauma, infection. • **Contact dermatitis** – itch but no bite history. • **Hordeolum** – focal lump, often painful	• Remove stinger if present. • Cold compresses. • Topical corticosteroids. • **Oral antihistamines.** • Refer for allergy testing if recurrent.	• Usually self-limiting. • Monitor for secondary infection or systemic allergic reaction. • Consider going to an allergist/GP – also be wary of anaphylaxis reaction
Basal Cell Carcinoma (periocular) ☺ 	• Most common periocular malignancy (80–90% of skin malignancies in white populations). • Median diagnosis age: 67 years, M:F = 3:2. • **Risk factors:** fair skin, chronic UV exposure (e.g. outdoor workers). **Clinical signs:** • Ulceration. • Lack of tenderness. • Induration (firmness). • Irregular borders. • Destruction of lid margins. • **Madarosis** (lash loss). **Differentials:** • Chalazion (painless, mobile nodule). • Sebaceous carcinoma (aggressive, yellowish, recurrent). • Squamous cell carcinoma (faster-growing, may ulcerate). • Actinic keratosis.	**Optometric role:** • No pharmacological treatment. • Photograph lesion, document size, shape, location. • Routine non-emergency referral to ophthalmologist. • Educate patient: low-risk cancer, sun protection (hat, sunglasses, sunscreen). **Ophthalmologist:** • Biopsy for confirmation. • Surgery or radiotherapy (up to 98% cure rate). • Mohs micrographic surgery for tissue-sparing excision	• Slow-growing, rarely metastasizes (<0.1%). • If untreated, can invade orbit and be vision/life-threatening. • Prognosis is excellent with timely intervention.
Ectropion 	• Outward rotation of eyelid margin, usually lower lid. • Leads to exposure keratopathy, epiphora, dry eye, redness. • Prevalence: 4% in those >50 y.o., bilateral in 70%. **Causes:** • Involutional (age-related, most common). • Paralytic (e.g., CN VII palsy). • Cicatricial (scarring). • Mechanical (tumour, swelling). **Differentials:** • Conjunctival chemosis. • Lid retraction (due to thyroid eye disease). • Floppy eyelid syndrome. • Bell's palsy (if facial asymmetry present).	• Advise to avoid lid rubbing ➔ increases lid laxity • Lid taping at night. • Lubricants (drops during day, ointment at night). • Therapeutic contact lenses. **Referral**: • Routine referral to ophthalmologist if symptomatic. • **Indications for surgery**: exposure keratopathy, epiphora, irritation, recurrent conjunctivitis, cosmesis.	• Progressive without treatment. • Prognosis excellent with surgical correction --- **Forward distraction or pinch test** pull the lower lid away from the globe, if 6mm = canthal tendon laxity

		<table><tr><th>Ectropion</th><th>Entropion</th></tr><tr><td>Involutional (age-related)</td><td>Involutional (age-related)</td></tr><tr><td>Cicatricial</td><td>Cicatricial</td></tr><tr><td>Paralytic</td><td>Spastic</td></tr><tr><td>Mechanical (e.g. tumour, swelling)</td><td>Congenital</td></tr><tr><td>Congenital</td><td>Epiblepharon</td></tr></table>	 **Lid snap-back test** pull lower lid down and release and lif should snapback (if not then **poor orbicularis tone**)
Entropion 	• Inward rotation of **eyelid margin** → lashes rub on cornea. • **Symptoms**: FB sensation, watering, redness, photophobia. • Risk of corneal abrasion, ulceration. • Prevalence: ~2% of elderly. **Causes**: • Involutional (weak lid retractors). • Cicatricial (conjunctival scarring). • Congenital, spastic (blepharospasm). **Differentials**: • Trichiasis (isolated lash misdirection without lid inversion). • Blepharospasm. • Epiblepharon (seen in children).	• Lid taping to cheek. • Epilation of misdirected lashes. • Therapeutic contact lenses. • Lubricants. **Referral**: • Routine referral to ophthalmologist for surgery (if irritation, recurrent bacterial conjunctivitis, reflex tear hypersecretion, superficial keratopathy, risk of ulceration and microbial keratitis) • Surgery = horizontal and vertical tightening procedures.	• Will not resolve spontaneously. • High risk of corneal damage. • Prognosis is good with timely surgery
Trichiasis 	• Inward misdirection of lashes, usually lower lid. • **Symptoms**: irritation, FB sensation, watering, red eye. • May be asymptomatic in elderly (reduced corneal sensation). • May cause: inferior punctate epitheliopathy, corneal ulceration and pannus • Unlike entropion, **lid position** is normal **Causes**: • Entropion, trachoma, SJS, pemphigoid, HZO. • Distichiasis (lashes from meibomian orifices). **Differentials**: • Entropion. • Distichiasis. • Blepharitis-associated lash misalignment.	• Epilation (warn of regrowth). • Lid taping (if entropion). • Treat underlying cause (blepharitis). • Lubricants, therapeutic contact lens. **Referral**: • If extensive or recurrent: refer for **electrolysis** (destruction of hair follicle with electric current at the lash root), **cryotherapy** (nitrous oxide to freeze lash roots), RF ablation, or lid surgery.	• Mild cases manageable in optometry. • Surgery for severe or recurrent. • Prognosis depends on underlying cause
Chronic Dacryocystitis 	• Chronic inflammation of lacrimal sac due to nasolacrimal duct obstruction. • **Symptoms**: recurrent epiphora, medial canthus swelling, tenderness, discharge. • May express mucous or pus from lacrimal sac with pressure. **Differentials**: • Acute dacryocystitis. • Conjunctivitis. • Carcinoma of lacrimal sac. • Canaliculitis.	• **Palliation**: warm compress, massage. → this helps with drainage • Topical **chloramphenicol** if active infection suspected. **Referral**: • Ophthalmologist or oculoplastic specialist for possible surgical treatment. • Routine referral for dacryocystorhinostomy (DCR) if persistent. ○ Lacrimal 'bypass' surgery – lacrimal sac is joined to nasal mucosa by creating a new opening in the bone between these structures	• Chronic but non-urgent. • Surgical success rates are high. • Prognosis excellent if surgically corrected

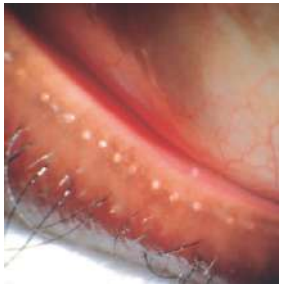
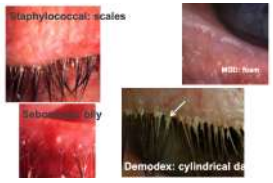
Acute Dacryocystitis 	• Bacterial infection of lacrimal sac, secondary to nasolacrimal duct blockage. • Unilateral painful, red, tender swelling at medial canthus. • May be febrile or present with preseptal cellulitis. • Common pathogens: S. aureus, S. pneumoniae, H. influenzae. **Differentials**: • Preseptal cellulitis. • Orbital cellulitis. • Chalazion. • Chronic dacryocystitis (less acute, no fever).	• Urgent referral for oral systemic antibiotics. • Prophylactic topical antibiotics (e.g., chloramphenicol) for secondary conjunctivitis. • **DO NOT probe,** irrigate or massage during acute infection ➡ because this is an active infection • Topical treatment is not appropriate – but can be prescribed as a prophylactic	• Risk of cellulitis and orbital spread. • Excellent prognosis with antibiotics and drainage
Dermatochalasis (Pseudoptosis)  	• Redundant or lax upper/lower eyelid skin and deeper tissue (including muscle and orbital septum). • Common in elderly, typically bilateral. • Orbital fat may **prolapse** into the eyelid due to septum weakening. • Patients may report needing to raise eyebrows to see, visual fatigue, or visual field impairment. • May obscure red reflex during retinoscopy. **Differentials**: • True ptosis (levator dehiscence, myasthenia gravis). • Brow ptosis. • Blepharochalasis (inflammatory swelling). • Orbital masses.	• Visual field testing (SAP or tangent screen) to assess functional impact. • Pupil occlusion testing (e.g. via red reflex such as retinoscopy). • Lid taping with transparent tape (interim). • Non-urgent referral to eyelid surgeon for blepharoplasty	• Surgery indicated for cosmetic, functional vision improvement or fatigue. • Prognosis excellent post-blepharoplasty.
Superior Limbic Keratoconjunctivitis (SLK)  	• Rare, **chronic** inflammatory condition, usually in middle-aged women. • Often bilateral. • Linked to dry eye, thyroid eye disease, and contact lens wear. • **Symptoms fluctuate:** irritation, FB sensation, photophobia, blurred vision. **Signs**: • 2–3mm velvety strip on superior tarsal conjunctiva. • Superior bulbar conjunctival chemotic band. • Pannus at superior limbus/cornea. • Fluorescein, rose bengal or lissamine staining of 3 sites. **Differentials**: • Thyroid eye disease. • Dry eye alone. • Conjunctival intraepithelial neoplasia. • Giant papillary conjunctivitis.	• Treat dry eye: preservative-free lubricants, superior punctal plugs. • Cold compresses. • Topical corticosteroids, topical mast stabiliser, mucolytic (acetylcysteine) • **Fluorometholone** 0.1% qid (short course) – 4x daily for few weeks • Sodium cromoglycate 2% q4h (every four hours) ➡ topical mast cell stabiliiser • 10% **acetylcysteine** qid (if mucus filaments; not commercially available). • If refractory: Refer for cyclosporine (off-label), silver nitrate wash, electrocautery, cryotherapy, bandage CL, or surgery	• Long-term, recurring condition. • Aim to reduce friction and inflammation. • Prognosis varies with severity and response to treatment.
Pingueculitis 	• Pinguecula = gelatinous, elevated conjunctival lesion near limbus **(3 and 9 o'clock).** • Benign degenerative conjunctival stroma change, usually bilateral. • May be clear, white, yellow**; round or triangular.** • Caused by UV, dust, wind exposure. • Often asymptomatic, but can become inflamed (pingueculitis). **Differentials**:	• UV protection: wraparound sunglasses, hat. • Lubricants, cold compresses. • **Steroid**: Topical **fluorometholone** 0.1% qid × 1 week OR • **NSAID**: Topical **ketorolac** 0.5% qid × 1 week	• Benign. Reassure patient. • May recur. Preventive sun protection is key.

	• Pterygium (extends onto cornea). • Conjunctival intraepithelial neoplasia (persistent, irregular). • Episcleritis.		
Pterygium 	• **Fibrovascular** growth from conjunctiva onto cornea. • Common in equatorial regions, surfers, outdoor workers (UV + wind exposure). • Symptoms: FB sensation, tearing, dryness, irritation, or blurred vision. • Can cause astigmatism or obscure visual axis if advanced. • Often asymptomatic early. **Differentials**: • Pinguecula (no corneal involvement). • Conjunctival intraepithelial neoplasia. • Limbal dermoid (congenital, yellow-white mass). • Conjunctival melanoma (pigmented, irregular).	• Visual hygiene (sunglasses, hats). • Lubricants (drops/ointments). • Short-course steroids (e.g. **fluorometholone** 0.1% qid × 7 days). • Topical NSAIDs (e.g. **ketorolac** 0.5% qid × 7 days). • Surgical referral if progressive, affecting vision or cosmetic concern	• Surgery may be needed for cosmetic/functional reasons. • Recurrence common—use conjunctival autograft to reduce this. • UV protection prevents recurrence.
Subconjunctival Haemorrhage 	• Bright red patch under conjunctiva, often spontaneous → non-blanching (does not make vessels pale) . • Usually **asymptomatic** or mild irritation only. • No pain, no inflammation, no infection and pressure on deeper ocular tissues. • Between the conjunctiva and sclera there isn't much innervation • **Causes**: Valsalva (e.g., coughing, sneezing, vomiting), trauma, HTN, blood thinners. **Differentials**: • Conjunctivitis (discharge, diffuse redness). • Episcleritis (localised redness + mild tenderness). • Scleritis (deep, painful redness). • Trauma-related globe rupture.	• **Cold compress in first 24 hrs** if actively bleeding. • Warm compresses 15–20 min several times/day × 3–5 days once bleeding stopped. • Lubricants if symptomatic. • **Avoid** aspirin unless GP involved. **Refer for blood tests if:** • Recurrent. • Prolonged. • Associated with other spontaneous bleeds	• Excellent prognosis—resolves in 1–2 weeks. • **Educate patient:** harmless appearance; reassure unless recurrent or traumatic.

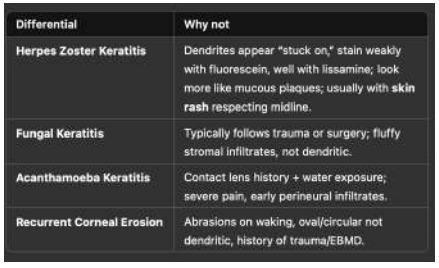
Dry Eye Disease + Meibomian Gland Dysfunction (Week 4)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Aqueous-deficient Dry Eye (ADDE) 	• Caused by **inadequate tear production.** • Common causes: Sjogren's syndrome, Graft-vs-Host Disease, age-related lacrimal gland dysfunction, neurotrophic keratitis, and long-term topical therapy (medicamentosa). • Symptoms: dryness, grittiness, FB sensation, blurred vision, photophobia. • Often worse in windy environments or during visual tasks. **Differentials**: • Evaporative DED (look for MGD signs). • Allergic conjunctivitis (itching predominates). • Conjunctivochalasis. • Neuropathic pain (disproportionate symptoms to signs).	**Diagnosis tools:** • DEQ-5 or OSDI for symptom grading. • Schirmer test / phenol red thread (tear production). • NIBUT / fluorescein TBUT. • Osmolarity testing → with an osometer • Staining with fluorescein/lissamine green. **Treatment:** 1. **Tear replacement:** ○ Unpreserved artificial tears qid or more. ○ Lipid-based drops if mixed etiology. ○ Autologous serum drops (vitamin A, EGF, etc). 2. **Tear conservation:** ○ Punctal plugs (collagen or silicone). ○ Moisture chamber glasses.	• Chronic but manageable. • Goal = restore tear film homeostasis. • With treatment, symptom relief usually precedes improvement in clinical signs.

		○ Night ointments. 3. **Tear stimulation:** ○ Oral secretagogues (e.g., pilocarpine). ○ Nasal neurostimulation. 4. **Anti-inflammatory:** ○ Cyclosporine A (off-label). ○ Corticosteroids (short-term, low dose). 5. **Environmental/Diet:** ○ Humidifiers, avoid air con. ○ Omega-3 supplements. ○ Lifestyle counseling	
Evaporative Dry Eye (EDE)	• Caused by increased evaporation of tears due to lipid layer instability. • Most common subtype: up to 86% of DED cases involve EDE. • **Major contributor: Meibomian Gland Dysfunction (MGD).** • Other causes: anterior blepharitis, rosacea, ectropion, contact lens wear, environmental exposure. • Symptoms: burning, dryness, blurry vision, worse with screen use. **Differentials**: • ADDE. • Ocular allergy. • Exposure keratopathy. • Neurotrophic keratitis.	**Diagnosis tools:** • NIBUT or TBUT (reduced). • Lipid layer interferometry. • Gland expressibility, meibography. **Treatment**: 1. **Lid hygiene:** warm compresses (5-10 min), lid scrubs (daily). 2. **In-office treatments:** ○ Thermal pulsation (Lipiflow). ○ Manual expression. ○ Intense Pulsed Light (IPL). 3. **Topical:** ○ Lipid-based artificial tears (e.g., Systane Balance, Optive Advanced). ○ Topical antibiotics or corticosteroids if inflammation present. 4. **Oral:** ○ Tetracyclines (doxycycline 50–100mg/day for 6–12 weeks). 5. **Adjunctive:** ○ Omega-3 supplementation. ○ Avoid preserved drops. ○ Blinking exercises, environment control	• Long-term maintenance essential. • Prognosis excellent with compliance. • Recurrence is common if the underlying cause is not addressed.
Blepharitis 	• Chronic inflammation of eyelid margins. • **Divided into:** • **Anterior blepharitis:** ○ Staphylococcal: crusting, madarosis, ulceration. ○ Seborrheic: greasy scales, hyperemia. • **Posterior blepharitis (MGD):** ○ Gland plugging, foam in tear meniscus, thickened secretions. • Symptoms: itchy, red eyes, crusty lids, FB sensation, CL intolerance. • **Worsens in the morning.** **Differentials**: • Dry eye (frequently coexists). • Conjunctivitis. • Demodex infestation. • Rosacea. • Meibomian gland carcinoma (unilateral, non-responsive).	**Diagnosis**: • Slit lamp for crusts, plugs, foam, telangiectasia. • Demodex signs (cylindrical dandruff). **Non-Pharmacological:** • Warm compresses, lid scrubs, massage. • Eyelid debridement (Blephex). • Avoid makeup/eye irritants. **Pharmacological**: • Topical antibiotics: ○ **Chloramphenicol** 1% ointment tid 3x daily ○ **Azithromycin** 1.5% drops (compounded) 2x daily ○ FML 0.1% or Hycor 1% for inflammation. • Oral: ○ **Systemic antibiotics:** Doxycycline/minocycline for MGD.	• Lifelong condition—manage expectations. • Review 1–2 weeks initially, then regularly. • Address co-existing DED and Demodex.

		○ Omega-3 supplements. ● Additional: ○ Dandruff shampoo for seborrheic components (scalp). ○ GP/dermatologist referral if rosacea is suspected. ○ Write GP letters for continuity of care ○ Treat demodex if present with tea tree oil	
	**Staphylococcal Blepharitis:** ● Caused by Staphylococcus bacteria. ● **Signs**: Lid margin swelling, crusting (scales at lash base), lash misdirection, madarosis, recurrent styes/chalazia, conjunctival hyperemia, marginal keratitis, phlyctenulosis, pannus, and papillary conjunctivitis OBJ. **Seborrheic Blepharitis:** ● Associated with seborrheic dermatitis. ● **Signs**: Lid margin hyperemia, greasy/oily lid deposits, conjunctival hyperemia OBJ. **Demodex-Associated Blepharitis:** ● Caused by Demodex mite infestation. ● **Signs**: Lid margin hyperemia, cylindrical dandruff, trichiasis, and madarosis OBJ. **Posterior Blepharitis (Meibomian Gland Dysfunction - MGD)** ● Involves inflammation of the inner part of the eyelid and meibomian glands. ● **Signs**: Plugged gland orifices, thick/opaque secretions, difficult oil expression, foamy tear film, conjunctival hyperemia, evaporative dry eye (EDE), corneal staining, marginal keratitis, scarring, pannus, and papillary conjunctivitis OBJ. **Mixed Blepharitis** ● Involves features of both anterior and posterior blepharitis, commonly seen in practice		
Demodex 	● Mite infestation (Demodex folliculorum and brevis). ● Lives in hair follicles, sebaceous glands. ● Present in everyone, symptomatic when overpopulated. ● **Signs**: ○ Cylindrical dandruff around lash base (pathognomonic). ○ Lid margin hyperemia, irritation, trichiasis, madarosis. ● **Symptoms**: itch, dryness, FB sensation, often worse in morning. ● Associated with chronic blepharitis and rosacea. **Differentials**: ● Staphylococcal blepharitis (flat scales). ● Phthiriasis (larger lice visible at lash base). ● Seborrheic dermatitis.	**In-office:** ● Blephex debridement. ● 20–50% tea tree oil (TTO) lid scrubs (requires training). **Home therapy:** ● 5% TTO ointment (compounded). ● Lid scrubs: Blephadex, Sterilid, Ocusoft Plus. ● TTO shampoo for facial wash. **Emerging therapy:** ● Lotilaner 0.25% (XDEMVY – not yet available in Australia). **Precautions**: ● Avoid undiluted TTO near eyes due to irritation.	● Chronic with frequent recurrence. ● Good prognosis if mites are eradicated. ● Patient education and compliance critical for success <u>Intense Pulse Light Therapy → MGD</u> ● IPL works by delivering thermal energy & thermocoagulation of the blood vessels ● Is indicated for treatment of DED & MDG ● Contradicted in px with darker skin tone ● Can reactive herpes simplex lesions, hence should be avoided or conducted with adjuvant prophylaxis in those with a history of HSV infection ● ADR includes redness, swelling & a burning sensation ● 3-6 treatments every 2-4 weeks is required for a full effect

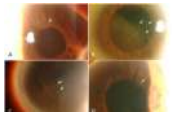
Viral Keratitis and Conjunctivitis Readings (Week 5)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Adenoviral Keratoconjunctivitis (EKC)	**Key Clinical Features:** • Initial unilateral red, watery, gritty eye. • "Started RE first, now LE but not as bad" is pathognomonic (25% start unilaterally). • Second eye often involved 2–5 days later but with less severity. • Lid swelling, chemosis, follicular conjunctivitis, subconjunctival haemorrhages, and preauricular lymphadenopathy. • Associated systemic features: mild fever, malaise, headache. • Severe photophobia due to corneal involvement. **Corneal Involvement:** • 70% of patients develop corneal signs: ○ Starts with epithelial vesicles (non-staining in Stage 0). ○ Progresses to fluorescein-staining vesicles (Stage 1). ○ Coalescence into deeper epithelial lesions (Stage 2). ○ Development of Subepithelial Infiltrates (SEIs) (Stage 3–5): ■ Immune response to infected cells. ■ Can lead to visual loss, photophobia, and glare. ■ SEIs may persist for months to years. **Differentials**: • **HSV epithelial keratitis:** Presents with dendritic ulcers with terminal bulbs. • **Thygeson's keratitis:** Bilateral chronic disease, white eye. • **Allergic conjunctivitis:** Seasonal, itching prominent. • **Bacterial conjunctivitis:** Mucopurulent discharge, less follicular reaction. • **Chlamydia conjunctivitis:** Chronic follicular conjunctivitis.	**Investigations:** • Clinical diagnosis usually sufficient. • PCR swabs if doubt — tests for HSV, VZV, CMV, EBV, adenovirus, enterovirus. • Genotyping to trace outbreaks. • Do NOT routinely test unless atypical presentation. **Pharmacological Management:** • No antiviral is officially effective for adenovirus. • Topical steroids (controversial): • Consider in SEIs or pseudomembranes: ○ Pred Forte qid (Grade 3–5 keratitis with visual symptoms). ○ FML/Flucon qid for SEIs. ○ Predsol minims tds. • Topical cyclosporin (e.g., 0.05–2%) for persistent SEIs. • Topical betadine 0.25%: Low-cost antiviral alternative. • Topical cidofovir (1%) has shown mixed results, often poorly tolerated. • Topical lubricants: Especially helpful for dry eye relief (oily or gel forms preferred). **Non-Pharmacological:** • Cold compresses. • Eye baths with salty boiled water. • Sunglasses for glare. • Strict hygiene measures: • Isolate for 10 days. • Avoid work in healthcare/childcare. • Do not share towels/pillows/drops. • Dispose of tissues properly. **Referral** – Must refer if: • Corneal involvement. • Pseudomembranes. • Symptoms persist >3 weeks. • Evidence of cross-infection.	**Complications:** • Chronic viral dry eye (6 months) due to goblet cell destruction. • Persistent SEIs can last up to 2 years. • Vision loss — reduced contrast, glare, night vision. • Pseudomembranes can scar conjunctiva. • Contagion can result in widespread office outbreaks: ○ Adenovirus survives 28–49 days on surfaces. ○ Resists alcohol; needs 2.4% glutaraldehyde for disinfection. **Prognosis**: • Generally self-limiting. • Vision typically recovers after 6 months. • SEIs and dry eye symptoms can last 1–2 years. • Proper supportive care and hygiene are critical to prevent spread. • Long-term sequelae include scarring, chronic irritation, and psychosocial frustration due to prolonged symptoms.
Herpes Simplex Virus (HSV) Eye Disease	Epidemiology & Natural History: • Endemic globally; HSV-1 seroprevalence is high. • 1/3 of infected individuals develop recurrent disease. • Latent in trigeminal ganglion — reactivates with stress, fever, trauma, UV, surgery, menstruation. • Reactivation is more common in younger patients (<20). **Disease Manifestations:** • Epithelial keratitis: ○ Dendritic ulcers: branching, terminal bulbs, central staining with fluorescein, edge staining with rose	**Workup**: • PCR swab (gold standard). • Corneal sensitivity test. • Fluorescein staining: dendritic pattern, stromal involvement. • Rose Bengal or lissamine green: edge staining. • Debridement can be therapeutic and diagnostic.	

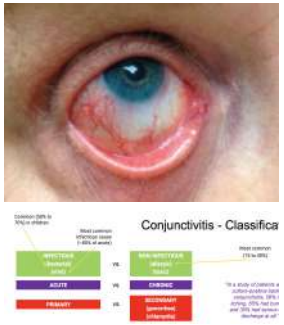
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bengal.
○ Geographic ulcers: larger, amoeboid, due to steroid misuse.
○ Marginal keratitis: near limbus, peripheral infiltrates.
• Stromal keratitis:
○ Immune stromal keratitis: no ulceration.
○ Necrotizing keratitis: ulceration, scarring.
• Endotheliitis (disciform keratitis): central corneal edema with keratic precipitates (KP), blurred vision.
• Neurotrophic keratitis: persistent epithelial defects, due to reduced corneal sensation.
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Treatment by Subtype:
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<tr><th>HSV Subtype</th><th>Antiviral</th><th>Steroid</th></tr>
<tr><td>Epithelial keratitis</td><td>Topical Aciclovir 3%, 5x/day for 10–14 days OR Valaciclovir 500 mg BD for 7 days</td><td>Contraindicated during active epithelial disease</td></tr>
<tr><td>Stromal keratitis (non-necrotising)</td><td>Aciclovir 5x/day + Valaciclovir 500 mg OD</td><td>Prednisolone acetate 1% + phenylephrine 0.12% 4–6x/day tapered over 10+ weeks</td></tr>
<tr><td>Stromal keratitis (necrotising)</td><td>Valaciclovir 1g TDS x 7–10d, then prophylactic dose</td><td>Pred acetate 1% + phenylephrine 0.12% BD slowly tapered</td></tr>
<tr><td>Endothelial keratitis</td><td>Valaciclovir 1g TDS x 7–10d</td><td>Pred acetate 1% + phenylephrine 0.12% 4–6x/day tapered</td></tr>
<tr><td>Prophylaxis</td><td>Aciclovir 400 mg BD or Valaciclovir 500 mg OD</td><td>Not used prophylactically</td></tr>
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<table>
<tr><th>Differential</th><th>Why not</th></tr>
<tr><td>Herpes Zoster Keratitis</td><td>Dendrites are smaller, "stuck-on", stain poorly with fluorescein but well with lissamine; rash respects midline.</td></tr>
<tr><td>Fungal Keratitis</td><td>Slower onset, history of trauma/vegetative matter, "fluffy" stromal infiltrates.</td></tr>
<tr><td>Acanthamoeba Keratitis</td><td>History of CL use or water, pain > signs, ring infiltrates, perineural infiltrates.</td></tr>
<tr><td>Recurrent Corneal Erosion</td><td>History of trauma or EBMD; circular/oval lesions on waking.</td></tr>
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<td>Herpes Zoster Ophthalmicus (HZO)
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• Key Features:
○ Painful vesicular rash on forehead, respecting midline. ➡ scalp
• Ocular signs:
○ Dendritic keratitis (smaller, limbal, “stuck-on”).
○ Pseudodendrites (poor fluorescein, good lissamine staining).
○ Episcleritis, scleritis, iritis.
○ Neurotrophic ulcers in late stages.

Risk Factors for Recurrence:
• Age, immunocompromised, previous chickenpox.
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Systemic Therapy:
• Oral antivirals mandatory:
○ Valaciclovir 1g TDS x 7 days
○ Alternatives: Famciclovir, Aciclovir
○ Continue longer if rash or inflammation persists

Ocular Treatment:
• Topical steroids (FML or Predsol) for keratitis/uveitis
• Topical betadine for vesicles
• Lubricants: preservative-free, frequent
• Tarsorrhaphy, bandage CL, or autologous serum for neurotrophic ulcers

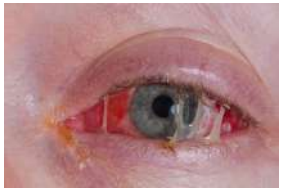
Workup:
• PCR swab if diagnosis uncertain.
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• Follow up every 3 months for 18 months (risk of lipid keratopathy)
• Post-herpetic neuralgia (PHN):
○ Treat aggressively: Amitriptyline, Gabapentin, Pregabalin
• Prognosis:
○ Best with early antiviral treatment.
• Complications like stromal melt, PHN, and neurotrophic ulcers can lead to permanent visual impairment if untreated.
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<td>CMV Endotheliitis (Cytomegalovirus Retinitis)
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• Diffuse corneal stromal edema and keratic precipitates (KPs).
• Often subtle onset.
• Differentials: HSV or VZV endotheliitis.
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• AC tap with PCR confirms CMV.
• Treatment:
○ Oral valganciclovir
○ Topical Maxidex (high dose)
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• Requires ongoing follow-up.
• Prognosis depends on early diagnosis and suppression.
• Vision can be preserved with antiviral and anti-inflammatory therapy.
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<td>Thygeson’s Keratitis</td>
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• Bilateral, chronic, white eye with coarse epithelial lesions.
• May fluorescein stain.
• No stromal infiltrates.
• Recurrent/remitting over 2–4 years.
• Often asymmetrical.
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• Topical steroids: FML/Predsol 1–2x/day.
• Topical cyclosporin 1% or tacrolimus (0.03–0.1%) 1–3x/day for 3 months.
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• Good prognosis.
• Requires long-term low dose anti-inflammatory control.
• No vision loss unless complications occur.
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<td>CLARE (Contact Lens-Associated Red Eye)</td>
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• Typically in contact lens wearers, especially overnight lenses.
• Acute onset redness and discomfort.
• Multiple, superficial stromal infiltrates.
• Etiologies:
○ Hypoxia
○ Thimerosal/preservative hypersensitivity
○ Inflammatory, not infectious.
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• Discontinue lens use immediately.
• Preservative-free lubricants.
• Avoid thimerosal-containing solutions.
• Review contact lens hygiene and fit.
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• Excellent prognosis with lens discontinuation.
• Recurrences common if causative factor not addressed.
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Conjunctival Diseases (Week 6)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Allergic Conjunctivitis ☺  	• **Epidemiology**: Affects up to 40% of the population. Most common form is SAC. Often underdiagnosed as many do not present for care. • Associated with **rhinitis** (inflammation of internal areas of nose – nasal dripping) or hay fever • Other comorbid conditions ➔ asthma + eczema • Etiology: **Type I hypersensitivity** reaction. Triggered by: ○ **SAC**: Seasonal pollens (trees, grass, weeds) ○ **PAC**: Perennial indoor allergens (dust mites, pet dander, mold) • Associated Conditions: Asthma, eczema, allergic rhinitis • **Symptoms**: ○ Hallmark: Itching ○ Redness, tearing, foreign body sensation, burning ○ Often bilateral ○ Symptoms intermittent or persistent depending on allergen exposure • **Differentials**: ○ Vernal keratoconjunctivitis (VKC) ○ Atopic keratoconjunctivitis (AKC) ○ Giant papillary conjunctivitis (GPC) ○ Dry eye disease ○ Viral conjunctivitis	**Nonpharmacological**: • Allergen **avoidance** (air filters, mite-proof bedding, avoid eye rubbing) • Cold compresses, ocular lubricants (especially **preservative-free ➔ because they are hypersensitive**) • Skin prick or patch testing via GP/allergist for allergen identification **Pharmacological**: • **Mild:** OTC antihistamines or mast cell stabilizers (e.g., olopatadine, ketotifen) • **Moderate:** ○ Combination drops (antihistamine + mast cell stabilizer) ○ Intranasal corticosteroids (e.g., fluticasone) especially with rhinitis ■ Available as OTC (eg. Rhinocort, Nasonex nasal sprays, Beconase) ○ Consider preservative-free formulations • **Severe**: ○ Short course of topical steroids (e.g., fluorometholone 0.1%) with IOP monitoring ○ Calcineurin inhibitors (e.g., ciclosporin, tacrolimus – off-label) ○ Immunotherapy (sublingual or subcutaneous) – not PBS-covered --- Topical antihistamine + mast cell stabilisers ***(eg. Levocabastine 0.05%** twice to 4 times daily)* • Combination drugs include *olopatadine + ketotifen* for PAC	• **Prognosis**: Excellent with proper avoidance and treatment. Persistent symptoms or vision-threatening signs require specialist referral ➔ ophthalmologist to px corticosteroids and immunotherapy. • Oral route ➔ adverse effects • **General Management:** ○ Education on eye rubbing (risk of keratoconus) ○ Reinforce long-term use of lubricants in dry eye comorbidity ○ Stepwise treatment based on symptom severity ○ Annual reviews for chronic cases
Atopic keratoconjunctivitis (AKC) 	• Epidemiology: Rare; peak incidence 30–50 years • **Etiology**: Type I and IV hypersensitivity, often with atopic dermatitis **Symptoms**: • Chronic, bilateral, persistent ocular itch • Mucous discharge, blurred vision, photophobia • Eye rubbing common **Signs**: • Thickened, fissured eyelids • Conjunctival hyperemia with large papillae • Corneal scarring, neovascularisation, conjunctivalization **Differentials**: • VKC (typically pediatric, seasonal) • GPC (CL-related) • Severe SAC/PAC • Herpes simplex keratitis (important to exclude prior to steroid use)	**Nonpharmacological**: • Avoid eye rubbing • Allergen avoidance (dust, mold, pets) • Cold compresses **Pharmacological**: • Year-long therapy typically required • **First-line:** Mast cell stabilizers + antihistamines (e.g., olopatadine, ketotifen) • **Add-ons:** ○ NSAIDs (e.g., ketorolac, diclofenac) ○ Short bursts of topical steroids (e.g., FML 0.1%) ○ Topical cyclosporin A (effective but not PBS-covered) ○ Systemic steroids (short-term) if severe • **Referral:** GP for allergist co-management; essential in moderate-severe cases	**Prognosis**: Guarded. Risk of corneal scarring and vision loss if poorly managed. **General Management:** • Regular ophthalmic monitoring • Emphasis on lid and ocular hygiene • Long-term steroid sparing agents preferred • Co-manage with dermatology/allergy/immunology if systemic atopy is significant



Bacterial Conjunctivitis



Epidemiology:
- Common in children (50-70% of infectious conjunctivitis in kids)
- Less common than viral in adults

Etiology:
- Common pathogens: Staphylococcus aureus, Streptococcus pneumoniae, Haemophilus influenzae
- Rare: Neisseria gonorrhoeae, Chlamydia trachomatis

Symptoms:
- Red eye, mucopurulent discharge
- Gritty, burning sensation
- Usually starts unilateral ➔ bilateral
- Mild to moderate eyelid edema

Signs:
- No preauricular lymphadenopathy (unlike viral)
- Papillae on palpebral conjunctiva

Differentials:
- Viral conjunctivitis (preauricular lymphadenopathy, watery discharge)
- Allergic conjunctivitis (itching, bilateral, clear discharge)
- Gonococcal conjunctivitis (hyperacute, severe discharge, emergency)
- Chlamydial conjunctivitis (chronic, follicles, sexual history)
- Dry eye syndrome

- Mild to moderate cases: Often self-limiting (resolve in 5–10 days)

Pharmacological:
- Topical antibiotics:
  - **First line:** chloramphenicol 0.5% drops 4x/day for 7 days (plus ointment at night)
  - **Alternatives:** fusidic acid (for staph), or ciprofloxacin/tobramycin in CL wearers
- Severe or hyperacute:
  - Refer for urgent care
  - May require systemic antibiotics (e.g., for gonococcal)
  - If severe: 6-8 times daily tapering to 4 times daily for 1-2 weeks

Nonpharmacological:
- Hygiene: hand washing, separate towels, no contact lens wear
- Warm compresses to remove crusting

Investigations:
- Swabs for culture if:
- Recurrent, atypical, severe, neonatal
- Treatment failure
- Suspicion of gonorrhoea/chlamydia (also requires sexual health workup)

Aminoglycosides	Fluoroquinolones	Macrolides	Sulfonamides
Gentamicin ointment 4x/d for 1wk Gentamicin 4x/d for 1wk Tobramycin ointment 3x/d for 1wk	Besifloxacin 3x/d for 1wk Ciprofloxacin ointment 3x/d for 1wk Ciprofloxacin 4x/d for 1wk Gatifloxacin 3x/d for 1wk Levofloxacin 4x/d for 1wk Moxifloxacin 3x/d for 1wk Ofloxacin 4x/d for 1wk	Azithromycin 2x/d for 2d then daily for 5d Erythromycin 4x/d for 1wk	Sulfacetamide ointment 4x/d + bedtime for 1wk Sulfacetamide every 2-3hr for 1wk

OR chloramphenicol
OR combination (antibiotic + steroid) drugs
OR oral antibiotics
OR povidone-iodine 1.25% four times a day

- **Prognosis**: Excellent in typical cases. Severe cases (e.g. gonococcal) can threaten vision.
- **General Management:**
- Infection control precautions: avoid eye rubbing, hand hygiene
- Educate on signs of worsening (pain, photophobia, reduced vision ➔ refer urgently)
- Though without treatment it can resolve in 7-10 days ➔ there is a risk of contagion
- Return to school/work policy: 24–48 hours after starting antibiotics and no discharge

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Other Broad Spectrum Antibiotics
- Aminoglycoside (tobramycin 0.3%)
- Fluoroquinolone (ciprofloxacin 0.3%, ofloxacin 0.3%)
- Propamidine 0.1%

Ointment at bedtime for a supplemental therapy (eg. chloramphenicol 1%, tobramycin 0.3)
- For young children, ointment 3 to 4 times daily as the main therapy is preferred

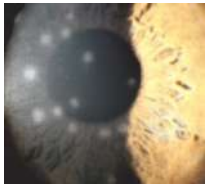
Adenoviral Conjunctivitis & Adenoviral Keratoconjunctivitis ☺

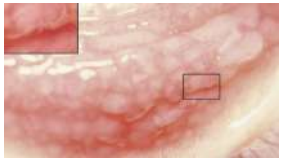
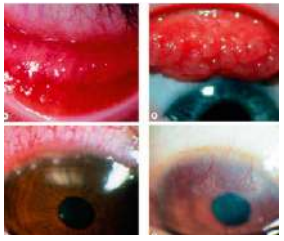
- Causative agent: Human adenovirus (non-enveloped, double-stranded DNA virus)
- **Epidemiology**: Most common cause of infectious conjunctivitis (up to 75%)

Forms:

- **Supportive therapy:**
  - Non-preserved lubricants
  - Cold compresses
  - Saline rinses (0.9% NaCl)
- **Topical antihistamines** (levocabastine 0.05%, azelastine

Prognosis:
- Self-limiting in most cases (resolves within 2 weeks)
- EKC may lead to long-term SEIs ➔ visual disturbance

	• **Pharyngoconjunctival fever (PCF):** Mild follicular conjunctivitis + URTI symptoms (fever, pharyngitis); Serotypes B3, B7, B11 • **Epidemic keratoconjunctivitis (EKC): Severe**, lid oedema, intense redness, preauricular lymphadenopathy; Serotypes D8, D19, D37 Signs: • Follicular conjunctivitis, watery discharge • Ipsilateral preauricular lymphadenopathy • Pseudomembranes or true membranes • Subepithelial corneal infiltrates (SEIs) in EKC (~7–10 days post onset) **Differentials**: Bacterial conjunctivitis, allergic conjunctivitis, HSV keratitis, chlamydial conjunctivitis	0.05%) • **Antivirals (e.g. aciclovir) not effective** • In-office Povidone Iodine (off-label): 0.4–1.0% PVP-I + 0.1% dexamethasone 4x/day – accelerates clinical and microbiological resolution [66] • Infectious period: 10–14 days after symptom onset • **Precautions**: Strict infection control measures, avoid contact lens wear • **To differentiate:** preauricular lymph node palpation → swollen lymph nodes indicates cancer, bacterial infections, as well as **viral infection**	Advice to patients: • Use separate towels and pillows • Discard makeup/contacts • Sunglasses for photophobia • Infectivity: High – 10–50% risk of contagion → should not go to school or childcare • Remove sticky discharge and secretions from eyelids with clean cotton wool pad soaked in warm tap water • Don't share towels
EKC Nummular Infiltrates  	• **NB:** EKC is a form of adenoviral infection ○ True or pseudomembranes develop on mucosal surfaces → can lead to conjunctival scarring or symblepharon (when conjunctiva sticks to thin membrane that covers the white part of eye) • Appear 7–10 days after onset • Described as nummular (coin-shaped), subepithelial infiltrates with a dense center and lighter periphery • Caused by immune reaction to viral antigen	• Topical corticosteroids (e.g. Fluorometholone 0.1%, Loteprednol 0.5%, Prednisolone phosphate 0.5% → unpreserved) QID then **taper** (over 2-4 weeks) • Loteprednol to be compounded, **four times a day** until no further resolution of infiltrates • Cyclosporin A 1%: For corticosteroid-dependent patients to taper steroids	• Prognosis: May fade over months to years; risk of scarring • Relapses common after stopping steroids (steroid-dependence may require prolonged therapy)
Membranous Conjunctivitis 	• **True membrane**: Firmly adherent; bleeds upon removal; involves subepithelial vascular integration • **Pseudomembrane**: Loosely adherent coagulated exudate; no bleeding • Manifest as smooth bands of material overlaid on top of red, meaty bottom or top conjunctiva **Causes**: • EKC • Bacterial: β-hemolytic streptococci, N. gonorrhoeae, C. diphtheriae • Severe allergic conjunctivitis • Stevens-Johnson Syndrome • Chemical/thermal burns	• Remove membrane using topical anaesthetic and forceps • **Steroid eyedrops** ○ Prednisolone acetate 1% QID ○ Fluorometholone 0.1% QID ○ **Previously:** Flarex 4x/day • Phenylephrine 2.5% for vasoconstriction if bleeding → cotton bud soaked • **PCR or AdenoPlus for diagnosis before steroid use** • Topical antibiotic prophylaxis	• Prognosis: Risk of symblepharon if untreated
Severe Bacterial Conjunctivitis 	**Signs and Symptoms:** • Acute, red eye with intense hyperaemia • Profuse purulent discharge • Eyelid oedema and conjunctival chemosis • Commonly bilateral, especially in neonates or sexually transmitted cases • May include true conjunctival membranes that bleed on removal if caused by pathogens such as Neisseria gonorrhoeae or Corynebacterium diphtheriae • Marked discomfort or irritation, but pain is less severe compared	**Pharmacological Treatment:** 1. **Topical antibiotics:** • Chloramphenicol 0.5% eyedrops four times daily • For contact lens wearers or Pseudomonas risk: fluoroquinolones (e.g. ciprofloxacin 0.3% or ofloxacin 0.3%) • In severe, hyperacute, or resistant infections: fortified antibiotics (e.g. cefazolin + tobramycin) 2. **If true membrane is present:** • Consider removal under topical anaesthesia	General Management: • Isolate highly contagious cases (e.g. gonococcal or severe bacterial conjunctivitis with membrane) • Practice strict infection control – gloves, disinfection of instruments, surfaces • Review at 24-48 hours, particularly if membrane, pseudomembrane or severe inflammation is present

	to keratitis **Hallmark Signs:** • True membrane formation: adherent to conjunctiva and bleeds on removal • Pseudomembrane: loosely adherent fibrinous exudate, easily peeled, no bleeding • Purulent discharge is key to distinguishing bacterial from viral conjunctivitis **Important Differentials:** • Viral conjunctivitis (EKC) – watery discharge, preauricular lymphadenopathy, follicles • Allergic conjunctivitis – itchiness, bilateral, no purulence • Gonococcal conjunctivitis – rapid progression, severe inflammation, true membrane formation, hyperacute • Membranous conjunctivitis due to Stevens-Johnson syndrome or chemical burns – consider systemic symptoms and exposure history • Corneal involvement (keratitis) – pain, photophobia, visual loss; must be ruled out via slit lamp and fluorescein staining	• Use cotton bud soaked in phenylephrine 2.5% for hemostasis if bleeding • Initiate topical **corticosteroids** (e.g., Fluorometholone alcohol 0.1% or Prednisolone phosphate 0.5%), particularly if severe inflammation/scarring risk – but only after ruling out HSV or gonococcus 3. Cycloplegic drops (e.g. Atropine 1% twice daily) if pain from significant inflammation or AC reaction is present 4. Prophylactic antibiotics may be continued with corticosteroids **Nonpharmacological / Workup:** • Swab and culture especially if: • Hyperacute, sexually transmitted cause suspected (e.g. N. gonorrhoeae) • Membranous conjunctivitis • Recalcitrant to first-line therapy • Notify public health unit for notifiable STIs (gonorrhoea, chlamydia) • Avoid contact lenses during treatment • Use separate hygiene items, reinforce hand hygiene • Irrigation with normal saline may assist in removing debris	Educate patient on: • Avoid touching/rubbing eyes • Discard eye makeup/contact lenses • Maintain hygiene with towels/pillows Prognosis: • Excellent in most cases with timely treatment • Risk of conjunctival scarring or symblepharon if true membranes not removed or severe inflammation persists • Vision is typically not affected, but corneal involvement or secondary keratitis can impact prognosis • Recurrence rare if underlying cause (e.g., blepharitis) is treated
Hyperacute Bacterial Conjunctivitis 	• Causative organism: **Neisseria gonorrhoeae** Presentation: • Rapid onset • Profuse purulent discharge • Eyelid swelling • Can penetrate intact cornea → risk of keratitis and perforation	• Urgent referral for: ○ Ceftriaxone 1g IM x 5 days ○ Topical fortified gentamycin 1.3% hourly if corneal involvement • Concurrent Chlamydia treatment (e.g., Azithromycin) • Swab and culture **required**	• Prognosis: Excellent with prompt treatment; vision-threatening if delayed • Notifiable disease – reported via lab • Requires urgent referral to ophthalmologist → ideally in hospital setting
Chronic Bacterial Conjunctivitis 	• Definition: Conjunctivitis persisting >4 weeks Possible causes: • Resistant bacteria • Lid disease (e.g., blepharitis, canaliculitis) • Secondary to adult inclusion conjunctivitis	• Culture with sensitivity testing (ensure washout of prior antibiotics) • Treat lid/lacrimal system disease • Consider STIs if unilateral or recurrent	• Prognosis: Good with targeted therapy; treat underlying sources
Adult Inclusion Conjunctivitis 	• Causative organism: Chlamydia trachomatis (Serotypes D–K) • Transmission: Sexual **Signs**: • **Follicles** in lower fornix (follicles are small nodules without a prominent vessel, whereas papillae caused by inflammation + leaky blood vessels) • Unilateral persistent red eye • Mild discharge	• Systemic treatment mandatory (gram negative affinity): ○ Azithromycin 1g PO single dose (20mg/kg for children) ○ Doxycycline 100mg PO BID x 7 days ○ Alternatives in pregnancy: erythromycin/amoxicillin • **Topical**: Lubricants, combo antibiotic-steroid (supportive only) • Partner treatment required	• Prognosis: Good ocular outcome; infertility risk if untreated • Notifiable disease (lab-confirmed)

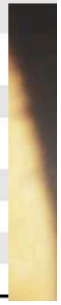
Trachoma 	• Causative organism: Chlamydia trachomatis (Serotypes A, B, Ba, C) • Transmission: Eye-to-eye (poor hygiene, flies) • Inward turning of eyelids and eyelashes → leading to rubbing to the cornea Course: • Chronic follicular conjunctivitis in childhood • Progresses to cicatricial trachoma → eyelid distortion, trichiasis, blindness	• SAFE strategy: ○ **Surgery** (for trichiasis) ○ **Antibiotics**: Azithromycin 20mg/kg single dose ○ **Facial cleanliness** ○ **Environmental** improvements	• Prognosis: Preventable cause of blindness; rare in Australia except in remote Indigenous communities
Ophthalmia Neonatorum	• Onset: Within first 28 days of life (4 weeks) • Also called **neonatal conjunctivitis** **Causes**: • Chlamydia trachomatis • Neisseria gonorrhoeae • HSV, Staphylococcus aureus	• Swab for culture and PCR mandatory • Empiric systemic treatment: ○ Gonorrhoea: Ceftriaxone IM ○ Chlamydia: Erythromycin or Azithromycin PO • Topical therapy supportive only • Possible requirement for notification to the national database depending on the infecting agent → pathology lab	• Prognosis: Vision-threatening if untreated • Notifiable disease • Referral to pediatric infectious disease specialist recommended • **Concurrent treatment of mother and her sexual partners is also important**

Contact Lens Complications (Week 6)

	Corneal Infiltrates (Sterile keratitis)	Bacterial Keratitis
Aetiology	Inflammation	Infection
Symptoms / Pain	Mild to moderate (foreign body sensation)	Increasing, may be severe
Photophobia	Mild	Moderate/severe
Vision	Usually not affected	Usually compromised
Epiphora	Mild	Intense
Conjunctival injection	Minimal to moderate, possibly sectorial (localised)	Moderate to severe, most likely diffuse (meaty)
Location and depth	Usually mid-peripheral to peripheral, subepithelial or anterior stromal only	Random with deeper involvement, tend to be more paracentral / central, anterior to mid-stromal, raised edges
Size	Usually 0.1 - 2mm; tends to remain small	>1mm; enlarges over 24-36 hr
Fluorescein Staining	None (intact) or minimal	Moderate to extensive epithelial defect
Anterior chamber reaction	None or minimal (only if severe)	Moderate. Flare and occasional hypopion
Stromal oedema	None or minimal	Moderate
Infiltration	Focal and slight diffuse	Focal and significant diffuse
Shape	Circular	Any, most likely irregular (may also have satellite lesions)
Purulent discharge	Mild (if any) corneal suppuration	Severe, progressive corneal suppuration
Lid oedema	None	Usual
Symptoms on cessation of lens wear	Improve	Escalation

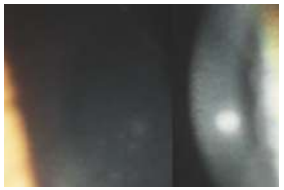
Modified from Carnt et al , Clin Exp Optom 2017; 100: 482-493.

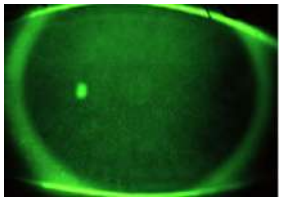
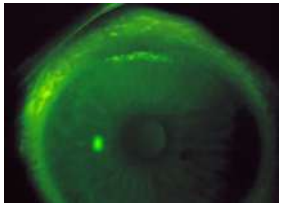
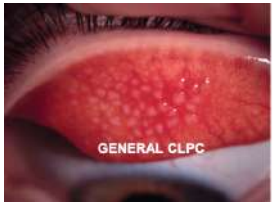
	Corneal infiltrates (Sterile keratitis)	Bacterial Keratitis
Symptoms	Minimal	Moderately severe
Conjunctival injection	Minimal, possibly sectorial	Moderate to severe, most likely diffuse
Location	Usually mid-peripheral to peripheral, subepithelial	Random with deeper involvement, tend to be more central
Size	Usually 1-1.5mm; tends to remain small	>1.5mm; enlarges over 24-36 hr
Fluorescein Staining	None or minimal	Moderate to extensive
Anterior chamber reaction	None or minimal	Moderate
Stromal oedema	None or minimal	Moderate
Number of infiltrates	Tend to be multiple (>1)	Tends to be 1 only
Shape	Round or oval	Any, most likely irregular
Purulent discharge	Negative	Positive



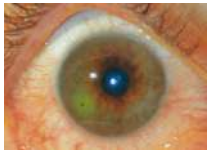
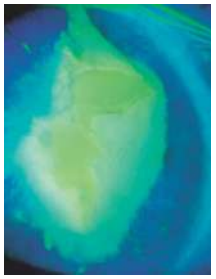
Can be induced by infection of the cornea (microbial keratitis) or can present as sterile where a stimulus has induced an inflammatory reaction in the cornea
Infiltrates are focal areas of corneal inflammation made up of white blood cells

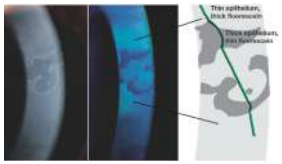
Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Contact Lens Related Infiltrates	• Pathophysiology: Corneal inflammatory reaction caused by stimulus from the contact lens or lens surface (often sterile, but infection must be ruled out). • Signs: Focal infiltrates (white blood cells in cornea), red eye, variable photophobia and tearing. • Symptoms: Mild to moderate discomfort; generally resolve with lens removal.	• Discontinue lens wear immediately • Review in 24-48 hours • Topical corticosteroids: **Fluorometholone** 0.1% QID may hasten resolution • Antibiotics: Optional, consider if diagnostic uncertainty • Palliative: Ocular lubricants, saline rinse, cold compress	• Prognosis: Typically self-limiting; recurrence ~7% if risk factors persist. • Prevention: Address modifiable risk factors (overnight wear, solution type, water exposure, case hygiene)

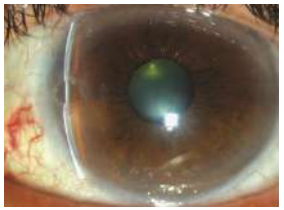
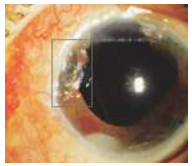
	Differentials: Infectious keratitis, CLARE, CLPU, EKC, marginal keratitis		
Sterile Infiltrates 	• Cause: Inflammatory response to stimuli (CL wear, staph blepharitis, conjunctivitis, EKC). • Signs: Small (<2mm), round, subepithelial infiltrates often mid-peripheral, minimal/no staining, no AC reaction. • Symptoms: Redness, foreign body sensation, photophobia, often mild [obj]. **Differentials**: Infectious infiltrates, marginal keratitis, CLPU, phlyctenulosis.	• Topical steroids: Fluorometholone 0.1% QID (consider loteprednol/prednisolone for penetration) • Prophylactic antibiotics: Ciprofloxacin 0.3% or tobramycin 0.3% QID (if CL wear) • Cease lens wear, treat associated blepharitis • Review: 24-48h	• Prognosis: Generally good; mild stromal scarring may occur; recurrence possible.
Infectious Infiltrates (Microbial Keratitis)	• Rapidly progressing infiltrates with overlying epithelial defect, discharge, pain, and AC reaction. • Signs: Central/paracentral infiltrate, >1mm, enlarges over 24–36h, epithelial defect, hypopyon, purulent discharge • Differentials: Sterile infiltrates, CLPU, viral keratitis	• Urgent: Fortified antibiotics or fluoroquinolone monotherapy • Cycloplegia for pain • Hospitalisation if severe	• Potential for scarring and vision loss; early treatment critical.
CLPU (Contact Lens Induced Peripheral Ulcer)  	• Cause: Sterile inflammatory reaction to exotoxins from gram-positive bacteria (e.g., S. aureus) • Signs: Small, dense, round subepithelial infiltrate, <1mm, minimal AC reaction, sectorial redness • Symptoms: None to moderate pain; vision unaffected **Differentials**: Microbial keratitis (needs exclusion), marginal keratitis	• Cease lens wear • **Topical steroid:** Fluorometholone 0.1% QID • **Prophylactic antibiotic:** Ciprofloxacin 0.3% QID (avoid chloramphenicol) • Mandatory 24h review to exclude MK	• Heals with a circular scar; resolution in 7–26 days. Recurrence ~33%, warn against extended wear
CLARE (Contact Lens Acute Red Eye) 	• Cause: Sterile inflammatory response to gram-negative bacterial exotoxins after extended wear • Signs: Circumferential limbal and bulbar redness, diffuse subepithelial infiltrates, no epithelial defect • Symptoms: Redness and tearing, improve with lens removal **Differentials**: EKC, microbial keratitis	• Cease lens wear • Topical steroids: **Fluorometholone** 0.1% QID to speed resolution • No antibiotics required unless diagnostic uncertainty	• Prognosis: Resolution in 1–3 weeks; recurrence up to 75% if extended wear resumed
Contact Lens-Related Infiltrates (other than CLARE & CLPU)	• Infiltrative keratitis excluding CLPU/CLARE • Often from mechanical trauma, solution toxicity, environmental factors	• Cease lens wear • Lubricants, saline rinse, cold compresses • Topical corticosteroids may hasten recovery (e.g., fluorometholone 0.1% QID)	• Self-limiting; recurrence possible (7%); identify and address risk factors
Corneal Erosion (CL-Related)	• Breach of epithelial integrity, usually traumatic or CL-related • Symptoms: Acute pain, tearing, blepharospasm • **Signs**: Document size, depth, possible delayed stromal glow **Differentials**: Microbial keratitis, infiltrates	• **Topical antibiotics:** Ciprofloxacin 0.3% QID until epithelium heals + 3 days • **Pain management:** Topical NSAIDs, oral paracetamol, cyclopentolate 1% BID if needed • Steroids and patching contraindicated • Review 24h	• Resolution in 1–7 days depending on size

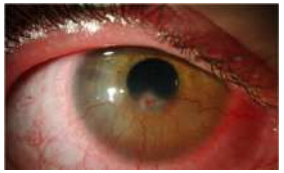
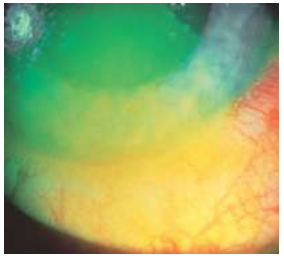
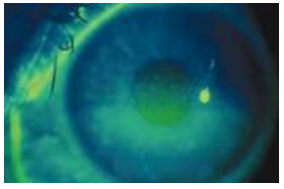
			
Contact Lens Solution Toxicity 	• Caused by preservatives (e.g., PHMB, Polyquad), especially with silicone hydrogels • **Signs**: Punctate staining, especially in annular pattern • **Symptoms**: Often asymptomatic, mild discomfort	• Discontinue lens and solution • Switch to peroxide-based system or daily disposables --- • Irritation with saline (NaCl) • Short course low strength topical corticosteroid, fluorometholone 0.1% four times daily for 1-2 days until the condition is resolved • Topical NSAID 4x daily if steroid is contraindicated	• Excellent if identified and solution/lens switched
Superior Epithelial Arcuate Lesion (SEAL) 	• Superior arcuate epithelial defect due to mechanical trauma from lens edge • Associated with tight-fitting, thick-edged silicone hydrogel lenses	• Cease lens wear • Refit lens with lower modulus, better edge design --- • Prophylactic antibiotic (e.g. gentamicin 0.3% or ciprofloxacin 0.3% four times a day until re-epithelialised)	• Heals quickly; recurrence with same lens fit
CLPC (Contact Lens Papillary Conjunctivitis) 	• Allergic/inflammatory reaction to lens deposits/mechanical trauma • Symptoms: Itching, mucous discharge, reduced lens tolerance • Signs: Enlarged papillae on upper tarsal conjunctiva	• Cease lens wear (initial 4 weeks) • Topical anti-allergy/steroid: ○ Olopatadine 0.1% BID ○ Fluorometholone 0.1% BID ○ Combination most effective • Monitor for IOP rise with steroids --- **Mild**: mast cell stabilisers (e.g. sodium cromoglycate or lodoxamide four times a day for 4-6 weeks, reduce to twice a day maintenance dosage before & after CL removal and recommence lens wear) or combo drugs (e.g. olopatadine 0.1% or ketotifen 0.025% twice daily before & after lens removal) **Severe**: topical NSAIDs (e.g. ketorolac 0.5% four times a day for 4 weeks) and/or short course topical steroids (e.g. fluorometholone alcohol 0.1% four times a day for 1 week) with or without mast cell stabilisers Increase CL replacement such as using daily disposables, better lens cleaning & decrease wear time	• Good with compliance and lens change; recurrence possible

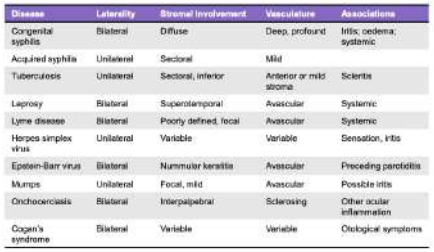
Corneal Conditions (Week 6)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Corneal Abrasions ☺ 	• **Superficial** epithelial injury without stromal involvement. • **Symptoms**: Acute pain, photophobia, tearing, foreign body sensation, blurred vision, lid swelling. • **Signs**: Clear-edged epithelial defect, staining with fluorescein, no infiltrate unless infected. • **Differentials**: Microbial keratitis, recurrent corneal erosion (RCE), chemical burn, foreign body, dystrophy, penetrating trauma	• **First aid:** Identify cause, irrigate, evert eyelid. • **Antibiotics**: Chloramphenicol 0.5% QID or ointment until 3 days after closure. Use fluoroquinolone in CL wearers. • Epithelial debridement for loose flaps. • **Pain control:** Topical NSAIDs (ketorolac 0.5% QID), oral paracetamol or ibuprofen. • Avoid: Pressure patching, topical steroids OBJ. • Review: 24h. Healing in 1–5 days expected.	• Prognosis excellent; slower healing if central/irregular/older patient/diabetic. • Risk of RCE post-healing (~10–12%)—warn patient
Corneal Foreign Body 	• Categorise as metallic vs non-metallic; embedded, superficial or penetrating. • **Signs**: Pain, photophobia, tearing, rust ring if metal. • **Differentials**: Penetrating FB, microbial keratitis.	• Removal: Topical anaesthetic to both eyes, irrigation or use of bent 25G needle/spud under slit lamp. • Post-removal: Irrigate, document size, apply fluorescein, measure VA. • Treat as abrasion: Topical antibiotics, NSAIDs, optional bandage CL or cycloplegia OBJ. • Rust ring: Remove with Alger brush. • Review: Day 1, 3, 5 or 7.	• Prognosis: Good if treated promptly. • Complication: RCE (~10%), stromal scarring if deep
Penetrating Foreign Body	• **Signs**: Positive Seidel, shallow AC, distorted pupil, hypopyon, cataract, low IOP. • Commonly work-related trauma (grinding, nails, glass). • Differentials: Non-penetrating FB, abrasion, microbial keratitis.	• First aid: Shield, no pressure patch, immobilise eye. • Immediate referral to corneal specialist. • Imaging: CT if IOFB suspected	• Prognosis: Vision-threatening. Risk of endophthalmitis—urgent care essential.
Superficial Foreign Body	• Non-embedded, rests on epithelium or conjunctiva. • **Symptoms**: Pain, FB sensation, lacrimation. • **Differentials**: Abrasion, microbial keratitis.	• Removal: Irrigation or moistened cotton bud under anaesthetic. • Post-care: Fluorescein, VA, antibiotics, NSAIDs. Consider fungal/acanthamoeba risk if organic/water exposure • Irrigate after FB & rust ring removal & use fluorescein to record size, location & depth of injury • Topical antibiotics, NSAIDs, bandage CL, cycloplegia to treat abrasion post FB removal ○ Review at 24 hours, 3 days, 5 or 7 days	• Good prognosis; scarring unlikely unless central/deep.
Recurrent Corneal Erosion (RCE) 	• Due to defective epithelial adhesion—traumatic (50%) or dystrophic (50% from EBMD, etc.) • **Symptoms**: Sudden severe pain on waking, FB sensation, photophobia, blurred vision. • **Signs**: Inferior epithelial defect, microcysts, negative staining. • **Differentials**: HSV, Meesman dystrophy, MK, abrasion, FB	**Acute**: Same as abrasion (antibiotics, pain relief). **Long-term:** • Lubricants 4–8x/day + ointment at bedtime for 3–6 months • Treat MGD/blepharitis • **Doxycycline 50mg BID** + topical FML 0.1% TID x 8 weeks • Diamond burr polishing • Hypertonic saline 5% ointment • Bandage CL (monitor for MK) • Autologous serum • Surgery (stromal micropuncture if >1 episode/month; PTK if widespread)	• Prognosis: ~40% recurrence over 4 years with standard treatment. Long-term management is crucial.
Epithelial Basement Membrane Dystrophy (EBMD)	• AKA Cogan's dystrophy / map-dot-fingerprint dystrophy • Most common dystrophy; bilateral; no gender preference; variable inheritance	• Asymptomatic: No treatment • Acute (with RCE): Same as RCE (abrasion management + long-term prevention)	• Prognosis: Good with appropriate long-term care; recurrence common without maintenance.

	• 90% asymptomatic; presents with RCE if symptomatic **Signs**: • Microcysts (epi) • Dots (desquamated cells) • Fingerprints (folds) • Maps (Bowman's membrane changes) **Differentials**: RCE, Meesman dystrophy	• Surgical options: ○ Epithelial debridement (30–50% success) ○ Stromal micropuncture (30–50%) ○ PTK (80% success)	
Fuch's Dystrophy ☺ 	• Hereditary, primarily autosomal dominant endothelial dystrophy (3rd most common dystrophy). • More prevalent in women (5:1). • Bilateral, asymmetric. • **Symptoms**: Blurred vision on waking that clears after 30 min (progresses to constant blur). • **Signs**: Central endothelial guttae, stromal striae/folds, hazy stroma, epithelial microcysts. • **Differentials**: Pseudophakic bullous keratopathy, posterior polymorphous dystrophy, ICE syndrome.	• Monitoring: Pachymetry & endothelial cell counts. • Topical: Hypertonic saline 5% (6–8x/day + ointment at night). • Drying: Warm air from hairdryer. • Bandage CL: If bullae present. • Surgery: Indicated if VA <6/12 or blur doesn't clear by midday. Options include DSAEK, DMEK, DLEK, PK	• Prognosis: Slowly progressive. Surgical outcomes excellent; graft survival dependent on donor quality and patient comorbidities.
Bullous Keratopathy 	• Severe endothelial failure → irreversible stromal and epithelial oedema. • **Causes**: Fuchs', post-cataract surgery, trauma, infection, uveitis, posterior polymorphous dystrophy, high IOP. • **Symptoms**: Pain (especially when bullae rupture), blurred vision (worse in AM), photophobia. • **Signs**: Bullae (~1mm^2 raised epithelial blisters with fluid), hazy stroma, epithelial oedema.	• Topical hypertonic agents: Ointment more effective than drops (reduce oedema ~24%). • Bandage CL: Pain relief. • Topical honey drops: Anecdotal benefit. • Treat ruptured bullae: As per corneal abrasion. • Surgical: Corneal transplant if vision/comfort unacceptable	• Prognosis: Pain managed conservatively; surgical correction definitive if vision-threatening.
Keratoconus 	• Progressive corneal ectasia → thinning and bulging. • Onset: Adolescence to middle age. • Bilateral, asymmetric. • Symptoms: Gradual reduction in vision. **Signs**: • Irregular astigmatism • Vogt striae, Fleischer ring, Munson sign • Scissor reflex, distorted keratometry mires • Advanced: scarring, corneal hydrops • **Differentials**: Pellucid marginal degeneration, keratoglobus, post-surgical ectasia.	• Diagnostic: Refraction, topography, tomography, keratometry. • Spectacles initially, then specialty CLs (RGP, hybrid, sclerals). • CXL (collagen cross-linking): Early stage to halt progression (UVA + riboflavin). • Surgery: DALK, ICR, PK if CLs fail. • Avoid refractive surgery (LASIK contraindicated)	• Prognosis: Variable. Early CXL stabilises disease. Visual rehab often possible with CLs; surgery for advanced.
Corneal Hydrops 	• Acute complication of keratoconus (~5%). • Rupture of Descemet's membrane → stromal and epithelial oedema. • Symptoms: Sudden vision drop, pain, photophobia. • Signs: Marked oedema, microcysts, protrusion.	**Self Limiting:** 8-10 weeks **Topical**: • Hypertonic agents • Broad-spectrum antibiotic prophylaxis • Corticosteroids (reduce scarring) • Referral: For surgery if CL intolerance or poor visual outcome	• Prognosis: Vision may deteriorate due to scarring; good resolution possible if managed early and effectively.

Photokeratitis 	• UV-induced epithelial damage. • Causes: Welding (arc eye), snow blindness, tanning lamps. • Onset: 6–12 hours post-exposure. • Symptoms: Severe pain, photophobia, FB sensation, tearing, lid swelling (often wakes patients at night). • Signs: Widespread interpalpebral punctate staining. **Differentials**: Toxic keratitis, Thygeson's, EKC (epidemic keratoconjunctivitis), Dry eye, FB	**Supportive**: • PF lubricants QID to Q8H • Ointment/gel at night • Cold compresses • Sunglasses **Optional**: • Chloramphenicol 0.5% QID • Cyclopentolate 1% BID • Oral analgesics: Paracetamol 1g or ibuprofen 400mg QID	
Dellen 	• Localised peripheral thinning due to tear film disruption and dehydration. • Often adjacent to an elevated area (e.g., pinguecula, limbal mass). • Symptoms: Mild FB sensation, photophobia, dryness. • Signs: Focal corneal depression (no epithelial staining centrally), possibly superficial stromal haze. **Differentials**: Corneal infiltrate, neurotrophic ulcer, peripheral ulcerative keratitis.	• Mainstay: Lubrication (ointment QID + at bedtime). • If ongoing: Add viscous daytime gel (e.g., Genteal Gel), consider punctal plugs. • Prophylactic antibiotics: Chloramphenicol 1% ointment TID. • Identify and treat cause (e.g., remove pinguecula). • Cease CL wear if contact lens-induced. • Review: Within a week to confirm resolution	• Prognosis: Excellent; resolves quickly once treated. No scarring unless chronic. 
Sterile Corneal Infiltrates	• Inflammatory reaction, not infectious. • Causes: CL wear, staph blepharoconjunctivitis, conjunctivitis, EKC. • **Symptoms**: Redness, photophobia, FB sensation; often milder than MK. **Signs**: • Peripheral subepithelial infiltrates • <1.5mm, multiple • Little/no overlying staining • Minimal AC reaction **Differentials**: Bacterial keratitis, phlyctenulosis, marginal keratitis	• **Topical steroids:** Fluorometholone 0.1% QID (others: pred phosphate, loteprednol) • **Prophylactic antibiotics:** Ciprofloxacin or tobramycin or gentamicin 0.3% QID in CL wearers • Cease CL wear • Review: 24–48 hours • Address associated conditions (e.g., blepharitis)	• Prognosis: Good. Mild scarring possible; recurrence in high-risk patients.
Marginal Keratitis 	• Hypersensitivity to staphylococcal exotoxins. • Associated with staph blepharitis, rosacea. • Symptoms: Irritation, photophobia, unilateral red eye. • Signs: ○ Peripheral, interpalpebral infiltrates (round, 1mm) ○ Usually inferior (4 & 7 o'clock) or superior (2 & 10 o'clock) ○ Clear zone between limbus and infiltrates • May ulcerate, rarely AC reaction **Differentials**: MK, phlyctenulosis, CLPU, PUK	• Mild: Warm compress, lid hygiene, chloramphenicol QID. • Moderate/severe: Add steroid (FML 0.1% or pred phosphate 0.5% QID). • If recurrent or rosacea/post blepharitis: Oral doxycycline 50–100mg BID x 2–4 weeks then taper. • Review: 24 hours; check response and rule out infection	• Prognosis: May scar. Manage underlying cause (blepharitis, rosacea). Recurrence likely if untreated 
Phlyctenulosis	• Delayed hypersensitivity to staph or TB antigen. • Common in children. • Raised nodules **Symptoms**: Photophobia, tearing, FB sensation. **Signs**:	• Topical steroid: Fluorometholone or pred phosphate QID. • Topical antibiotic: Chloramphenicol or fluoroquinolone QID. • Lid disease: Lid scrubs, warm compresses, topical antibiotic ointment (chloramphenicol 1%). • Oral doxycycline (100mg BID x 4–6 weeks) if resistant.	• Prognosis: Good if underlying cause managed; corneal phlyctens may scar if recurrent or untreated. Topical corticosteroids (eg. fluorometholone alcohol 0.1%) four times daily until resolved

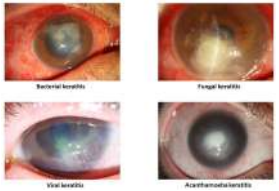
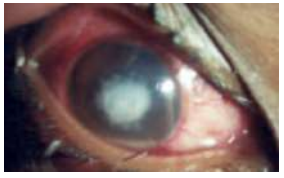
	• Small, pink-white nodules at limbus or conjunctiva • Surrounding hyperaemia • Rarely corneal involvement **Differentials**: Marginal keratitis, pinguecula, pyogenic granuloma, microbial keratitis	• TB screening: Mantoux, CXR, sputum culture if TB suspected.	Topical antibiotic (eg. ciprofloxacin 0.3%) four times daily until resolved Blepharitis: oral tetracycline (eg. doxycycline 100mg) twice daily for 4 to 6 weeks
Chemical Burns 	• Alkalis worse than acids due to deeper penetration. • Common sources: Cement, lime, airbag, ammonia, lye. • Symptoms: Pain, tearing, blurred vision, photophobia. **Signs**: • Conjunctival chemosis, epithelial defect • Limbal ischaemia • Corneal haze/opacification **Differentials**: Thermal burn, photokeratitis, toxic keratitis Bizrah M, Yusuf Am Ahmad S. An update on chemical eye burns. Eye 2019; 33: 1362-1377.	• Immediate irrigation: 20–30 mins with saline/Ringer's. • Evert lids, remove particles. • Topical meds: ○ Steroid (pred phosphate 1% QID) ○ Antibiotic prophylaxis (e.g., chloramphenicol) ○ Cycloplegic (cyclopentolate 1%) ○ PF lubricants • Vitamin C, doxycycline (to inhibit collagenase) • Refer: Ophthalmology if grade II or worse. **GRADE 1 BURNS:** • Prophylactic antibiotic treatment (eg. chloramphenicol 0.5% or ciprofloxacin 0.3% or gentamicin 0.3%) four times daily • Topical corticosteroid eye drops (eg. fluorometholone alcohol 0.1%, prednisolone phosphate unpreserved 0.5%) four times daily until epithelium is healed • Cycloplegics (eg. atropine 1% or cyclopentolate 1% unpreserved) twice daily for pain relief • Bandage CL can be used in severe cases • Long-term treatment: preservative free ocular lubricants **GRADE 2+ BURNS** • Hourly corticosteroids with minims of prednisolone phosphate 0.5% to reduce inflammation • Cycloplegics (eg. atropine or cyclopentolate 1%) twice daily to manage pain & complications • Topical antibiotics (eg. chloramphenicol) four times daily to prevent corneal infection • Sodium citrate 10% & sodium ascorbate 10% every 2 hours + oral vitamin C 500 mg every 6 hours to reduce inflammation and minimise loss of stromal tissue • Oral tetracyclines (eg. doxycycline 100mg) twice daily are used for the antiinflammatory effect ○ Treatment will continue until corneal epithelium begins to heal & then taper according to clinical appearance	• Prognosis: Depends on extent/limbal damage. Limbal stem cell deficiency → scarring, symblepharon, blindness.  Chemical Burns: Roper-Hall classification Grade / Corneal Findings / Limbal Ischemia / Prognosis I / Corneal epithelial damage / None / Excellent II / Corneal haze, iris details visible / <1/3 / Good III / Total epithelial loss, Stromal haze obscures iris details / 1/3 – ½ / Guarded IV / Opaque cornea; iris and pupil are not visible / >1/2 / Poor Chemical Burns: Dua classification  Grade / Prognosis / Clinical findings / Conjunctival involvement I / Very good / 0 clock hours of limbal involvement / 0% II / Good / ≤3 clock hours of limbal involvement / ≤30% III / Good / >3-6 clock hours of limbal involvement / >30-50% IV / Good to guarded / >6-9 clock hours of limbal involvement / >50-75% V / Guarded to poor / >9-12 hours of limbal involvement / >75-<100% VI / Very poor / Total limbus (12 clock hours) involved / Total conjunctiva (100%) involved 
Toxic Keratitis 	• Caused by preservatives (e.g., BAK), contact lens solution, or contaminated drops. • Symptoms: Pain, FB sensation, redness, photophobia. **Signs**: • Diffuse punctate epithelial staining • Epithelial sloughing • Bilateral **Differentials**: EKC, photokeratitis, HSV keratitis, CL solution toxicity.	• Cease offending agent immediately • Lubricants: PF, QID or more • Topical steroid: Low-dose (e.g., FML 0.1% QID) for inflammation • Cycloplegic: If photophobia present • Review: 2–3 days	• Prognosis: Excellent if cause eliminated early; may scar if delayed. • Mild steroids or topical NSAIDs (eg. minims prednisolone phosphate 0.5%) four times daily • Consider preservative-free solutions to decrease chances of

			worsening toxicity induced by preservatives • Epithelial damage: prophylactic antibiotic (eg. minims chloramphenicol 0.5%) • Ocular lubricants (preservative-free) to promote maintenance of a healthy epithelial surface (eg. Hylo-Fresh, Refresh)
"Christmas Eye" 	• Rural SE Australia; linked to tree beetle (Orthoperus sp.) toxin. • Sudden onset in summer. • Symptoms: Extreme pain (wakes from sleep), photophobia, tearing, blurry vision. **Signs**: • Inferior corneal epithelial defect • Hyperaemia • Lid oedema **Differentials**: HSV, corneal abrasion, RCE, MK.	• Prophylactic antibiotic: e.g., chloramphenicol • Cycloplegia: For pain • Oral analgesia • Lubricants • Topical NSAIDs contraindicated (may worsen outcome) • Review: 24–48h	• Prognosis: Recurrence possible over summer; resolves with supportive care.
Interstitial Keratitis 	• Inflammation of corneal stroma with intact epithelium. • Common causes: Syphilis (congenital), HSV, TB, Lyme. • Symptoms: Red eye, photophobia, pain, blurry vision. **Signs**: • Stromal haze/opacification • Ghost vessels • No epithelial staining • Often bilateral	• Refer: Systemic workup (syphilis serology, TB, etc.) • **Steroid**: Topical prednisolone phosphate 1% every 1-2 hours for active disease → as well as treatment of the underlying condition • Treat underlying cause: Penicillin for syphilis, acyclovir for HSV • Cycloplegic: For AC inflammation	• Prognosis: Variable. May lead to permanent stromal scarring. Early treatment improves outcomes. 
Filamentary Keratitis 	**Definition and Cause:** Filamentary Keratitis is characterised by the presence of mucus filaments adherent to the corneal epithelium. These filaments are composed of degenerating epithelial cells and mucus. The condition is most commonly **associated with underlying dry eye disease** but can also occur in the context of: • Neurotrophic keratopathy • Superior limbic keratoconjunctivitis (SLK) • **Post-ocular surgery** • Systemic conditions such as Sjögren's syndrome or thyroid eye disease **Signs**: • Presence of grayish, thread-like filaments on the corneal surface, which may stain with fluorescein or rose bengal. • Often located on the superior cornea • Conjunctival hyperaemia • Reduced tear break-up time and other signs of dry eye • Filaments can be visualized during slit lamp exam as being anchored at one end to the corneal surface	**Workup:** • Careful slit lamp examination to detect filaments and associated epithelial changes • Schirmer's test, TBUT (tear break-up time) • Staining with fluorescein and/or lissamine green or rose bengal • Evaluate for underlying dry eye or SLK **Non Pharmacological Treatment:** • Removal of filaments using forceps under topical anaesthesia • Lubrication: Frequent use of non-preserved artificial tears (e.g., 0.3-1% carboxymethylcellulose or hyaluronic acid), ointment at night • Punctal occlusion (temporary or permanent) to conserve tears • Lid hygiene and warm compresses if MGD or blepharitis is present • Scleral contact lenses in refractory cases to protect the ocular surface	**General Management:** • Treat any underlying condition contributing to FK (e.g., dry eye, SLK, thyroid disease) • Patient education on chronic nature of the condition and importance of compliance with lubricants and therapy • Address environmental factors (humidifiers, screen breaks) • Regular follow-up to monitor for recurrence **Prognosis**: • Chronic and often recurrent condition • Symptoms can be significantly alleviated with consistent treatment, especially if underlying causes are addressed

Disease	Laterality	Stromal involvement	Vasculature	Associations
Congenital syphilis	Bilateral	Diffuse	Deep, profound	Iritis; oedema; systemic
Acquired syphilis	Unilateral	Sectoral	Mild	
Tuberculosis	Unilateral	Sectoral, inferior	Anterior or mid stroma	Sclerits
Leprosy	Bilateral	Superotemporal	Avascular	Systemic
Lyme disease	Bilateral	Poorly defined, focal	Avascular	Systemic
Herpes simplex virus	Unilateral	Variable	Variable	Sensation, iritis
Epstein-Barr virus	Bilateral	Nummular keratitis	Avascular	Preceding parotiditis
Mumps	Unilateral	Focal, mild	Avascular	Possible iritis
Onchocerciasis	Bilateral	Interpalpebral	Sclerosing	Other ocular inflammation
Cogan's syndrome	Bilateral	Variable	Variable	Otological symptoms

	Symptoms: • Foreign body sensation • Pain and discomfort, especially during blinking (due to mechanical traction of filaments) • Photophobia • Epiphora • Blurred vision **Differential Diagnoses:** • Dry Eye Disease (evaporative or aqueous-deficient) • Superior Limbic Keratoconjunctivitis (SLK) • Mucous Fishing Syndrome • Thygeson's Superficial Punctate Keratitis • Herpetic keratitis (HSV) • Neurotrophic keratitis • Exposure keratopathy	**Pharmacological Treatment:** • Hypertonic saline (5% NaCl) eye drops and ointment: helps reduce corneal edema and improve symptoms • Topical acetylcysteine 5% (mucolytic): reduces filament formation by breaking down mucus; used 3-4x/day (needs to be compounded) • Topical corticosteroids (e.g., fluorometholone 0.1%): may be indicated if there is significant inflammation (used cautiously and typically short term) • Topical cyclosporin A 0.05-0.1%: may reduce inflammation and promote ocular surface healing in chronic cases • Autologous serum eyedrops: helpful in refractory dry eye and FK cases due to their epitheliotropic factors **Avoid**: • Contact lens use (can exacerbate symptoms) • Preserved eye drops	• Visual prognosis is generally good, but untreated FK can lead to persistent epithelial defects and risk of secondary infection

Microbial Keratitis (Week 6)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Bacterial Keratitis 	• Etiology: Infection of the cornea by bacteria such as Pseudomonas aeruginosa, Staphylococcus aureus, Streptococcus pneumoniae. • Risk Factors: Contact lens wear, trauma, ocular surface disease, surgery. • Symptoms: Rapid onset pain, photophobia, vision loss, discharge, red eye. **Signs**: • Central/paracentral stromal infiltrate >1 mm, enlarging in 24–36 hours • Epithelial defect • AC reaction (flare, cells, hypopyon) • Mucopurulent discharge • Lid oedema **Differentials**: Sterile infiltrates, fungal keratitis, Acanthamoeba keratitis.	• Empirical therapy (no culture) appropriate initially unless: ulcer >2 mm, deep/central, immunocompromised, post-op, atypical organism suspected. • Antibiotics: ○ Fluoroquinolone monotherapy (ciprofloxacin/ofloxacin 0.3%) ■ e.g. q15min for 6h, then q30min, taper as improves ○ OR Fortified cephalosporin + aminoglycoside **(e.g., cefazolin + tobramycin)** hourly alternating • **Cycloplegic**: Cyclopentolate 1% or atropine 1% BID for pain and AC reaction • Review: 24h follow-up mandatory • Avoid topical steroids initially unless under specialist guidance (e.g., SCUT trial findings suggest caution)	• Prognosis: Variable; poor if delay in treatment, central lesion, or severe inflammation. • Scarring common; endophthalmitis rare (~0.5%) but sight-threatening. • Oxflacin has better corneal penetration and lower dosage
Fungal (Myotic) Keratitis 	• Etiology: Filamentous fungi (Fusarium, Aspergillus) and yeast (Candida). • Risk Factors: Ocular trauma (especially with organic matter), CL wear, steroids, surgery, diabetes OBJ. • Symptoms: Gradual onset pain, blurred vision, photophobia, discharge. **Signs**: • Feathery-edged infiltrates • Satellite lesions • Raised sloughing edges • Delayed epithelial defect • Hypopyon	• Diagnostics: Culture (Sabouraud), smears (KOH, Giemsa, PAS), biopsy • First-line: ○ Topical **natamycin** 5% hourly while awake, then QID for ≥3 weeks (esp. Fusarium) ○ Topical amphotericin B 0.15–0.3% for yeast • Voriconazole is NOT effective as monotherapy for Fusarium • Adjuncts: ○ Cycloplegics (for pain, AC inflammation) → atropine or cyclopentolate 1% ○ Antibiotics if secondary bacterial infection	• Prognosis: Generally poorer than bacterial keratitis. • MUTT trial: Natamycin superior to voriconazole in Fusarium cases • 16% required keratoplasty or had perforation Yeast (Canida) infection → amphotericin B (0.15% to 0.3%)

	• Neovascularisation in chronic disease **Differentials**: Bacterial keratitis, Acanthamoeba, HSV stromal keratitis.	suspected ○ Surgery: Penetrating keratoplasty if perforation or poor response • Antiseptics: PVP-I, PHMB may be considered if antifungals not accessible	
Acanthamoeba Keratitis 	• Etiology: Free-living amoeba found in water, soil. • Risk Factors: Contact lens wear, exposure to water (e.g., swimming, tap water). • **Symptoms**: Severe pain (disproportionate), blurred vision, photophobia. Early Signs: • Punctate epitheliopathy • Pseudodendrites (mimicking HSV) • Perineural infiltrates Later Signs: • Ring infiltrate • Stromal ulceration • Anterior uveitis, hypopyon • Reduced corneal sensitivity is common **Differentials**: HSV keratitis, fungal keratitis, bacterial keratitis	**Diagnostics**: • Confocal microscopy • Smear (calcofluor white) • Culture (non-nutrient agar with E. coli) • PCR **Topical Therapy (Dual):** • Biguanides: PHMB 0.02–0.06% or chlorhexidine 0.02–0.2% • Diamidines: Propamidine 0.1% or hexamidine 0.1% • Regimen: Hourly (day and night) x 48h, then hourly day x 72h, taper over months • Duration: Avg 6 months (0.5–29 months) • **Steroids**: Only after 2 weeks of treatment (pred 0.5% QID) • Adjunct: ○ Oral NSAIDs or systemic steroids for scleritis/limbitis ○ Continue anti-amoebic 4 weeks after steroid cessation	• Prognosis: Poor if diagnosis/treatment delayed (>3 weeks). ○ Recurrence possible due to cyst survival. ○ Requires close monitoring due to treatment toxicity and disease progression

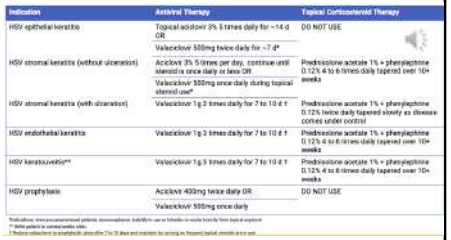
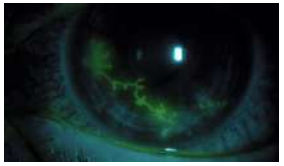
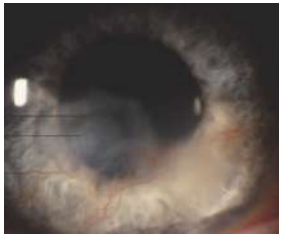
Myopia + Amblyopia (Week 6)

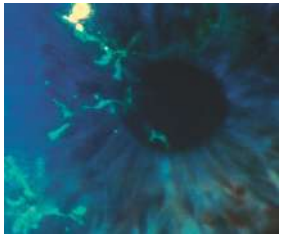
Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Myopia	• **Definition**: A refractive error where the spherical equivalent is ≤ -0.50 D when ocular accommodation is relaxed . • **Progressive Myopia:** No standardised definition. Typically begins in childhood and continues into late teens. Progression changes with age. Adult-onset myopia can also occur. **Signs**: • Distance blur (especially at school) • Squinting, eye strain • Onset during childhood **Risk Factors:** • Genetic: Parental myopia increases risk • Environmental: Less outdoor time, intense near work **Differentials**: • Pseudomyopia • Accommodative spasm • Latent hyperopia (especially in young children)	• Pharmacological Treatment – Atropine ○ Mechanism: Non-selective muscarinic antagonist; possibly works via scleral remodelling rather than accommodation. ○ Concentrations: 0.01% to 0.05% commonly used. Commercial (e.g. Eikance) or compounded. • Dosing: ○ Typically nightly ○ Lower concentrations (0.01%) have fewer side effects but reduced efficacy • Side Effects: ○ Photophobia, reduced accommodation, irritation, systemic effects (dry mouth, tachycardia) ○ Precautions: Avoid in preterm infants, children with Down syndrome, brain injury, spastic paralysis . • Punctal Occlusion: Recommended post-instillation to minimise systemic absorption.	• Review Schedule: Monitor every 3–6 months with refraction and axial length. • Treatment Outcome Metrics: ○ Axial length growth < 0.20 mm/year considered controlled. ○ Use Emmetropic Progression Ratio for relative treatment success. • Discontinuation: ○ Tapering reduces rebound risk. • Monitor post-treatment progression. • Prognosis: ○ Early and consistent intervention can significantly slow progression.

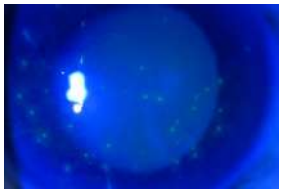
		• Evidence-Based Trials: ○ LAMP: 0.05% most effective; 5-year study showed best control of axial elongation and refractive progression . ○ CHAMP: Showed concentration-dependent effect (0.02% > 0.01%). ○ MOSAIC: 0.01% atropine vs placebo in Irish population – supports efficacy across ethnicities. • Non-Pharmacological Interventions: • Optical: ○ Spectacle lenses: HAL (Stellest), DIMS lenses ○ Contact lenses: Dual-focus or multifocal soft lenses ○ Orthokeratology (overnight lens reshaping) • Behavioural: ○ Increased outdoor time (2+ hrs/day) ○ Limit screen and near work ○ Combination Therapies: ■ Atropine + Ortho-K / DIMS / SCLs — better outcomes ○ Red light therapy (emerging)	○ High myopia (> -6.00D) increases risk of: • Retinal detachment, glaucoma, myopic maculopathy, cataracts --- Atropine 0.01% eye drops or pirenzepine 2% gel or cyclopentolate: 2 times daily (poorer compliance) • Currently, atropine is the most effective form of myopia control ○ Higher dose of atropine = less myopia progression (dose-dependent relationship) ○ Cessation of treatment resulted in faster myopia progression and faster axial elongation compared to the untreated eye Timolol eye drops or 7-methylxanthine (7-MX) oral tablets
Amblyopia	• Definition: Reduced best-corrected visual acuity not due to pathology, with at least one amblyogenic factor during the critical period of development . • Types: ○ Anisometropic (unequal refractive error) ○ Strabismic (constant unilateral strabismus) ○ Deprivation (e.g., cataract, ptosis) ○ Mixed • Signs: ○ Reduced vision in one or both eyes despite correction ○ Eye preference or squinting ○ Lack of fixation • **Differentials**: ○ Pathological causes of VA loss ○ Refractive error that improves immediately with lenses ○ Optic neuropathies ○ Media opacities (cataract, corneal opacity)	**1. Atropine Penalisation** • Concentration: **1% atropine** ○ Mechanism: Cycloplegia and mydriasis in the sound eye to force use of amblyopic eye ○ Dosage: Daily or weekend-only ○ Outcomes: Equivalent to 2–6 hours of patching daily . • Side Effects: ○ Photophobia, blurred near vision, eye pain ○ Systemic: dry mouth, restlessness, tachycardia **2. Patching:** • Occlude sound eye for 2–6 hours/day • Useful when atropine not suitable or less effective **3. Plano Lens with Atropine:** • Used when atropine alone insufficient • Plano lens further blurs sound eye • Evidence-Based Support: ○ PEDIG Trials: Atropine and patching are equally effective in moderate amblyopia . ○ Gunton et al: Weekend-only atropine has similar outcomes to daily.	• Monitoring: • Continue for 1–2 years • Review every 6–8 weeks • Track VA, refraction, compliance • Prognosis: ○ Best results in 3–7 years age range ○ Most improvement within 6 months ○ Recurrence possible; long-term monitoring essential • Moderate amblyopes show excellent recovery; severe cases may be slower, less complete • Psychosocial & Practical Considerations: • Atropine often better tolerated (fewer compliance and stigma issues) • Prefer holidays for treatment in school-aged children • Educate families: instillation technique, manage expectations, support adherence

Viral Corneal Infections (Week 7)

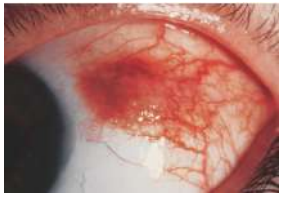
<table>
<tr><th>Condition</th><th>Information (Signs, Symptoms and Differentials)</th><th>Treatment and Workup (pharmacological and nonpharmacological treatments)</th><th>Other Information (General Management, Prognosis)</th></tr>
<tr><td>Herpes Simplex Infection</td><td colspan="2">HSV-1 is the most common ocular strain.
Transmission: Direct contact with infected secretions.
Triggers: UV light, stress, trauma, fever, immunosuppression.
Often recurrent and unilateral (but may alternate eyes).
Differentials: EKC, Thygeson’s, toxic keratitis, Acanthamoeba, bacterial keratitis, zoster ophthalmicus.
General Features:
Reduced corneal sensation
History of cold sores, stress or immune suppression
Recurrent episodes affecting eyelid, conjunctiva, or cornea</td><td></td></tr>
<tr><td>HSV Blepharoconjunctivitis
  </td><td>Common primary ocular HSV presentation.
Symptoms: Pain, lid swelling, watery discharge, vesicular rash on eyelids.
Signs: Vesicles on lids, follicular conjunctivitis, preauricular lymphadenopathy.
Differentials: Varicella zoster, allergic conjunctivitis, bacterial conjunctivitis.</td><td>Topical acyclovir 3% ointment 5x/day for 7 days (on lid margin and conjunctiva)
Oral acyclovir: 400 mg 5x/day if moderate/severe or systemic signs present
Avoid topical steroids unless specifically advised (risk of exacerbation)</td><td>Prognosis: Good; usually self-limiting. Risk of recurrence.</td></tr>
<tr><td>Herpes Simplex Keratitis (Condensed Summary)
<table><tr><th>HSV category</th><th>Common nomenclature</th></tr><tr><td>Epithelial keratitis</td><td>Dendritic keratitis
Geographic keratitis
Marginal keratitis / limbitis</td></tr><tr><td>Stromal keratitis without ulceration (non-necrotising)</td><td>Interstitial keratitis
Immune stromal keratitis</td></tr><tr><td>Stromal keratitis with ulceration (necrotising)</td><td>Necrotizing keratitis</td></tr><tr><td>Endothelial keratitis</td><td>Disciform keratitis</td></tr><tr><td>Uveitis</td><td>Uveitis / trabeculitis
Keratouveitis</td></tr></table></td><td>Herpes Simplex Keratitis (HSK) presents across a spectrum of corneal disease forms:
Epithelial keratitis: dendritic, geographic, marginal keratitis/limbitis
Stromal keratitis
Non-necrotising (interstitial, immune-mediated)
Necrotising (stromal ulceration)
Endothelial keratitis: disciform
Keratouveitis (uveitis/trabeculitis)
Epithelial Keratitis
Most common form (50–80%)
Presents with dendritic ulcers (with terminal bulbs, ulcerated base stains with fluorescein, heaped edges with lissamine green)
May also appear as geographic ulcers (progression or misdiagnosed/steroid-treated dendrites)
Symptoms: FB sensation, irritation, redness, blurred vision (not typically painful)
Reduced corneal sensation, especially in recurrent cases
Loss of corneal sensation may be detected using cotton wisp test
No stromal involvement (must be ruled out)
Stromal Keratitis (Non-necrotising)
~20–48% of recurrent HSV cases
Intact epithelium; stromal haze or infiltrates
Immune-mediated, not active viral replication</td><td>Epithelial Keratitis Treatment
Topical Antiviral: Aciclovir 3% ointment, 1cm strip to lower fornix 5 times/day until healed (~10 days)
Oral Antivirals (if topical not tolerated):
Aciclovir 400 mg 5x/day
Valaciclovir 500 mg 2x/day
Famciclovir 250 mg 3x/day
No corticosteroids — they worsen epithelial HSV keratitis
Review 24–48h, then at 7 days
Counsel regarding 15% recurrence within 1 year
Debridement (Optional)
Moistened cotton tip or cellulose sponge to remove viral-laden edges
Requires topical anaesthesia
Evidence mixed – some clinicians still use it
Stromal Keratitis (Non-necrotising)
Topical steroid: Prednisolone acetate 1% + phenylephrine 0.12% q2h to qid, taper slowly over 10+ weeks
Concurrent topical aciclovir 3% 5x/day until steroid down to once daily or less
Often replace topical with oral antivirals (preferred)
Long-term prophylaxis strongly indicated
Stromal Keratitis (Necrotising)
Oral antivirals (e.g., valaciclovir 1g tds x 7–10 days)</td><td>General Management
Educate on chronic nature and recurrence
Avoid corticosteroids unless stromal or endothelial involvement and always with antiviral cover
Avoid contact lenses during active infection
Refer stromal, necrotising, endothelial, or keratouveitis cases to cornea or uveitis specialist
Prognosis
Epithelial disease: excellent prognosis if treated; self-limiting but slower resolution without antivirals
Stromal disease: guarded prognosis; risk of permanent vision loss due to scarring
Keratouveitis: can be recurrent and cause glaucoma
Recurrence rates:
9% (epithelial) with prophylaxis vs 14% without
14% (stromal) with prophylaxis vs 28%</td></tr>
</table>

	• Risk of blinding complications • Symptoms: photophobia, discomfort, decreased vision • Look for corneal vascularisation, stromal infiltrates **Stromal Keratitis (Necrotising)** • Severe pain, ulceration • Associated with viral replication and necrosis • Often accompanied by epithelial defects **Keratouveitis** • Inflammation extends to anterior chamber: cells/flare, hypopyon, raised IOP • Often recurrent • May mimic anterior uveitis **Differential Diagnoses** • Adenoviral keratitis (EKC): may resemble HSV dendrites • Thygeson's SPK • Herpes Zoster Ophthalmicus (HZO): dendrites without terminal bulbs, coarser, non-ulcerative • Bacterial marginal keratitis • Fungal keratitis • Sterile infiltrates • Neurotrophic keratopathy in chronic HSV	• Topical steroids as above • Refer to ophthalmologist • Consider steroid-sparing agents if recurrent **Keratouveitis** • Oral antivirals at therapeutic dose • Topical steroids with ophthalmology supervision • Monitor IOP closely, manage with glaucoma meds if needed **Prophylactic Antiviral Therapy** • For recurrent or stromal disease: • Aciclovir 400 mg bd • Valaciclovir 500 mg daily • Famciclovir 250 mg daily or 125 mg bd • Continue for long durations; protection only lasts while on medication • Not routinely recommended for epithelial disease alone	without • Herpes simplex virus epithelial keratitis – oral antivirals are **not** indicated as first line therapy if topical treatment can be tolerated 
HSV Epithelial Keratitis 	• Most common form of HSV keratitis. • Symptoms: Pain, photophobia, FB sensation, blurred vision. • Signs: ○ Dendritic ulcer (branching, central staining with terminal bulbs) ○ Geographic ulcer (enlarged dendrite with scalloped edges) ○ Reduced corneal sensitivity • **Differentials**: Thygeson's, Acanthamoeba, toxic keratitis, EKC, bacterial keratitis, herpes zoster.	• Topical antiviral: ○ Aciclovir 3% ointment 5x/day for 10–14 days OR ○ Ganciclovir 0.15% gel 5x/day then taper • Debridement: Only for non-responders • NO steroids in epithelial disease	• Prognosis: Good if promptly treated. Recurrence possible; may progress to stromal disease.
HSV Stromal Keratitis (without ulceration) 	• Immune-mediated inflammation of stroma (not direct viral replication). • May follow epithelial keratitis. • **Symptoms**: Decreased vision, photophobia, redness, minimal pain. • **Signs**: Stromal haze, neovascularisation, intact epithelium, AC reaction possible. • **Differentials**: Interstitial keratitis (e.g., syphilis, TB), HZO, fungal keratitis, immune ring (Acanthamoeba).	• **Topical corticosteroids** (e.g., pred acetate 1% q2h) • **Topical antiviral cover:** Aciclovir ointment or ganciclovir gel • Oral antiviral (e.g., acyclovir 400 mg BID) for steroid cover • Taper steroids slowly over weeks to months depending on response	• Prognosis: Guarded; recurrent inflammation can cause scarring and vision loss. Recurrence likely.
Varicella zoster virus disease	• Reactivation of VZV in the trigeminal ganglion → Herpes Zoster Ophthalmicus (HZO). • Symptoms: Pain, redness, vesicular skin rash (V1 dermatome), malaise, fever. **Signs**: • Hutchinson's sign (lesion on tip of nose = nasociliary involvement	• Oral antiviral within 72h of onset: ○ Acyclovir 800 mg 5x/day for 7–10 days ○ Valacyclovir 1 g TID x 7–10 days • Topical lubricants for keratitis • Topical steroids only under specialist supervision if stromal or uveal involvement • Pain relief: Oral analgesics, NSAIDs	• Prognosis: Early treatment reduces post-herpetic neuralgia and severity. • Complications: Stromal scarring, neurotrophic keratopathy, glaucoma.

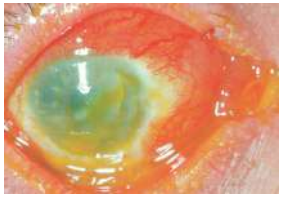
	➔ high risk of ocular disease) • Vesicles ➔ pustules ➔ crusting • May involve conjunctiva, cornea, uvea, retina **Differentials**: HSV blepharoconjunctivitis, EKC, allergic conjunctivitis, other causes of keratitis.		
Herpes zoster ophthalmicus  	• **Reactivation** of VZV in the ophthalmic (V1) division of the trigeminal nerve. • Prodrome: Malaise, fever, headache, fatigue, periorbital burning. • Dermatologic signs: ○ Vesicular rash in V1 dermatome (forehead, scalp, upper eyelid, nose). ○ Hutchinson's sign: Tip of nose involvement suggests nasociliary branch ➔ high risk of ocular involvement. • Ocular signs (may involve multiple tissues): ○ Conjunctivitis ○ Keratitis (epithelial, stromal, neurotrophic) ○ Episcleritis/scleritis ○ Uveitis ○ Cranial nerve palsies (III, IV, VI) ○ Secondary glaucoma • **Differentials**: HSV, contact dermatitis, EKC, cellulitis, staph blepharitis, orbital involvement.	• Systemic antivirals within 72 hours of rash onset: ○ Acyclovir 800 mg 5x/day x 7–10 days ○ OR Valacyclovir 1 g TID • Topical therapy based on ocular involvement: ○ PF lubricants ○ Antibiotic ointment for vesicular lesions ○ Steroids for stromal keratitis/uveitis under supervision ○ Cycloplegics for uveitis • **Pain control:** Paracetamol, NSAIDs, possibly gabapentin if neuralgia	• **Prognosis**: Early antivirals reduce ocular complications and post-herpetic neuralgia. • Risk of recurrence: Reduced with long-term antiviral prophylaxis if recurrent keratitis or uveitis. • Potential for postherpetic neuralgia – IMPORTANT to diagnose these promptly and get high dose oral antivirals .
Herpes zoster ocular manifestations 	• Ocular tissues affected include: • Lids: Vesicles, ptosis, preseptal cellulitis • Conjunctiva: Follicular conjunctivitis • Cornea: ○ Punctate epithelial keratitis ○ Pseudodendrites (no terminal bulbs, no staining centrally) ○ Stromal keratitis (infiltrates, scarring) ○ Neurotrophic keratopathy (reduced corneal sensation, non-healing epithelium) • Uvea: Anterior uveitis, iris atrophy • Sclera: Episcleritis or scleritis • Retina: Acute retinal necrosis in immunocompromised • Optic nerve: Optic neuritis	• Tailored to tissue involved: antivirals, steroids, cycloplegics, IOP-lowering agents. • Monitor for secondary complications like glaucoma and neurotrophic ulcer.	• Multidisciplinary care (e.g., ophthalmologist, GP, neurologist) may be required in complicated cases.
VZV Dermatitis / Varicella zoster conjunctivitis	• Seen in primary varicella (**chickenpox**) or as part of HZO. • Common in children or immunocompromised. • Signs: ○ Lid vesicles ○ Follicular conjunctivitis ○ Mild pseudomembrane formation ○ Chemosis	• Supportive care: PF lubricants, cold compresses • Topical antibiotic ointment (e.g., chloramphenicol) if secondary infection suspected • Systemic acyclovir if ocular involvement or systemic disease • Avoid steroids in primary VZV unless directed --- • IDEAL TREATMENT WITHIN 72 HRS OF RASH ONSET	• Prognosis: Usually self-limiting. Monitor for signs of progression to HZO in older patients.

	• **Differentials**: HSV blepharoconjunctivitis, bacterial conjunctivitis, allergic conjunctivitis.	• Oral antivirals: MIC for systemic antivirals to be effective against VZV is ~8x, hence higher dosage required for treatment ○ Acyclovir 800 – 1000 mg 5 times a day for minimum of 7 days ○ Valaciclovir 1 mg 3x a day ○ Famciclovir 500 mg 3x a day • Corneal involvement: topical steroids (eg. prednisolone acetate 1%) four times daily until lesions have resolved, tapering slowly • Ocular lubricants (ointments)	
Thygeson's Keratitis 	• Chronic, bilateral, idiopathic keratitis with remitting–relapsing pattern. • Onset in 3rd–4th decade; affects both sexes equally. • Superficial punctate that looks like **bread crumbs** • **Symptoms**: Mild photophobia, FB sensation, tearing, blurred vision. Often asymptomatic between attacks. • Signs: ○ Elevated coarse grey-white epithelial lesions centrally ○ Minimal conjunctival injection ○ No stromal involvement ○ Usually bilateral • **Differentials**: HSV epithelial keratitis, toxic keratitis, adenoviral keratitis, EBMD, dry eye.	Mild cases: • Preservative-free lubricants QID or more Moderate-severe: • Topical corticosteroids (e.g., fluorometholone 0.1% BID–QID) • OR topical cyclosporine A 0.05% or lifitegrast (steroid-sparing) • Taper steroids slowly; monitor IOP with long-term use • Considerations of weekly use of corticosteroids or use of ciclosporin eyedrop or bandage contact lenses may be indicated → frequent	• Prognosis: Benign but chronic; each flare may last 2–3 months. • Recurrences are common over many years. • No permanent vision loss if treated appropriately.

Episcleritis/Scleritis (Week 7)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Episcleritis  	• Symptoms: None to mild; FB sensation, itch, mild tenderness; often unilateral. • Onset: Acute, often sectoral redness. • VA: Normal. • Common in: Young adults, especially females; ~25% bilateral. • Forms: ○ Simple (sectorial or diffuse) – most common. ○ Nodular – localized raised lesion, mobile, mildly tender. • Vascular signs: ○ Blanching with phenylephrine 2.5%. ○ Mobile superficial episcleral vessels. • Differential Diagnosis: ○ Conjunctivitis (more discharge, **follicles/papillae**). ○ Scleritis (deep vessels, severe pain, doesn't blanch). ○ Pinguecula, foreign body, ingrown lash, HZO, contact lens-related red eye, ocular surface neoplasia.	• Topical corticosteroids (first-line moderate-severe cases): ○ Prednisolone phosphate 0.5% or FML 0.1%, 4-5x/day ○ Wean slowly; review every 7–10 days. • Oral NSAIDs: ○ Ibuprofen 400mg QID ○ Indomethacin 25mg QID ○ Diclofenac 50mg TID ○ Naproxen 250–500mg BID ○ Switch NSAIDs if no response in 1 week • .Non-Pharmacological ○ Lubricants (unpreserved) – comfort ○ Cold compresses – may not be effective	• Self-limiting: Resolves in ~3 weeks; therapy accelerates resolution. • Recurrence: ~12% per year; may switch eye. • Complications: Rare (10–15%), but monitor. • Prognosis: Excellent. Watch for conversion to scleritis (~5%). |	Conjunctivitis	Episcleritis
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Epidemiology	Common	Uncommon
Redness	General or sectorial, conjunctival vessels	Sectorial, episcleral vessels
Symptoms	Irritation, foreign body sensation	Irritation to mild pain
Recurrence	Nil	Common, up to 10 times per year
Discharge	Abundant, watery or 'pussy'	None
Unilateral / bilateral	Bilateral typically, 2nd eye delayed onset	Unilateral or bilateral, may recur in other eye if unilateral
Follicles / Papillae	Yes	No
Age	Any age	Young adult	
Scleritis	• Symptoms: Severe boring pain **(deep and penetrating)**, radiating to temple/jaw, worse at night, worsened with eye movement, poor response to analgesics. • VA: May be reduced.	**Workup** • ANA, RF, ESR, CRP, ACE, ANCA, FTA-ABS, TB screening, uric acid • Imaging (B-scan for posterior scleritis)	• Complications: Very common (~60%) • Recurrence: High (~25% in fellow eye within 1 year)





- Vessels: Non-blanching with phenylephrine 2.5%; deep, immobile, blue-violet hue.
- Types:
  - **Diffuse anterior** – most benign
  - **Nodular anterior** – risk of progression to necrosis
  - **Necrotising with inflammation** – severe, scleral thinning, uveal show, 25% 5-year mortality
  - **Scleromalacia perforans** – asymptomatic, seen in longstanding RA
  - **Posterior** – presents as amelanotic mass, proptosis, disc swelling, etc.
- Differential Diagnosis:
  - Episcleritis (**non tender upon palpation, vessels blanch under phenylephrine)**
  - Anterior uveitis
  - Orbital cellulitis (pain, proptosis)
  - Conjunctivitis (no deep pain)
  - Keratitis

	Episcleritis	Scleritis
Epidemiology	Uncommon	Rare
Redness (use red-free)	Sectorial episcleral (fine) vessels	Sectorial or diffuse, deep vessels involved, avascular areas
Symptoms	Irritation to mild pain	Severe boring pain (especially at night)
Recurrence Complications	Common (1-2 per year) Never	Moderate (depends on systemic condition) Frequent
Systemic disease	Rarely associated	Common (50% of patients)
Unilateral / bilateral	Both	Both
Classification	Nodular, simple	Diffuse, nodular, necrotising, scleromalacia, posterior
Age Gender	Younger (18-40s) Female more often	Older (25-60s) Female more often
Phenylephrine 2.5%	Vessels blanch	Deep vessels remain red
Onset	Acute	Gradual
Colour	Salmon (pink) or bright red	Bluish red or purple
Palpation	Non tender	Tender
Prognosis	Excellent	Variable

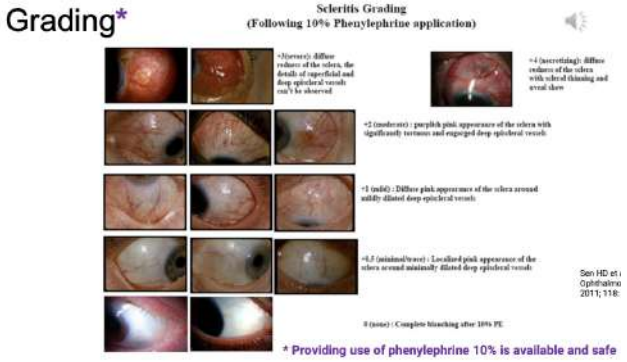
- Review of Systems: RA, SLE, gout, sarcoid, HZO/HSV, TB, syphilis.

First-line: Oral NSAIDs
- Ibuprofen 400–800mg QID
- Indomethacin 25mg QID
- Naproxen 500mg BID

Second-line: Oral corticosteroids (if NSAIDs fail)
- Prednisone 1mg/kg/day

Third-line: Immunosuppressants
- Methotrexate, ciclosporin (especially for necrotising/posterior)
- Adjunct topical corticosteroids: Only in anterior scleritis
- Cycloplegia as needed

Avoid: Topical steroids alone (47% failure rate in non-necrotising scleritis)



- Visual risk: High, especially necrotising type
- Prognosis: Variable; guarded in necrotising/posterior forms

Limbitis (Marginal)





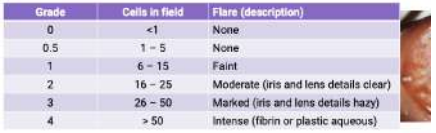
- Symptoms: Redness near limbus (the corneal-scleral junction), photophobia, mild discomfort.
- Signs:
  - Sectoral limbal injection
  - Possible epithelial defect (dendrite) in HSV
  - Corneal infiltrate (marginal keratitis)
  - Adjacent conjunctival hyperaemia
- Differential Diagnosis:
  - HSV epithelial keratitis (dendrites with terminal bulbs)
  - Marginal keratitis (staph hypersensitivity, sterile)
  - HZO (non-ulcerative dendrites)
  - Phlyctenulosis
  - Episcleritis/scleritis

- Workup
  - Cotton wisp test (corneal sensation – ↓ in HSV)
  - Slit lamp – fluorescein and lissamine green staining
  - History of herpes
  - Consider viral PCR if uncertain

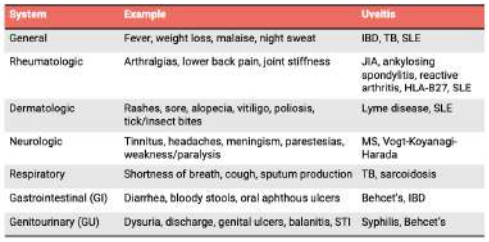
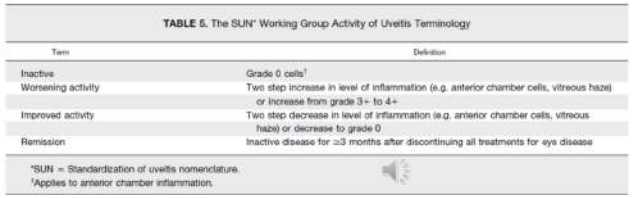
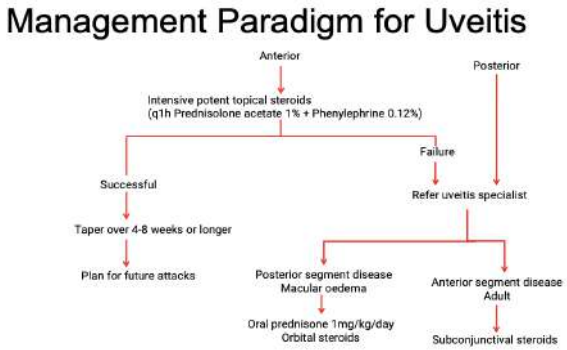
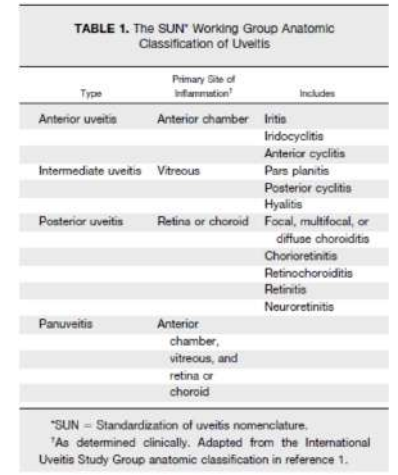
Pharmacological Treatment
- HSV-related limbitis:
  - Topical aciclovir 3% ointment 5×/day
  - Oral antivirals: Acyclovir 400mg 5×/day or Valaciclovir 500mg TID
  - No steroids until epithelium healed
- Marginal keratitis:
  - Topical steroids (FML/Predsol) + antibiotic cover (e.g., chloramphenicol)
  - Avoid steroids in HSV unless concurrent antiviral used

- HSV limbitis:
  - Risk of recurrence ~15%/year; consider prophylaxis in stromal cases
  - Avoid steroids during epithelial involvement
- Marginal keratitis:
  - Resolves in 2–3 weeks with appropriate therapy
  - May recur with chronic blepharitis
- Prognosis: Good with correct diagnosis; HSV needs vigilant monitoring due to recurrence and stromal involvement risk.

Uveitis (Week 7)

Condition	Information (Signs, Symptoms and Differentials)	Treatment and Workup (pharmacological and nonpharmacological treatments)	Other Information (General Management, Prognosis)
Uveitis 	• **Types (SUN Classification):** ○ **Anterior Uveitis:** Iritis, iridocyclitis, anterior cyclitis. ○ **Intermediate:** Vitritis, pars planitis. ○ **Posterior:** Choroiditis, chorioretinitis, retinitis, neuroretinitis. ○ **Panuveitis:** Involves anterior chamber, vitreous, and retina/choroid. • Onset & Course: ○ Acute vs Chronic ○ Recurrent vs Persistent (>3 months) ○ Sudden vs Insidious • Symptoms: ○ Red eye (especially around limbus = ciliary flush) ○ Photophobia (key symptom!) ○ Blurred vision ○ Pain ○ Tearing ○ Floaters (if posterior involvement) • Signs: ○ Anterior chamber cells and flare (graded per SUN) ○ Keratic precipitates (KPs) ○ Mutton-fat KPs in granulomatous uveitis ○ Hypopyon (layered WBCs) ○ Posterior synechiae (iris–lens adhesions) ○ Band keratopathy ○ Cataract (especially anterior subcapsular) ○ Vitreous cells/haze (intermediate/posterior) • ↑/↓ IOP ○ Low IOP in acute anterior uveitis ○ High IOP in herpetic, Posner-Schlossman, trabeculitis Causes: • Idiopathic (~50%) • HLA-B27 associated (e.g. Ankylosing spondylitis, Reactive arthritis) • Infectious: HSV, VZV, CMV, TB, Syphilis, Leprosy • Autoimmune/Inflammatory: Sarcoidosis, JIA, Psoriatic arthritis, IBD • Others: Lens-induced, trauma, Fuchs', Posner-Schlossman **Differential Diagnoses:** • Conjunctivitis • Episcleritis/scleritis (no AC cells) • Herpetic eye disease • Keratitis → the cornea is involved, white-grey opacity, seen on the fluorescein stain • Endophthalmitis • Fuchs' heterochromic iridocyclitis (quiet eye, iris atrophy) • Masquerade syndromes (e.g. lymphoma) • Trauma	Only indicated if: • Bilateral • Recurrent • Granulomatous • Unresponsive to treatment • Associated systemic symptoms Investigations: • FBC, ESR, CRP • ANA, ENA, RF • HLA-B27 typing • Syphilis serology (RPR, FTA-ABS) • Quantiferon Gold or Mantoux (TB) • ACE levels (sarcoidosis) • Urinalysis (TIN) • CXR, CT chest (sarcoid, TB) • Aqueous/vitreous biopsy (if suspect lymphoma) Topical corticosteroids (Pred Forte 1%): • Every 1–2 hours initially (or loading every 15 mins × 2 hours if severe) • Taper only when AC cells and flare resolve • Typical taper: reduce by 1 drop/week over 4–8 weeks • Watch for IOP rise, PSC cataract, HSV reactivation Cycloplegics (to relieve pain and prevent synechiae): • Cyclopentolate 1%: BID to TID (mild-moderate) • Atropine 1%: OD–QID (severe) • Add Phenylephrine 2.5–10% to break synechiae Oral corticosteroids: • Prednisone 1 mg/kg/day if topical not enough Periocular injections: Triamcinolone, dexamethasone if posterior Systemic immunosuppressants (if refractory or systemic disease): • Methotrexate • Cyclosporine • Interleukin receptor antagonists Antivirals (if HSV/VZV suspected): • Acyclovir 400 mg 5x/day (HSV) • Valaciclovir 1 g TID (VZV) --- Potent, penetrating corticosteroid eye drops: • Prednisolone acetate 1% + phenylephrine 0.12% • Dexamethasone 0.1% • Loteprednol 0.5% (requires compounding) suspension	• Review: ○ 24–48h initially ○ Weekly while tapering ○ Watch for posterior complications (CMO, ERM) • Complications: ○ Glaucoma (steroid or trabeculitis) ○ Cataract ○ Posterior synechiae → pupil block ○ CME ○ Band keratopathy • Prognosis: ○ Variable ○ Anterior uveitis: Good with aggressive early therapy ○ Posterior uveitis: Sight-threatening • Recurrence risk: ~24% per year • Warn patients to return if red, photophobic eye returns **Grading anterior chamber cells & flare**  **TABLE 2.** The SUN* Working Group Descriptors of Uveitis | Category	Descriptor	Comment
---	---	---
Onset	Sudden	
	Insidious	
Duration	Limited	≤3 months duration
	Persistent	>3 months duration
Course	Acute	Episode characterized by sudden onset and limited duration
	Recurrent	Repeated episodes separated by periods of inactivity without treatment ≥3 months in duration
	Chronic	Persistent uveitis with relapse in <3 months after discontinuing treatment	 *SUN = Standardization of uveitis nomenclature.

Grade	Cells in field	Flare (description)
0	<1	None
0.5	1 – 5	None
1	6 – 15	Faint
2	16 – 25	Moderate (iris and lens details clear)
3	26 – 50	Marked (iris and lens details hazy)
4	> 50	Intense (fibrin or plastic aqueous)

	 	(shake well before use) • Prednisolone phosphate 0.5% (unpreserved minims) eyedrops for milder presentations • Dosage depends on severity of inflammation ○ Every 1-2 hours until 2- step improvement according to SUN grading ○ Then 4 times daily until fully resolved ○ Taper slowly (over 5-6 weeks) once cells & flare are gone 	 
HLA-B27 Positive Conditions	• Typical features: ○ Acute, recurrent, unilateral anterior uveitis ○ Sudden onset, photophobia, pain, and redness ○ May alternate between eyes ○ Hypopyon may be present • **Differentials**: ○ Endophthalmitis (more severe vision loss, history of surgery) ○ HSV/VZV uveitis (elevated IOP, iris atrophy) ○ Sarcoid (bilateral granulomatous uveitis) ○ Fuchs' uveitis (no pain, heterochromia) ○ Traumatic uveitis ○ Behçet's (oral/genital ulcers) • Associated Conditions: ○ Ankylosing spondylitis ○ Reactive arthritis (Reiter's) ○ Psoriatic arthritis ○ IBD (Ulcerative colitis, Crohn's) • Systemic Symptoms to Elicit: ○ Back pain (inflammatory, improves with activity) ○ Joint stiffness ○ Diarrhoea or gut inflammation ○ Genital ulcers/discharge (Reiter's) ○ Skin lesions (psoriasis)	Workup • HLA-B27 typing • ESR, CRP • Review of systems (dermatologic, GI, rheumatologic, GU) • Refer to GP or rheumatologist for systemic disease management Pharmacological Management • Same as Anterior Uveitis (Pred Forte, Cycloplegics) • Oral NSAIDs often helpful for systemic symptoms • May need immunomodulatory therapy if recurrent or systemic	• Recurrence: Common; annual or every couple years • Systemic disease: May predate or follow uveitis • Prognosis: Good if treated early; recurrence manageable • Referral: ○ GP for systemic workup ○ Rheumatologist, gastroenterologist, dermatologist depending on findings
Hyphema	• Causes:	Workup	• Non-Pharmacological:



- Trauma (most common)
- Blood thinners (e.g. aspirin, warfarin)
- Ocular surgery
- Spontaneous in bleeding disorders
- Juvenile xanthogranuloma (kids)

- Symptoms:
  - Blurred vision
  - Eye pain
  - Photophobia
  - Hx of blunt trauma or surgery
- Signs:
  - Visible blood in AC (document height)
  - Elevated IOP
  - Secondary uveitis (AC cells/flare)
  - Iridodialysis, angle recession
  - Seidel sign to rule out perforation
- **Differential Diagnosis:**
  - Uveitis with hypopyon
  - Pigment dispersion
  - Intraocular tumour bleed
  - Endophthalmitis (with layered hypopyon, pain, history)

- Check IOP daily
- Avoid gonioscopy acutely
- DFE and B-scan if posterior segment cannot be visualised
- History of bleeding disorders or anticoagulant use

Pharmacological Treatment

- Topical corticosteroids:
- Pred Forte q1h or QID depending on AC inflammation
- Taper once cells, flare, and blood resolve
- Cycloplegia:
  - Cyclopentolate 1% BID–TID
  - Atropine 1% OD–BID if severe
- IOP control (if elevated after 24 hours):
  - Timolol 0.5% BID
  - Brimonidine 0.2% TID
- Avoid prostaglandins (pro-inflammatory)
- Analgesia:
  - Paracetamol only
  - Avoid NSAIDs or aspirin

NO ANTIPLATELET MEDICATIONS (EG. ASPIRIN, NSAIDs)

Cycloplegics to prevent formation of secondary synechiae (eg. cyclopentolate 1% 2-3 times daily, atropine 1% 1-2 times daily)

Secondary uveitis: topical steroids (eg. prednisolone acetate 1% + phenylephrine 0.12%) dosed according to severity of inflammation
Taper once hyphema and cells/flare have resolved

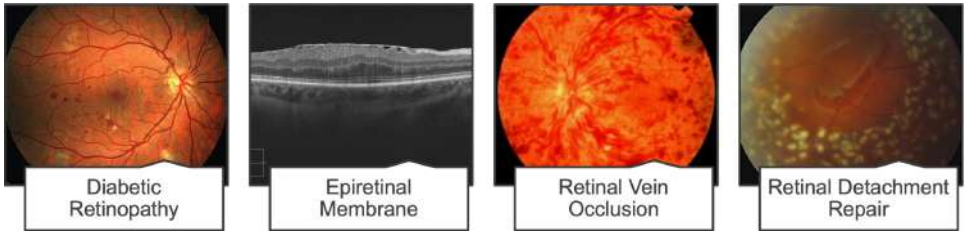
If IOP is elevated after 24 hrs: anti-glaucoma meds (eg. timolol 0.5% twice daily or Alphagan 0.2% 2-3 times daily)

  - Bed rest with head elevated
  - Eye shield (not patch)
  - Avoid exertion, straining, bending
  - Hospitalisation if severe bleed for enforced rest
- Monitoring:
  - Daily IOP and vision checks
  - Warn patient: return if vision worsens or pain increases
- Complications:
  - Rebleed (higher risk days 2–5)
  - Corneal blood staining
  - PAS, secondary glaucoma
  - Angle recession glaucoma (months–years later)
- Prognosis:
  - Poorer if rebleed, high IOP, or corneal blood staining
  - Most resolve without sequelae if managed well
- Follow-up:
  - Once resolved, perform gonioscopy to assess angle recession
  - Long-term IOP monitoring (annual)

Cataract Surgery (Week 8-10)

Ocular Comorbidities

- Preoperative comorbidities may adversity affect the outcome of cataract surgery

Posterior Segment	Anterior Segment
• Ensure all conditions are stable prior to referral for surgery • Advise px of risks associated with these conditions 	• Abnormal tear film or corneal surface impacts the keratometric & topographic measurements during preoperative workup ○ Can cause miscalculations for the IOL which can impair visual function & quality post-surgery **EBMD**: epithelial irregularities can produce irregular astigmatism which causes higher order aberrations & fluctuating visual disturbance • Increased risk of RCE • Treatment: epithelial debridement, excimer laser ablation & anterior stromal puncture

• All conditions have an increased risk of pseudophakic cystoid macular oedema (CME) post surgery	**Corneal guttata (Fuch's):** surgical trauma (eg. cataract surgery) causes a loss of endothelial cells → impacts whether or not cataract surgery is performed along with keratoplasty • If CCT > 640um, <1000 cells/mm^2 or microcystic oedema present → perform cataract surgery with keratoplasty **Uveitis**: inflammation must be inactive or at a level of control >3 months prior to surgery • Risk factor of CME • Topical & systemic anti-inflammatory meds should be started prior to surgery & use more frequently & for a longer duration post-surgery **DED**: impaired ocular surface can affect topographic measurements

Antibiotic Prophylaxis

RECOMMENDATION IS INTRACAMERAL INJECTION + POST-OPERATIVE ANTIBIOTIC PROPHYLAXIS	
Intraoperative Antibiotic Prophylaxis (New)	**Post-operative Antibiotic Prophylaxis (Traditional)**
• Involves giving a single dose of cefazolin intracamerally (injected into AC) at the end of surgery ○ Has produced a 5x reduction in endophthalmitis rates (0.07% vs. 0.34%)	• Involves prescribing **chloramphenicol** 0.5% eye drops 4 times daily for 7 days after surgery

Post-Operative Inflammation & Pain

Post-operative Topical Eye Drops (Traditional)	**Intra-operative (Alternative methods)**
• Corticosteroids & NSAIDs effective at treating post-operative inflammation in cataract surgery Standard: corticosteroid + NSAID 3 times daily for 1 month	• Single dose of dexamethasone intraocular suspension 9% intracamerally at the end of surgery • Intracanalicular dexamethasone insert that delivers a tapered dose for up to 30 days

Post-Operative Follow Up

RECOMMENDATION IS FIRST WITHIN 48 HRS & SUBSEQUENT FOLLOW UP USUALLY 2-4 WEEKS

- Exact frequency is determined by surgical technique, operative & post-operative complications
- Inform px of need for follow up & potential changes to eyes & vision
- Px are usually discharged by hospital on the day of surgery
  - Community optometrists perform follow up visits

Low Risk Px	**High Risk Px**
• Involves px without high risks or signs/ symptoms of possible complications following small-incision cataract surgery • First visit: within 48 hrs • Subsequent visits: frequency & timing depends on refraction, visual function & medical condition of the eye	• Involves px who are functionally monocular,intra-operative complications & high risk of immediate postoperative complications • First visit: within 24 hrs • Subsequent visits: more frequent follow ups

Post-Operative Complications

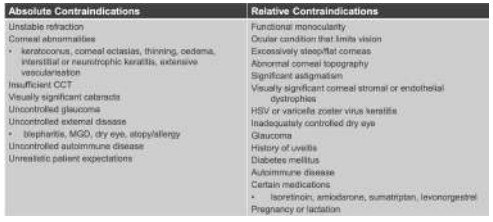
Complication	What?	Treatment
Endophthalmitis	• Is inflammation of the vitreous & aqueous humour ○ Occurs due to gram positive bacteria invading & infecting the eye during or post-surgery	Intravitreal antibiotic (eg. ceftazidime or vancomycin)

	• Is rare with incidence <1% • 2 stages: ○ Acute: within 72 hrs post-surgery ■ Pain, redness, severe AC reaction, hypopyon, inflammation cells in the vitreous ○ Chronic: weeks after surgery ■ Persistent inflammation Risk factors: posterior capsular rupture, prolonged surgical time, wound leak	Vitrectomy to remove vitreous gel
Posterior capsular opacification (PCO)	• Is the most common post-surgery complication • Involves proliferation & metaplasia of epithelial cells within the capsular bag ○ Metaplasia: replacement of one mature cell type with another in the same tissue Risk factors: younger age, uveitis, ocular inflammation	Refer for Nd:YAG capsulotomy
Raised IOP	Day 1 post-surgery: IOP spikes >30mmHg due to retained viscoelastic material >1 week post-surgery: px most likely a steroid responder	Day 1 post-surgery: topical & oral anti-glaucoma agents >1 week post-surgery: taper/ discontinue steroid & prescribe topical anti-glaucoma agents • Px can continue NSAID
Corneal oedema	• Occurs due to corneal decompensation • Is self-resolving within days- weeks post-surgery • Possible causes: ○ Dense cataract ○ Complicated surgery ○ Pre-operative endothelial dysfunction (Fuch's) ○ Descemet's split extending from corneal incision	Initial post-surgery: increase topical steroids to 2 hourly >2 weeks post-surgery: • Hypertonic saline 5% 4 times daily • Referral to opthal
Cystoid macular oedema	• Occurs in 1-3.5% of uncomplicated cases at ~6 weeks post-surgery Risk factors: diabetes, uveitis, previous RVO, ER, use of prostaglandin analogue	Prophylaxis NSAID post-surgery
Retinal detachment	• Occurs due to PVD induced by loss of internal volume from removal of the lens • Warn px at risk of signs & symptoms of retinal detachment Risk factors: history of previous RD in other eye, long AL, males, high myopes	

Refractive Surgery (Week 8-10)

Keratorefractive surgery: reshapes the cornea to correct the refractive error

Refractive lens surgery: involves removing the natural lens and replacing it with an artificial intraocular lens (IOL)

KERATOREFRACTIVE SURGERY


Absolute Contraindications	Relative Contraindications
Unstable refraction	Functional monocularity
Corneal abnormalities	Ocular condition that limits vision
• keratoconus, corneal ectasias, thinning, oedema, interstitial or neurotrophic keratitis, extensive vascularisation	Excessively steep/flat corneas
Insufficient CCT	Abnormal corneal topography
Visually significant cataracts	Significant astigmatism
Uncontrolled glaucoma	Visually significant corneal stromal or endothelial dystrophies
Uncontrolled external disease	HSV or varicella zoster virus keratitis
• blepharitis, MGD, dry eye, atopy/allergy	Inadequately controlled dry eye
Uncontrolled autoimmune disease	Glaucoma
Unrealistic patient expectations	History of uveitis
	Diabetes mellitus
	Autoimmune disease
	Certain medications
	• isoretinoin, amiodarone, sumatriptan, levonorgestrel
	Pregnancy or lactation

Technique	What?	Advantages	Disadvantages
Surface treatment technique (PRK, LASEK, Epi-LASIK)	• Use if preoperative corneal thickness <500um • No risk of flap complications • Better for contact sports, armed forces • Less risk of dry eye		
Lamellar treatment techniques (LASIK)	• Rapid visual recovery • Minimal post-surgery pain • High precision		• Dry eye due to damage to nerves • Flap complications • Mechanical microkeratome due to post-surgery complications
Incisional techniques	Astigmatic: used to correct astigmatism where incision is made in the steep axis to effectively flatten the meridian Radial keratotomy: corrects myopia -1.00D to -4.00D • Long-term instability		
Intrastromal implantation	Intrastromal Corneal Ring Segment Implantation: a crescent shaped PMMA implant used to treat LASIK corneal ectasia, pellucid degeneration, KC Intracorneal Inlays for Presbyopia: involves refractive intracorneal inlays, non-refractive inlays, pinhole intracorneal inlays		
Other	Refractive Lenticule Extraction (ReLEx): minimally flapless laser vision correction Small Incision Lenticule Extraction (SMILE): predictable & good safety outcomes SmartSight: rotational eye tracking with femtosecond laser allows for more accurate astigmatism correction • Tissue saving algorithm removes less tissue		

REFRACTIVE LENS SURGERY

Technique	Range of Applicability	Borderline Applicability
Surface Techniques (PRK, LASEK, epi-LASIK)	Myopia up to -6D Astigmatism up to 5D	Myopia up to -8D Astigmatism up to 6D Hyperopia up to +4D
Lamellar Techniques (LASIK)	Myopia up to -8D Astigmatism up to 5D Hyperopia up to +3D	Myopia up to -10D Astigmatism up to 6D Hyperopia up to +4D
SMILE	Myopia up to -12D Astigmatism up to -5D	-
Phakic IOL	Myopia up to -20D Hyperopia of +4D/>	Myopia of -5D/> Hyperopia of +3/>
Refractive Lens Exchange	Better for hyperopes, low myopes, and for astigmatism	

Phakic Intraocular Lenses (PIOL)	• Involves placing an implant without removing the natural lens ○ Can be implanted in the AC or between the iris & natural lens
Refractive Lens Exchange (RLE)	• IOL is inserted after the natural lens is removed • More suitable for px > 50 years • Is essentially cataract surgery

Post-Operative Follow Up, Medications & Findings

Surface Treatments (PRK, LASEK, epi-LASIK	LASIK
1 – 3 days post-op Blurred vision Photophobia Pain	1 Day post-op Mild-mod discomfort
1 months Prolonged visual recovery	48 hours Rapid visual recovery

	1 Day	1 Week	1 Month	3 Month	9-12 months as needed
Interval history	✓	✓	✓	✓	✓
Uncorrected VA	✓	✓	✓	✓	✓
Manifest refraction & BCVA			✓	✓	✓
Cycloplegic refraction				✓	✓
Anterior Segment exam	✓	✓	✓	✓	✓
Topography				✓	✓
DFE				✓	✓
IOP Measurement		✓	✓	✓	✓

Table 3. Post-operative Management following Various Laser Refractive Surgeries.

Management	PRK	LASIK	LASEK	Epi-LASIK
Topical broad-spectrum antibiotics	Until epithelium heals	First week	First week	First 2 weeks
Topical corticosteroids	First 3 months	First 2 to 4 weeks	First 2 to 4 weeks	First 3 months
Topical NSAID	First 24 to 48 hours	Optional	Optional	First 24 to 72 hours
Non-preserved artificial tears	First 3 to 6 months	First 1 to 2 month	First 1 to 2 months	First 2 to 3 months
Therapeutic soft contact lens	Until epithelium heals	Optional	First 1 to 3 days	First 3 to 4 days
Punctual plugs	Optional	Optional	Optional	Optional
Oral analgesics	First 48 to 72 hours	Optional	Optional	First 24 to 48 hours

Measures or dosages should be adjusted according to surgeon's clinical judgement.

SMILE Medications [40]
Topical broad-spectrum AB
Non-preserved artificial tears

Refractive Surgery Complications

Complication	What?	Treatment
Dry eye	● Is the most common complication ● Visual fatigue & glare sensitivity ● LASIK: settles between 6-12 months ● SMILE: at 3 months, Schirmer & TBUT is better than femto-LASIK	Typical dry eye management
Transient light sensitivity syndrome	● Is a rare inflammatory complication ● Occurs 2-4 weeks post-surgery ● Slit lamp examination will appear normal → normal VA, clear & quiet eye ● Unusual photosensitivity	Weak penetrating steroid (eg. FML 4 times daily) ● Can use a stronger steroid if px is slow to respond
Corneal haze	● Is an anterior stromal opacity ○ Occurs due to surgery-induced damage of the basement membrane that results in an abnormal deposition of collagen fibres in the sub-epithelial area ■ Decreased corneal transparency ● Appears within the first month after PRK & when px stops steroid use ● Reduced VA, BCVA	**First line:** intraoperative mitomycin-C (MMC) **Mild-moderate:** FML/ Flarex 4 times daily with a slow taper **Severe**: Maxidex/ Pred Forte 4 times daily to every 2 hours with a slow taper
Epithelial ingrowth	● Occurs 1 month after lamellar procedures (SMILE & SmartSight) ● Small area: asymptomatic or mild discomfort ● Large area (>2mm): blurred vision, irregular astigmatism, corneal melting ● White, milky deposits at the peripheral interface	Weak penetrating steroid (eg. FML 4 times daily) with a long taper Referral to opthal
Diffuse lamellar keratitis	● Is an autoimmune reaction to surgery due to infiltration of inflammatory cells between the flap & stromal bed ● Usually occurs in LASIK ● Occurs within 1-7 days post-surgery ● Visual impairment depends on severity ● Blurred vision, redness, discomfort/ FB sensation, photophobia ● Conjunctival hyperaemia, white cells in the flap interface, tearing & pain ○ 'Sands of the Sahara' syndrome: accumulation of fine white infiltrates that manifest clinically as grainy, corneal opacifications	Stage 1 & 2: intensive topical steroids (eg. Prednefrine Forte hourly) taper according to resolution of white cells over 3-4 weeks Stage 3: referral to opthal for a flap lift, irrigation & repositioning of flap

	Table 5. Staging of Diffuse Lamellar Keratitis (DLK) Stage — Description 1 — Infiltrates in the periphery of the flap without involvement of the central cornea. This stage most commonly presents the day after surgery. 2 — White granular cells involving the visual axis. This is more frequently seen on day 2 or 3 and is the result of central migration of cells. It leads to the "Sands of Sahara" syndrome. 3 — Usually appears 48 to 72 hours after surgery with a 1- or 2-line loss of visual acuity. Stage 3 has been referred to as "threshold" DLK because permanent scarring could occur if not appropriately treated. 4 — Severe lamellar keratitis. Stromal melting and permanent scarring is the likely result.	
Corneal ectasia	• Progressive thinning & weakening of the central cornea • Occurs when CCT <300um • Onset can be months or years post–surgery • Reduced vision, glare, ghosting, diplopia • Myopic shift, corneal thinning, topographic asymmetric inferior corneal steepening	RGP lenses Referral for penetrating keratoplasty, cross-linking, intrastromal rings

Common Differential Questions

Condition	Key Features & Differentiation	Condition	Key Features & Differentiation
Acute Anterior Uveitis	- Inflammation of the uvea (iris and ciliary body) - **Painful, photophobia**, blurred vision - **Circumcorneal redness**, **no discharge** - Pupil may be irregular or sluggish to light - Often associated with systemic conditions (e.g., HLA-B27 diseases)	**Herpes Simplex Conjunctivitis**	- **Unilateral red eye, watery discharge** - Vesicular periocular rash (if zoster) - May have **dendritic corneal ulcers** on fluorescein staining - Can cause recurrent episodes
Acute Angle Closure Glaucoma	- **Severe eye pain, headache, nausea, vomiting** - **Sudden vision loss, halos around lights** - Mid-dilated, **fixed pupil** - Cornea appears hazy (due to edema) - High intraocular pressure	**Keratitis**	- **Painful, photophobia, vision loss** - Corneal haze or ulcer - **Severe conjunctival injection** - Common causes: **bacterial, viral (HSV), fungal, contact lens use**
Blepharitis	- **Lid margin inflammation, crusting, itching** - **Burning, foreign body sensation** - No significant conjunctival injection - Often chronic and associated with seborrhea or meibomian gland dysfunction	**Staphylococcal Conjunctivitis**	- **Purulent discharge, crusting of lids** - **Red, swollen conjunctiva** but no significant pain - Often bilateral - Responds to topical antibiotics
Episcleritis	- **Localized, sectoral redness**, usually self-limited - **Mild discomfort but not painful** - No vision loss, no photophobia - Blanches with topical phenylephrine	**Subconjunctival Hemorrhage**	- **Bright red patch on sclera**, sharply demarcated - No pain, no vision changes, no discharge - Resolves spontaneously over 1–2 weeks - Common causes: trauma, sneezing, hypertension
Foreign Body	- **Sudden onset, unilateral discomfort, irritation** - Foreign body sensation, **tearing, reflex blepharospasm** - May see foreign body or corneal abrasion - History of trauma or exposure	**Trachoma**	- **Chronic conjunctivitis due to Chlamydia trachomatis** - Follicular conjunctivitis with **tarsal scarring** - **Trichiasis (in-turned lashes) → corneal damage → blindness** - Common in poor hygiene settings

Microorganism	Ocular Condition(s)	Key Clinical Features

Acanthamoeba	**Acanthamoeba keratitis** (Protozoan)	- **Severe pain out of proportion to findings** - Ring-shaped corneal ulcer - Associated with **contact lens use, contaminated water**
Adenovirus	**Viral conjunctivitis** (Epidemic Keratoconjunctivitis)	- **Highly contagious, bilateral** - **Watery discharge, preauricular lymphadenopathy** - Pseudomembrane formation in severe cases - Associated with **upper respiratory infections**
Chlamydia trachomatis	**Trachoma & Adult Inclusion Conjunctivitis**	- **Trachoma (serotypes A-C): chronic follicular conjunctivitis, scarring, trichiasis → blindness** - **Inclusion conjunctivitis (serotypes D-K): mucopurulent discharge, neonatal conjunctivitis**
Herpes Simplex Virus (HSV-1)	**Herpes simplex keratitis & conjunctivitis**	- **Unilateral red eye, watery discharge** - **Dendritic corneal ulcers** on fluorescein staining - **Recurrent episodes** triggered by stress or immunosuppression
Neisseria gonorrhoeae	**Gonococcal conjunctivitis (hyperacute bacterial conjunctivitis)**	- **Profuse purulent discharge, severe redness** - **Rapid onset, can cause corneal perforation** - Neonatal gonococcal conjunctivitis (Ophthalmia neonatorum)
Molluscum Contagiosum	**Chronic follicular conjunctivitis**	- **Painless, dome-shaped umbilicated nodules on eyelid margin** - May cause **chronic irritation and follicular conjunctivitis** - Seen in immunocompromised patients
Pseudomonas aeruginosa	**Bacterial keratitis (contact lens-related)**	- **Rapidly progressive corneal ulcer with hypopyon** - **Associated with contact lens wear** - **Greenish discharge, corneal perforation risk**
Staphylococcus aureus	**Bacterial conjunctivitis, blepharitis, hordeolum**	- **Purulent discharge, eyelid crusting** - **Blepharitis: chronic lid margin inflammation** - **Hordeolum (stye): localized, painful eyelid swelling**
Varicella Zoster Virus (VZV)	**Herpes Zoster Ophthalmicus (HZO)**	- **Dermatomal rash (V1 distribution) with eye involvement** - **Hutchinson's sign (vesicles on nose indicate ocular involvement)** - May cause **keratitis, uveitis, or retinal necrosis**

Differential Diagnosis of Ocular Symptoms

Burning

- **More Common.** Blepharitis, meibomitis, dry eye syndrome, conjunctivitis (infectious, allergic, mechanical, chemical).
- **Less Common.** Corneal defects (usually marked by fluorescein staining of the cornea), inflamed pterygium or pinguecula, episcleritis, superior limbic keratoconjunctivitis, ocular toxicity (medication, makeup, contact lens solutions), contact lens-related problems.

Crossed Eyes in Children

- ESODEVIATIONS (eyes turned in)
- EXODEVIATIONS (eyes turned out)

Decreased Vision

1. Transient visual loss (vision returns to normal within 24 hours, usually within 1 hour).
    - **More Common.** Few seconds (usually bilateral): Papilledema. Few minutes: Amaurosis fugax (transient ischemic attack; unilateral), vertebrobasilar artery insufficiency (bilateral). Ten to 60 minutes: Migraine (with or without a subsequent headache).
    - **Less Common.** Impending central retinal vein occlusion, ischemic optic neuropathy, ocular ischemic syndrome (carotid occlusive disease), glaucoma, sudden change in blood pressure, central nervous system (CNS) lesion, optic disc drusen, orbital lesion (vision loss may be associated with eye movement).

2. Visual loss lasting >24 hours
    - **Sudden, painless loss**
        - More Common. Retinal artery or vein occlusion, ischemic optic neuropathy, giant cell arteritis, vitreous hemorrhage, retinal detachment, optic neuritis (pain with eye movement in >50% of cases), sudden discovery of pre-existing unilateral visual loss.
        - Less Common. Other retinal or CNS disease (e.g., stroke), toxins (e.g., methanol poisoning), ophthalmic artery occlusion (may also have extraocular motility deficits and ptosis).
    - **Gradual, painless loss (over weeks, months, or years)**
        - More Common. Cataract, refractive error, open angle glaucoma, chronic angle closure glaucoma, chronic retinal disease [e.g., age-related macular degeneration (ARMD), diabetic retinopathy].
        - Less Common. Chronic corneal disease (e.g., corneal dystrophy), optic neuropathy/atrophy (e.g., CNS tumor).
        - Painful loss: Acute angle closure glaucoma, optic neuritis (may have pain with eye movements), uveitis, endophthalmitis, corneal hydrops (keratoconus).
3. Post Traumatic visual loss: Eyelid swelling, corneal irregularity, hyphema, ruptured globe, traumatic cataract, commotio retinae, retinal detachment, retinal or vitreous hemorrhage, lens dislocation, traumatic optic neuropathy, cranial neuropathies, CNS injury, sympathetic ophthalmia (rare).

    NOTE: Although a diagnosis of exclusion, remember to consider nonphysiologic vision loss.

Discharge

- **[RED EYE (BELOW)]**

Distortion of Vision

- **More Common.** Refractive error [including presbyopia, acquired myopia (e.g., from cataract, diabetes, pregnancy, ciliary spasm or ciliary body rotation, medications, retinal detachment surgery), acquired astigmatism (e.g., from anterior segment surgery, periorbital or eyelid edema/mass [e.g., chalazion, orbital trauma]), macular disease (e.g., central serous chorioretinopathy, macular edema, ARMD, and others associated with choroidal neovascular membranes [CNVMs]), corneal irregularity, intoxication (e.g., ethanol, methanol), pharmacologic (e.g., scopolamine patch).
- **Less Common.** Keratoconus, topical eye drops (e.g., miotics, cycloplegics), retinal detachment, migraine (transient), hypotony, CNS abnormality (including papilledema), nonphysiologic.

Double Vision (DIPLOPIA)

1. Monocular (diplopia remains when the uninvolved eye is occluded)
    - More Common. Refractive error, incorrect spectacle alignment, corneal opacity or irregularity (including corneal or refractive surgery), cataract, iris defects (e.g., iridectomy).
    - Less Common. Dislocated natural lens or lens implant, macular disease, retinal detachment, CNS causes (rare), nonphysiologic.
2. Binocular (diplopia eliminated when either eye is occluded)

- Typically intermittent: Myasthenia gravis, intermittent decompensation of an existing phoria.
- Constant: Isolated sixth, third, or fourth nerve palsy; orbital disease (e.g., thyroid eye disease; idiopathic orbital inflammation [orbital pseudotumor], tumor); cavernous sinus/superior orbital fissure syndrome; status-post ocular surgery (e.g., residual anesthesia, displaced muscle, muscle surgery, restriction from scleral buckle, severe aniseikonia after refractive surgery); status-post trauma (e.g., orbital wall fracture with extraocular muscle entrapment, orbital edema); convergence/divergence insufficiency; internuclear ophthalmoplegia; vertebrobasilar artery insufficiency; other CNS lesions; spectacle problem.

Dry Eyes

- **[DRY EYE SYNDROME (ABOVE)]**

Eyelash Loss

- Trauma, burn, cutaneous neoplasm (e.g., sebaceous gland carcinoma), eyelid infection or inflammation, radiation, chronic skin disease (e.g., alopecia areata), Vogt–Koyanagi–Harada syndrome, thyroid disease, trichotillomania.

Eyelid Crusting

- **More Common.** Blepharitis, meibomitis, conjunctivitis.
- **Less Common.** Canaliculitis, nasolacrimal duct obstruction, dacryocystitis.

Eyelids Drooping (PTOSIS)

- See: ptosis

Eyelid Swelling

1. Associated with inflammation (usually erythematous).
    - **More Common.** Hordeolum, blepharitis, conjunctivitis, preseptal or orbital cellulitis, trauma, contact dermatitis, herpes simplex or zoster dermatitis.
    - **Less Common.** Ectropion, corneal abnormality, urticaria or angioedema, blepharochalasis, insect bite, dacryoadenitis, erysipelas, eyelid or lacrimal gland mass, autoimmunities (e.g., discoid lupus, dermatomyositis).
2. Noninflammatory: Chalazion; dermatochalasis; prolapse of orbital fat (retropulsion of the globe increases the prolapse); eyelid or lacrimal gland mass; eyelid laxity; foreign body; cardiac, renal, or thyroid disease; superior vena cava syndrome.

Eyelid Twitch
Orbicularis myokymia (related to fatigue, excess caffeine, medication, or stress), corneal or conjunctival irritation (especially from an eyelash, cyst, or foreign body/suture), dry eye, blepharospasm (bilateral), hemifacial spasm, serum electrolyte abnormality, tourettes, tic douloureux, albinism/congenital glaucoma (photosensitivity), anemia (rare).

Eyelids Unable to Close (LAGOPHTHALMOS)
- Severe proptosis, ectropion or eyelid laxity, severe chemosis, eyelid scarring, eyelid retractor muscle scarring, seventh cranial nerve palsy, status-post facial cosmetic or reconstructive surgery.

EYES "Bulging" (PROPTOSIS)
- **[ORBITAL DISEASE (ABOVE)]**

Eyes "Jumping" (OSCILLOPSIA)
- Acquired nystagmus, internuclear ophthalmoplegia, myasthenia gravis, vestibular function loss, opsoclonus/ocular flutter, superior oblique myokymia, various CNS disorders.

Flashes of Light
- More Common. Retinal break or detachment, posterior vitreous detachment, migraine, rapid eye movements (particularly in darkness), oculodigital stimulation, dysphotopsias caused by intraocular lens.
- Less Common. CNS (particularly occipital lobe) disorders, vestibulobasilar artery insufficiency, optic neuropathies, retinitis/uveitis, entoptic phenomena, drug-related, hallucinations, iatrogenic (e.g., post laser photocoagulation).

Floaters
- **[SPOTS IN FRONT OF THE EYES (BELOW)**

Foreign Body Sensation
- Dry eye syndrome, blepharitis, conjunctivitis, trichiasis, corneal or conjunctival abnormality (e.g., cyst, corneal abrasion or foreign body, recurrent erosion, superficial punctate keratopathy), contact lens-related problem, episcleritis, pterygium, pinguecula.

Glare
- Cataract, pseudophakia, posterior capsular opacity, corneal irregularity or opacity, altered pupillary/iris structure or response, status-post refractive surgery, posterior vitreous detachment, pharmacologic (e.g., atropine).

Hallucinations (Formed Images)
- Posterior vitreous detachments, retinal detachment, optic neuropathies, blind eyes, bilateral eye patching, Charles Bonnet syndrome, psychosis, parietotemporal lesions, other CNS causes, medications.

Halos Around Lights
- Cataract, pseudophakia, posterior capsular opacity, acute angle closure glaucoma or corneal edema from another cause (e.g., aphakic or pseudophakic bullous keratopathy, contact lens overwear), corneal dystrophies, status-post refractive surgery, corneal haziness, discharge, pigment dispersion syndrome, vitreous opacities, drugs (e.g., digitalis, chloroquine).

Headache
- **SEE 10.26, HEADACHE.**

Itchy Eyes
- Conjunctivitis (especially allergic, vernal, and viral), blepharitis, dry eye syndrome, topical drug allergy or contact dermatitis, giant papillary conjunctivitis, or contact lens-related problems.

Light Sensitivity (PHOTOPHOBIA)
1. Abnormal eye examination
   - **More Common.** Corneal abnormality (e.g., abrasion or edema), anterior uveitis.
   - **Less Common.** Conjunctivitis (mild photophobia), posterior uveitis, scleritis, albinism, total color blindness, aniridia, mydriasis of any etiology (e.g., pharmacologic, traumatic), congenital glaucoma.
2. Normal eye examination: Migraine, meningitis, retrobulbar optic neuritis, subarachnoid hemorrhage, trigeminal neuralgia, or lightly pigmented irides.

Night Blindness
- **More Common.** Refractive error (especially undercorrected myopia), advanced glaucoma or optic atrophy, small pupil (especially from miotic drops), retinitis pigmentosa, congenital stationary night blindness, status-post panretinal photocoagulation, drugs (e.g., phenothiazines, chloroquine, quinine).
- **Less Common.** Vitamin A deficiency, gyrate atrophy, choroideremia.

Pain
1. Ocular
   - Typically mild to moderate: Dry eye syndrome, blepharitis, infectious conjunctivitis, episcleritis, inflamed pinguecula or pterygium, foreign body (corneal or conjunctival), corneal disorder (e.g., superficial

punctate keratopathy), superior limbic keratoconjunctivitis, ocular medication toxicity, contact lens-related problems, postoperative, ocular ischemic syndrome, eye strain from uncorrected refractive error (asthenopia).
    - Typically moderate to severe: Corneal disorder (e.g., abrasion, erosion, infiltrate/ulcer/keratitis, chemical injury, ultraviolet burn), trauma, anterior uveitis, scleritis, endophthalmitis, acute angle closure glaucoma.
2. Periorbital: Trauma, hordeolum, preseptal cellulitis, dacryocystitis, dermatitis (e.g., contact, chemical, varicella zoster, or herpes simplex), referred pain (e.g., dental, sinus), giant cell arteritis, tic douloureux.
3. Orbital: Sinusitis, trauma, orbital cellulitis, idiopathic orbital inflammatory syndrome, orbital tumor or mass, optic neuritis, acute dacryoadenitis, migraine or cluster headache, diabetic cranial nerve palsy, postinfectious neuralgia (herpetic).
4. Asthenopia: Uncorrected refractive error, phoria or tropia, convergence insufficiency, accommodative spasm, pharmacologic (miotics).

Red Eye

1. Adnexal causes: Trichiasis, distichiasis, floppy eyelid syndrome, entropion or ectropion, lagophthalmos, blepharitis, meibomitis, acne rosacea, dacryocystitis, canaliculitis.
2. Conjunctival causes: Ophthalmia neonatorum in infants, conjunctivitis (bacterial, viral, chemical, allergic, atopic, vernal, medication toxicity), subconjunctival hemorrhage, inflamed pinguecula, superior limbic keratoconjunctivitis, giant papillary conjunctivitis, conjunctival foreign body, symblepharon and associated etiologies (e.g., ocular cicatricial pemphigoid, Stevens–Johnson syndrome, toxic epidermal necrolysis), conjunctival neoplasia.
3. Corneal causes: Infectious or inflammatory keratitis, contact lens-related problems (SEE 4.20, CONTACT LENS-RELATED PROBLEMS), corneal foreign body, recurrent corneal erosion, pterygium, neurotrophic keratopathy, medicamentosa, ultraviolet or chemical burn.
4. Other: Trauma, postoperative, dry eye syndrome, endophthalmitis, anterior uveitis, episcleritis, scleritis, pharmacologic (e.g., prostaglandin analogs), angle-closure glaucoma, carotid–cavernous fistula (corkscrew conjunctival vessels), cluster headache.

"Spots" in front of the Eyes

1. Transient: Migraine.
2. Permanent or long-standing
    - More Common. Posterior vitreous detachment, intermediate or posterior uveitis, vitreous hemorrhage, vitreous condensations/debris.
    - Less Common. Hyphema/microhyphema, retinal break or detachment, corneal opacity/dystrophy or foreign body.

    NOTE: Some patients are referring to a blind spot in their visual field caused by a retinal, optic nerve, or CNS disorder.

Tearing

1. Adults
    - Pain present: Corneal abnormality (e.g., abrasion, foreign body or rust ring, recurrent erosion, edema), anterior uveitis, eyelash or eyelid disorder (e.g., trichiasis, entropion), conjunctival foreign body, dacryocystitis, dacryoadenitis, canaliculitis, trauma.
    - Minimal/no pain: Dry eye syndrome, blepharitis, nasolacrimal duct obstruction, punctal occlusion, lacrimal sac mass, ectropion, conjunctivitis (especially allergic and toxic), crocodile tears (congenital or seventh nerve palsy), emotional state.
2. Children: Nasolacrimal duct obstruction, congenital glaucoma, corneal or conjunctival foreign body, or other irritative disorder.

Differential Diagnosis of Ocular Signs

Anterior Chamber/Anterior Chamber Angle

Hyphema

Traumatic, iatrogenic (e.g., intraocular surgery or laser), iris neovascularization, herpes simplex or zoster iridocyclitis, blood dyscrasia or clotting disorder (e.g., hemophilia), anticoagulation, Fuchs heterochromic iridocyclitis, intraocular tumor (e.g., juvenile xanthogranuloma, retinoblastoma, angioma).

Hypopyon

Infectious corneal ulcer, endophthalmitis, severe iridocyclitis (e.g., HLA-B27 associated, Behçet disease), reaction to an intraocular lens (sterile hypopyon), retained lens particle, device contaminant after cataract surgery (toxic anterior segment syndrome), intraocular tumor necrosis (e.g., pseudohypopyon from retinoblastoma), retained intraocular foreign body, tight contact lens, chronic corneal edema with ruptured bullae, severe inflammatory reaction from a recurrent corneal erosion, drugs (e.g., rifampin).

Blood in Canal of Schlemm on Gonioscopy

Compression of episcleral vessels by a gonioprism (iatrogenic), Sturge–Weber syndrome, arteriovenous fistula (e.g., carotid–cavernous sinus fistula [c-c fistula]), superior vena cava obstruction, hypotony.

Cornea/Conjunctival Findings

Conjunctival Swelling (Chemosis)

Allergy, any ocular or periocular inflammation, postoperative, drugs, venous congestion (e.g., c-c fistula), angioneurotic edema, myxedema.

Conjunctival Dryness (Xerosis)

Vitamin A deficiency, postcicatricial conjunctivitis, Stevens–Johnson syndrome, ocular cicatricial pemphigoid, exposure (e.g., lagophthalmos, absent blink reflex, proptosis), radiation, chronic dacryoadenitis, Sjogren syndrome.

Corneal Crystals

SEE 4.14, CRYSTALLINE KERATOPATHY.

Corneal Edema

1. Congenital: Congenital glaucoma, congenital hereditary endothelial dystrophy, posterior polymorphous corneal dystrophy (PPMD), birth trauma (forceps injury).
2. Acquired: Postoperative edema, aphakic or pseudophakic bullous keratopathy, Fuchs endothelial dystrophy, contact lens overwear, traumatic, exposure-related, chemical injury, acute increase in intraocular pressure (e.g., angle-closure glaucoma), corneal hydrops (decompensated keratoconus), herpes simplex or zoster keratitis, iritis, failed corneal graft, iridocorneal endothelial (ICE) syndrome, PPMD.

Dilated Episcleral Vessels (Without Ocular Irritation or Pain)

Underlying uveal neoplasm, arteriovenous fistula (e.g., c-c fistula), polycythemia vera, leukemia, ophthalmic vein or cavernous sinus thrombosis, extravascular blockage of ophthalmic/orbital venous outflow.

Enlarged Corneal Nerves

- Most Important. Multiple endocrine neoplasia type IIb (medullary carcinoma of the thyroid gland, pheochromocytoma, mucosal neuromas; may have marfanoid habitus).
- Others. Acanthamoeba keratitis, chronic keratitis, keratoconus, neurofibromatosis, Fuchs endothelial dystrophy, Refsum syndrome, trauma, congenital glaucoma, failed corneal graft, leprosy, ichthyosis, idiopathic, normal variant.

Follicles on the Conjunctiva

SEE 5.1, ACUTE CONJUNCTIVITIS AND 5.2, CHRONIC CONJUNCTIVITIS.

Membranous Conjunctivitis

(Classic teaching is that removal of the membrane is difficult and causes bleeding). Streptococci, pneumococci, chemical burn, ligneous conjunctivitis, Corynebacterium diphtheriae, herpes simplex virus, ocular vaccinia. (Compare with “Pseudomembranous Conjunctivitis.”)

Pseudomembranous Conjunctivitis

(Classic teaching is that removal of the pseudomembrane is easy without bleeding.) See earlier for causes of membranous conjunctivitis, as well as adenovirus (rarely causes true membrane formation), ocular cicatricial pemphigoid, Stevens–Johnson syndrome, superior limbic keratoconjunctivitis, gonococci, staphylococci, chlamydia in newborns, and others.

Opacification of the Cornea in Infancy

Congenital glaucoma, birth trauma (forceps injury), congenital hereditary endothelial or stromal dystrophy (bilateral), PPMD, developmental abnormality of the anterior segment (e.g., Peters anomaly), metabolic abnormalities (bilateral; e.g., mucopolysaccharidoses, mucolipidoses), interstitial keratitis, herpes simplex virus, corneal ulcer, corneal dermoid, sclerocornea.

Pannus (Superficial Vascular Invasion of the Cornea)

Ocular rosacea, tight contact lens or contact lens overwear, phlyctenule, chlamydia (trachoma and inclusion conjunctivitis), superior limbic keratoconjunctivitis (micropannus only), staphylococcal hypersensitivity, vernal keratoconjunctivitis, herpes simplex or zoster virus, chemical burn, ocular cicatricial pemphigoid, aniridia, molluscum contagiosum, leprosy.

Papillae on the Conjunctiva

SEE 5.1, ACUTE CONJUNCTIVITIS AND 5.2, CHRONIC CONJUNCTIVITIS.

Pigmentation/Discoloration of the Conjunctiva

Racial melanosis (perilimbal), nevus, primary acquired melanosis, melanoma, ocular and oculodermal melanocytosis (congenital, blue-gray, not conjunctival but episcleral), Addison disease, pregnancy, radiation, jaundice, resolving subconjunctival hemorrhage, conjunctival or subconjunctival foreign body, pharmacologic (e.g., chlorpromazine, topical epinephrine), cosmetic (e.g., mascara/makeup deposits, tattoo).

Symblepharon (Fusion of the Palpebral Conjunctiva with the Bulbar Conjunctiva)

Ocular cicatricial pemphigoid, Stevens–Johnson syndrome, chemical burn, trauma, drugs, long-standing conjunctival or episcleral inflammation, epidemic keratoconjunctivitis, atopic conjunctivitis, radiation, congenital, iatrogenic (postsurgical).

Whorl-Like Opacity in the Corneal Epithelium (Verticillata)

Amiodarone, chloroquine, Fabry disease and carrier state, phenothiazines, indomethacin.

Eyelid Abnormalities

Eyelid Edema

SEE “EYELID SWELLING” in Chapter 1, Differential Diagnosis of Ocular Symptoms.

Eyelid Lesion

SEE 6.11, MALIGNANT TUMORS OF THE EYELID.

Ptosis and Pseudoptosis

SEE 6.1, PTOSIS.

Fundus Findings

Bone Spicules (Widespread Pigment Clumping)

SEE 11.28, RETINITIS PIGMENTOSA AND INHERITED CHORIORETINAL DYSTROPHIES.

Bull's-Eye Macular Lesion

Age-related macular degeneration (ARMD), Stargardt disease or fundus flavimaculatus, albinism, cone dystrophy, rod-cone dystrophy, chloroquine or hydroxychloroquine retinopathy, adult-onset foveomacular vitelliform dystrophy, Spielmeyer–Vogt syndrome, central areolar choroidal dystrophy. SEE 11.32, CHLOROQUINE/HYDROXYCHLOROQUINE TOXICITY.

Choroidal Folds

Orbital or choroidal tumor, idiopathic orbital inflammatory syndrome, thyroid eye disease, posterior scleritis, hypotony, retinal detachment, marked hyperopia, scleral laceration, papilledema, postoperative.

Choroidal Neovascularization (Gray-Green Membrane or Blood Seen Deep to the Retina)

More Common. ARMD, ocular histoplasmosis syndrome, high myopia, idiopathic polypoidal choroidal vasculopathy, angioid streaks, choroidal rupture (trauma).

Less Common. Drusen of the optic nerve head, tumors, retinal scarring after laser photocoagulation, posterior uveitis (e.g., VKH, MFC, serpiginous choroiditis), idiopathic.

Cotton–Wool Spots

SEE 11.5, COTTON–WOOL SPOT.

Embolus

SEE 10.22, TRANSIENT VISUAL LOSS/AMAUROSIS FUGAX; 11.6, CENTRAL RETINAL ARTERY OCCLUSION; 11.7, BRANCH RETINAL ARTERY OCCLUSION; 11.33, CRYSTALLINE RETINOPATHY.

- Platelet–fibrin (dull gray and elongated): Carotid disease, less common cardiac.
- Cholesterol (sparkling yellow, usually at an arterial bifurcation): Carotid disease.
- Calcium (dull white, typically around or on the disc): Cardiac disease.
- Cardiac myxoma (common in young patients, particularly in the left eye; often occludes the ophthalmic or central retinal artery behind the globe and is not seen).
- Talc and cornstarch (small yellow-white glistening particles in macular arterioles; may produce peripheral retinal neovascularization): Intravenous (i.v.) drug abuse.
- Lipid or air (cotton–wool spots, not emboli, are often seen): Results from chest trauma (Purtscher retinopathy) and fracture of long bones.
- Others (tumors, parasites, other foreign bodies).

Macular Exudates

- More Common. Diabetes, choroidal (subretinal) neovascular membrane, hypertension.
- Less Common. Macroaneurysm, Coats disease (children), peripheral retinal capillary hemangioma, retinal vein occlusion, papilledema, radiation retinopathy.

Normal Fundus in the Presence of Decreased Vision

Retrobulbar optic neuritis, cone degeneration, Stargardt disease or fundus flavimaculatus, other optic neuropathy (e.g., Leber hereditary optic neuropathy, tumor, alcohol, or tobacco), rod monochromatism, amblyopia, cancer-associated retinopathy (CAR), melanoma-associated retinopathy (MAR), nonphysiologic visual loss.

Optociliary Shunt Vessels on the Disc

Orbital or intracranial tumor (especially meningioma), prior central retinal vein occlusion, chronic papilledema (e.g., pseudotumor cerebri), chronic open angle glaucoma, optic nerve glioma.

Retinal Neovascularization

1. Posterior pole: Diabetes, prior central retinal vein occlusion.
2. Peripheral: Sickle cell retinopathy, prior branch retinal vein occlusion, diabetes, sarcoidosis, syphilis, ocular ischemic syndrome (carotid occlusive disease), pars planitis, Coats disease, retinopathy of prematurity, embolization from i.v. drug abuse (talc retinopathy), chronic uveitis, others (e.g., leukemia, anemia, Eales disease, familial exudative vitreoretinopathy).

Roth Spots (Retinal Hemorrhages with White Centers)

- More Common. Diabetes, leukemia, septic chorioretinitis (e.g., secondary to bacterial endocarditis).
- Less Common. Pernicious anemia (and rarely other forms of anemia), sickle cell disease, scurvy, systemic lupus erythematosus, other connective tissue diseases.

Sheathing of Retinal Veins (Periphlebitis)

- More Common. Syphilis, sarcoidosis, pars planitis, sickle cell disease.
- Less Common. Tuberculosis, multiple sclerosis, Eales disease, viral retinitis (e.g., human immunodeficiency virus, herpes), Behçet disease, fungal retinitis, bacteremia.

Tumor

SEE 11.36, CHOROIDAL NEVUS AND MALIGNANT MELANOMA OF THE CHOROID.

Intraocular Pressure

Acute Increase in Intraocular Pressure

Acute angle-closure glaucoma, inflammatory open angle glaucoma, malignant glaucoma, hyphema, glaucomatocyclitic crisis (Posner–Schlossman syndrome), postoperative (SEE “POSTOPERATIVE COMPLICATIONS,” in this chapter), suprachoroidal hemorrhage, c-c fistula, spontaneous closure of cyclodialysis cleft, retrobulbar hemorrhage, other orbital disease.

Chronic Increase in Intraocular Pressure

SEE 9.1, PRIMARY OPEN ANGLE GLAUCOMA.

Decreased Intraocular Pressure (Hypotony)

Ruptured globe, phthisis bulbi, retinal/choroidal detachment, iridocyclitis, severe dehydration, cyclodialysis cleft, ocular ischemia, drugs (e.g., glaucoma medications), postoperative (SEE “POSTOPERATIVE COMPLICATIONS,” in this chapter), traumatic ciliary body shutdown.

Iris

Iris Heterochromia (Irides of Different Colors)

1. Involved iris is lighter than normal: Congenital Horner syndrome, most cases of Fuchs heterochromic iridocyclitis, chronic uveitis, juvenile xanthogranuloma, metastatic carcinoma, Waardenburg syndrome.
2. Involved iris is darker than normal: Ocular melanocytosis or oculodermal melanocytosis, hemosiderosis, siderosis, retained intraocular foreign body, ocular malignant melanoma, diffuse iris nevus, retinoblastoma, leukemia, lymphoma, ICE syndrome, some cases of Fuchs heterochromic iridocyclitis.

Iris Lesion

1. Melanotic (brown): Nevus, melanoma, adenoma, or adenocarcinoma of the iris pigment epithelium.
   NOTE: Cysts, foreign bodies, neurofibromas, and other lesions may appear pigmented in heavily pigmented irides.
2. Amelanotic (white, yellow, or orange): Amelanotic melanoma, inflammatory nodule or granuloma (e.g., sarcoidosis, tuberculosis, leprosy, other granulomatous disease), neurofibroma, patchy hyperemia of syphilis, juvenile xanthogranuloma, medulloepithelioma, foreign body, cyst, leiomyoma, seeding from a posterior segment tumor.

Neovascularization of the Iris

Diabetic retinopathy, ocular ischemic syndrome, after central or branch retinal vein or artery occlusion, chronic uveitis, chronic retinal detachment, intraocular tumor (e.g., retinoblastoma, melanoma), other retinal vascular disease.

Lens

SEE ALSO 13.1, ACQUIRED CATARACT.

Dislocated Lens (Ectopia Lentis)

SEE 13.10, SUBLUXED OR DISLOCATED CRYSTALLINE LENS.

Iridescent Lens Particles

Drugs, hypocalcemia, myotonic dystrophy, hypothyroidism, familial, idiopathic.

Lenticonus

1. Anterior (marked convexity of the anterior lens): Alport syndrome (hereditary nephritis).
2. Posterior (marked concavity of the posterior lens surface): Usually idiopathic, may be associated with persistent fetal vasculature.

Neuro Ophthalmic Abnormalities

Afferent Pupillary Defect

1. Severe (2+ to 3+): Optic nerve disease (e.g., ischemic optic neuropathy, optic neuritis, tumor, glaucoma); central retinal artery or vein occlusion; less commonly, a lesion of the optic chiasm or tract.
2. Mild (1+): Any of the preceding, amblyopia, dense vitreous hemorrhage, advanced macular degeneration, branch retinal vein or artery occlusion, retinal detachment, or other retinal disease.

Anisocoria (Pupils of Different Sizes)

SEE 10.1, ANISOCORIA.

Limitation of Ocular Motility

1. With exophthalmos and resistance to retropulsion: SEE 7.1, ORBITAL DISEASE.
2. Without exophthalmos and resistance to retropulsion: Isolated third, fourth, or sixth cranial nerve palsy; multiple ocular motor nerve palsies (SEE 10.10, CAVERNOUS SINUS AND ASSOCIATED SYNDROMES [MULTIPLE OCULAR MOTOR NERVE PALSIES]), myasthenia gravis, chronic progressive external ophthalmoplegia and associated syndromes, orbital blow-out fracture with muscle entrapment, ophthalmoplegic migraine, Duane syndrome, other central nervous system (CNS) disorders.

Optic Disc Atrophy

- More Common. Glaucoma; prior central retinal vein or artery occlusion; prior ischemic optic neuropathy; chronic optic neuritis; chronic papilledema; compression of the optic nerve, chiasm, or tract by a tumor or aneurysm; prior traumatic optic neuropathy.
- Less Common. Syphilis, retinal degeneration (e.g., retinitis pigmentosa), toxic or metabolic optic neuropathy, Leber hereditary optic atrophy, Leber congenital amaurosis, radiation neuropathy, lysosomal storage disease (e.g., Tay–Sachs), other forms of congenital or hereditary optic atrophy (nystagmus almost always present in congenital forms).

Optic Disc Swelling (Edema)

SEE 10.15, PAPILLEDEMA.

Optociliary Shunt Vessels

SEE "FUNDUS FINDINGS" in this chapter.

Paradoxical Pupillary Reaction (Pupil Dilates in Light and Constricts in Darkness)

Congenital stationary night blindness, congenital achromatopsia, optic nerve hypoplasia, Leber congenital amaurosis, Best disease, optic neuritis, dominant optic atrophy, albinism, retinitis pigmentosa. Rarely amblyopia.

Orbit

Extraocular Muscle Thickening on Imaging

- More Common. Thyroid orbitopathy (often spares tendon), idiopathic orbital inflammatory syndrome.
- Less Common. Tumor (e.g., lymphoma, metastasis, or spread of lacrimal gland tumor to muscle), c-c fistula, superior ophthalmic vein thrombosis, cavernous hemangioma (usually appears in the muscle cone without muscle thickening), rhabdomyosarcoma (children).

Lacrimal Gland Lesions

SEE 7.6, LACRIMAL GLAND MASS/CHRONIC DACRYOADENITIS.

Optic Nerve Lesion (Isolated)

- More Common. Optic nerve glioma (especially children), optic nerve meningioma (especially adults).
- Less Common. Metastasis, leukemia, idiopathic orbital inflammatory syndrome, sarcoidosis, increased intracranial pressure with secondary optic nerve swelling.

Orbital Lesions/Proptosis

SEE 7.1, ORBITAL DISEASE.

Pediatrics

Leukocoria (White Pupillary Reflex)

SEE 8.1, LEUKOCORIA.

Nystagmus in Infancy

SEE ALSO 10.21, NYSTAGMUS.

Congenital nystagmus, albinism, Leber congenital amaurosis, CNS (thalamic) injury, spasmus nutans, optic nerve or chiasmal glioma, optic nerve hypoplasia, congenital cataracts, aniridia, congenital corneal opacities.

Postoperative Complications

Shallow Anterior Chamber

1. Accompanied by increased intraocular pressure: Pupillary block glaucoma, capsular block syndrome, suprachoroidal hemorrhage, malignant glaucoma.
2. Accompanied by decreased intraocular pressure: Wound leak, choroidal detachment, overfiltration after glaucoma filtering procedure.

Hypotony
Wound leak, choroidal detachment, cyclodialysis cleft, retinal detachment, ciliary body shutdown, pharmacologic aqueous suppression, overfiltration after glaucoma filtering procedure.

Refractive Problems
Progressive Hyperopia
Orbital tumor pressing on the posterior surface of the eye, serous elevation of the retina (e.g., central serous chorioretinopathy), posterior scleritis, presbyopia, hypoglycemia, cataracts, after radial keratotomy or other refractive surgery.
Progressive Myopia
High (pathologic) myopia, diabetes, cataract, staphyloma and elongation of the globe, corneal ectasia (keratoconus or after corneal refractive surgery), medications (e.g., miotic drops, sulfa drugs, tetracycline), childhood (physiologic).

Visual Field Abnormalities
Altitudinal Field Defect
- More Common. Ischemic optic neuropathy, hemi or branch retinal artery or vein occlusion, optic neuritis.
- Less Common. Glaucoma, optic nerve or chiasmal lesion, optic nerve coloboma.

Arcuate Scotoma
- More Common. Glaucoma.
- Less Common. Ischemic optic neuropathy (especially nonarteritic), optic disc drusen, high myopia, optic neuritis.

Binasal Field Defect
- More Common. Glaucoma, bitemporal retinal disease (e.g., retinitis pigmentosa).
- Rare. Bilateral occipital disease, tumor or aneurysm compressing both optic nerves or chiasm, chiasmatic arachnoiditis, nonphysiologic.

Bitemporal Hemianopsia
- More Common. Chiasmal lesion (e.g., pituitary adenoma, meningioma, craniopharyngioma, aneurysm, glioma).
- Less Common. Tilted optic discs.
- Rare. Nasal retinitis pigmentosa.

Blind Spot Enlargement
Papilledema, glaucoma, optic nerve drusen, optic nerve coloboma, myelinated (medullated) nerve fibers off the disc, drugs, myopic disc with a crescent, multiple evanescent white dot syndrome (MEWDS), acute idiopathic blind spot enlargement syndrome ([AIBSE] may be on spectrum with MEWDS).
Central Scotoma
Macular disease, optic neuritis, ischemic optic neuropathy (more typically produces an altitudinal field defect), optic atrophy (e.g., from tumor compressing the nerve, toxic or metabolic disease), and rarely, an occipital cortex lesion.
Constriction of the Peripheral Fields Leaving a Small Residual Central Field (Tunnel Vision)
Glaucoma, retinitis pigmentosa or other peripheral retinal disorders (e.g., gyrate atrophy), chronic papilledema, status-post panretinal photocoagulation or cryotherapy, central retinal artery occlusion with cilioretinal artery sparing, bilateral occipital lobe infarction with macular sparing, nonphysiologic visual loss, medications (e.g., phenothiazines), vitamin A deficiency, and carcinoma, melanoma, or autoimmune-associated retinopathy.
Homonymous Hemianopsia
Temporal, parietal, or occipital lobe lesion of the brain (e.g., stroke and tumor more common; aneurysm and trauma less common), optic tract or lateral geniculate body lesion. Migraine may also cause a transient homonymous hemianopsia.

Vitreous
Vitreous Opacities
Asteroid hyalosis, vitreous hemorrhage, inflammatory cells from vitritis or posterior uveitis, snowball opacities of pars planitis or sarcoidosis, normal vitreous strands from age-related vitreous degeneration, tumor cells, foreign body, hyaloid remnants, synchysis scintillans, and rarely, amyloidosis or Whipple disease.

Week 1: Optometric Prescribing

1.1. Optometric Prescribing

OBA List

Appendix A: diagnostic drugs (e.g. Alcaine) – no endorsement needed

Appendix B: therapeutic drugs – endorsement required

Appendix B

Board-approved list of topical schedule 2, 3 and 4 medicines that optometrists with a scheduled medicines endorsement are qualified to administer, obtain, possess, prescribe, supply or use for the purposes of the practice of optometry[5]

Schedule 2 Pharmacy Medicine

Anti-infectives
Dibromopropamidine
Propamidine

Decongestants/ anti-allergics
Antazoline
Azelastine
Ketotifen
Levocabastine
Lodoxamide
Naphazoline
Pheniramine
Sodium Cromoglycate

Miotics, mydriatics and cycloplegics
Phenylephrine ≤2.5%

Schedule 3 Pharmacist Only Medicine

Anti-infectives
Chloramphenicol

Schedule 4 Prescription Only Medicine

Anti-infectives
Aciclovir
Azithromycin
Bacitracin
Cephazolin
Ciprofloxacin
Framycetin
Ganciclovir
Gentamicin
Gramicidin
Neomycin
Ofloxacin
Polymyxin
Tetracycline
Tobramycin

Anti-inflammatories
Cyclosporin
Dexamethasone
Diclofenac
Fluorometholone
Flurbiprofen
Hydrocortisone
Ketorolac
Lifitegrast
Loteprednol
Prednisolone

Decongestants/ anti-allergics
Olopatadine

Anti-glaucomas
Apraclonidine
Betaxolol
Bimatoprost
Brimonidine
Brinzolamide
Dorzolamide
Latanoprost
Pilocarpine
Tafluprost
Timolol
Travoprost

Miotics, mydriatics and cycloplegics
Atropine
Cyclopentolate
Homatropine
Pilocarpine
Phenylephrine
Tropicamide

Local anaesthetics
Amethocaine
Lignocaine
Oxybuprocaine
Proxymetacaine

- **Anti-infectives:** antiviral (aciclovir), rest are antibacterials
- **Anti-inflammatories:** steroids (end in "one" + loteprednol)
- **Immunomodulatory drugs:** cyclosporin and lifitegrast
- **Anti-allergic (olopatadine):** available as S3 drugs (ketotifen, azelastine)

Schedule 2	**Pharmacy medicine** – medicine available on shelf at supermarkets and pharmacies	Panadol, ibuprofen, cold and flu preparations
Schedule 3	**Pharmacist only medicine** – medicines available from pharmacist without prescription, but behind pharmacy	Limited topical antibiotics, limited oral antifungals, acid reflux medicines, sleep aid medicines, emergency counter	contraception, antibiotic eye ointment (chloramphenicol)
Schedule 4	**Prescription only medicine** – must be prescribed by a healthcare professional. Supplies in hospitals or pharmacies with a prescription.	Antibiotics, anti hypertensives, anticoagulants, cholesterol medicines
Schedule 8	**Controlled drugs** – special rules for producing, supplying, distributing, owning and using. Prescribed by authorised healthcare professionals with a special prescribing permit.	Morphine, oxycodone, fentanyl, methadone

S2 = OTC pharmacy (e.g. Panadol)
S3 = Pharmacist only (e.g. limited antibiotics)
S4 = Prescription (e.g. antibiotics, steroids)
S8 = Controlled (e.g. opioids)

PBS (Pharmaceutical Benefit Scheme)

- Part of **National Medicines Policy**
  - **Unrestricted**: medicines prescribed through PBS without PBS restrictions for therapeutic use
  - **Restricted**: prescribed through PBS if prescribed satisfies the patient's clinical condition (e.g. Aciclovir is a restricted treatment of HSV keratitis)
  - **Authority**: requires approval by PBS prior to prescription
    - Authority required: requires phoning PBS or obtaining approval online
    - Authority required (STREAMLINED): no call is required but optometrist must include four digit authority code listed beside item on PBS
- Subsidises the cost of medical conditions, (e.g. 60-day prescription for dry-eye and glaucoma)
- Patient needs to be stable on current treatment
- **Green '60D'** and **'OP'** logo on the PBS Optometrist's list
- Streamlined authority code for unit-dose tear supplements **(15559)** introduced for 60D) prescriptions (e.g. Hylo-fresh, Hylo-forte, Cationorm, Viscotears, Systane)
- Streamlined authority codes for 30 day dispensing
  - **6172** for unit dose tear supplements (Cationorm, Viscotears, Systane)
  - **4105** for multi dose preservative free (Hylo-fresh, Hylo-forte)

Prescribing Repeats

- There are no limits to repeats for private prescribing
- PBS prescribing involves a maximum number of repeats
  - Medicines to treat **acute conditions** usually have no repeats (NR) or 1 repeat
  - Medicines to treat chronic conditions usually have multiple repeats
  - Irrespective of the condition, best to avoid providing repeats for **corticosteroid eye drops** –

Off-Label Prescribing

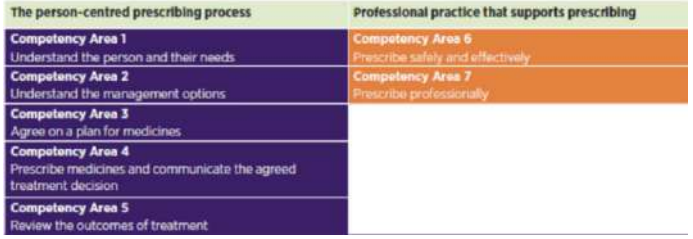
- Only considered when TGA approved medicines are not available
- Medicine administered for an indication, route of administration or px group that is unapproved
- Very common in children and pregnancy, and requires seeking review and evidence to support your decision
- **E.g. 5% betadine eye rinses → acute adenoviral conjunctivitis**
- MUST obtain and document patient consent; only when TGA approved medicines are not available
- There should be evidence supporting the appropriateness of the recommendation
- X% = X grams in 100 milliliters of solution
  - E.g. 3mg/mL is 0.003g in 1mL or 0.3g in 100mL of solution or 0.3%

Quality Use of Medicines (QUM) – one of the four arms of the National Medicines Policy

- **Judicious** use – selecting management options wisely
- **Appropriate** use – choosing most appropriate medicine, considering risks, cost and benefit
- **Safe** use – using medicines safely/effectively
- **Efficacious** use – achieve goals of therapy, delivering beneficial changes in actual health outcomes

2021 Prescribing Competencies Framework – 7 competency areas

- **Areas 1-5;** person-centred prescribing process – understand the person and their needs
- **Area 6-7:** professional practice that supports prescribing



Scenario-based practice: chloramphenicol for central staining; prednefrin forte for inflammation

Resources To Consult

- Product Information (**PI**) Leaflet – find out more about special populations or drug interactions
  - Scientific, objective account of a medicine's usefulness and limitations

    - Describes both active and inactive ingredients – how the medicine acts, metabolised and eliminated
    - Clinical trials
    - Indications, contraindications, precautions, known adverse reactions
- **TGA** (Therapeutic Goods Association) – responsible for evaluating, assessing and monitoring products that are defined as therapeutic good
- Consumer Medicine Information (**CMI**) – based on PI, written specifically for the consumers
- Information and instructions on how to use your medicine safely and effectively
- National Prescribing Services (**NPS**) Medicine Wise – working towards **QUM** (Quality Use Medicines) in Australia\
- Provides practical tools and information about medicines, health conditions and medical tests (including access to **CMI** on many approves drugs)
- Australian Register of Therapeutic Goods (**ARTG**) – a searchable engine that contains information on therapeutic goods that can be supplied in Australia (this includes their schedule)
- Australian Medicines Handbook (**AMH** Online)
    - Detailed profiles of individual drugs (prescribing guidelines, drug interactions, off-label uses and PBS status)
- eTG complete Therapeutic guidelines
    - Condition based treatment algorithms (management of clinical presentations based on best evidence)
    - Therapeutic management – first, second line drugs and when to escalate or refer
    - Non pharmacological treatment guidelines

1.2. Special Populations Prescribing

Children

- Paediatric population has 3 times more adverse drug reactions than hospitalised adults **(Cohen, 1998)**
- 15% of children prescriptions have errors (8% overdoses, 7% underdoses)

Challenges:

- **Age:** different/changing pharmacokinetic at various ages and stages of development
- **Dosing**: individualised doses based on patient's age, weight (mg/kg), body surface area (mg/m2), and clinical condition
- **Availability:** dosage forms and concentrations appropriate for administration to children or people with dexterity issues. Dosage formulations are compounded and lack stability, compatibility, or bioavailability data.
- **Lack of information:** little published information or regulatory-approved labeling regarding dosing, pharmacokinetics, safety, efficacy, and clinical use of drugs in paediatric population.
- **Other common problems:** Need for precise dose measurement and appropriate drug delivery systems

Paediatric Drug Overdose:

- Children cannot verbalise well hence high incidence of accidental drug overdose in children
- Safe drug practices + educating parents/carers
    - **Safe storage:** keep drug bottles away from infants
    - **Promoting child-resistant containers**
    - **Selecting appropriate packaging**

Elderly

- Reduced renal function and high likelihood of polypharmacy, likely to display **unintended noncompliance**
    - Use non-pharmacological treatments when possible
    - Prescribe lowest feasible dose (often < ½ adult dose*)
    - **Minimise polypharmacy:** prescribe the smallest number of drugs, with the simplest dose regimens
    - Prescribe from a limited range of drugs – get familiar with their effects in the elderly
    - Regularly **review drop instillation technique** – demonstrate frequently to improve compliance
    - Simple verbal and written instructions for every medication, to improve compliance
    - Presenting symptoms may be a result of existing medications
    - Regularly review chronic treatment; possible to stop medications or modify formulations if ocular surface health changes
    - If necessary, make sure the carer understand the treatment regimen

Pregnancy

- Drugs during pregnancy only if the expected benefits to the mother is thought to be greater than the risk to the fetus
- Counsel pregnant women to avoid exposure to all unnecessary drugs and chemicals
- Few drugs conclusively teratogenic in humans, but no drugs is safe beyond all doubt in early pregnancy
- Seek advice from pregnancy drug information centres before prescribing

Breastfeeding

- Prescribe only drugs clearly indicated for breastfeeding women, as for many drugs there is insufficient or inadequate evidence available to provide appropriate advice
- Use drugs well-investigated and widely used during pregnancy
- Taking the drug after a feed decreases the amount of the drug in breast milk; this may reduce effects on the infant and reassure the mother
- Seek advice from pregnancy drug information centres before prescribing

Statistics

- Most women use at least one drug during pregnancy
- Background risk to all couples for major birth defect is 2-5%
- Take into account ***need for drug, stage of pregnancy, route of administration***

Resources:

- ***AMH, PI, LactMed, eTG***
- Consult breastfeeding guidelines (WHO, Lactation Risk Categories)

Aboriginal and Torres Strait Islander

- **Closing the Gap (CTG) PBS Co-Payment Program**
- Optoms able to register eligible ATSI for PBS co-payment
    - w/o concession = $7.50 for majority of PBS meds, w/ concession = free for majority

Adverse Effects

- Noxious and/or undesirable effects associated with the use of a drug at doses normally used
- Classified as ***predictable versus unpredictable***
- Predisposition factors: age, gender, dose, polypharmacy, history, genetics
- Unknown in drugs new to market

Medicine Interactions

- Occurs when 2 medicines used together may increase risk of **toxicity** or when a medicine may affect the likely benefit of another through the following mechanisms:
    - Increase exposure to other medicine by increasing absorption or reducing metabolism or excretion = **toxicity**
    - Reduce exposure to other medicine by reducing absorption or increasing metabolism or excretion = **reduced therapeutic efficacy**
    - Increase likelihood of adverse effects if two medicines have similar adverse effects
    - Antagonise pharmacological effect of other medicine = **reduced therapeutic efficacy**
- Can be pharmacodynamic (overlap of pharmacological actions) or pharmacokinetic (changes in absorption, distribution, metabolism, excretion)
- Risk factors for medicine interactions
    - **Px factors:** extremes of age, frailty, female gender, poor cognition, comorbidities, genetic disposition
    - **Medicine factors**: polypharmacy, recent change in medicines, medicines with low therapeutic index, limited elimination pathways

You do not need to be certain, just suspicious! Report all suspected reactions to prescriptions.

Black Triangle Scheme - TGA

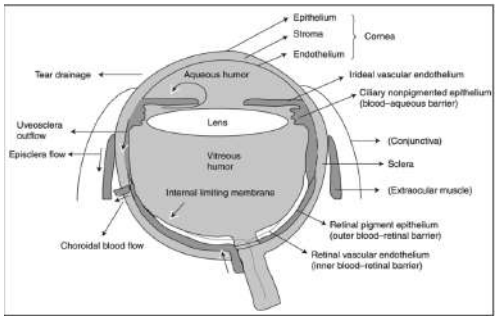
- Provides simple means for practitioners and px to **identify certain types of new medicine →** encourages reporting
    - Will appear on PI and CMI for 5 years
- Black Triangle ophthalmic products:
    - **Ciclosporin** 0.1% eye drops (IKERVIS)
    - **Ciplosporin** 0.09% eye drops (CEQUA)
    - **Cenegermin** (rbe) 0.02% eye drops (OXERVATE)
    - **Lifitegrast** 5% eye drops (XIIDRA)
    - **Atropine** 0.01% (eikance)

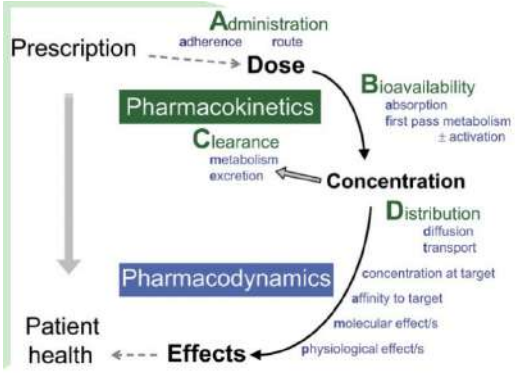
National Notifiable Diseases Surveillance System (NNDSS)

- Collects information state by state on more than **60 communicable diseases**
- Reporting is by health practitioners when they detect one of the notifiable diseases in one of their patients

- Anonymous data is collected e.g. age, gender, postcode, species, etc.

1.3. Ophthalmic Drug Formulation + Delivery





Prescription to Patient Health

PharmacoKINETICS *= body ➜ drug (movement of drug through body)*
PharmacoDYNAMICS *= drug ➜ body (effects of drug on body)*

Pharmacokinetics

- **ABCD:** administration, bioavailability, clearance, distribution
- Study of absorption, distribution, metabolism and elimination ➜ drug delivery is mainly absorption and distribution
- Ocular **drug *absorption*** dependent on:
  - Molecular properties of the drug, **viscosity** of vehicle, functional status of the tissue forming the barrier to penetration
- **Avascularity** of eye structures:
  - Responsible for providing a clear pathway of light through to the retina e.g tear film, cornea, aqueous humour, lens, vitreous
  - **Advantage ➜** allows for ocular drug penetration without significant absorption by the systemic circulation
- Drug *distribution* or *bioavailability* can be predicted by the relationships between the eye's compartments and various barriers (e.g blood-aqueous barrier)

Barriers to delivery of ocular medicines

- Rapid tear turn-over
- Cornea is part lipophilic & hydrophilic
- **Iris**: contains lipophilic epithelial pigment + sphincter (cholinergic) & dilator (adrenergic)
- CB major role in drug removal from the eye – contains drug metabolizing enzymes
- **Lens** = natural barrier to molecular transport
- Blood-ocular barrier (tight- junctions ciliary & RPE, nonfenestrated & iridal capillaries)

Routes of Administration

Topical:

- Most common route of administration to eye, in the form of eye drops
- Used to deliver pharmacological agents to *conjunctiva, cornea, a/c and p/c*
- Rarely used to deliver drugs to vitreous cavity bc **limited penetration➜** <5% will reach intraocular sites ➜ not adequate drug concentration
  - Due to *natural barriers* to ocular drugs such as blood-aqueous barrier and inner+outer blood-retinal barriers
  - Endothelial pump, iris vascular endothelium and lens epithelium also have role in preventing diffusion of drugs from one part to another part in the eye
- **Rapid tear washout**
- **Poor px compliance**

Intraocular Injection:

- Best method to deliver drugs to treat *vitreous cavity or inner retina*
- e.g anti-VEGF injections for advanced macular degeneration ➜ note* drug transport between aqueous and vitreous humour will change after cataract removal
- Some drugs more readily cross the blood-retina barrier such as systemic agents *hydroxychloroquine* and *sildenafil* ➜ can cause toxic effect

  ➜ ***Intravitreal injection:***
  - Intravitreal triamcinolone used for treating posterior eye inflammatory disease
  - Chosen route of delivery for treatment of many vascular and inflammatory diseases of the macula e.g wet AMD and diabetic macular oedema
  - **Increased risk of retinal detachment, hemorrhage, endophthalmitis and cataracts**
  - **Rapidly diluted**
  - **Repeat procedures necessary**

  ➜ ***Periocular*** (through orbit)

  ➜ ***Intracameral route*** (through anterior chamber)

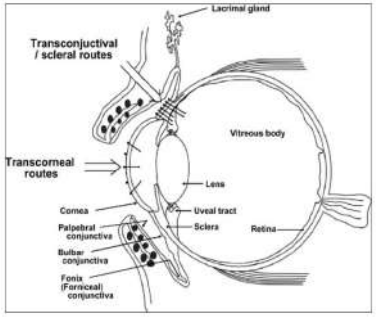
Systemic Administration:

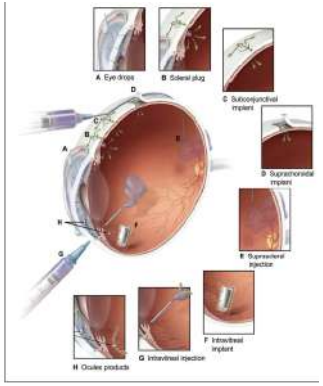
- Through the body's circulation (IV, PO, IM, SC, transdermal)
- **limited/variable penetration**
- **Potential for systemic toxicity**

Intraocular Implants:

- **Increased risk of retinal detachment and intravitreal hemorrhage**
- **Invasive**

Sites and methods for ocular drug delivery





- *Injectable drugs* can be given by periocular routes ➜ subconjunctival, retrobulbar, peribulbar and posterior sub-tenon infections
  - Minimise dosing frequency + sustained drug delivery to anterior and/or posterior segment during a prolonged period of time
- *Hydrophilic drugs* can be given subconjunctivally ➜ don't have to penetrate the conjunctival epithelium + can access the eye's cavity more readily, bypassing the conjunctival circulation
  - Injection into subconjunctival space ➜ drug penetrates sclera which is more permeable than cornea
  - Unlike cornea and conjunctiva, scleral permeability is not dependent on drug lipophilicity but mainly on molecule size

Periocular & intraocular injections

- **Periocular injections**
  - **ADV**: easily deliver high concentrations with smaller quantities of drugs, esp to deeper ocular structures
  - The relative positions of periocular injections for subconjunctival, sub-Tenon's and retrobulbar injections
    - This can be performed through the **eyelid** or directly through the **conjunctiva**
- **Retrobulbar injections**
  - Early 1920s dev + used for cataract extraction ➜ superseded by topical anaesthetic for cataract surgery

- Still common for drug delivery to eye
- Injections paced **around the globe** and not directly into muscle cone
  - Risk of perforation and complications decreases ➔ further away from globe

- **Intracameral injections**
  - Drug directly to anterior chamber of eye
- **Intravitreal injections**
  - Directly into vitreous cavity
  - anti-VEGF drugs used for NV AMD can also be used to deliver intravitreal corticosteroids in posterior uveitis or intravitreal antivirals in treatment of CMV retinitis
  - Also solid drug delivery systems e.g intravitreal drug implant

Topical administration

- Solutions & suspensions • Sprays • Ointments • Lid scrubs • Gels • Solid Delivery Devices • Continuous Flow Devices
- **Most commonly** used route of admin for ocular drugs
- Convenient, simple, cost-effective, noninvasive and px self-administered
- Choice of formulation may depend on ➔ **drug characteristics, desired site of action, nature of disease, px characteristics, frequency of dosage and length of treatment**

Solution	Suspension	Ointment
Most common	Common	Less common
Short contact time	Short contact time	Difficult instillation
Contamination of multi-dose bottles	Contamination of multi-dose bottles	Interferes with vision
Dropper size reduced (now 25µm to 56µm)	For poorly soluble drugs; micronised; must be resuspended by shaking	Alters bioavailability of subsequent topical drugs (instil last)
Can be formulated / compounded as sprays or unit dose	Instillation challenging in special populations (children, older age)	
Instillation challenging in special populations (children, older age)		
Systemic uptake greatest (~15%/min clearance)		
Chloramphenicol	Dexamethasone	Chloramphenicol

- Topical ophthalmic drugs are solutions **[water soluble salt of the active drug + excipients]**
- **Suspensions:** medicines with poorly soluble active ingredients ➔ active ingredient is reduced to micron size or micronised before it is suspended in a solution
  - Active ingredient sits at bottom of bottle and must be shake to unsuspend before instillation
  - Most topical corticosteroid drugs on your list are suspensions e.g **prednisolone acetate, dexamethasone, fluorometholone and fluorometholone acetate**
- In theory, insoluble drugs in suspension could be reformulated as solutions using cyclodextrins by drug manufacturers
- Ointments offer increased retention time and bioavailability at the cost perhaps of reduced vision and difficult instillation

- **Solutions**: most common; easy but short retention
- **Suspensions**: for poorly soluble drugs (e.g. pred acetate) ➔ does not dissolve well in water
- **Ointments**: longer retention but blur vision
- **Gels/Polymers:** increase residence time (e.g. Timoptic-XE)

Ocular Medicine Absorption

- As most drugs are formulated as eye drops, first barrier is **tear film**
  - **Tear pH = 7.4; tear volume ~ 7uL**
    - Single drop of topical drug ~ 30 L ➔ excess fluid (TIP: remember ~30 drops in 1 mL or 1,000)
  - **Quick tear turnover rate (2-3 mins)** + continuous secretion of tear film limits contact time of topically applied eyedrops
    - Most drugs cleared completely within 5-6 mins
    - Excess gets drained down **nasolacrimal passage** and over lid and skin ➔ systemic absorption through lacrimal sac and blood vessels of nasal mucosa/swallowing
    - **Increasing drop volume = higher systemic absorption NOT drug penetration**
  - Tear osmolarity ~290 mOsmol/kg
  - Reflex tearing ➔ reduces med absorption + increases chance of systemic absorption
    - Commonly used ophthalmic drugs are **alkaloids** or weak bases which are chemically unstable in neutral or alkaline solutions ➔ hence formulated as slightly acidic to maintain stability
      - E.g **pilocarpine** produces even more irritation immediately upon instillation with even more lacrimation and dilution

1) Double DOT technique [Don't Open eye Technique and Digital Occlusion of the Tear duct]

- Minimises medicine entry to **nasolacrimal** passage and avoid first-pass hepatic metabolism
- Wait 5 mins between drops as to not dilute the drop (ocular medicine resembles intravenous admin more than oral)

1. Close the eye
2. Apply digital pressure to lacrimal sac for 2-3 minutes with finger ➔ reduces systemic absorption by 2/3
3. Wait 5 minutes between drops



2) Tissue press technique

- Easier for px to administer

Corneal Uptake	Conjunctival/Scleral Absorption
• Corneal thickness ~ 500um • 5-layers: epithelium (2), Bowman's, stroma (3,4), Descemet (5), endothelium (5) • Corneal route: major functional barrier vs. major site of absorption ○ Epithelium = lipophilic (stops penetration of hydrophilic drugs) vs. stroma = hydrophilic (hydrated collagen prevents/slows diffusion of lipophilic agents) • E.g sodium fluorescein is hydrophilic and does not penetrate epithelium unless epithelium not intact • Uptake limited by ○ Short residence time (tearing) ○ Small surface area (cf gut) ○ Properties of the cornea (lipophilic vs hydrophilic) • Conjunctiva / sclera route: larger surface but vascularised (<20% drug penetrates to iris & CB)	• Conjunctiva / sclera extensively vascularised (systemic absorption) • <20% medicine reaching iris / CB via this route • 'Non-productive' route as drug penetrates blood circulation instead of intraocular segments • Sector dilation experiments is evidence of some medicine absorption (remember this technique for uveitis) • Apply filter paper or cotton but moistened with 2.5% phenylephrine to conjunctive for a 5-10 secs ➔ sector mydriasis + some transscleral drug absorption

Formulation - medicine characteristics

- Bioavailability: amount of drug present at desired effect site
- Bioavailability can be **increased** by:
  - Modifying **formulation** (suspensions, emulsions, gels)
  - Inserting **reservoirs** (inserts, micro-polymer systems, liposomes)
  - Increasing **tissue permeability** (iontophoresis, penetration enhancers, absorption promoters)
- Bioavailability can be **affected** by the following drug properties:

- **Active ingredients:** can affect other drugs uptake (e.g. anti-glaucoma drugs, NSAIDs)
- **Preservatives**: aim to prevent bacterial growth in multi-dose bottles. Can irritate and damage the ocular surface. Action is non selective (against all cells). Many types available
- Vehicles: agent other than active ingredient or preservative added to provide tonicity, buffering and viscosity

Prodrugs

- Improve bioavailability and reduce adverse drug reactions by **enhancing lipophilicity**
  - **Prodrug**: medicine that is metabolised or converted within the body after administration into a more pharmacologically active drug
  - Drug metabolism can be used to design prodrugs, which are inactive drugs → active from after tissue penetration (e.g. through the cornea) has occurred
- Used to increase penetration of topical ocular drugs through corneal (or other) barrier
- Metabolite more active than the initial parent form
- Must metabolise before reaching effector site
- **Example**: latanoprost, travoprost have an ester-linked group which is cleaved off after corneal penetration and the free acid has the more potent effect in the aqueous humour

Preservatives

- Act **non selectively on ALL cells**
- Medicines must be preserved to allow for multi-use and adequate shelf life
- Ideal preservative is non-toxic, stable, compatible, broad-spectrum and has a rapid kill rate
  - Surfactants or detergents such as benzalkonium chloride or BAK are most commonly used
    → ionically charged substances that kill bacteria by disruption plasma membrane
    → great stability, good antimicrobial properties and long shelf life
- Agents like chlorhexidine are thought to have less effect on corneal permeability and tear film integrity than BAK
- Chemical preservatives also include mercurials like thimerosal, products containing chlorobutanol, iodine and alcohols; can be bacteriostatic or bactericidal
- Oxidative preservatives like hydrogen peroxide can also be **bactericidal** at the right concentration
  - Penetrating cell membrane + interfering w/ cell function
- Newer oxidative preservatives like Purite (in Refresh Tears) and Sodium perborate (in GenTeal) are important innovation AKA 'vanishing preservatives'
- Purite dissipates into water and sodium chloride when exposed to light
- Sodium perborate is converted to hydrogen peroxide than oxygen and water once into the eye
- SofZia is the most recently introduced low toxicity preservative, which can be found in the anti-glaucoma preparation of travoprost, Travatan Z
  - Becomes inactive when exposed to cationic ions in tear film

Table 1. Classes and representative molecules of common preservatives in ocular therapeutic formulations [2–7].

Chemical class	Component(s)	Proprietary name (if applicable)	Commercial formulation examples
Quaternary ammoniums (detergents)	Benzalkonium chloride		Many
	Polyquaternium-1	Polyquad (Alcon)	Travoprost PQ
	Polyquaternium-42		–
	Cetrimide		Some artificial tears
Oxidizing agents	Sodium perborate	GenAqua (Novartis), Dequest (TheraTears)	Some artificial tears
	Stabilized Oxochloro Complex (SOC)	Purite (Allergan, an AbbVie company), OcuPure (AMO)	Brimonidine P
Ionic buffers	Borate, sorbitol, propylene glycol, and zinc	SofZia (Alcon)	Travoprost Z
Amidines	Chlorhexidine		Some contact lens solution
Parabens	Methylparaben		Some artificial tears
	Propylparaben		
Alcohols	Chlorobutanol		Some artificial tears
	Phenylethanol		
Mercury-based	Thimerosal		Pilocarpine
	Phenylmercuric nitrate/acetate		

Unit Dose Dispensers

- Allow repeated and frequent admin of topical therapy without exposure risks with preserved solutions
  - Exist in dispensers where tip is broken before drop instillation + discarded after use
  - Minims systems also exist where a small screw top device for single use is used
- Volumes range from 0.1-0.6ml, all classes of drugs can exist in unit dose formulation
  - **Common**: systane ultra, cellufresh or antibiotics (chloramphenicol or gentamicin minims), anti-histamine (Zaditen), anti-inflammatory (prednisolone sodium phosphate), dilating drop (tropicamide)

Formulation: vehicles, other excipients

- **Excipients:** things other than active ingredients and preservatives
- Characteristics of excipients:
  - **Stability**: protect against physical degradation (e.g. Oxidation); required for drug storage (i.e. adequate shelf life, typically 18 mths). Acid solutions are typically more stable than neutral or alkaline pH. Eg: Ascorbic acid
  - **Buffers**: maintain pH close to normal (7.4 for the eye) and adjust osmolarity of the solutions (salts and sugars can also be used for this). Potassium bicarbonate
  - **Tonicity agents**: maintain osmolarity (290 mOsm = 0.9% saline). Dextran.
  - **Viscosity enhancers**. Delay washout from tear film & increase bioavailability. Polyionic molecules bind corneal surface. Petroleum / oil base longer retention time. Dextran. Carbomers.
  - **Wetting agents and Solubilisers**. Polysorbate

Viscosity agents

- Increasing solution viscosity → prolongs ocular residence time of drugs, hence increasing bioavailibility
- Common agents:

Ingredient	Characteristics
Polyvinylpyrrolidone (PVP)	Non ionic Very low toxicity
Polyvinyl alcohol (PVA)	Water soluble viscosity enhancer Commonly used at 1.4% May promote wound healing
Hydroxypropyl methylcellulose (HPMC)	Prolong tear film wetting 0.5% HPMC better retention than 1.4% PVA
Carboxymethylcellulose (CMC)	Similar to HPMC Greater adherence to mucins
Sodium hyaluronate	High-molecular weight polymer Non Newtonian fluid - susceptible to shear thinning during blink

- Less frequent agents (newer to market):

Ingredient	Characteristics
Polyionic vehicles	Compounds that have 2 or more regions that vary in their lipophilic and binding natures Poloxamer 407
Polyacrylic acids	Display pseudoplastic properties Carbopol gels contained in anti-glaucoma drops Betoptic-S (Carbomer 934P) and Azopt (carbomer 974P)
Cation exchange resine (Amberlite)	Biphasic lipid-water or water-lipid combinations Can dissolve and deliver hydrophilic and lipophilic compounds. Eg Betoptic

→ used primarily in antiglaucoma preparations in an effort to increase eyedrop retention, drug absorption and resulting in a decreased dosage requirements

- **Polymeric gel-forming systems**
  - Polymeric gels classified into: **preformed gels** and **in-situ forming gels**
    - Preformed gels are simple, more viscous solutions that do not modify after administration whereas in-situ forming gels or gel-forming systems are viscous liquids that undergo a solution to gel phase transition after exposure
  - Large molecules
  - Reversible phase transition: aqueous drop reversibly gels on contact with precorneal tear film
  - Increased contact time, increased drug absorption, increase therapeutic effect
  - Delivery system and trigger (temperature, pH, ion, lysozyme) varies

- Xanthan gum, Gelrite

Ophthalmic ointments

- Provide extended residence time – increasing surface effects and bioavailability
- Cleared as slowly as 0.5% per min (32x slower than drops)
- **Ointments**: usually mixtures of white petrolatum (~60%), liquid mineral oil (~40%) + water-miscible agent (e.g. lanolin)
  - Mineral oil = allows melting at body temp, lanolin = allow absorption of water and water-soluble drugs
- **Indications**: increased retention time e.g. nighttime therapy, for antibiotic therapy with patching, in children
- **Disadvantages**: transient blurry vision, difficult administration

Increasing pre-ocular retention/bioavailability: The Future

- **Cyclodextrins and Colloidal systems: liposomes, nanoparticles**
  - Conventional penetration enhances e.g BAK disrupt ophthalmic barrier
- **Cyclodextrins**
  - Natural cylindrical oligonucleotides with a hydrophilic outer surface and a lipophilic inner surface which complex with the drug and naturally enhance its penetration and bioavailability
  - Can form water-soluble complexes with lipophilic drugs, which 'hide' in the cavity
  - Used to improve solubility, stability, or irritability of various compounds
  - Several eye drop products containing cyclodextrins have already been registered around the world including products as corticosteroids, choloramphenicol, diclofenac, cyclosporine and sulfanomide carbonic anhydrase inhibitors eyedrop formulations
- **Colloidal systems**
  - liquid-retentive drug delivery systems where the drug is contained in a carrier
  - Like liposomes and nanoparticles, prolong contact time and reduce the need for frequent dosing
  - Lipisomes are spherical microscopic vesicles composed of one or more concentric lipid bilayers surrounding aqueous compartments ➡ gives them an amphiphilic character and therefore the ability to entrap both hydrophilic and hydrophobic compounds or drugs
  - Because of their lipid composition, lipisomes are non-toxic and biodegradable but they often suffer from issues of poor stability and reproducibility, limiting their current use
  - Nanoparticles are polymeric colloidal 25 particles that contain drug-entrapped macromolecular material. Nanoparticles are biodegradable particles of less than 1um in diameter in which a drug is dissolved, entrapped or encapsulated
  - Drugs can be integrated in the matrix of the nanoparticle or attached to the surface, can be nanospheres or nanocapsules
  - Lots of experimental data is available on use of colloidal systems for ophthalmic drugs but these are to date rarely commercialised

Drug release systems

- Solid delivery systems
  - **Soft contact lenses (e.g. bandage lens)**
    - Absorb drugs from solution and slowly release medication when placed on eye
    - Most useful type of cl for drug delivery e.g. bullous keropathy, severe dry eye, corneal erosions
  - **Collagen shields (e.g. dry eye)**
    - Absorb drugs from solution and slowly release medication when placed on eye
    - Thin membranes of porcine or bovine scleral collagen shaped like a CL that contain a solution of water-soluble drug trapped in the collagen matrix and dissolve over time
  - **Cotton pledgets (e.g. acute anterior uveitis)**
    - using the cotton tip of an applicator to form a small long ball of cotton and soaking it with the drug solution before placing the pledget in the inferior conjunctival fornix
    - sometimes used preoperatively as a method of delivery for achieving effective dilation
    - Manage posterior synechiae during episode of acute anterior uveitis where they can be used to deliver a more sustained dose of phenylephrine to the iris
    - Little evidence of discomfort or infection
- 1st order kinetics: drug concentration dependent
- Drug release systems aim: overcome disadvantages of other topical means of admin
- Commercial use slow-release drug systems
  - Ocusert (pilocarpine) [no longer available] and Lacrisert (hydroxypropyl cellulose)
  - Both approach zero order kinetics - independent of drug concentration
  - Instilled in conjunctival fornix to manage anterior eye conditions
  - Many drug release systems under development for intravitreal delivery (e.g. dexamethasone (Surodex), triamcinolone, fomivirsen, ganciclovir (Vitrasert), fluocinolone acetonide (Retisert), anti-VEGF, etc.)
  - Expensive, poor patient acceptance

Compounded ophthalmic preparations

- Indications
  - Commercial product or therapeutic alternative does not exist
  - Liquid formulation is needed (e.g. patient cannot swallow a tablet or capsule)
  - Allergy or sensitivity to an excipient in the commercial product (e.g. colouring agent or preservative)
  - Different dose or concentration required (e.g. fortified antibiotics)
  - Product discontinued or out of stock or not marketed in Australia
  - Unstable product must be prepared fresh to maintain potency and stability
  - Specific formulation required for a trial
- Important to provide px with clear instructions and possible adverse effects and NEVER attempt to compound yourself

Examples

Active ingredient	Uses
Adrenaline (epinephrine)	Open-angle glaucoma
Amphotericin	Fungal infections
Atropine 0.01%	Slow progression of myopia
N-acetyl cysteine	Mucolytic agent in dry eye therapy; alkali burns
Cefazolin with gentamicin or vancomycin	Treat/prevent bacterial infections
Cocaine (with or without homatropine)	Local anaesthetic (with or without pupillary dilation); diagnosis of Horner's syndrome
Cyclosporin	Reduce inflammation; dry eye
Dexamethasone without preservative	Reduce inflammation; dry eye
Ethylenediaminetetraacetic acid	Chelate calcium plaques
Oestradiol	Dry eye
Foscarnet	Cytomegalovirus anterior uveitis
Hypertonic saline 5%	Corneal oedema
Sodium bicarbonate	Removing mucus
Tacrolimus	Reduce inflammation for dry eye; allergy
Testosterone	Dry eye
Vancomycin	Bacterial infection
Voriconazole	Fungal infections
Zinc and adrenaline	Astringent with vasoconstrictor

Official list of medicines optometrists may prescribe (Appendix B)
Drug Classes Optometrists Can Prescribe:
Anti-infectives:
Topical antibiotics: chloramphenicol, fusidic acid, tobramycin, etc.
Antivirals: topical aciclovir, oral valaciclovir (HSV, HZV only)
Oral antibiotics for specific use (e.g. doxycycline for meibomianitis)

Anti-inflammatory Agents:

Corticosteroids: prednisolone acetate, fluorometholone, loteprednol
NSAIDs: ketorolac, diclofenac, flurbiprofen
Immunomodulators: cyclosporin, lifitegrast

Anti-allergic Agents:
Olopatadine (antihistamine/mast-cell stabiliser)

Anti-glaucoma Medications:
Beta-blockers: timolol
Prostaglandins: latanoprost, travoprost
Carbonic anhydrase inhibitors: brinzolamide, dorzolamide
Alpha agonists: brimonidine

Mydriatics, Cycloplegics & Miotics:
Atropine, tropicamide, cyclopentolate, phenylephrine, pilocarpine

Ocular Lubricants:
Hypromellose, carbomer, sodium hyaluronate, paraffin-based gels

Additional Notes:
- PBS prescribing only applies to a subset of these drugs
- Some require streamlined authority (e.g. preservative-free unit-dose lubricants)
- Ophthalmologist consultation needed for some oral antibiotics (e.g. ciprofloxacin)

Week 2: Ophthalmic Medications

2.1. Antibacterials

- Mode of action
- Spectrum
- Indications
- Contraindications
- Support Antimicrobial Stewardship (AMS)

Bacterial Cell Wall and Ribosomes (Review)

Parameter	Gram-positive bacteria	Gram-negative bacteria
Cell Wall	A single-layered, smooth cell wall	A double-layered, wavy cell-wall
Cell Wall thickness	The thickness of the cell wall is 20 to 80 nanometres	The thickness of the cell wall is 8 to 10 nanometres
Peptidoglycan Layer	It is a thick layer/ also can be multi-layered.	It is a thin layer/ often single-layered.
Teichoic acids	Teichoic acids are present.	Teichoic acids are not present.
Lipopolysaccharide	Lipopolysaccharide is not present.	Lipopolysaccharide is present.
Outer membrane	The outer membrane is not present.	The outer membrane is mostly present.
Lipid content	The Lipid content is very low.	The Lipid content is 20% to 30%.
Resistance to Antibiotic	These are very susceptible to antibiotics.	These are very resistant to antibiotics.

- **Bacteria cell walls** are complex, semi rigid structure that give bacteria their various shapes and protect them from the outside environment
  - Main component of bacterial cell wall is peptidoglycan
- **Gram-positive bacteria** have thicker peptidoglycan layer/cell wall than gram negative
  - Do not have additional outer membrane
- **Gram-negative bacteria** – thin layer peptidoglycan layer or cell wall, negatively charged out membrane made of lipopolysaccharides (LPS)
  - Lipid portion of the LPS is toxic to humans when released in the host – referred to as "endotoxin"
- **Ribosomes** present in cells of prokaryotes (bacteria and archaea) and eukariotes (amoeba, fungi, animals, plants) are responsible for protein synthesis
  - Bacterial ribosomes are smaller and less dense than the ribosomes of eukariotics cells (which have a 80S ribosomes) ➔ called 70S ribosomes
  - Two subunits ➔ 30S smaller subunit, larger 50S subunit
- Unit of measurement for ribosomes is the **Svedberg unit** ➔ measure of the rate of sedimentation in centrifugation rather than size ➔ fragments do not add up (e.g. 70S made of 50S and 30S)
- Antibiotics work by inhibiting protein synthesis by attaching to or attacking the 30S and 50S subunits (e.g. gentamicin, erythromycin, chloramphenicol)
- Bacteria produces **toxins**, two main types:

Exotoxins	Endotoxins
• Proteins that are produced and secreted by certain types of bacteria • When released by bacteria, diffuse into surrounding medium or tissue • Most commonly gram-positive bacteria as part of their growth and metabolism • **Example**: Gram positive Staphylococcus aureus • Act a specific sites ➔ characteristic, severe diseases	• Lipid portions of lipopolysaccharides (LPS) • Protein complexes that form part of the structure of the cell wall of gram negative bacteria • Liberated when the bacteria lyses or dies • Throughout body and produce lower grade reactions (fever, weakness, aching)

  - Some gram negative bacteria can produce both endotoxins and exotoxins
    - Pseudomonas Aeruginosa produces LPS/endotoxin, as well as at least three exotoxins ➔ exotoxin A is the most virulent

Gram Staining

- Thin film of material containing microorganisms spread over surface of a slide
- Allowed to air dry before fixed (passing through flame with methyl alcohol)
- Smear is covered with basic purple dye crystal violet
- Stain is washed off and intensified with "mordant" iodine
- Alcohol wash decolourises gram-negative bacteria which becomes colourless, whereas gram-positive bacteria retains the purple stain, and the safranin counterstain turns the gram-negative bacteria pink

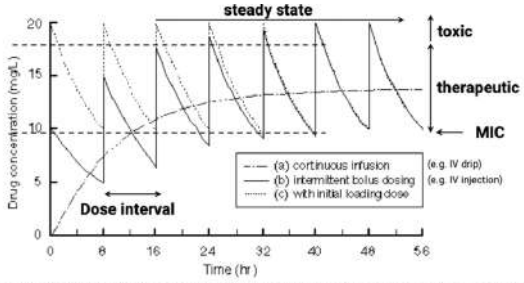
Antibiotics

- Substance that kills or halts the growth of micro-organism
- **Bactericidal:** kills bacteria
- **Bacteriostatic:** prevent them from reproducing or slow their growth
- Bactericidal medicines tend to act on bacterial cell wall – complete disintegration of the bacterium and cell death
- Bacteriostatic medicines will act on bacterial protein formation where they can prevent proper functioning – without causing bacterium death

Terminology

- **Prophylactic Therapy:** medication (or treatment) designed and used to prevent a disease from occurring
- **Empirical Therapy:** when antibiotics are given to a person before the specific bacterium causing an infection is known. Treatment is commenced based on the most likely causative organism.
- **Directed Therapy:** directed use of antibiotics to treat an infection where the pathogen has been identified
- **Minimal Inhibitory Concentration (MIC):** lowest drug concentration that inhibits bacterial growth. Determined through *in vitro* testing of concentrations and agents.
  - Low MIC – indication of a better antimicrobial agent
- **Post Antibiotic Effect (PAE):** persistent suppression of bacterial growth by an antibiotic, even in the absence of host defense mechanisms
  - Potentially effective at concentrations below MIC

Pharmacokinetics



Loading dose: An initial higher dose of a drug given at the beginning of a course of treatment to ensure that therapeutic level is reached quickly.

- Aim of drug therapy ➔ above minimum inhibitory concentration (MIC) and below toxic level ➔ "therapeutic range"

- Loading is often considered for the eye in management of bacterial keratitis → important to reach effective drug concentration quickly and halt the progression of the disease

Antibiotic Creed (MIND ME)

- **M** microbiology guides therapy wherever possible
  - Choice of antibiotic is based either of the "most likely" organism as in empirical treatment or that a diagnostic specimen is taken
- **I** indications shoulds be evidence-based
- **N** narrowest spectrum therapy required
- **D** dosage individualised to the patient and appropriate to the site and type of infection
  - E.g. age, pregnancy and lactation
- **M** minimise duration of therapy → less risk of adverse reactions
- **E** ensure oral therapy is used where clinically appropriate
  - Associated with less adverse reactions and less costly than IV administration
  - Topical therapy should generally be avoided to treat systemic infections

- Topical ophthalmic preparations of many of the medicines on your list are not available commercially (e.g. azithromycin, bacitracin, cephazolin, gramicidin, neomycin, polymyxin and tetracycline are not)

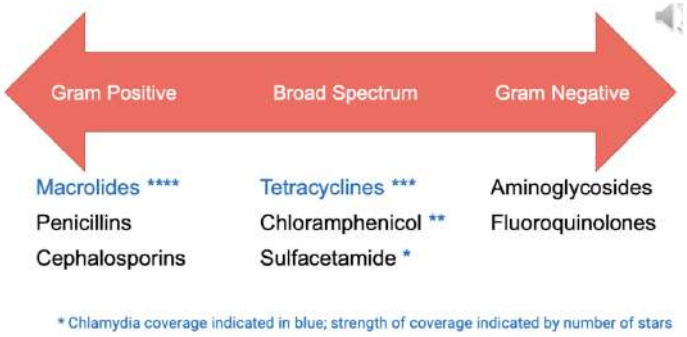
Antibiotics: Mode of Action

- Three groups:
  - Those that act of bacterial cell wall synthesis
  - Those that act on bacterial nucleic acid synthesis
  - Those that act on bacterial protein synthesis

Antibiotic Mode of Action

Bactericidal	Bacteriostatic
Penicillins	Tetracyclines
Cephalosporins	Chloramphenicol
Aminoglycosides	Sulphonamides
Fluoroquinolones	
Macrolides*	

- Medicines that act on the cell wall or nucleic acid synthesis are largely **bactericidal** whereas medicines that are protein synthesis inhibitors are largely **bacteriostatic** (exception: aminoglycosides are largely bactericidal)



- Inhibitors of cell wall synthesis such as penicillins and cephalosporins tend to be mostly effective against gram positive bacteria
- Aminoglycosides and fluoroquinolones are relatively broad spectrum, but more effective against gram-negative bacteria
- Macrolides are used not primarily because of their effectiveness against gram positive bacteria → good effectiveness against the gram-negative intracellular bacteria Chlamydia

Approved Optometrist Rx Anti-infective Medicines — Commercially Available

Chloramphenicol (S3, PBS Restricted)	Protein Synthesis Inhibition	Broad-spectrum
Framycetin (PBS) **Tobramycin** (PBS Restricted) • Management of suspected pseudomonas eye infection of preoperative use on ophthalmic surgery	Protein Synthesis Inhibition Cell Wall Disruption	**Gram-ve** >>> Gram+ve
Ciprofloxacin (PBS Authority) **Ofloxacin** (PBS Authority) • Both above: management of bacterial keratitis, treated by ophthalmologist	DNA Gyrase Inhibition	**Gram-ve** > Gram+ve
Aciclovir (PBS Restricted - HSK)	DNA Polymerase Inhibition	Anti-Viral
Dibromopropamidine (S2) **Propamidine** (S2)		Antiseptic Antiprotozoal

Not Commercially Available

Tetracycline	Protein Synthesis	Broad-spectrum
Azithromycin	Inhibition	
Bacitracin **Cephazolin**	Cell Wall Synthesis Inhibition	Gram +ve
Gramicidin **Polymyxin**	Cell Wall Disruption	Gram +ve
Gentamicin **Neomycin**	Protein Synthesis Inhibition Cell Wall Disruption	**Gram -ve** > Gram+ve
Ganciclovir	DNA Polymerase Inhibition	Anti-Viral

2.2. Anti-bacterials Part 2

- Focus on aminoglycosides (gentamicin, tobramycin), macrolides, tetracyclines, chloramphenicol.
- Aminoglycosides: bactericidal, protein synthesis inhibition (30S), effective against gram-negatives.
- Gentamicin & tobramycin used topically, not orally due to toxicity.
- Macrolides (erythromycin): 50S inhibitors, safer, used in children.
- Tetracyclines: broad spectrum, anti-inflammatory in blepharitis (e.g. doxycycline).
- Chloramphenicol: broad spectrum, commonly used for conjunctivitis.
- Can cause aplastic anaemia (rare), avoid in infants under 2.
- Compounding is essential for many agents not commercially available (e.g. cephazolin).
- Bacteriostatic vs bactericidal choice depends on site/severity of infection.
- Adverse effects vary with class — eg. ototoxicity (aminoglycosides), GI upset (macrolides).

2.3. Anti-bacterials Part 3

- Focus on **fluoroquinolones**: ciprofloxacin, ofloxacin (DNA synthesis inhibitors).
  - Mode of action: block DNA gyrase/topoisomerase IV → halt replication/transcription.
  - Broad spectrum: gram-negative and gram-positive activity.
- Bactericidal; risk of resistance means they're reserved for serious infections.
- Ciprofloxacin may cause white corneal precipitates; ofloxacin better corneal penetration but ↑ perforation risk.
- Rare systemic effects: tendon rupture, CNS symptoms (headache, dizziness).
- Risk of photosensitivity; caution with NSAIDs (CNS toxicity).
- Pregnancy/breastfeeding: low concern due to minimal systemic absorption.
- Topical dosage in keratitis: q15min loading → q1h then taper.

- Oral ciprofloxacin used for deep corneal ulcers.
- Should be reserved for severe conjunctivitis only; chloramphenicol preferred otherwise.

2.4. Anti-bacterials and Anti-infectives

- Antivirals: aciclovir, valaciclovir (HSV, HZV), topical ganciclovir (e.g. for dendritic ulcers).
- Aciclovir inhibits viral DNA polymerase, selective for infected cells.
- Fungal keratitis: natamycin (preferred), amphotericin B, voriconazole (compounded).
- Natamycin effective for filamentous fungi like Fusarium.
- Acanthamoeba keratitis: treat with PHMB, chlorhexidine (protozoal agents).
- Polypharmacy often required due to multiple microbial forms.
- Importance of lab diagnosis (PCR, culture) for accurate treatment.
- Topical antivirals not effective for HZO keratitis ➔ use systemic therapy.
- Emerging resistance in immunocompromised patients warrants systemic monitoring.

Table 5.17 – Classification of antiviral drugs

Class	Drug	Primary indications
guanine analogues	aciclovir, famciclovir	herpes simplex, shingles
	valaciclovir	herpes simplex, shingles, CMV
	ganciclovir, valganciclovir	CMV
neuroaminidase inhibitors	oseltamivir, peramivir, zanamivir,	influenza A and B
HCV NS3/4A inhibitors, HCV NS5A inhibitors, HCV NS5B nucleotide inhibitors	ledipasvir, pibrentasvir, velpatasvir, glecaprevir, voxilaprevir, sofosbuvir	hepatitis C
other antivirals	amantadine	influenza A
	adefovir, entecavir, peginterferon alfa-2a	hepatitis B
	cidofovir, foscarnet, letermovir, maribavir	CMV
	palivizumab	respiratory syncytial virus
	casirivimab and imdevimab, molnupiravir, nirmatrelvir and ritonavir, remdesivir, sotrovimab, tixagevimab and cilgavimab	coronavirus disease 2019 (COVID-19)
	ribavirin	hepatitis C, respiratory syncytial virus

Modified from AMH

Fungus

- Are eukaryotes containing a distinct nucleus with the cell's DNA
- Can be uni or multicellular
- **Yeast**: round, unicellular & slightly larger than bacteria
- **Molds**: multicellular & consists of long branching filaments (hyphae) that intertwine & branch
- Antifungals consists of 4 major classes:
  - **Polyenes**: bind the ergosterol in the fungal cell membrane, increasing cell permeability
  - **Azoles**: impairs biosynthesis of ergosterol in fungal cell membrane & increases cell permeability through the fungal cytochrome P450 enzymes ➔ fungistatic
  - **Pyrimidines**: block thymidine synthesis in fungal organisms which impairs DNA & protein synthesis ➔ fungistatic
  - **Echinocandins**: inhibit glucan synthesis, causing abnormal fungal cell wall
- Antifungals are not listed on OBA ➔ not commercially available as ophthalmic preparations
- Modern treatment of fungal keratitis would involve a combination of polyene & azole
- Fungal infections of the eye required **prolonged treatment**

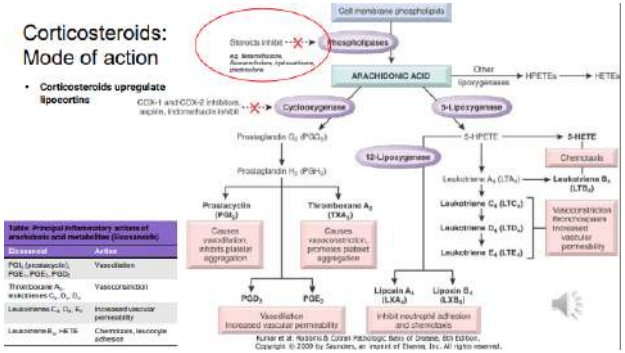
2.5. Corticosteroids and Immunomodulatory Medicines

- Corticosteroids: inhibit phospholipase A2, reduce prostaglandin/leukotriene production.
  - **Act higher up the pathway than NSAIDs.**
  - They inhibit phospholipase A2 via lipocortin induction, which blocks the formation of arachidonic acid itself.
  - Result: Reduces both prostaglandins and leukotrienes.
  - Cortico- derived and related to the adrenal cortex ➔ mimic effects of adrenal hormones (above kidneys)
- Five optometrist-approved corticosteroids: dexamethasone, prednisolone, loteprednol, fluorometholone, hydrocortisone.
- NSAIDs (Non-Steroidal Anti-Inflammatory Drug): ketorolac, flurbiprofen, diclofenac (anti-inflammatory but less potent than steroids).
  - **Block COX enzymes** ➔ reduce prostaglandin synthesis ➔ reduce pain, inflammation, fever.
- These inhibit both COX-1 and COX-2 (non-selective)
- Immunomodulators: cyclosporin (PBS authority), lifitegrast (not available).
- Potency: dexamethasone > prednisolone acetate > fluorometholone > hydrocortisone.
- Formulation impacts penetration: acetate (lipid soluble) > phosphate (water soluble).
- AE: increased IOP, cataracts (long-term), delayed healing, infection masking.
- Contraindicated in active HSV, fungal infections, glaucoma.
- Short-term use generally safe; monitor for IOP rises.
- Safe in pregnancy and breastfeeding with caution.

Approved Optometrist Rx Anti-inflammatory Medicines

Dexamethasone (PBS) Prednisolone (PBS restricted) Loteprednol * Fluorometholone Hydrocortisone (PBS)	Corticosteroid	Potency +++ to — Potency
Diclofenac * Flurbiprofen * Ketorolac § (see note)	NSAID	
Ciclosporin (PBS authority) Lifitegrast *	Immunosuppressive	

*Not commercially available – compound Rx
§ Note: **Nepafenac** is an NSAIDs suspension eyedrop commercially available in Australia but not on OBA list



Corticosteroids: Mode of Action

- When cells are activated or damaged by a stimuli (eg. microbe), cell membrane arachidonic acid (AA) is rapidly converted by the action of enzymes to produce cell-derived mediators (prostaglandins (PG) & leukotrienes (LT))
  - Lipid mediators known as AA metabolites or eicosanoids
- AA derived mediators are synthesised by 2 major classes of enzymes:
  - Cyclooxygenase (COX-1 & COX-2): generates PGs, prostacyclin & thromboxane
  - Lipoxygenases: producers LT & lipoxins
- PGs are involved in the pathogenesis of pain & fever in inflammation
  - Produces local effects on blood vessels & plasma proteins

Steroid-Induced IOP

3 proposed explanations:

1. Trabecular meshwork have alpha & beta steroid responsive receptors
   - Alpha receptor is responsible for producing the ECM
   - Beta receptor naturally inhibits the alpha receptor
2. Steroids inhibit the normal activity of beta receptors, leading to an overactivity of ECM producing alpha receptors
   - Steroids also inhibit the mechanisms that degrade ECM that help keep the channels open
   - Results in accumulation of ECM & debris in the trabecular meshwork
3. Steroids also cause cross-linking of action which laters the trabecular cytoskeleton
   - Cross-linking makes any blockage stronger/ less permeable to flow
   - Steroids inhibits phagocytosis by trabecular meshwork cells, reducing elimination of blockages

Corticosteroid: Tapering

- Is required due to the possible rebound effect with corticosteroids of the relapse of the underlying disease

- Involves gradually decreasing the dose rather than abruptly stopping
- Is required after chronic treatment with topical steroids since corticosteroids reduce blood leukocyte elements
    - This removes mature leukocytes, leaving immature WBC to proliferate when therapy stops
        - Immature WBC produces large quantities of antibodies to any residual antigens ➜ can generate antigen-antibody reaction, leading to recurring, serious & necrotising inflammation
    - Tapering schedule depends on the length of treatment
        - Slow taper: weeks to months
        - Fast taper: less than 1.5 weeks

A potential tapering schedule (assumes initial four times a day dosage for 1 week):

- 1 drop three times per day X 3 days
- 1 drop twice a day X 2 days
- 1 drop once a day X 1 day,
- 1 drop every other day
- then discontinue



Role in inflammation	Mediators
Vasodilation	**Prostaglandins** Nitric oxide Histamine
Increased vascular permeability	Histamine and serotonin C3a and C5a (by liberating vasoactive amines from mast cells, other cells) Bradykinin **Leukotrienes C4, D4, E4** PAF Substance P
Chemotaxis, leukocyte recruitment and activation	TNF, IL-1 Chemokines C3a, C5a **Leukotriene B4** (Bacterial products, e.g., N-formyl methyl peptides)
Fever	IL-1, TNF **Prostaglandins**
Pain	**Prostaglandins** Bradykinin
Tissue damage	Lysosomal enzymes of leukocytes Reactive oxygen species Nitric oxide

Important adverse effects of corticosteroids

Glucocorticoid effects
Avascular necrosis of the femoral & humeral head
Dermatological effects (eg skin atrophy, purpura, ecchymoses, striae, hirsutism)
GI effects (eg dyspepsia, peptic ulceration, GI bleeding)
Growth retardation – monitor in children
Immunosuppression, risk of infections
Metabolic effects (eg hyperglycaemia, hypertriglyceridaemia)
Myopathy
Ocular effects, particularly increased IOP & cataracts
Osteoporosis
Adrenal suppression (see tapering)
Psychological & sleep disturbances (eg euphoria, depression, paranoid psychosis)
Weight gain & redistribution of fat
Mineralocorticoid effects
Hypertension
Hypokalaemic alkalosis
Sodium-retaining effects (oedema)



Most of these AEs relate to long-term treatment and do not preclude short-term use

Comparison of ocular corticosteroids

Drug and form	Potency (relative to hydrocortisone)	Penetration[1] (Ranked relative effect within the corneal stroma)	IOP rise Mean peak aqueous concentration (ng/ml)	Trade Name	Time to peak (minutes)
Prednisolone acetate 1%, phenylephrine hydrochloride 0.12% suspension	4x high	1 +++	++++ 669.9	Prednefrin Forte	120
Dexamethasone (alcohol) suspension, 0.1%	25x high	3 ++	++++ 31	Maxidex	91-120
Fluorometholone (acetate) suspension, 0.1%	40x high	2 ++	+++ Not available	Flarex (not available)	Not available
Loteprednol (etabonate), 0.5%	20x high	4 ++	?++? Not available	Lotemax (not available)	Not available
Fluorometholone (alcohol) suspension, 0.1%	40x mid	5 -	+++ 5.1	FML Liquifilm	30-60
Prednisolone (sodium phosphate) solution, 0.5%	4x mid	6 +	+++ 25.6	Minims Prednisolone	90-240
Hydrocortisone (acetate) ointment 1%	1x low	7 ++	++ Not available	Hycor	Not available

Steroid Responders

- 1st group – 1.6mm Hg elevation (< 5 mmHg) from the outset with no continuous rise during course of tx – NON responders (66% of the population)
- 2nd group – 10mm Hg elevation (6 to 15mm Hg) rising continuously during course of tx (30-40% of population) = STEROID RESPONDER
- 3rd group: >16mm Hg elevation rising continuously during course of tx (5% of population)

Sustained Drug Delivery Devices for the Eye – Hui

- Ocular drug delivery is challenging due to barriers (cornea, blood-ocular).
- <5% of topical drug reaches target tissues.
- Devices for post-op inflammation: Dexycu (intracameral), Dextenza (punctal plug).
- Allergy: ketotifen-releasing contact lenses (TheraVision).
- IOP control: Ocusert (pilocarpine), Durysta (bimatoprost implant), Lacrisert (HPC insert).
- Posterior inflammation: Ozurdex (biodegradable dexamethasone), Retisert (fluocinolone), Vitrasert (ganciclovir).
- Retisert has risk of IOP elevation, cataracts, spontaneous pellet dislocation.
- Ozurdex better safety profile but requires repeat dosing.
- Future: nanocarriers, microneedles, and hydrogel-based implants under development.

Cochrane Review – CD001211

- Systematic review comparing antibiotics vs placebo in acute bacterial conjunctivitis.
- Small benefit in clinical and microbiological remission with antibiotics.
- Many cases self-resolve without treatment.
- Antibiotics may reduce symptom duration by 1–2 days.
- No major safety concerns with topical antibiotics.
- Recommends careful consideration before prescribing to avoid resistance.

Phu – Review of Systems (SOVS)

- 7 B's for systemic history: Body/Back, Brain/Senses, Beat/Breath, Belly/Bum, Blemishes, Blood, Bugs/Drugs.
- Review of systems is essential for identifying systemic associations (e.g., uveitis, arcus).
- Ask layperson-friendly screening questions, tailored to signs (e.g. "tired?" for pallor).
- Case 1 (AAU): consider MSK (e.g. joint pain), respiratory (e.g. TB), GI (IBD).
- Refine systemic history for differentials (RA vs OA, active TB vs latent).
- HLA-B27 not always diagnostic; used with clinical context.
- Case 2 (glaucoma): assess systemic risk (diabetes, HTN, hyperlipidaemia).
- Consider neurologic signs (e.g. migraine, intracranial mass) in glaucoma-like presentations.
- Review of systems prevents diagnostic overshadowing and supports holistic care.

Tutorial Notes (Week 2)

- Prescribing mindset: Diagnose ➜ Check OBA list ➜ PBS ➜ Commercial availability ➜ Consider compounding.
- Learn when to use empirical vs directed therapy.
- Practice writing prescriptions for real cases (e.g. child, aplastic anaemia, ointments).
- **Case practice**: red eye, differentials like conjunctivitis, uveitis, keratitis.
- Differential exclusion based on signs/symptoms (e.g. itchiness, discharge, pain).
- **Cochrane Review**: antibiotics mildly improve resolution in conjunctivitis.
- **Infection control:** PPE, no-touch technique, equipment disinfection.
- Workup indications: chronic/recurrent conjunctivitis, neonatal, suspected chlamydia/gonorrhoea.
- **Important tools:** AMH, eTG, Optometry Australia infection control guide.

Week 3: Ophthalmic Medications (Part 2)

3.1. Anti-allergic Medicines 2025

- Covers pharmacological options for allergic eye disease.
- Differentiates Type I hypersensitivity as immediate, IgE-mediated reactions.
- Histamine effects via H1 and H2 receptors: vasodilation, itch, mucus secretion, bronchospasm.

Histamines

Tissue	Histamine Receptor	Effects / Function
Vascular system		
Capillary (microcirculation)	H_1 and H_2	Dilation, increased permeability
Arteriole (smooth muscle)	H_1 and H_2	Dilation (allergic response)
Smooth muscle		
Bronchial, bronchiolar	H_1	Contraction (bronchospasm)
Gastrointestinal	H_1	Contraction
Exocrine glands		
Gastric	H_2	Gastric acid secretion (HCl)
Epidermis	H_1	Triple response (flush, flare, wheal)
Adrenal medulla	H_1	Adrenaline and noradrenaline release
Central nervous system	H_1	Motion sickness, nausea, alertness
	H_3	Decrease release of neurotransmitters
Intestine	H_4	Unknown
Blood leukocytes		

Approved Optometrist Rx Anti-allergy Drugs

Antazoline Levocabastine Pheniramine	Antihistamines	OTC
Cromoglycate Lodoxamide	Mast cell stabilisers	OTC
Azelastine Ketotifen Olopatadine	Antihistamines + Mast cell stabilisers	OTC OTC S4
Naphazoline Tetryzoline*	*** Do not use ***	Vasoconstrictors (Decongestants)	OTC

+ anti-inflammatories (corticosteroids, NSAIDs, immunosuppressants) on your list

- Anti-allergic strategies:
    - Mast cell stabilisers prevent histamine release.
    - Antihistamines block H1 receptors.
    - Steroids and NSAIDs reduce inflammation.
        - **Steroids** are used in more severe cases of inflammation – where there is a potential for tissue damage. It is also used in chronic conditions such as macular oedema or optic neuritis. Also used for autoimmune inflammatory conditions, Sjorgen's syndrome or ocular pemphigoid.
        - **NSAIDs** are less potent → preferred for short-term and post-operative, there is also less risk of increased IOP (keep in mind if the px has glaucoma), and side effects like cataracts and immune suppression
- Common agents:
    - Mast cell stabilisers: Cromoglycate, Lodoxamide.
    - Antihistamines: Azelastine, Levocabastine.
    - Combinations: Ketotifen, Olopatadine (S4).
    - Oral antihistamines:
        - Sedating: Diphenhydramine, Promethazine.
        - Less sedating: Cetirizine, Loratadine, Fexofenadine.
    - Adverse effects of sedating agents: drowsiness, confusion, dry mouth, urinary retention.
    - Less sedating options preferred due to reduced CNS penetration.
    - Vasoconstrictors (e.g., Naphazoline, Tetryzoline) are discouraged due to rebound redness.
    - Clinical considerations for pregnancy, breastfeeding, children, and elderly.
    - DO NOT USE VASOCONSTRICTORS OR DECONGESTANTS FOR MANAGEMENT OF OCULAR ALLERGIES

Hypersensitivity Reactions

Type I (immediate) hypersensitivity: associated with the release of histamine

- IgE antibody is produced by activated B-lymphocytes after exposure to an allergen
- IgE is bounded to mast cells & basophils, allowing for the release of vasoactive amines & other mediators from mast cells
- Inflammatory cells are finally recruited → produces effects such as vascular dilation, oedema, smooth muscle contraction, mucus production, tissue injury & inflammation

Allergen: protein to which the immune system responds with a type I hypersensitivity reaction

Histamine: chemical mediator that occurs naturally & is stored in almost all body tissues at the cytoplasmic granules of the mast cell

- Highest concentrations in skin, lung & GIT
- Mast cells degranulate through stimulation by the immune system, tissue injury or pharmacologically
    - Mast cells release histamines which bind to histamine receptors prevent at tissues including blood vessels & nerves
- **Induces vessel dilation & nerve stimulation → itching**

Mast cell stabilisers: prevents release of histamine

Antihistamines: blocks the availability of histamine receptors in various tissues

- There are 4 types of histamine receptors
    - Antihistamines target H1 receptors
- **Steroids prevent degranulation** by inhibiting the recruitment & maturation of the mast cells
- Antihistamines & steroids prevent released histamine from having an effect or reduces its impact
    - Steroids prevent the synthesis of proinflammatory mediators following mast cell & other cell membrane degradation
- Majority of anti-allergic medications on OBA list are S2
    - **Olopatadine is the only S4 medicine** → is a combination drug containing both a mast cell stabilisers & H1 selective antihistamine

3.2. Mydriatics and Cycloplegics

Comparison of ocular anticholinergics

Medicine / use	Peak effect[1]	Duration[1]	Systemic adverse effects
Atropine 1%			
mydriasis	30-40 minutes	7-10 days	+++
cyclopegia	1-3 hours	7-14 days	
Cyclopentolate 0.5-1%			
mydriasis	30-60 minutes	1 day	++
cyclopegia	25-75 minutes	6-24 hours	
Tropicamide 0.5-1%			
mydriasis	20-40 minutes	6 hours	Rare
incomplete cycloplegia	30-40 minutes	2-6 hours	
Homatropine 2% (NOT COMMERCIALLY AVAILABLE)			
Mydriasis	20-30 minutes	6 hours – 4 days	+
Cycloplegia	30-90 minutes	10 hours – 2 days	



- Mydriatics dilate pupils via sympathomimetic action (phenylephrine).
- Effects of the sympathetic nervous system (SNS), which is responsible for the "fight or flight" response.
- Cycloplegics inhibit accommodation by blocking muscarinic receptors (atropine, cyclopentolate).
- Indications:
    - Refraction in children.
    - Uveitis management.
        - Uveitis, particularly anterior uveitis, often causes iritis (inflammation of the iris), which leads to pain due to spasm of the ciliary muscle. Cycloplegic drugs, such as atropine, tropicamide, or cyclopentolate, help paralyze the ciliary muscle, preventing painful spasms.
    - Anterior segment examination.
- Drug profiles:
    - Tropicamide: rapid onset, short duration.
    - Cyclopentolate: intermediate.
    - Atropine: long-lasting, risk of systemic toxicity.
- Side effects: photophobia, blurred vision, systemic anticholinergic symptoms.
- Cautions in Down syndrome, children, narrow angles.
- Contraindications in angle-closure risk.
    - Cycloplegics, such as atropine, tropicamide, or cyclopentolate, work by paralyzing the ciliary muscle and causing pupillary dilation (mydriasis). This dilation reduces the outflow of aqueous humor through the trabecular meshwork in the angle of the eye.
- Use of phenylephrine can raise IOP, cause rebound miosis.

3.3. NSAID Medicines

- Non-steroidal anti-inflammatory drugs inhibit COX enzymes, reducing prostaglandin synthesis.

- Used in pain relief, post-operative inflammation, allergic conjunctivitis.
- Common agents:
    - Ketorolac, Diclofenac, Bromfenac, Nepafenac.
- Side effects: burning/stinging, delayed corneal healing, epithelial defects.
- Less systemic side effects than steroids but also less potent.
- Careful use in dry eye patients, combination therapy often employed.

3.4. Pain Management for Optometrists – Part 1

Analgesics: ability to modulate pain sensation

- Are classified into:
    - Anaesthetics: interrupts the signal between a peripheral source of pain & the central pain processing centre
        - Not indicated for pain management long term
    - Peripherally acting agents (topical NSAIDs): prevents sensitisation & discharge of nociceptors
    - Centrally acting agents (opioids): interrupts pain message & emotional message, making px indifferent to pain
- Caffeine & tricyclic antidepressants can potentiate the effect of analgesics

Narcotic: substance that induces stupor, coma or insensibility to pain
Opiate: compounds derived from opium (eg. morphine, codeine, heroin)
Opioids: compound that bind opioid receptors
Palliative: aims to reduce suffering & pain
Analgesic: medicine that relieves pain
Euphoria: emotional & mental state of profound wellbeing
Tolerance: reduction of subject's reaction to a medicine

Drug	Type	Onset of action	Duration	Duration of initial sting	Punctate epithelial damage
Tetracaine	ester	20 seconds	20 minutes	30 seconds	+++
Oxybuprocaine	ester	20 seconds	10-20 minutes	10 seconds	++
Proxymetacaine	ester	20 seconds	10-20 minutes	10 seconds	++
Lidocaine	amide	2-5 minutes	20-30 minutes	30 seconds	+

Pain Management

Pain: unpleasant sensory & emotional experience associated with actual or potential tissue damage

- **Acute pain:** less than 3 months duration
- **Chronic pain:** lasting for more than 3 months or persisting beyond the course of an acute disease or after tissue healing
    - Pain often becomes the disease
- Can be classified as:
    - Nociceptive: stimulation of nerve endings
    - Neuropathic: due to injury to the nerves themselves
    - Psychogenic: emotional

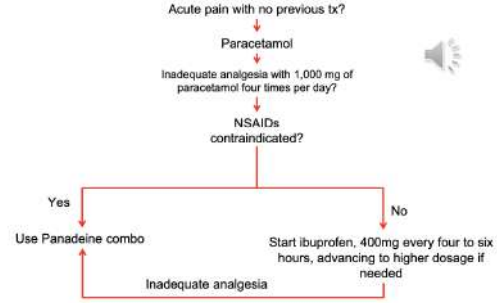
Mild pain: regular paracetamol or NSAID
Moderate pain: opioid in combination with non-opioid agents such as paracetamol
Severe pain: potent opioid with non-opioid

- WHO analgesic ladder:
    - Step 1: non-opioids (e.g., paracetamol).
    - Step 2: weak opioids.
    - Step 3: strong opioids.
- Pain scales: VAS, NRS, Wong-Baker.
- Consider psychological aspects (fear, anxiety).
- Communication skills in pain assessment.
- Placebo and nocebo effects in optometric practice.
- Ethics of prescribing analgesics.

3.5. Pain Management for Optometrists – Part 2

- Focus on ocular pain conditions: corneal abrasions, uveitis, post-surgical pain.
- Local anaesthetics: tetracaine, oxybuprocaine, proparacaine – not for long-term use due to epithelial toxicity.
- **Systemic analgesics:**
    - Paracetamol: first-line.
    - NSAIDs: anti-inflammatory action.
    - Codeine: weak opioid, often in combo with paracetamol. ➜ these are used in combination because they have a synergistic effect and because they have different mechanism of action
        - Paracetamol is an analgesic and antipyretic (pain reliever and fever reducer). It works primarily in the central nervous system by inhibiting an enzyme called COX (cyclooxygenase), but its exact mechanism is not fully understood. It primarily affects prostaglandin synthesis in the brain, helping to reduce pain and fever.
        - Paracetamol unlike NSAIDs do not have an anti-inflammatory effect ➜ NSAIDs are reserved for pain with an inflammatory component (it is usually the cause of the pain). Paracetamol is the first choice to avoid side effects ➜ NSAIDs have kidney risk
    - Codeine is an opioid analgesic. It works by binding to opioid receptors in the brain and spinal cord, modulating pain perception by altering the way the body processes and responds to pain. Codeine is converted into morphine in the liver, which then provides pain relief through this action.
- Use of topical NSAIDs and steroids for post-op pain and inflammation.
- Risk management:
    - Avoid overprescribing.
    - Recognise red flags for serious pathology.
- When to refer for pain: suspected scleritis, angle-closure glaucoma, orbital cellulitis.

Ocular Pain Management Paradigm



Neurotrophic Pain Management

- **Tricyclic antidepressants (TCA) & anti-epileptic** medicines can be used for px who complaints or symptoms are out of proportion to the clinic signs
    - There is injury to the nerve themselves, not just tissue damage
    - TCAS:
        - Amitriptyline (Endept, Entrip) 10-25 mg at night, titrate up to maximum 150mg
        - Duloxetine (painful diabetic neuropathy) (Andepra, Cymbalta, Depreta, Dytrex, Tixol) 60mg once daily
    - Anti-epileptic medicines:
        - Gabapentin (Gapentin, Gabapentin, Neurontin, Nupentin, Gabacor) 100- 300 mg at night, increase dose graduate according to response (up to 1.8- 3.6g daily in 3 doses usual)
        - Pregabalin (Lypralin, Lyrica, Lyzalon) 75 mg at night, increase to 150mg daily in 1-2 doses (maximum 300 mg twice daily)
        - Carbamazepine (trigeminal neuralgia) (Tegretol) 50-100 mg twice daily
- All TCAs appear equal
    - Dyloxetine has been shown to be effective in painful diabetic neuropathy
- Indications for use include post-herpetic neuralgia, neuropathic dry eye, post refractive surgery
- Topical anaesthetic agents can help to determine whether the pain is:
    - Peripherally mediated: disappears with topical anaesthetic use

    - Mediated centrally: remains present following anaesthesia
- Referral to a pain specialist and/or an ophthalmologist is indicated when neuropathic pain treatment is required

NIHMS644976 – Dry Eye Disease

- Explores pathophysiology of dry eye disease: tear film instability, inflammation, neuro-sensory abnormalities.
- Diagnosis via tear breakup time, osmolarity, ocular surface staining.
- Importance of meibomian gland function.
- Treatment tiers:
    - Artificial tears.
    - Anti-inflammatory drops (cyclosporine, steroids).
    - Punctal plugs.
        - Punctal plugs are small devices inserted into the puncta (the tiny openings in the corners of the eyelids that drain tears) to help treat dry eye disease (DED). Their primary function is to block the tear drainage system, thereby increasing the amount of moisture on the eye's surface and improving tear retention. This helps alleviate symptoms of dry eye, such as burning, itching, grittiness, and blurred vision.
    - Oral omega-3 supplementation.
- Highlights emerging therapies and clinical trials.

OPTM 6413 RED EYE PROTOCOLS

- Clinical triage for red eye emergencies: pain, photophobia, vision loss, trauma, chemical burns.
- Examination approach:
    - History: symptom onset, trauma, CL use.
    - Slit lamp: fluorescein staining, grading inflammation.
    - Tonometry, dilation if indicated.
- Key red flags:
    - Chemical injury, corneal ulcer, uveitis, scleritis.
- Special tests:
    - SUN grading for anterior chamber cells/flare.
    - Preauricular lymph node assessment.
- In-office management:
- Pseudomembrane removal.
- Posterior synechiae breaking.
    - Posterior synechiae refer to the adhesion or fusion between the iris and the lens of the eye. This condition typically occurs as a result of inflammation in the anterior chamber, such as in uveitis (especially anterior uveitis). Synechiae can lead to complications such as pain, increased intraocular pressure, and impaired vision.
    - Cycloplegics (e.g., atropine, cyclopentolate) are commonly used to dilate the pupil and reduce the risk of further synechiae formation. In some cases, this medication may help break the adhesion by separating the iris from the lens.

Uveitis is an inflammation of the uvea, which is the middle layer of the eye. The uvea consists of three parts: the iris (the colored part of the eye), the ciliary body (which produces aqueous humor and controls the lens), and the choroid (a layer of blood vessels behind the retina that supplies nutrients). Uveitis can affect any of these structures, leading to symptoms such as pain, redness, light sensitivity (photophobia), blurred vision, and in severe cases, vision loss.

- Review schedule guidance:
    - Daily review for microbial keratitis.
    - Frequent for uveitis and CL-related issues.
- Referral decisions based on severity, scope of practice.

Tutorial Notes (Week 3)

Focus: therapeutic prescribing and anterior eye disease.
Learning objectives:

- Write legal prescriptions.
- Avoid resistance in antimicrobial use.

Referral decisions for corneal ulcers.
Case-based learning:

- Bacterial keratitis in a CL wearer.
- Differentials excluded: AK, HSV, fungal.

EMQ and discussion: ulcer size, position, symptom severity.
Script writing practice:

- Fluoroquinolone or aminoglycoside hourly.
- No chloramphenicol.

Extension:

- Steroid use post-infection control.

Resources recommended:

- AMH, lectures on anti-allergics, NSAIDs, mydriatics.
- Cochrane reviews on DED, blepharitis, allergic conjunctivitis.

Week 4: Ocular Surface Disease

4.1. Ocular Lubricants

- Over the counter (S2)
- PBS listing restricted: severe dry eye syndrome, including Sjogren's syndrome
- Unpreserved preparations restricted and authority required (streamlined) (severe dry eye in patients sensitive to preservatives in multi-dose eye drops)
- Can be catergorised by:
    - Viscosity agent (type or individual agent)
    - Preservative used
    - Broad category (drops, emulsions, gels, ointments)
    - Never versus older type

Formulation

- **Demulcents**: cellulose derivatives (HPMC, CMC), dextran 70, gelatin, liquid polyols (PG, PEG400, glycerin, polysorbate), polyvinyl alcohol (PVA), povidone
    - Another way to describe viscosity
- **Viscosity agents:** demulcents in high concentrations (e.g. CMC 1.0%, HPMC 0.3%), carbopol 980, HP-Guar, petrolatum, mineral oil
- **Preservatives**: thimerosal, chlorobutanol, sorbate, benzalkonium chloride (BAK), polyquaternium, sodium chlorite (purite), sodium perborate
- **Electrolytes**: bicarbonate (foil packaging), potassium, calcium, magnesium, zinc, phosphate
    - Added to ocular lubricant to mimic tear film
    - Bicarbonate reacts with air → therefore need to be dispensed from foil packaging
    - Potassium – contribute to maintenance of corneal thickness
- **Lipids**: soy lecithin, triglycerides, phosphatidylcholine, perfluorohexyloctane
- **Osmolarity**
- **pH**

- **Emulsions**: lipid-containing eye drops are emulsions → contain mixture of 2 liquids that should not normally mix in (lipids in a water-based solution
    - Lipid preparations are available as sprays that are applied to closed eyelids
        - Sprayed droplets migrate down to the lid margin, collecting at the edge & are mixed with the natural term film layer during a normal blink
        - Sprays allow for ease of installation & lack of significant blurriness
- **Cationic emulsions:** designed to interact electrostatically with tear mucins & negatively charge corneal cells (eg. Cationorm)
    - Aims to improve retention time & ocular drug absorption

Viscosity Agents

Viscosity agents

Ingredient	Characteristics	Brands
Polyvinyl alcohol (PVA)	Water soluble viscosity enhancer Commonly used at 1.4% May promote wound healing	Liquifilm Forte, Liquifilm Tears, Murine Revital Eyes, Murine Tears, PVA Forte, PVA Tears, Vistil, Vistil Forte, Refresh
Hydroxypropyl methylcellulose (HPMC)	Prolong tear film wetting 0.5% HPMC better retention than 1.4% PVA	Genteal, In a Wink, Methopt, Poly-Tears, Tears Naturale, Xailin Hydrate
Carboxymethylcellulose (CMC)	Similar to HPMC Greater adherence to mucins	Optive, Refresh Contacts, Refresh Tears Plus, Refresh Liquigel, Cellufresh, Refresh Plus, Celluvisc, Optifresh Plus, Optifresh Tears, Optive Sensitive, Theratears, Xailin Fresh
Sodium hyaluronate	High-molecular weight polymer Non Newtonian fluid - susceptible to shear thinning during blink	Hylo-Forte, Hylo-Fresh, Luxyal, Luxyal monodose, Xailin HA, Systane Hydration
Polyhydric alcohols	Can be used in combo with a gelling agent Includes propylene glycol (PG) & polyethylene glycol 400 (PEG400)	Systane, Systane Ultra, Blink Intensive Tears, Systane Gel Drops, Systane Hydration
Polyacrylic acids	Display pseudoplastic properties Carbomers (various numbers)	Genteal Gel, HPMC PAA, PAA, Viscotears, Xailin Gel

Preservatives

- Two categories:
  - Detergents (surfactants and chemical toxins)
  - Oxidising preservatives

Preservative	Class	Advantages	Disadvantages	Medication examples
SofZia® (sorbic acid)	Oxidative	Modified into harmless elements upon instillation; smaller amounts of conjunctivo-corneal inflammation compared with BAK	Newer agent requiring more studies to understand ocular safety profile of the preservative independent of active ingredients	Travatan Z®
Sodium perborate (GenAqua®)	Oxidative	Catalyzed into hydrogen peroxide, water and oxygen upon instillation; activity against *Aspergillus*; less toxicity than BAK	Few studies documenting ocular tolerability and side-effect profile	Genteal®
Stabilized oxychloro complex (SOC/Purite®)	Oxidative	Dissociates into water, oxygen, sodium and chlorine free radicals	As with sofZia, more studies are needed to assess ocular side effects independent of active ingredients	Alphagan-P®, Refresh Tears®
Polyquaternium-1 (Polyquad®)	Detergent	Less toxicity to corneo–conjunctival surface than BAK	Superficial corneal epithelial damage reduces density of conjunctival goblet cells	Tears Naturale II®, Opti-Free® Express Disinfecting Solution
Chlorobutanol	Detergent	Toxic effects take longer to manifest than BAK; doesn't affect stability of lipid component of tear film; extensive antimicrobial activity	Causes keratitis and irritation to ocular surface; decreased amount of mitoses to corneal epithelial cells; unstable when stored at room tempeature	TobraDex® Ointment
Cetrimonium chloride	Detergent	Excellent antiseptic qualities	Causes keratinization and inflammatory infiltrates at the limbus and within the conjunctival stroma and epithelium	Civigel®
Benzalkonium chloride	Detergent	Excellent antimicrobial efficacy; disruption of corneal cell–cell junctions allow medicinal entry to anterior chamber; well-established familiarity in industry	Breakdown of corneal epithelium; apoptosis of ocular surface cells; accumulation in surface tissues; tear-film instability	Timoptic®, Azopt, Lumigan®, Xalatan
Edetate disodium	Chelating agent	Inactivates trace amounts of heavy metals	Few studies documenting chronic side effects	Acular®, Betagan®

BAK: Benzalkonium chloride

Lipid Eye Drops/Sprays

- Lipid-containing eye drops are emulsions (two liquids that don't mix)
  - Example of liquids: mineral oil, caster oil, glycerol and soy lecithin
- Sprays → closed eyelids → 10cm away
- Perflurohexyloctante → semifluorinated alkanes → non blurring agent → NovaTears → 100% Perflurohexyloctante
- Considerations:
  - Electrolytes of potassium, calcium, magnesium and zinc
  - Vitamin A, vitamin E
  - Low osmolarity
  - Charge distribution + emulsion technology (no charge, anionic, cationic)
  - pH

Ocular Lubricants				
Mode of action	Indications	Adv Effects	Dosage	Practice Pts
Provide lubrication	Dry eye, exposure keratopathy (e.g., facial nerve palsy), neurotrophic keratitis, relief of mild ocular irritation (e.g., viral or allergic conjunctivitis, after dust, wind, or sun exposure, contact lenses)	Ocular irritation, the frequency of which differs slightly with the	**Eye drops:** 1 drop every 1 to 12 hours as required **Eye ointment, gel**: use at night as required, or throughout the day if symptoms are severe **Eye spray:** 1-2 sprays on closed eyelids 3 to 4 times daily or as required	Encourage regular use of eye drops (up to hourly) • Damage to ocular surface (e.g. staining) can be reduced by switching to preservative-free drops • Safe to use pregnancy & breastfeeding

Comparison of ocular lubricants

Brand®	Preservative	Lubricant
Drops (multidose) with Preservative		
Blink Intensive Tears	Oxychloro-complex (Purite®)	0.25% polyethylene glycol-400 (macrogol 400), 0.2% sodium hyaluronate
Genteal, In a Wink Moisturising	Sodium perborate	0.3% Hypromellose **(PBS-R)**
Liquifilm Tears, PVA Tears	Benzalkonium chloride	1.4% polyvinyl alcohol
Methopt	Benzalkonium chloride	0.5% Hypromellose **(PBS-R)**
Murine Tears	Benzalkonium chloride	0.5% polyvinyl alcohol, 0.6% povidone
Optive	Oxychloro-complex (Purite®)	0.5% carmellose, 0.9% glycerin **(PBS-R)**
Optive Fusion	Oxychloro-complex (Purite®)	0.5% carmellose, 0.9% glycerin, 0.1% sodium hyaluronate
Optive Gel Drops	Oxychloro-complex (Purite®)	1% carmellose, 0.9% glycerin
Optix	Oxychloro-complex (Purite®)	0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol **(PBS-R)**
Poly-Tears, Tears Naturale	Polyquaternium	0.3% hypromellose, 0.1% dextran 70 **(PBS-R)**
Refresh Tears Plus	Oxychloro-complex (Purite®)	0.5% carmellose **(PBS-R)**
Refresh Liquigel	Oxychloro-complex (Purite®)	1% carmellose **(PBS-R)**
Revive Tears	Sorbic acid	0.3% hypromellose **(PBS-R)**
Systane	Polyquaternium	hydroxypropyl guar, 0.4% polyethelene glycol-400 (macrogol 400), 0.3% propylene glycol **(PBS-R)**
Systane Ultra	Polyquaternium	0.175% hydroxypropyl guar, 0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol, 1.4% sorbitol

PBS-R: Benefit restricted for the treatment of severe dry eye syndrome, including Sjogren's syndrome

- BAK → most irritant and only prescribed in mild cases where drops are used infrequently

Comparison of ocular lubricants

Brand®	Preservative	Lubricant
Drops (multidose) Preservative-free		
Evolve Carmellose	None	0.5% carmellose **(PBS-A)**
Evolve Hypromellose	None	0.3% Hypromellose **(PBS-A)**
Hylo-Forte	None	0.2% sodium hyaluronate **(PBS-A)**
Hylo-Fresh	None	0.1% sodium hyaluronate **(PBS-A)**
Murine Dry Eyes	None	0.15% sodium hyaluronate
Systane Ultra Preservative-free	None	0.175% hydroxypropyl guar, 0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol, 1.4% sorbitol
Lipid drops (multidose) Preservative-free		
Systane Complete Preservative-free	None	0.15% hydroxypropyl guar, 1% mineral oil, 0.6% propylene glycol, 0.7% sorbitol

Brand®	Preservative	Lubricant
Drops (single use)		
Blink Intensive Tears	None	0.25% macrogol 400, 0.2% sodium hyaluronate
Cellufresh	None	0.5% carmellose **(PBS-A)**
Optifresh Tears, Refresh Plus	None	0.5% carmellose
Celluvisc	None	1% carmellose **(PBS-A)**
Optifresh Plus	None	1% carmellose
Optive Sensitive	None	0.5% carmellose, 0.9% glycerin
Optive Fusion	None	0.5% carmellose, 1% glycerin, 0.1% sodium hyaluronate
Refresh	None	1.4% polyvinyl alcohol, 0.6% povidone
Systane	None	Hydroxypropyl guar, 0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol **(PBS-A)**
Systane Hydration	None	0.175% hydroxypropyl guar, 0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol, 0.15% sodium hyaluronate
Systane Ultra	None	0.175% hydroxypropyl guar, 0.4% macrogol 400, 0.3% propylene glycol, 1.4% sorbitol
TheraTears	None	0.25% carmellose
Eye gel (multidose)		
Genteal Gel, HPMC PAA	Sodium perborate	0.3% hypromellose, 0.2% carbomer 980 **(PBS-R)**
Optifresh Eye Gel, PAA, Viscotears	Cetrimide	0.2% carbomer 980 **(PBS-R)**
Systane Gel Drops	Polyquaternium	0.75% hydroxypropyl guar, 0.4% polyethylene glycol-400 (macrogol 400), 0.3% propylene glycol, 1.4% sorbitol
Eye gel (single use)		
TheraTears	None	1% carmellose
Viscotears Gel PF	None	0.2% carbomer 980 **(PBS-A)**
Eye ointment		
Refresh Night Time	None	Paraffin **(PBS-R*)**
Poly Visc	None	Paraffin, lanolin **(PBS-R*)**
VIA-POS	None	0.0138% retinal palmitate, paraffin, lanolin

PBS-R: Benefit restricted for the treatment of severe dry eye syndrome, including Sjogren's syndrome (*for eye ointment: condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription)
PBS-A: Authority Required (**STREAMLINED**) for patients with severe dry eye syndrome; patient must be sensitive to preservatives in multi-dose eye drops

- Single use drops are particularly required when frequent administration is required in order for the treatment to be effective (sometimes up to hourly)
- Lanolin → in many skin ointments → common irritant
- Carmellose → another name for carboxymethylcellulose or CMC

Brand®	Preservative	Lubricant
Lipid drops		
Cationorm	None	0.5% heavy mineral oil, 0.5% light mineral oil, glycerin **(PBS-A)**
Systane Balance	Polyquaternium	0.05% hydroxypropyl guar, 1% mineral oil, 0.6% propylene glycol, 0.7% sorbitol
Systane Complete	Polyquaternium	0.15% hydroxypropyl guar, 1% mineral oil, 0.6% propylene glycol, 0.7% sorbitol
Optive Advanced	Oxychloro-complex (Purite®)	0.5% carmellose, 1% glycerin, polysorbate 80
NovaTears	None	100% perfluorohexyloctane **(PBS-A)**
Lipid drops (single use)		
Cationorm	None	0.5% heavy mineral oil, 0.5% light mineral oil, glycerin
Optive Advanced	None	0.5% carmellose, 1% glycerin, polysorbate 80
Lipid sprays		
Tears Again	Phenoxyethanol	Phospholipid liposomes **(PBS-A)**
Optrex Actimist	Phenoxyethanol	Phospholipid liposomes
Murine Eye mist	None	Phospholipid liposomes

- New category of drop aimed at MBG and abnormal tear film lipid layer → not generally available on PBS
- Phenoxyethanol is used in cosmetic products → linked to eczema and contact dermatitis

Preserved Lubricants on PBS

Tear supplements					
Active ingredients	Product	Restrictions	Maximum quantity*		Repeats
			30-day	60-day	
Carbomer-980 0.2% eye gel, 10 g	Optifresh Eye Gel, PAA, Viscotears	**Restricted:** severe dry eye syndrome, including Sjögren's syndrome. **Restricted:** Condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription.	1	2	5
Carmellose sodium 0.5% eye drops, 15 mL	Refresh Tears Plus		1	-	5
Carmellose sodium 1% eye drops, 15mL	Refresh Liquigel		1	-	5
Carmellose sodium 0.5% + glycerol 0.9% eye drops, 15 mL	Optive		1	-	3
Dextran-70 0.1% + hypromellose 0.3% eye drops, 15 mL	Poly-Tears, Tears Naturale		1	2	5
Hypromellose 0.3% w/w eye drops, 10 mL	Genteal, In A Wink Moisturising, Revive Tears		1	2	5
Hypromellose 0.5% eye drops, 15 mL	Methopt		1	2	5
Hypromellose 0.3% + carbomer-980 0.2% eye gel, 10 g	HPMC PAA, Genteal Gel		1	-	5
Polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 15 mL	Optix, Systane		1	2	5

= not available on 60-day prescription

Unpreserved Ocular Lubricants on PBS

Unpreserved tear supplements

Active ingredients	Product	Restrictions	Maximum quantity 30-day	Maximum quantity 60-day	Repeats
Carbomer-980 0.2% eye drops, 30 x 600 mg ampoules	Viscotears Gel PF	**Restricted:** severe dry eye syndrome in patients sensitive to preservatives in multi-dose eye drops. **Authority required (STREAMLINED):** Optometrists have 3 authority codes that are streamlined. You do not need to contact PBS to obtain an authority number. Write the pre-approved code on the prescription. **4105** - Hylo-Fresh & Hylo-Forte **6172** - all other unit-dose tear supplements **15559** - **(new from 1 Sept 2024)** unit dose tear supplements (60-day prescription). The condition must be stable for the prescriber to consider the listed maximum quantity of this medicine suitable for this patient AND patient must be sensitive to preservatives in multi-dose eye drops	3	6	5
Carmellose sodium 0.5% eye drops, 10 mL	Evolve Carmellose		1	2	5
Carmellose sodium 0.5% eye drops, 30 x 0.4 mL ampoules	Cellufresh		3	6	5
Carmellose sodium 1% eye drops, 30 x 0.4 mL ampoules	Celluvisc		3	6	5
Hyaluronate sodium 0.1% eye drops, 10 mL	Hylo-Fresh5		1	2	5
Hyaluronate sodium 0.2% eye drops, 10 mL	Hylo-Forte		1	2	5
Hypromellose 0.3% w/v eye drops, 10 mL	Evolve Hypromellose		1	2	5
Liquid paraffin + glycerol + tyloxapol + poloxamer-188 + trometamol hydrochloride + trometamol + cetalkonium chloride eye drops, 10 mL	Cationorm		1	2	5
Perfluorohexyloctane 100% eye drops, 3 mL	Novatears		1	2	5
Polyethylene glycol-400 0.4% + propylene glycol 0.3% eye drops, 30 x 0.8 mL unit doses	Systane		2	4	5
Soy lecithin 1% + tocopherol 0.002% + vitamin A palmitate 0.025% eye spray, 100 actuations	Tearsagain		2	4	5

Lubricant Ointments on PBS

Topical ocular lubricant ointments

Active ingredients	Product	Restrictions	Maximum quantity 30-day	Maximum quantity 60-day	Repeats
Paraffin 1 g/g eye ointment, 2 x 3.5 g	Poly Visc, Refresh Night Time	**Restricted:** Condition must be stable for prescriber to consider the listed maximum quantity on 60-day prescription.	1	2	5
Paraffin 1 g/g eye ointment, 3.5 g	Poly Visc		2	4	5

Lubricant Components

- Demulcents: CMC, HPMC, HA
- Viscosity agents: Carbopol, PVA
- Electrolytes: Mimic tear composition
- Lipids: Address evaporative dry eye
- Preservatives:
  - BAK = toxic
  - Alternatives: Purite, Polyquad, Sodium perborate

Forms

- Drops, gels, ointments, sprays
- Single-dose (preservative-free) vs multi-dose

PBS Eligibility

- For severe DED and Sjogren's
- Streamlined authority for unpreserved tears (e.g. Hylo-Forte)

4.2. Other Preparations

Eyelid Scrubs

- NOT good in practice because:
  - Noncompliance and misunderstanding of instructions by patient
  - Indications that the likelihood of reaction to the lid scrub solution is more likely with an in-house preparation of diluted baby shampoo
- Difference in conjunctival response

Eyelid Cleansers

Eyelid cleansers

Name	Constituents
Systane LID WIPES	PEG-200 Hydrogenated Glyceryl Palmate, Disodium Laureth Sulfosuccinate, Cocoamido Propyl Amine Oxide, PEG-80 Glyceryl Cocoate, Benzyl Alcohol and Edetate Disodium
OcuSoft LidScrub (OcuSoft)	Cocamidopropyl hydroxysultaine, sodium trideceth sulfate, Disodium lauroamphodiacetate
Oust Demox (OcuSoft)	PEG-80 Sorbitan Laurate, methyl gluceth-20, PEG-120 methyl glucose dioleate, polysorbate 20, melaleuca alternifolia (tea tree), leaf oil, cocosnucifera (coconut) oil, decyl glucoside, 1,2-hexanediol, caprylyl glycol, disodium cocoamphodiacetate, panthenol, polyaminopropyl biguanide
Thera-Tears SteriLid Foam	Sorbitan laurate, sodium trideceth sulfate, cocamidopropyl betaine, sodium lauroamphoacetate, PEG-150 distearate, sodium laureth-13 carboxylate, linalool oil, honohydrate, panthenol, allantoil, sodium chloride, tea tree oil
Blephadex (foam, wipes)	Coconut (cocos nucifera) oil, glycerol, aloe barbadensis gel, sodium lauryl sulfate, cocamidopropyl betaine, DMDM hydantoin, lauryl glucoside, melaleuca alternifolia (tea tree) leaf oil
Clear Eyes by Murine® (wipes)	Polysorbate 20, disodium phosphate, lauryl glucoside, sodium chloride, sodium phosphate, PVP, disodium EDTA, chamomilla recutita (Matricaria) flower extract, glycerine

- Remove lid debris without irritating the bulbar conjunctiva
- Preparations come in moistened towelettes
- Closed eyes rinsed with water and patted dry → aim of lid scrubbing is to aid the removal of oils, debris or desquamated skin from eyelid margins and surrounding regions, most particularly from the meibomian gland orifices
- Tea tree oil → terpinen-4-ol
- Linolool → strong antimicrobial property

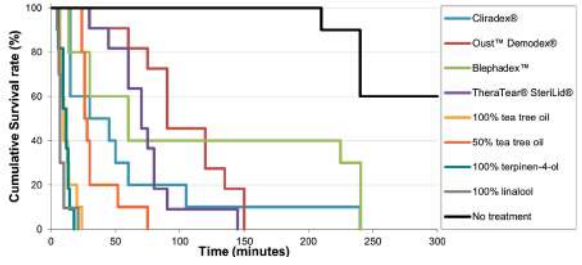
Demodex

- Parasite that is found around the face, eyelashes, eyelids, meibomian glands
- May develop into irritation, dryness, itchiness and signs of blepharitis
  - Treatment with natural antiseptic tea tree oil
  - Lid care products → SteriLid, Ocusoft Lid Scrub Plus, Blephadex, TTO ointment
  - In-office treatment with concentrated tea oil ard 20%
- Tea Tree Oil (50%) most effective (in vitro)
- Other effective agents: Cliradex, Blephadex, SteriLid
- Lotilaner (XDEMVY):
- Antiparasitic for Demodex
- Targets GABA receptors unique to mites

Saline

- Physiological saline (sodium chloride 0.9% bottles or ampoules) are available OTC
- Designed for irritation of the eye and other surfaces or for rehydrating

Eyelid cleansers efficacy against Demodex

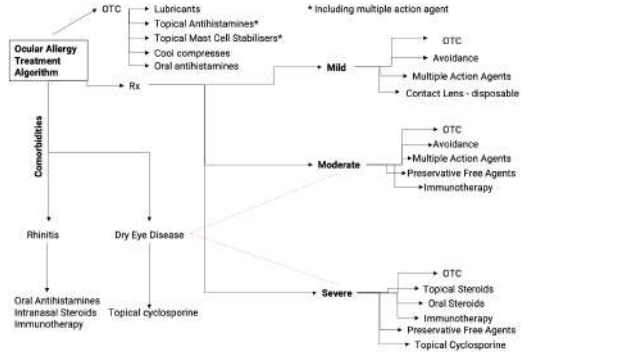


- In order of most efficacious to least efficacious
  - 50% tea tree oil, Cilradex, Blephadex, Sterilid, Oust Demodex
- USA accepted → lotilaner ophthalmic solution 0.25% → XDEMVY to treat demodex blepharitis

Lid Hygiene – warm compresses, wipes, and clea

Product name	Manufacturer	Product type	Comments	Distributor
Aslo Eyelid & Eyelash Cleanser	Aslo	Lid cleanser	Perfect for maintaining daily eyelid and eyelash hygiene, these cleansers are ideal for anyone who wears false lashes, uses eye cosmetics, works in dusty environments, or just enjoys the sensation of clean eyes. With a formulation of carefully selected ingredients, they are gentle enough for sensitive eyes and the delicate skin surrounding them.	Aslo hello@aslo.com.au
Aslo Self-Heating Eye Mask	Aslo	Moist heat warm compress	Designed to hydrate and warm the delicate eye area, these masks reach 40ºC for 30 minutes and are great for styes. Each box contains 10 individually packed self-heating eye masks, perfect for travel and planes. They are disposable for hygienic purposes.	Aslo hello@aslo.com.au
Allergy 30's Wipes	Cynacon OcuSoft	Lid cleanser	Seasonal allergic conjunctivitis.	Good Optical (03) 9645 1066
Avenova®	Novabay	Hypochlorous acid spray	0.01% hypochlorous acid spray solution that kills a broad spectrum of bacteria and successfully helps relieve chronic eye conditions like MGD, blepharitis and dry eye, without stinging or irritation. No bleach, and no impurities.	Designs For Vision 1800 225 307
Blephadex Eyelid Applicators	Bluphadex	Eyelid applicator	In-practice manual debridement treatment whilst simultaneously depositing Blephadex solution.	Optimed 1300 657 720
Blephadex Eyelid Foam	Bluphadex	Foam bottle	Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex.	Optimed 1300 657 720
Blephadex Eyelid Wipes	Bluphadex	Wipes	Tea tree oil, coconut, aloe barbadensis formulation in foam designed for cleaning lashes, eyelids and treatment of demodex.	Optimed 1300 657 720
Blephadex Pro with Manuka Honey	Bluphadex	Wipes	Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex. With added benefits of manuka honey (UMF 10+).	Optimed 1300 657 720
Blephadex Warming Eyelid Wipes	Bluphadex	Wipes	Warming wipe is activated with water droplet. Tea tree oil, coconut, aloe barbadensis formulation designed for cleaning lashes, eyelids and treatment of demodex.	Optimed 1300 657 720
BlephEx	BlephEx	In-practice mechanical debridement	In-practice procedure to remove bacterial biofilm, scurf, debris from lashes and lid margin by means of microblepharoexfoliation (MBE).	Optimed 1300 657 720

Product name	Manufacturer	Product type	Comments	Distributor
Blephex Lid Cleanser	BlephEx	Lid cleanser	Suitable for maintaining Blephex Procedural Tips. Contains Manuka honey and Dead Sea salt to fight inflammation on the biofilm and offers anti-demodex properties.	Optimed 1300 657 720
Bruder Hygienic Eyelid Solution	Bruder	Spray	Pure 0.02% hypochlorous acid solution in amber bottle (preservative free).	Optimed 1300 657 720
Bruder Hygienic Eyelid Wipes	Bruder	Wipes	No Rinse, leave on formula. Gentle enough for daily use. Low residue, non-soapy formula. No stinging or burning.	Optimed 1300 657 720
Bruder Moist Heat Eye Compress	Bruder	Compress	Available in double eye and single eye. Patented Medibead technology to deliver uniform moist heat.	Optimed 1300 657 720
D.E.R.M®	Eyeeco	Moist heat therapy	D.E.R.M®, Dry Eye Relief Mask is an entry level solution for immediate dry eye relief. The mask contains silica beads, which naturally attract moisture from the air and release it when microwaved.	Designs For Vision 1800 225 307
Dry Eye Compress	Ocusci	Moist heat therapy Cold therapy	→ 123% improvement in DE Symptoms - 5 days of use → Machine-washable cover with Silver+ antimicrobial → Just 25 seconds in the microwave → Patented Hydrobiock coating - delivers 10 minutes of moist, therapeutic heat → Custom logo options	OphthalmoPro 1300 578 288
Dry Eye Compress (USB)	Ocusci	USB powered warm compress	→ Delivers therapeutic heat at your convenience → USB powered, plug into any PC or portable battery → 3 temperature settings and 3 timer settings → Automatic shut off	OphthalmoPro 1300 578 288
Eyecloud®	Eyeeco	Moist heat therapy Cold therapy	A controlled moist heat therapy system designed to provide 600, 20 minute treatments. The expanded mask coverage and pressure control design evenly distributes therapy to the eyelids and surrounding skin. Moist heat therapy: → Stimulates tear production → Promotes blood circulation → Returns oil along eyelids to healthy state. Cold treatments reduce: → Eyelid and undereye puffiness → Inflammation and itchiness associated with allergies.	Designs For Vision 1800 225 307
Eye Doctor Essential	Body Doctor	Warm compress	Microwaveable. Brandable for practices.	Good Optical (03) 9645 1066
Eye Doctor Spa Mask	Body Doctor	Warm compress for one eye	Great one eye mask for styes.	Good Optical (03) 9645 1066



Product name	Manufacturer	Product type	Comments	Distributor
Eye Doctor Mask Premium	Body Doctor	Warm/cool compress	Use hot or cold. Can heat in conventional oven. Brandable for practices.	Good Optical (03) 9645 1066
Eye Doctor Click n Go Mask	Body Doctor	Travel compress	Great for planes and camping. Up to 100 uses.	Good Optical (03) 9645 1066
Eyeglana InstaWarmth Mask Compress	Eyeglana	Compress	Disposable wafers utilise unique warming technology when removed from packet and slipped into the ultra-soft eye mask.	Optimed 1300 657 720
Eyeleve by Bruder Contact Lens Compress	Bruder	Compress	Patented Medibead technology to deliver uniform moist heat. Comfort stitch to reduce the pressure off the eyelid.	Optimed 1300 657 720
Eyelid & Lash Foaming Cleanser EVERYDAY	Heibron Laboratories	Cleanser	Designed for use in a general eyelid and eyelash hygiene regime. A perfect option for wearers of false lashes and makeup, as well as those who work in dusty and dirty environments - or those who just love the feeling and look of clean bright eyes.	Ocul Brands Pty Ltd (03) 9593 6501
Eyelid & Lash Foaming Cleanser FORTE	Heibron Laboratories	Cleanser	A higher strength formulation designed for the control of pre existing eyelid and eyelash conditions such as blepharitis, demodex, dry eye, sore and itchy eyes. With boosted tea tree oil to provide additional antibacterial help, this formulation is a front line tool for control of many common eyelid and eyelash conditions.	Ocul Brands Pty Ltd (03) 9593 6501
Eye Warmer	SilinFra	Compress	USB powered heating pad for the eyes.	Optimed 1300 657 720
Hypochlor Gel	Cynacon OcuSoft	Hypochlor Gel	Great results with Rosacea.	Good Optical (03) 9645 1066
Hypochlor Spray	Cynacon OcuSoft	Hypochlor Spray	Great results with Rosacea.	Good Optical (03) 9645 1066
I Lid 'n Lash®	I-Med Pharma	Eyelid Wipes	Clean and remove ocular debris and makeup from around the eyes. Used daily, I-LID 'N LASH® is effective in maintaining lid and lash health for patients with dry eyes or sensitive skin. Available in standard and 5% tea tree for demodex management.	Designs For Vision 1800 225 307
Nulids	Nulids	At home 'mi-croblepharoexfoliation' device	Works well with Zocular Gel.	Good Optical (03) 9645 1066

Product name	Manufacturer	Product type	Comments	Distributor
Optimel® Manuka+ Eyelid Cream	Maicare	Eyelid Cream	Helps restore skin's protective acid mantle. Dry eye conditions such as blepharitis and meibomian gland disease (MGD) are frequently present with various dermatological conditions, including seborrheic dermatitis, rosacea, and eczema. Nearly half of people suffering from MGD have had seborrheic dermatitis at some time. Skin pH in atopic dermatitis is often high and the resulting cascade of changes contribute to the development of atopic dermatitis leading to dryness and flaky skin. Optimel® Manuka+® Eyelid Cream is a low pH emollient cream containing Leptospermum spp honey and Jojoba oil to help restore skin's natural structure. The humectant properties of honey also help maintain a lasting skin softness.	Designs For Vision 1800 225 307
Opti-Soothe Moist Heat Mask	Scope	Moist heat mask	The Opti-Soothe Moist Heat Mask is reusable and designed to give you effective relief from dry eye and other related eye conditions.	Good Optical (03) 9645 1066
Opti-Soothe Preservative Free Eyelid Wipes	Scope	Eyelid wipe	Contains a unique and gentle formulation of tea tree oil, hyaluronic acid, chamomile, aloe vera. Preservative-free nature makes it ideal for daily eyelid hygiene. No need to rinse off formulation after use.	Good Optical (03) 9645 1066
Original Foam 50ml	Cynacon OcuSoft	Lid cleanser	General maintenance and cleanser.	Good Optical (03) 9645 1066
Original 30's Wipes	Cynacon OcuSoft	Lid cleanser	Demodex infestation	Good Optical (03) 9645 1066
Duo Foam 50ml	Cynacon OcuSoft	Lid cleanser	Demodex infestation	Good Optical (03) 9645 1066
Platinum Plus Foam 50ml	Cynacon OcuSoft	Lid cleanser	Anterior blepharitis	Good Optical (03) 9645 1066
Plus 30's Wipes	Cynacon OcuSoft	Lid cleanser	Anterior blepharitis	Good Optical (03) 9645 1066
Plus Foam 50ml	Cynacon OcuSoft	Lid cleanser	Anterior blepharitis	Good Optical (03) 9645 1066
Systane® LID WIPES	Alcon®	Daily eyelid cleanser	Non-medicated lid wipe for daily eyelid hygiene.	Alcon® 1800 224 153
Tranquileyes XL®	Eyseco	Moist heat therapy	Tested to sustain optimum temperature and humidity for 20 minute treatments. Moist heat therapy: → Stimulates tear production → Promotes blood circulation → Returns oil along eyelids to healthy state. Cold treatments reduce: → Eyelid and undereye puffiness → Inflammation and itchiness associated with allergies.	Designs For Vision 1800 225 307

Product name	Manufacturer	Product type	Comments	Distributor
TearRestore Thermal Mask	TearRestore Inc	Warm compress	Improve compliance, patient satisfaction and meibomian gland expression with the first open-eye warm compress. The TearRestore Thermal Mask delivers consistent heat, the ability to see, and natural meibomian gland expression (blinking) throughout compress. Reusable, self-heating, & durable heat packs. Each heat pack produces 60-90 treatments.	Good Optical (03) 9645 1066
Tranquileyes Tea Tree Eyelid & Facial Cleanser™	Eyseco	Foaming eyelid and facial cleanser	Infused with chamomile and shea butter, these facial cleansers gently remove all dirt and makeup, leaving the skin feeling smooth, silky, and refreshed. Combined with tea tree oil to help manage demodex and related ocular surface diseases. Available in Gentle (1%) or Advanced (2%) tea tree formulations.	Designs For Vision 1800 225 307
ZocuFoam	Okra LLC	Foaming cleanser	Penetrating foam with okra micelles, bid or qd. Each bottles lasts 3 months.	Good Optical (03) 9645 1066
ZocuKit for ZEST	Okra LLC	In-office procedure	Therapeutic deep lid cleansing (debridement) to immediately and dramatically improve ocular surface and periorbital inflammation.	Good Optical (03) 9645 1066
ZocuShield	Okra LLC	Okra gel	Leave-on gel for moderate-severe cases. Only one drop of gel needed per eye and each ZocuShield bottle lasts for 3 months.	Good Optical (03) 9645 1066
ZocuWipe	Okra LLC	Towelette	Leave-on formulation effective at just once a day.	Good Optical (03) 9645 1066

- Range → masks, cleansers, wipes, gels, sprays, creams and in-office/home mechanical devices
- Warm compresses available: microwavable, USB, masks and self-heating
    - Individually packaged (in-office for infection control), one eye (great for hordeolum)
- In office devices → for microblepharoexfoliation → Bleplex (in office) and Nulids (at home)
- Ocular lubricants not on AMH:
    - Sodium chloride 0.9% → designed for irrigation of eye and other surfaces
    - Saline → not viscous but non irritating, suggests equal performance to existing lubricating eyedrops

Hypochlorous Acid

- Natural disinfectant, used in wound care (dermatology and dentistry)
- Similar to bleach → lower concentrations
- Available in 0.01%, 0.015%, 0.02% as eyesprays → indications for daily used and helping prevent/manage blepharitis
- Also available as gel

Manuka Eye Drops

- **TGA** registered as medical device, unscheduled
- **MOA**: antibacterial, hyperosmotic
- **Dosage**: apply three time a day
- **Contraindications**: history of allergy to honey or bee products or gum foods, in peri-operative setting
- **Side effects:** expect temporary significant stinging and redness in all px (moreso with gel)
- Approved indications → dry eye diseases including MGD
- Stinging is common due to low levels of pH

Cyclodextrin-complexed Manuka honey (MGO TM)

- High concentration of methylglyoxal (MGO) → New Zealand Manuka Honey
- Comparable efficacy to 50% tea tree oil (TTO) → possible alternative to TTO → for demodex eradication and treatment of blepharitis

"Natural" product: Oculocin Propo

- **Ingredients**: propolis (1% to 1.5%), aloe vera, chamomile, phenols (0.5%), aloins & emodins, alfa-bisabolol
    - Don't seem to promote viscosity or mimic tear film
- **Indications**: dry eye, chronic (mild) conjunctivitis
- Claims to have antiviral, antibacterial and anti-inflammatory properties
- **Dosage**: 1 drop every 2 hours
- No data on efficacy of product

Blood Derivatives

- Autologous serum eye drops 20%
- Allogeneic serum eye drops
- Autologous plasma rich in platelet derived growth factors eye drops
- Complex preparation (Blood Service involvement)
- Risk of infection transmission
- High cost, cost-effectiveness unknown
- Issues with packaging and drop insertion
- Effectiveness and safety largely unknown
- Used to treat severe disorders of the ocular surface → severe dry eye disease, Sjogren's disease and neuropathic ulcers → conventional strategies failed
- Promote ocular surface health → epithelial growth and repair
    - Blood is collected from px, room temp for a few hours to achieve clotting, centrifuged, serum extracted and balanced salt solution added to achieved desired concentration of autologous serum tears

Topical Hyperosmotic Agents

Brand ®	Formulation	Preservative
Adsorbonac ophthalmic (Alcon)	2% or 5% NaCl solution	Thimerosal
Muro-128 ophthalmic (Bausch & Lomb)	2% or 5% NaCl solution with methylcellulose	Methylparaben Propylparaben
AK-NaCl (Akorn) Muro-128 ophthalmic (Bausch & Lomb)	5% NaCl ointment	-
Glucose-40 Ophthalmic (Cooper Vision)	40% ointment in petrolatum and lanolin	-

- Duration of action: up to 7 hours
- Side-effect: mild stinging (drops) & blurry vision (ointment)

- **Aim**: dehydrate the cornea to draw liquid out
- **Indications**: conditions with significant swelling (e,g, Fuch's dystrophy, postsurgical corneal decompensation)
- Agents provide corneal clarity → VA in early stages of corneal decompensation, delay need for keratoplasty
- NO commercial preparations of ophthalmic hyperosmotic agents available in Australia, not on PBS, can be compounded

Secretagogues

- Initially developed and used in diseases of mucous membranes such as gastric ulcers, cystic fibrosis, Sjogren's syndrome

Diquafosol

- Diquafosol 3% (Diquas, Santen Pharmaceuticals) ophthalmic solution commercially available in many Asian countries: Japan, South Korea, Thailand, Vietnam, 5mL bottle
- **Mechanism of action:** P2Y2 purinogenic receptor agonist: promotes tear fluid and protein secretion by conjunctival epithelium and goblet cells through elevation of intracellular calcium ion levels
- **Dosage**: one eye drop 6 times daily

Adverse Effects:

- Very common (>5%): Eye irritation
- Uncommon to common (0.1-5%): blepharitis, eye discharge, conjunctival hyperaemia,

Rebamipide

- Rebamipide 2%, unit dose (Mucosta, Otsuka Pharmaceuticals) ophthalmic suspension commercially available in many Asian countries: Japan, South Korea, Thailand, Vietnam, 5mL bottle

- **Mechanism of action:** quinolinone derivative that stimulates endogenous PGE2 mucous secretion and has possible anti-inflammatory action (↓NFκB activation, possible suppression of ROS
- **Dosage**: one eye drop 4 times daily
- Published reports of effectiveness in DED, superficial limbal keratopathy (SLK), lagophthalmos, lid wiper epitheliopathy (LWE), recurrent corneal erosion (RCE)

Adverse Effects:

- Bitter taste (9.7%)

Varenicline 0.03mg solution

- Commercial name TYRVAYA → treat dry eye disease
- Formulated as a nasal spray → highly selective nicotinic acetylcholine receptor (nAChR) agonist
- Stimulate trigeminal nerves within the nasal cavity → mediate afferent signals and activate lacrimal functional unit to produce tears

Lid Hygiene Product Overview

- Includes compresses, sprays, gels, mechanical debridement (Blephex)
- Ingredients: TTO, honey, aloe, chamomile
- Mask types: USB, microwavable, self-heating

4.3. Lid Diseases Part 1

Anterior Blepharitis

- Chronic lid inflammation
- Often coexists with DED
- Subtypes: Staph, Seborrheic, Demodex

Treatment

- Lid scrubs and warm compresses essential
- Commercial options:
- Tea Tree Oil (TTO) based: Cliradex, Blephadex
- Honey-based: Optimel
- In-office: Blephex, IPL

New Therapies

- Lotilaner 0.25% (Xdemvy) – FDA-approved for Demodex in US
- Selectively targets parasite GABA channels
- High efficacy in Phase 3 trials (collarette cure, mite eradication)

4.4. Lid Diseases Part 2

Ectropion

- Lid margin everts outwards
- Common in >50 y.o., bilateral in 70%
- Causes exposure keratopathy

Entropion

- Lid margin rolls inward
- Lashes touch cornea, can lead to ulcer
- Prevalence ~2% elderly
- Classified by cause: involutional, cicatricial, congenital, etc.

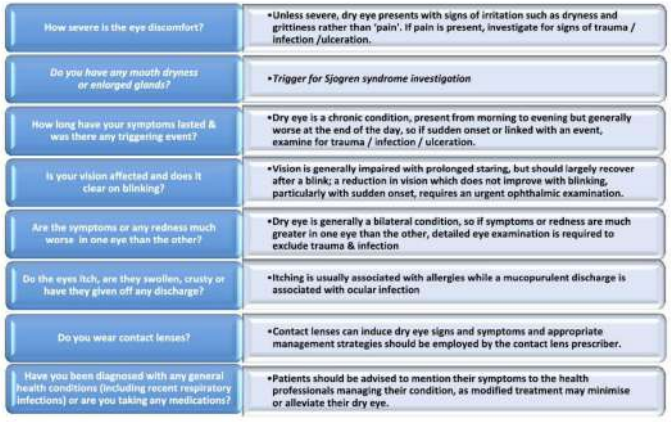
Management

- Non-urgent referral to ophthalmology
- Document with photos, advise sun protection
- Surgery often indicated

4.5. Dry Eye Disease Part 1

Diagnosis & Classification

ADDE	EDE
Sjogren Syndrome Dry Eye (SSDE)	MGD
Age-related Dry Eye	Anterior blepharitis
Graft Versus Host Disease (GVHD)	Rosacea
Neurotrophic Keratitis	Ectropion
Refractive Surgery	Thyroid Eye Disease
	Contact Lens Wear



- DED requires both symptoms and ≥1 sign (per TFOS DEWS II).
  - TFOS DEWS II stands for the Tear Film & Ocular Surface Society (TFOS) Dry Eye Workshop II. It is the second international workshop focused on advancing the understanding of dry eye disease (DED) and its impact on the ocular surface. The DEWS II report, published in 2017, provides an updated comprehensive consensus on the pathophysiology, diagnosis, and management of dry eye disease. It is considered one of the most important and widely referenced guidelines in the field of optometry and ophthalmology for dry eye disease.
- Diagnostic signs:
  - Non-invasive **TBUT <10s**
    - A non-invasive tear break-up time (TBUT) is a diagnostic test used to assess the stability of the tear film on the surface of the eye without the need for the application of dyes (such as fluorescein) that are typically used in traditional TBUT testing.
  - Osmolarity **≥308 mOsm/L** or interocular difference >8
  - Ocular surface staining: corneal (>5), conjunctival (>9), or lid margin staining
- Subtypes:
  - Evaporative DED – abnormal lipids, MGD
  - Aqueous-deficient DED – low tear volume

Feature	Evaporative Dry Eye	Aqueous-Deficient Dry Eye
Cause	Meibomian gland dysfunction (MGD), poor tear evaporation **(rapid tear evaporation but normal tear production)**	Insufficient tear production due to lacrimal gland dysfunction
Tear Film	Normal aqueous production, but **oily layer is insufficient**	Reduced aqueous production, leading to tear film instability
Main Issue	Increased tear evaporation	Insufficient tear production
Symptoms	Dryness, burning, irritation, increased tearing, blurred vision	Dryness, burning, foreign body sensation, irritation
Common Causes	Meibomian gland dysfunction (MGD), blepharitis, contact lens wear	Sjogren's syndrome, aging, systemic diseases, medications
Treatment Focus	Improving meibomian gland function and tear stability	Increasing tear production or conserving existing tears

Risk Factor Analysis

Factors include contact lens wear, medication use (e.g. anti-glaucoma drops), systemic diseases (Sjogren's), digital device use.

Tests

- Lid margin assessment:
  - Observe Marx's line (mucocutaneous junction)
  - Check for scarring, displacement, or gland dropout
- Lid wiper region: key site of frictional damage in dry eye

Tear Film Assessment

- TBUT (fluorescein and non-invasive)
- Osmolarity testing (TearLab)
- Phenol red thread test (tear volume)
- Questionnaires: DEQ-5, OSDI

Treatment Modalities

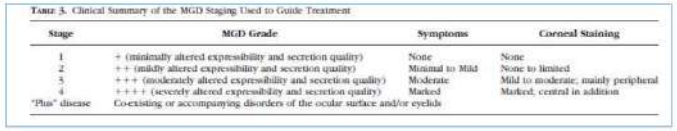


TABLE 3. Clinical Summary of the MGD Staging Used to Guide Treatment

Stage	MGD Grade	Symptoms	Corneal Staining
1	+ (minimally altered expressibility and secretion quality)	None	None
2	++ (mildly altered expressibility and secretion quality)	Minimal to Mild	None to limited
3	+++ (moderately altered expressibility and secretion quality)	Moderate	Mild to moderate; mainly peripheral
4	++++ (severely altered expressibility and secretion quality)	Marked	Marked; central in addition
"Plus" disease	Co-existing or accompanying disorders of the ocular surface and/or eyelids		

Categorised into:

1. Tear replacement (artificial tears, autologous serum)
2. Tear conservation (punctal plugs, moisture chambers)
3. Tear stimulation (secretagogues, nasal neurostimulation)
4. Anti-inflammatory therapy (steroids, ciclosporin, lifitegrast)
5. Surgical approaches (punctal occlusion)
6. Environmental/dietary modifications

Most px will suffer from a combination of both abnormal MG physiology (EDE) & tear underproduction (ADDE)

4.6. Dry Eye Disease Part 2

Blepharitis Overview – inflammation of the eyelids

- **Types**:
  - **Staphylococcal**: scales at lash base
  - **Seborrheic**: oily lids, greasy scales
  - **Demodex**: cylindrical dandruff along lash shaft
  - **MGD**: thick secretions, duct plugging
- Symptoms: itching, burning, gritty sensation, photophobia, CL intolerance
- Often presents with mixed subtypes

Staphylococcal	Seborrheic	Demodex	MGD
• Lid margin hyperemia • Lid margin swelling • **Lid margin crusting (scales at lash base)** • Lash misdirection • Madarosis • Recurrent styes / chalazia • Conjunctival hyperemia • Staining, marginal keratitis, phlyctenulosis, neovascularisation / pannus, papillary conjunctivitis	• Lid margin hyperemia • **Oily or greasy deposits on lid margins** • Conjunctival hyperemia	• Lid margin hyperemia • **Cylindrical dandruff** • Misalignment • Trichiasis • Madarosis	• **Thick and/or opaque secretions at gland orifices** • **Oil expression by finger pressure difficult** • Foam in lower tear meniscus • **Plugging of duct orifices** • Conjunctival hyperemia • EDE • Staining, marginal keratitis, scarring, neovascularisation / pannus, papillary conjunctivitis

Signs per Type

- MGD: foam in tear film, thick meibum
- Demodex: cylindrical dandruff, lash misdirection
- Staph: crusting, lash loss
- Seborrheic: oily lid margin, conjunctival hyperemia

Treatment Strategies

- Lid hygiene: scrubs (20 passes with commercial wipes), warm compresses
- In-office therapies: LipiFlow, IPL, electrical stimulation
- Evidence:
  - 2012 Cochrane review: lid hygiene effective; topical steroids and antibiotics inconclusive
    - The review concluded that topical antibiotics were shown to provide some symptomatic relief and could eradicate bacteria from the lid margin.
  - Need for better RCTs stratified by subtype

4.7. Dry Eye Disease Part 3

Anti-inflammatory Therapies

- Corticosteroids: short-term only
- Ciclosporin, Lifitegrast (potentially compounded), NSAIDs
- Macrolides (e.g. azithromycin for MGD): both oral and topical useful

Evidence Summary

- 138 SRs reviewed; only 26 reliable
- No conclusive evidence for any treatment
- Inconclusive support for:
  - IPL, Lipiflow, punctal occlusion
  - Artificial tears, dietary/environmental mods
  - Topical steroids, cyclosporin, secretagogues, autologous serum
- InflammaDry: MMP-9 positive → inflammatory DED → justify anti-inflammatory treatment



Eyelid warming methods (modified from Murakami DK et al. OVS 2015; 92: e327-e333)	
Dry – chemical	
EyeGiene (www.eyedetec.com)	Reusable mask. Heat is chemically produced in a packaged unit
Dry – microwave	
MediBeads (www.bruder .com)	Cloth mask enclosed with patented "medibeads"
Eye-ssntial (www.therapearl.com)	BPA / phthalate-free latex mask filled with polymer "pearls"
Rice bag	~1 cup of uncooked rice in rice bag or sock
MGDRx EyeBag (www.eyebagcompany.co.uk)	Cloth / silk mask filled with flaxseeds
BlephaMask	Organic cotton mask filled with grape seeds. Contains a red temperature testing strip
Wet - microwave	
Bundle method	Dampen and squeeze out 5 to 6 microfiber face clots. Fold each cloth; roll up the layers (see image). Soak and heat in microwave
Tranquileyes XR (www.eyeeco.com)	Dry and wet compress in a moisture goggle
Dry Eye Relief Mask (D.E.R.M.) (www.eyeeco.com)	Contains silica bead and is applied with a premoistened disposable cotton liner
Heated goggle - electronic	
Blephasteam (www.blephasteam.com)	10-min automated treatment. Plastic goggle with rings moistened by saline placed in each goggle chamber

Dry Eye Treatment Pyramid

Pyramid Strategy

- Base (Level 1): education, blinking, lubricants, diet, hygiene
- Level 2-3: anti-inflammatories, gland expression, moisture conservation
- Top (Level 4): surgery, amniotic membranes, serum tears

Jaiswal Dry Eye Webinar

Blink Training

- Full, light, rapid blinks (upper lid meets lower)
- Practice method: 8 keyword repetition → 24 efficient blinks per session
- Practice every 30 minutes for 1 week

Patient Communication

- Addressing lifestyle (screen time, sleep, water)
- Expectations management (chronic condition)
- Show blink video, provide home care sheets

2025 OPTM6413 Dry eye treatment handout Stapleton.pdf

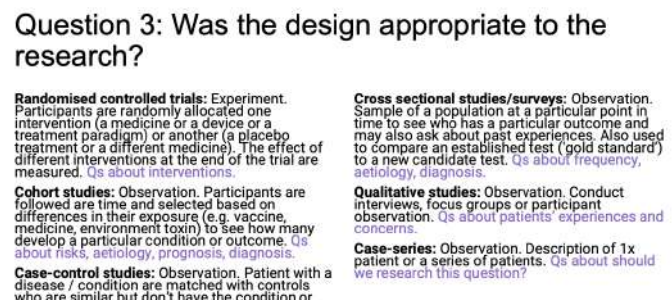
TFOS DEWS II Algorithm (4-Step)

1. Education, environment, diet, unpreserved tears, hygiene
2. Anti-evaporative, gland expression, prescription drugs
3. Serum drops, scleral lenses
4. Surgery, long-term steroids, amniotic membrane

Practical Notes

- Everyone starts at Level 1
- Most patients = mixed type DED
- Escalation/De-escalation based on response
- Treat underlying pathology, not just symptoms
- Scaffold steroids when starting immunomodulators

Tutorial Notes (Week 4)



Question 3: Was the design appropriate to the research?

Randomised controlled trials: Experiment. Participants are randomly allocated one intervention (a medicine or a device or a treatment paradigm) or another (a placebo treatment or a different medicine). The effect of different interventions at the end of the trial are measured. Qs about interventions.

Cohort studies: Observation. Participants are followed are time and selected based on differences in their exposure (e.g. vaccine, medicine, environment toxin) to see how many develop a particular condition or outcome. Qs about risks, aetiology, prognosis, diagnosis.

Case-control studies: Observation. Patient with a disease / condition are matched with controls who are similar but don't have the condition or disease. Data are collected on (past) exposure (e.g. risk factors) and analysed. Qs about risks, prognosis, diagnosis, rare AR/harm of interventions.

Cross sectional studies/surveys: Observation. Sample of a population at a particular point in time to see who has a particular outcome and may also ask about past experiences. Also used to compare an established test ('gold standard') to a new candidate test. Qs about frequency, aetiology, diagnosis.

Qualitative studies: Observation. Conduct interviews, focus groups or participant observation. Qs about patients' experiences and concerns.

Case-series: Observation. Description of 1x patient or a series of patients. Qs about should we research this question?

Table 2.5: Hierarchies of evidence: steps in the process of searching for the best evidence for different types of questions

Question	Step 1 (Level 1)	Step 2 (Level 2)	Step 3 (Level 3)	Step 4 (Level 4)
Does this intervention help? (intervention benefits and harms)	SR of RCTs	RCT	Non-randomised controlled cohort/ follow-up study	Case-series, Case-control studies or historically controlled study
What might happen if we do not add an intervention? What might happen over time? (prognosis)	SR of inception cohort studies	Inception cohort study	Cohort study or control arm of a RCT	Case-series or case-control studies or poor-quality prognostic cohort study
Is the diagnostic or monitoring test accurate? (diagnosis)	SR of cross-sectional studies (of test accuracy) with consistently applied reference standard and blinding (and among consecutive persons)	Single cross-sectional study (of test accuracy) with consistently applied reference standard and blinding (and among consecutive persons)	Non-consecutive studies, or studies without consistently applied reference standards	Case-control studies, or poor-quality or non-independent reference standard
Is this (early detection) test worthwhile? (screening)	SR of randomised trials	Randomised trial	Non-randomised controlled cohort/follow-up study	Case-series, case-control or historically controlled study
How common is the problem?	Local and current random sample of surveys (or censuses)	SR of surveys that allow matching to local circumstances	Local non-random sample	Case-series

Study Design	Definition
Case Series	A descriptive study that tracks patients with a similar condition or treatment over time, without a control group. Useful for identifying trends but lacks strong evidence.
Case-Control Study	A retrospective study comparing individuals with a specific condition (cases) to those without (controls) to identify potential risk factors or causes.
Clinical Guidelines	Evidence-based recommendations developed by experts to guide healthcare decisions for specific conditions, incorporating the best available research.
Cross-Sectional Study	A study that examines data at a single point in time to assess the prevalence of conditions, behaviors, or risk factors in a population.
Randomized Controlled Trial (RCT)	An experimental study where participants are randomly assigned to an intervention or control group to measure the effectiveness of treatments or interventions. Considered the gold standard for clinical research.
Prospective Cohort Study	A study that follows a group of people over time to assess how certain exposures or risk factors influence the development of outcomes.
Qualitative Study	A research approach focusing on understanding experiences, behaviors, or social phenomena through interviews, focus groups, or observations.
Retrospective Cohort Study	A study that looks back at past data to examine how exposures or risk factors influenced outcomes in a defined group of individuals.
Systematic Review	A comprehensive summary of existing research on a specific topic, using rigorous methods to identify, appraise, and synthesize findings from multiple studies.

Parallel group comparison	Each group receives a different treatment, with both groups being entered at the same time; results are analysed by comparing groups For example: Imagine testing two headache medications: • Group A takes **Medication X** • Group B takes **Medication Y** Both groups are studied **simultaneously**, and at the end, researchers compare the effectiveness of the two medications.
Paired (or matched) comparison	Subjects receiving different treatments are matched to balance potential confounding variables such as age and sex; results are analysed in terms of differences between subject pairs (pairing people who are similar but receive different treatments) For example: • If one treatment is being tested on older adults and another on younger adults, age could affect the outcome. • To avoid this, researchers match people of similar ages in each group and then compare their individual responses, rather than just comparing overall group averages.
Within subject comparison	Subjects are assessed before and after an intervention and results are analysed in terms of changes within the subjects
Single blind	Subjects did not know which treatment they were receiving
Double blind	Neither did the investigators
Placebo controlled	Control subjects receive a placebo (inactive pill) which should look and taste the same as the active pill. Placebo operations may also be used in trials of surgery
Factorial design	A study which permits investigation of the effects (both separately and combined) of more than one independent variables on a given outcome. For example: A 2x2 factorial design tested the effects of placebo, aspirin alone, streptokinase alone, or aspirin + streptokinase in acute heart attack

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Week 5: Readings

5.1. Petsoglou – Viral Keratitis & Conjunctivitis (2025)

Overview of Viral Keratitis and Conjunctivitis

- Definition of Virus: Small, parasitic agents that replicate in host cells and target specific cells using protein coats.
- Affected Ocular Structures: Primarily targets the corneal epithelium, stroma, and endothelium.
- Major Viral Culprits:
  - DNA Viruses: HSV-1, HSV-2, VZV, CMV, adenovirus.
  - RNA Virus: Coronavirus.

Adenoviral Keratoconjunctivitis (EKC)

- Etiology: Adenovirus, especially type D (8, 19, 37).
- Transmission: Highly contagious via contact, fomites, swimming pools.
- Stages of EKC Keratitis:
  - Stage 0-1: Vesicles, mild epithelial staining.
  - Stage 2: Coalesced epithelial lesions (day 5–10).
  - Stage 3: Subepithelial infiltrates (SEIs) due to immune response.
  - Stage 4: Expanded SEIs, vision loss.
  - Stage 5: Granular overlying epithelium, lasts up to 2 years.
- Complications:
  - Chronic dry eye, permanent corneal scars, visual disturbances, lawsuits from transmission.

Investigation & Management

- Diagnosis: Clinical, PCR when uncertain.
- Management:
  - Supportive: Lubricants, rest, hygiene.
  - Antivirals: Not usually effective for adenovirus.
  - Topical Betadine (0.25%): Some evidence of efficacy.
  - Steroids: Controversial; used cautiously in pseudomembranes, SEIs.
  - Topical Cyclosporin/Cidofovir: For persistent infiltrates.
- Infection Control:
  - Avoid work if symptomatic (esp. childcare, healthcare).
  - Disinfect tonometers, eye drops, and clinic surfaces.
  - Adenovirus remains viable for weeks on surfaces.

Herpes Simplex Virus (HSV) Eye Disease

- Biology: Latent virus that reactivates under stress, immune suppression, trauma.
- Clinical Manifestations:
  - Primary Blepharoconjunctivitis (often in children).
  - Recurrent Disease:
    - Epithelial: Dendritic/geographic ulcers.
    - Stromal: Immune or necrotising.
    - Endotheliitis: Disciform, linear.
  - Treatment:
    - Topical Aciclovir for epithelial ulcers.
    - Steroids only after epithelium heals or in stromal/endothelial disease (with antiviral cover).
    - Oral antivirals (Aciclovir, Valaciclovir) critical in recurrent or severe disease.
    - Prophylaxis: Long-term antivirals and/or low-dose steroids to prevent recurrence.

Herpes Zoster Ophthalmicus (HZO)

- Cause: Reactivation of VZV.
- Management Rules:

1. Oral antivirals (Valaciclovir) always.
2. Betadine for skin lesions.
3. Neurotrophic cornea needs aggressive support.
4. PCR useful in unclear cases.
5. Prevent lipid keratopathy with inflammation control.
6. Avoid steroids on epithelial defects.
7. Treat post-herpetic neuralgia early with Pregabalin/Gabapentin/Amitriptyline.

Other Conditions Covered

- CMV Endotheliitis: Seen in young, immunocompetent patients; treated with Valganciclovir.
- Thygeson's Keratitis: Chronic, bilateral punctate keratitis; responds to steroids or cyclosporin.
- CLARE (Contact Lens-Associated Red Eye): Inflammatory response to hypoxia or preservatives.

Tutorial 5 2025 – HSV Keratitis Case Discussion

Key Learning Objectives

- Evidence-based antiviral/corticosteroid prescribing.
- Patient communication about prognosis.
- Preserving vision in corneal infections.

Case Diagnosis: HSV Epithelial Keratitis

- Classic dendritic ulcer.
- Key Differentials Excluded:
  - HZO (no skin lesions or coarse dendrites).
  - Fungal keratitis (no trauma/vegetative matter).
  - Acanthamoeba (no CL wear or severe pain).
  - Recurrent corneal erosion (different history/presentation).

Work-Up Essentials

- Debridement of ulcer.
- PCR swab.
- Corneal sensitivity test.
- Lissamine green stain.
- Rule out penetrating injury (Seidel sign).

Condition	Antivirals	Steroids
Epithelial keratitis	Topical Aciclovir 3% 5x/day	❌ Do not use
Stromal (non-necrotising)	Aciclovir 5x/day OR Valaciclovir	Prednisolone acetate 1% (4-6x/day, taper slowly)
Stromal (necrotising)	Valaciclovir 1g tds **(three times a day)**	Prednisolone 2x/day, then taper
Endothelial keratitis	Valaciclovir 1g tds	Prednisolone 4-6x/day
Prophylaxis	Aciclovir 400mg bd OR Valaciclovir 500mg daily	❌ Not used for prophylaxis

Prognosis & Follow-up

Recurrence Risks:

- 1 in 6 for epithelial HSV.
- 1 in 3 for stromal HSV.

Prophylaxis:

- Indicated for stromal disease (reduces recurrence risk by 50%).
- Not routinely indicated for epithelial disease.

Week 6: Corneal Disease

6.1. Conjunctival Disease 2025 – Part 1

- Distinguishes between bacterial, viral, and allergic conjunctivitis.
- Bacterial conjunctivitis presents with purulent discharge, lid crusting, and conjunctival injection; treated with chloramphenicol or fusidic acid.
- Viral conjunctivitis (especially adenoviral) is highly contagious, causes watery discharge, follicles, preauricular lymphadenopathy; manage with hygiene, artificial tears, and cold compresses.
- Allergic conjunctivitis causes itching, watery eyes, and papillae on upper tarsal conjunctiva; treated with olopatadine or antihistamines.
- Diagnosis involves assessing discharge type, hyperemia pattern, and conjunctival reaction (follicles vs papillae).
- Red flags prompting referral: significant pain, reduced vision, corneal involvement, or lack of response to initial therapy.
- Emphasis on infection control practices: handwashing, avoiding towel sharing, and surface disinfection.

Proposed Allergic Conjunctivitis Management

		Level 1 MILD	Level 2 MODERATE	Level 3 SEVERE	Notes
Main factors	Itching	Mild Intermittent	Mild to severe Intermittent to persistent	Moderate to severe Persistent	If severe, consider alternative diagnosis (e.g., vernal, atopic, or GPC)
	Redness	Absent	Absent	Moderate to severe	If severe, consider alternative diagnosis (e.g. infectious conjunctivitis)
Supportive factors	Foreign body sensation, tearing, burning, and/or other symptoms	Absent	Absent	Moderate to severe	For dryness, inquire about oral antihistamines (may contribute to ocular dryness); consider diagnosis of comorbid dry eye disease
	Symptom duration	Days	Days to weeks	Weeks to months	
	Prior treatments	None	Non or OTC medications not tolerated or not effective	Previous therapy tried	
Treatment	First line	Cold compress and ocular lubricants	Antihistamine / mast cell stabilizer	Topical steroid	
	Alternatives	short-term topical OTC treatment or Antihistamine / mast cell stabilizer	Immunotherapy	Immunotherapy	
Follow-up	Clinic visit / IOP assessment	As needed	As needed	At 10-14 days then every 2-4 weeks through week 6 then every 3-6 months while using steroid	
	Complete ophthalmic exam with dilation	Yearly or as needed	Yearly or as needed	Yearly	

Bacterial conjunctivitis treatment: published literature

Aminoglycosides	Fluoroquinolones	Macrolides	Sulfonamides
Gentamicin ointment 4x/d for 1wk Gentamicin 4x/d for 1wk Tobramycin ointment 3x/d for 1wk	Besifloxacin 3x/d for 1wk Ciprofloxacin ointment 3x/d for 1wk Ciprofloxacin 4x/d for 1wk Gatifloxacin 3x/d for 1wk Levofloxacin 4x/d for 1wk Moxifloxacin 3x/d for 1wk Ofloxacin 4x/d for 1wk	Azithromycin 2x/d for 2d then daily for 5d Erythromycin 4x/d for 1wk	Sulfacetamide ointment 4x/d + bedtime for 1wk Sulfacetamide every 2-3hr for 1wk

OR chloramphenicol
OR combination (antibiotic + steroid) drugs
OR oral antibiotics
OR povidone-iodine 1.25% four times a day

Allergen	Control measures
Pollen and outdoor moulds	Limit outdoor activities during symptomatic period Monitor pollen levels using internet and media broadcast resources to plan outdoor activities Avoid rubbing eyes and nose and wash hands after being outdoors Wear closely fitting wrap around style sunglasses when outdoors Close windows and use air conditioning when in a vehicle and doors leading outside when in the home
House dust mites	Use protective pillow, mattress and duvet covers Regularly wash bedding at 60°C (at least) Vacuum and damp dust entire house on a weekly basis Remove or regularly clean carpets, upholstery, curtains and any other areas or objects that can gather dust Use a de-humidifier to reduce humidity in the home to between 35% and 50%
Animal dander	Avoid contact with animals Keep pets outdoors or none at all Regularly vacuum the home and clean areas that gather animal dander Avoid rubbing eyes or nose after being in contact with animals Wash hands after and clothes which have been in contact with animals

6.2. Conjunctival Disease 2025 – Part 2

Covers degenerative, neoplastic, and cystic conjunctival conditions.

- Pterygium: UV-related fibrovascular growth; may induce astigmatism or obscure vision; surgery considered if encroaching visual axis.
    - Lubrication: Artificial tears or lubricating eye drops to relieve dryness and irritation.
    - Anti-inflammatory medication: Topical corticosteroids (e.g., prednisolone acetate 1%) may be used for inflammation, especially in cases of irritation or mild inflammation. These should be used cautiously to avoid complications such as cataracts or elevated intraocular pressure (IOP).
    - UV protection: Advise wearing sunglasses with UV protection to prevent further UV damage, as UV exposure is a major risk factor.
- Pinguecula: Yellowish conjunctival thickening adjacent to limbus; usually asymptomatic, treated with lubricants.
    - Lubrication: Artificial tears can be used to alleviate irritation if the pinguecula causes dryness or a foreign body sensation.
    - Anti-inflammatory treatment: In cases of inflammation or irritation, low-potency corticosteroids (e.g., fluorometholone) can be used, but typically only for short durations to avoid side effects like cataract formation or elevated IOP.
- Conjunctival papilloma: Benign viral-induced lesion; consider surgical excision if symptomatic or cosmetically concerning.
- Conjunctival intraepithelial neoplasia (CIN)/SCC: Malignant potential; requires urgent referral and biopsy.
- Retention cysts/concretions: Clear fluid-filled or yellow-white masses; asymptomatic but removable if bothersome.
- Key management principle: determine lesion nature (benign vs malignant) and urgency of referral.

6.3. Contact Lens Complications 2025

- Focuses on inflammatory, mechanical, toxic, and infectious complications from contact lens wear.
- CLARE (Contact Lens Acute Red Eye): immune response to extended wear; sudden redness, photophobia, tearing; treated by discontinuing lenses and using lubricants.
- CLPU (Contact Lens Peripheral Ulcer): small peripheral infiltrate; not infectious but needs monitoring and cessation of lens wear.
- Infiltrative keratitis (IK): multiple small infiltrates; typically associated with overnight lens wear or solution reactions.
- Mechanical trauma: corneal abrasions, GPC from edge interactions or deposits; resolve by changing lens type/material and improving care.
- Toxic/allergic reactions: often due to preservatives (e.g. thimerosal, BAK) or solution incompatibilities; switch to preservative-free systems.
- Microbial keratitis (MK): red eye, pain, discharge, corneal infiltrate or ulcer; requires immediate empirical antibiotic therapy and urgent referral.
- Prevention revolves around patient education, hygiene, proper lens selection, and compliance.

Corneal Conditions 2025 – Part 1

- Corneal abrasions: acute epithelial injuries due to trauma (e.g., fingernails); present with pain, tearing, photophobia; treat with chloramphenicol drops, avoid patching.
- Topical NSAIDs (e.g. ketorolac): reduce pain, avoid overuse due to corneal melt risk.
- Cycloplegics (e.g. atropine, cyclopentolate): relieve ciliary spasm when there's anterior chamber inflammation.
    - In summary, cycloplegics work to reduce the pain and complications of anterior chamber inflammation by paralyzing the ciliary muscle, decreasing

inflammation-induced spasms, and preventing the formation of synechiae. This makes them an important part of managing conditions like iritis (anterior uveitis) and uveitis.

- Bandage contact lenses: useful for pain relief and promoting epithelial healing; ensure absence of infection before prescribing.
- Recurrent Corneal Erosion (RCE): often from prior abrasion or EBMD; manage with nighttime ointments, saline drops, and possible debridement.
- Corneal foreign bodies: assess for depth and penetration; perform Seidel test for perforation and remove using sterile technique.
- Severe trauma/penetration: needs urgent referral, imaging, and antibiotic prophylaxis.

6.4. Corneal Conditions 2025 – Part 2

- Explores chronic and degenerative corneal conditions.
- Dry Eye Disease (DED): classified into aqueous-deficient (e.g. Sjögren's) and evaporative (e.g. MGD); diagnosis with TBUT, Schirmer's, osmolarity.
- Treatment progression: artificial tears → lipid-based drops → cyclosporine/lifitegrast → punctal plugs.
- Neurotrophic keratitis: impaired healing due to trigeminal damage; managed with preservative-free lubrication, amniotic membranes, or cenegermin (NGF).
- Exposure keratopathy: occurs in Bell's palsy or lagophthalmos; managed with lubricants, moisture goggles, or surgery.
- Limbal stem cell deficiency: leads to persistent epithelial defects and vascularized pannus; managed by amniotic membrane or stem cell transplant.
- Emphasizes the link between lid disease (blepharitis, MGD) and ocular surface stability.

6.5. Microbial Keratitis 2025 (Isabelle Jalbert)

General Principles:

- Differentiates sterile vs infectious infiltrates based on symptoms, size, pain, AC reaction, discharge.
- Empirical treatment often initiated before culture results, but culture is essential in certain high-risk cases.

When to Culture:

- Large (>2mm), deep (>⅓ corneal thickness), or central ulcers.
- History suggests unusual organism (e.g. trauma with soil = fungal; hot tub exposure = Acanthamoeba).
- Poor response to initial treatment, post-surgery cases, monocular patients, or immunocompromised.

Sampling & Testing:

- Corneal scraping with Kimura spatula or blade; plate onto blood/chocolate/Sabouraud agar.
- Use Gram stain, Giemsa, calcofluor white for rapid diagnosis of fungi or Acanthamoeba.
- Confocal microscopy and biopsy may assist in diagnosis.

Bacterial Keratitis:

- Fluoroquinolone monotherapy (e.g. ciprofloxacin 0.3%) if small, non-central ulcers.
- Fortified therapy (cephazolin 5% + gentamicin 1.5%) for severe/central/deep ulcers or older patients (>60).
- Add cycloplegics for pain (e.g. atropine 1%).
- Close review after 24 hours; ensure no worsening.
- SCUT trial: Steroids at 48h showed no overall benefit, may help Pseudomonas ulcers, worsen Nocardia.

Fungal Keratitis:

- Common in trauma cases in tropical regions; rare but severe in developed countries.
- Filamentous fungi (Fusarium, Aspergillus); yeasts like Candida seen in immunocompromised.
- Insidious onset, possible delayed diagnosis; presents with feathery edges, satellite lesions, hypopyon
    - Satellite lesions are small secondary lesions or spots that appear around a primary lesion or area of infection.
- MUTT trial: Natamycin 5% superior to Voriconazole 1% for Fusarium ulcers.
- Long treatment course; cultures take weeks to grow.

Acanthamoeba Keratitis:

- Associated with contact lens wear, especially water exposure.
- Severe pain out of proportion to clinical signs.
- Ring infiltrates, perineural infiltrates, stromal opacities.
- Treated with biguanides (PHMB, chlorhexidine) and diamidines.
- Treatment is prolonged (months), steroids avoided early.

Myopia 2025 (Pauline Kang)

Understanding Myopia:

- Defined as ≤ -0.50 D spherical equivalent in relaxed accommodation.
- Progressive, typically starts in childhood, accelerates in early years, slows with age.
- High myopia increases risk of: retinal detachment, myopic maculopathy, glaucoma, cataract.

Mechanism:

- Eye growth is locally mediated and influenced by retinal defocus.
    - Myopic defocus → slows axial growth.
    - Hyperopic defocus → stimulates axial growth.
- Accommodative lag theory: leads to peripheral hyperopic defocus and elongation.
- Optical Interventions:
    - Spectacles: PALs, bifocals reduce lag (COMET, COMET2 studies).
    - Soft Contact Lenses: Dual-focus designs like MiSight.
    - Orthokeratology (Ortho-K): flattens central cornea to create myopic defocus.
    - DIMS lenses (e.g., Stellest): spectacle lenses designed to control peripheral defocus.

Atropine Treatment:

- Non-selective muscarinic antagonist, mechanism not fully understood.
- Low-dose Atropine (0.01–0.05%) used to slow axial elongation with fewer side effects.
- LAMP Study (Yam et al.): 0.05% most effective; photopic pupil size and accommodation affected dose-dependently.
- MOSAIC study: 0.01% in White children (Ireland).
- CHAMP trial: 0.01% and 0.02% over 3 years in 6–10 year-olds.

Side Effects & Clinical Practice:

- Local: photophobia, blurred near vision, allergic conjunctivitis.
- Systemic (rare): dry mouth, flushing, tachycardia; especially in small children.
- Use punctal occlusion or Double DOT to reduce systemic absorption.
- Contraindications: cardiac issues, asthma, developmental delay, children <3 months.

Monitoring:

- Assess axial length, refraction, VA, accommodation, pupil size every 6 months.
- Consider combination therapy for suboptimal control.
- Tapering strategy advised to reduce rebound effect when discontinuing.

Therapeutic Management of Amblyopia 2025 (Amanda Lea & Pauline Kang)

What is Amblyopia:

- Reduced BCVA in one or both eyes due to amblyogenic factors during the critical period of visual development.
- Must match the degree of refractive error/strabismus with vision loss.
- Diagnosis of exclusion: no pathology or retinal disease causing VA loss.

Causes:

- Refractive amblyopia: anisometropia (esp. hyperopia), bilateral high ametropia.
- Strabismic amblyopia: constant unilateral strabismus.
- Deprivation amblyopia: media opacity (e.g. cataract, ptosis).

First-line Treatment:

- Full-time spectacle wear correcting full cycloplegic prescription.

- Expect improvement over 4–6 months; monitor every 6–8 weeks.
- Continue correction until vision stabilizes.

Penalisation Therapy:

- Used if VA plateaus with correction alone.
- Atropine 1%: blurs near vision in the sound eye, forcing use of amblyopic eye.
- Patching: 2–6 hrs/day, especially effective in younger children.
- PEDIG studies: show similar outcomes between patching and atropine in moderate amblyopia.

Atropine Considerations:

- Contraindications: infants <3 months, Down syndrome, spastic CP, brain damage.
- Monitor for systemic side effects post-first dose (e.g. increased HR, drowsiness).
- Atropine + plano lens strategy may enhance penalisation (PEDIG 2015).

Compliance & Psychosocial Factors:

- Atropine better tolerated than patching in some families.
- Less stigma, better adherence especially in school-aged kids (weekend dosing recommended).
- Inform families about instillation technique, side effects, and follow-up.

When to Stop:

- After 2 consecutive visits with a stable VA.
- If VA remains unequal, re-check diagnosis, consider switching therapy method.
- Regular monitoring advised to catch regression.

Tutorial Notes (Week 6)

*Case presentation:

- Painful red eye
- Blurry vision (since yesterday)
- Photophobia “light hurts eyes”
- Cornea is clear

Diagnosis: Acute Anterior Uveitis

- To describe: grade the cells and flare (SUN)
- Hypopyon present or not

Differentials: angle closure attack, scleritis, episcleritis, conjunctivitis (viral), posterior uveitis (essential to exclude), bacterial or microbial keratitis

Differential	Reason For Exclusion
Posterior scleritis	Need to do dilated fundus exam (DFE) to rule out vitreous haze, retinitis, choroiditis, or pars planitis (snowballs/snowbanks)
Infectious keratitis	Cornea is clear (would expect corneal infiltrate if keratitis).
Acute angle closure	Open angle confirmed in workup + normal IOP.
attack	

Review of Systems for Uveitis

- General (Fever, weight loss, malaise, night sweats): Think IBD, TB, SLE.
- Rheumatologic (Arthralgias, back pain, stiffness): Think JIA, Ankylosing spondylitis, Reactive arthritis, HLA-B27, SLE.
- Dermatologic (Rashes, sores, alopecia, vitiligo, poliosis, tick/insect bites): Think Lyme disease, SLE.
- Neurologic (Tinnitus, headaches, meningism, weakness/paralysis): Think MS, Vogt-Koyanagi-Harada disease.
- Respiratory (Shortness of breath, cough, sputum): Think TB, Sarcoidosis.
- Gastrointestinal (Diarrhea, bloody stools, ulcers): Think Behçet’s, IBD.
- Genitourinary (Dysuria, discharge, genital ulcers, balanitis, STIs): Think Syphilis, Behçet’s.

First Medication (Corticosteroid):

- Prednisolone acetate 1% + phenylephrine 0.12%
- Dexamethasone 0.1%
- Loteprednol 0.5% (needs compounding).

(Note: Fluorometholone alcohol and hydrocortisone acetate ointments are NOT recommended due to their low potency.)

- Initial dosage:
  - Severe cases: every hour.
  - Mild cases: no less than 4 times daily.
  - Better to overtreat than undertreat.
  - Consider night-time dosing.

Second Medication (Cycloplegic):

- Atropine 1%, OR
- Cyclopentolate 0.5–1%.
- Cycloplegia helps:
  - Relieve pain (paralysis of ciliary body).
  - Prevent formation of posterior synechiae.
  - Stabilise blood-aqueous barrier.
- Adjust dosage for degradation by esterases in aqueous humour.

Posterior Synechiae Management

If posterior synechiae (iris sticking to lens) develops:

1. Break in-office:
   - Atropine 1% eyedrop + Phenylephrine 2.5% eyedrop.
2. If unsuccessful:

- Use a pledget technique:
  - Anaesthetise the eye.
  - Remove cotton from Q-tip head.
  - Moisten with 1% atropine.
  - Insert in lower fornix for 15 minutes.
  - Remove and check if synechiae have broken.

Monitoring and Modifying Uveitis Treatment

Treatment Step	Details
Initial Therapy	- Prednefrin Forte 1% every hour. - Cycloplegic 2–4x/day.
First Review	- Review at 24-48 hours. Check for worsening.
Assessment of Improvement	- Look for 2-step grade reduction in SUN cell/flare grading.
Maintenance	- Prednefrin Forte 1% four times daily until no cells and flare.
Tapering	- Begin tapering once no cells/flare. - Taper slowly: Reduce by one drop per month (or slower).
Cycloplegia	- Continue cycloplegic until beginning taper or cells/flare are resolved.

Complications and Prognosis

Possible Complications:

- Posterior synechiae formation.
- Cataract development.
- Secondary glaucoma (raised IOP).
- Persistent inflammation leading to vision loss.

Prognosis:

- Generally good if treated early and correctly.
- Need to manage recurrences if underlying systemic disease exists.

Follow-up Complications Scenarios

Irregular pupil:

- Caused by posterior synechiae.
- Management: Break adhesions using atropine, phenylephrine, or pledget technique.

Raised IOP (20mmHg at 3-week follow-up):

- Cause: Steroid-induced ocular hypertension.
- Management:
  - Monitor IOP closely.
  - Consider lowering steroid dosage cautiously.
  - Add IOP-lowering agents if necessary.

Week 7: Inflammatory Disease

7.1. Viral Infections 2025 – Prof Isabelle Jalbert

Viral Infections in the Eye

- Caused by DNA viruses (e.g., HSV, VZV, CMV, adenovirus) and RNA viruses (e.g., enterovirus, measles, mumps)
- Affect different ocular structures (e.g., conjunctiva, cornea, uvea, retina)
- May be primary or recurrent, latent or reactivated

Types of Viruses & Their Key Features

Herpes Simplex Virus (HSV)

- Primary infection: mild blepharoconjunctivitis
- Reactivation: dendritic or geographic keratitis, stromal keratitis, iritis
- Can lead to:
  - Neurotrophic keratopathy
  - Elevated IOP
  - Sectoral iris atrophy
- Treatment:
  - Topical antiviral (e.g. ganciclovir gel)
  - Oral antiviral (e.g. acyclovir 400mg 5x/day)
  - Topical steroids ONLY under antiviral cover

Varicella Zoster Virus (VZV)

- Causes Herpes Zoster Ophthalmicus (HZO)
- Dermatomal rash along trigeminal (esp. nasociliary) nerve
- Eye involvement: conjunctivitis, keratitis, episcleritis, uveitis
- Red flags:
  - Hutchinson's sign = nasociliary involvement → ↑risk ocular disease
- Treatment:
  - Oral antivirals: Acyclovir 800mg 5x/day for 7–10 days
  - Consider systemic steroids if severe

Cytomegalovirus (CMV)

- Common in immunocompromised (e.g. HIV/AIDS)
- Causes CMV retinitis:
  - White retinal infiltrates with hemorrhage
  - “Pizza pie” fundus
- May also cause:
  - Anterior uveitis
  - Endotheliitis
  - Corneal edema
- Treatment:
  - Oral valganciclovir or IV ganciclovir
  - Refer to ophthalmologist & infectious disease specialist

Adenovirus

- Most common cause of viral conjunctivitis
- Highly contagious
- Subtypes:
  - Pharyngoconjunctival fever
  - Epidemic keratoconjunctivitis (EKC)
- Signs:
  - Follicular conjunctivitis
  - Preauricular lymphadenopathy
  - Pseudomembrane formation
- Treatment:
  - Supportive care: cold compress, hygiene
  - Artificial tears
  - Betadine in-office (controversial)
  - Steroids for membranes ONLY if significant discomfort

Summary Table:

Virus	Common Ocular Disease	Treatment
HSV	Dendritic keratitis, uveitis	Acyclovir, topical steroids (with cover)
VZV	Herpes zoster ophthalmicus	Acyclovir 800mg 5x/day
CMV	Retinitis, anterior uveitis	Valganciclovir/ganciclovir
Adenovirus	EKC, conjunctivitis	Supportive, hygiene, ± steroid

When to Refer

- Sight-threatening keratitis or uveitis
- Immunocompromised
- Severe CMV retinitis or posterior involvement
- No improvement with standard therapy
- Diagnostic uncertainty

Clinical Pearls

- NEVER use topical steroids in viral eye disease without antiviral cover
  - An "antiviral cover" typically refers to the administration of antiviral medications to prevent or treat viral infections, particularly in cases where there's a risk of viral reactivation or spread. This term is often used in the context of managing herpes simplex virus (HSV) or varicella-zoster virus (VZV) infections, such as in conditions like herpes simplex keratitis or herpes zoster ophthalmicus. The idea is to provide antiviral drugs (like acyclovir or valaciclovir) as a preventive measure to avoid further viral activity, particularly in the eye, which can result in serious complications such as corneal scarring or loss of vision
- HSV/VZV = reactivation from latency in trigeminal ganglia
- Adenovirus is the most infectious, strict hygiene needed
- CMV = emergency in HIV-positive patients → blindness

7.2. Episcleritis, Scleritis 2025 – Prof Isabelle Jalbert

EPISCLERITIS

Presentation & Symptoms

- Acute, sectorial redness, mild irritation or foreign body sensation
- No discharge, VA normal
- May be bilateral (25%), recurs frequently (up to 10x/year)
- Simple (diffuse/sectorial) or nodular
- Looks worse than it feels

Anatomy Overview

- Affected layers: episclera + Tenon’s capsule
- Superficial vessels = radial, salmon pink
- Deep episcleral plexus = criss-cross, blue-red (only involved in scleritis)
- Conjunctival vessels = fine, interlacing

Workup

- Move conjunctiva:
  - Conjunctival vessels = highly mobile
  - Superficial episcleral = slightly mobile
  - Deep episcleral = immobile
- Phenylephrine 2.5% test:
  - Conjunctival vessels blanch quickly
  - Episcleral blanch after ~15 min
  - Scleral vessels do not blanch
- Use red-free filter to detect avascular (necrotic) areas

Differentiating Episcleritis vs Conjunctivitis

Feature	Episcleritis	Conjunctivitis
Discharge	None	Watery/mucopurulent
Pain	Mild irritation	FB sensation
Vessels	Episcleral	Conjunctival
Follicles/Papillae	Absent	Present
Age group	Young adults	Any age

Treatment

- Self-limiting (resolves ~21 days), but can treat to relieve symptoms
- Topical lubricants, cold compresses, NSAIDs – not effective
- Topical steroids for moderate–severe:
  - Prednisolone phosphate 0.5% or FML 0.1% 4–5x/day
  - Review 7–10 days; taper slowly to avoid rebound
- Oral NSAIDs:
  - Ibuprofen 400mg QID
  - Indomethacin 25mg QID
  - Diclofenac 50mg TID
  - Treat ~2 weeks
  - Switch NSAID if ineffective
- Risk of recurrence: ~12% in 1 year
- Consider 6-month NSAID course for frequent recurrences

Workup (if >5 episodes/year or systemic suspicion)

- ANA, RF, ESR, SUA, CRP
- FTA-ABS / RPR / VDRL (syphilis)

- ANCA (Wegener's), ACE (sarcoid)

SCLERITIS

Symptoms

- Severe, boring pain, radiates to face
- Worse at night, wakes patient from sleep
- Worsens with eye movement
- Gradual onset, VA often affected
- Recurrence common (same form)
- Blue sclera may indicate thinning

Differentiating Scleritis vs Episcleritis

Feature	Scleritis	Episcleritis
Pain	Severe, boring (deep)	Mild
Vessel depth	Deep scleral vessels	Superficial episcleral
Colour	Bluish-red, violaceous	Salmon pink
Blanching (phenylephrine)	No	Yes
Systemic association	Common (50%)	Rare (10%)
Complications	Frequent	Rare
Recurrence	Common	Less frequent

Scleritis Systemic Associations

- Connective tissue:
  - RA, SLE (most common)
- Infectious:
  - Syphilis, TB, HZO, HSV
- Others: gout, sarcoidosis

Classification (5 types)

1. Diffuse anterior: Most common, less severe
2. Nodular anterior: May progress to necrotising (20%), HZO common
3. Necrotising anterior with inflammation: Very painful, uveal show, high mortality (25%)
4. Scleromalacia perforans: Painless, seen in long-standing RA
5. Posterior scleritis: Rare; exudative RD, optic disc swelling, mimics tumour

Grading (Sen et al., 2011 using 10% phenylephrine)

- 0: none
- 1+: pink sclera, mild deep vessel dilation
- 2+: tortuous, purplish sclera
- 3+: severe redness, obscured vessels
- 4+: necrosis, uveal show

Treatment

- Topical steroids alone not effective
- Stepwise:

1. Oral NSAIDs (e.g. ibuprofen 600mg QID, indomethacin 25mg QID)
   - Try up to 3 NSAIDs before failure declared
2. Oral corticosteroids (e.g. prednisone 1mg/kg/day)
3. Immunosuppressants (methotrexate, ciclosporin) — 1st line in necrotising cases

- Subconjunctival steroid injections (60–100mg) possible
- Cycloplegics may help with pain
- Maintenance dose after initial control
- No tapering needed once resolved

Special Management

- Necrotising: systemic steroids/immunosuppressants; possible scleral graft
- Scleromalacia: lubricants, shields, systemic therapy
- Posterior: urgent referral, systemic immunosuppressants

7.3. Uveitis 2025

Definition

- Uveitis = inflammation of the uveal tract (iris, ciliary body, choroid)
- High vascularity → high inflammation potential
- Many causes: genetic, immune, infectious

Classification (SUN Working Group)

By Location:

- Anterior: iritis, iridocyclitis
- Intermediate: vitritis, pars planitis
- Posterior: choroiditis, retinitis
- Panuveitis: inflammation of all layers

By Clinical Course:

- Acute
- Recurrent: repeated episodes with inactivity between
- Chronic: continuous inflammation (>3 months)

By Duration:

- Limited: <3 months
- Persistent: >3 months
- Chronic = persistent + relapse <3 months after tx ends

By Onset:

- Sudden vs Insidious

Epidemiology

- Anterior uveitis: ~70% of all uveitis
- Acute anterior uveitis: ~75% of anterior cases
- Incidence: ~1/1000/year

Symptoms & Signs

Acute Anterior Uveitis

Symptoms:

- Redness (ciliary flush), photophobia, dull ache, blurred vision, tearing

Signs:

- Cells & flare (anterior chamber)
- Mutton fat KPs (granulomatous)
- Busacca nodules (iris surface)
- Koeppe nodules (pupil margin)
- Posterior synechiae, hypopyon
- Low IOP (↓ aqueous production)

Posterior Uveitis

- Blurred vision, floaters
- Signs: CME, epiretinal membrane, cataract

Grading – SUN Working Group

Anterior chamber cells (1x1 mm slit beam)

Grade	Cells
0	<1
0.5+	1–5
1+	6–15
2+	16–25
3+	26–50
4+	>50

Flare

Grade	Appearance
0	None
1+	Faint
2+	Moderate (iris/lens clear)
3+	Marked (iris/lens hazy)
4+	Intense (fibrin, plastic aqueous)

Vitreous Haze:

- No SUN consensus; Zierhut grading used

Table 5: Grading Scheme for Vitreous Haze[13]

Score	Description	Clinical findings
0	Nil	None
1	Minimal	Posterior pole clearly visible
2	Mild	Posterior pole details slightly hazy
3	Moderate	Posterior pole details very hazy
4	Marked	Posterior pole details barely visible
5	Severe	Fundal details not visible

Common Causes of Anterior Uveitis

- Idiopathic (~50%)

- HLA-B27 associated (ankylosing spondylitis, reactive arthritis, IBD)
- Sarcoidosis
- Herpetic (HSV/VZV)
- JIA
- Fuchs' heterochromic iridocyclitis
- Posner-Schlossman
- TINU (tubulointerstitial nephritis and uveitis)
- Syphilis
- Masquerade syndromes (e.g., lymphoma)

History Taking

- Present Illness
  - Onset, laterality, course, treatment response
- Past Ocular History
  - Past episodes, trauma, surgery, steroid use
- Medical History
  - Systemic diseases: IBD, JIA, SLE, AIDS, TB, syphilis
- Medications (immunosuppressants, antivirals
- Social History
  - Diet, sexual history, IV drug use

Review of Systems

System	Examples
General	Fever, malaise, night sweats
Rheumatologic	Joint stiffness, back pain
Derm	Rashes, alopecia, vitiligo
Neuro	Headache, paresthesia, tinnitus
Resp	Cough, SOB
GI	Diarrhoea, oral ulcers
GU	Dysuria, discharge

Diagnostic Signs for Differentiation

Sign	Associated Condition
↑ IOP	Herpetic, FHIC, Posner-Schlossman
Fibrin	HLA-B27, endophthalmitis
Hypopyon	HLA-B27, Behçet's, Rifabutin
Iris nodules	Sarcoidosis, TB, syphilis
Iris atrophy	Herpetic, FHIC
Band keratopathy	JIA (young), chronic uveitis
Quiet eye	FHIC, masquerade syndromes

Workup

Non-Specific Testing

- Indications:
  - Recurrent, bilateral, granulomatous, severe, resistant to treatment
  - If history doesn't indicate a specific cause

Test	Purpose
CBC, ESR, CRP	Inflammatory markers
ANA, RF	Autoimmune screening
ACE	Sarcoidosis
PPD	TB
VDRL, RPR, FTA-ABS	Syphilis
Lyme titre	Lyme disease
Chest X-ray	TB, sarcoid, lymphoma
Urinalysis	TINU

Specific Testing

- Toxoplasma IgG/IgM
- Toxocara ELISA
- HLA typing (e.g., B27, B51)
- Aqueous/vitreous biopsy (PCR)
- Imaging: OCT, FFA, B-scan, MRI (posterior uveitis)

Management

Topical Steroids

- Prednisolone acetate 1% + phenylephrine 0.12%: q1h → taper
- Alternatives: Dexamethasone 0.1%, loteprednol 0.5% (compounded)
- For mild: pred phosphate 0.5% minims
- Taper only when cells & flare = 0 (5–6 weeks+)
- Night dosing: hycor 1% ointment
- Caution: HSV history (add acyclovir prophylaxis), IOP rise, PSC cataract

Cycloplegia

- Atropine 1%: 1–4x/day for severe
- Cyclopentolate 1%: 2–3x/day for mild
- Break synechiae: add phenylephrine 2.5%
- Advise Px re: driving, sunlight, blurred vision

Breaking Posterior Synechiae

- Try atropine + phenylephrine
- If ineffective, try pledget:
  - Cotton bud soaked in cyclopentolate 1% + phenylephrine 2.5%
  - Insert into lower fornix 15–30 min
  - Repeat at 24h if needed

Monitoring

- 1st review: 24h
- Maintain hourly drops until 2-step SUN improvement
- Then reduce to QID maintenance until resolution
- Begin taper only after no cells/flare
- Cycloplegia: stop once taper starts

HLA-B27 Uveitis

- Bilateral, recurrent, alternating
- Associated: Ankylosing spondylitis, reactive arthritis, psoriatic arthritis, IBD
- Refer: rheumatology, GI, sexual health
- Treat systemic disease + uveitis
- Workup: HLA typing, ESR

Prognosis

Depends on:

- Cause/type/severity
- Prompt & effective treatment
- Recurrence common (24% annually)
- Warn Px to return if symptoms recur

Uveitis Management Paradigm

Type	First-line	Escalation
Anterior	Topical steroid + cycloplegic	Add oral steroid if severe
Posterior	Refer to ophthalmologist	Oral steroids, periocular/intravitreal injections, immunosuppressants

Posterior Uveitis

- Outside scope → REFER
- Tx: Oral prednisone (0.5–1mg/kg), IV or periocular steroids, immunosuppressants (e.g. methotrexate, ciclosporin)

Hyphema

- Causes: blunt trauma, blood thinners
- Workup:
  - Measure hyphema height
  - Check IOP
  - NO gonioscopy in acute phase
- Tx:
  - Bed rest upright, shield (no patch)
  - Cycloplegic: atropine or cyclopentolate

- Topical steroid if secondary uveitis
- IOP >24h: timolol 0.5% or brimonidine 0.2%
- Avoid NSAIDs/aspirin → use paracetamol
- Warn Px rebleed symptoms
- Gonioscopy/DFE post-resolution

7.3. Management of Episcleritis and Uveitis 2025

EPISCLERITIS – Summary Management

Key Features

- Acute, sectorial/diffuse redness
- Mild irritation or foreign body sensation
- No discharge, VA normal
- Recurs in 10–15%, typically self-limiting (~3 weeks)

Management

First-Line:

- Observation if mild (lubricants, cold compress = low evidence)
- Topical steroids:
  - Prednisolone sodium phosphate 0.5% or FML 0.1% qid to 5x/day
  - Review in 7–10 days
  - Slow taper if recurrences

Oral NSAIDs:

- Ibuprofen 400–800mg QID
- Indomethacin 25mg QID
- Diclofenac 50mg TID
- Trial different NSAID if no effect by 1 week

Recurrences:

- Frequent attacks: consider 6-month NSAID prophylaxis
  - e.g. Piroxicam 10mg daily

Investigations:

- Not routinely needed unless >5 episodes/year
- Review of systems + optional labs:
  - ANA, RF, ESR, CRP, ACE, ANCA, VDRL/RPR, SUA

UVEITIS – Summary Management

Key Features

- Inflammation of iris, ciliary body, choroid
- Anterior most common (~70%)
- Symptoms: photophobia, redness (ciliary), ache, blurred vision, tearing

Grading (SUN)

- Cells: graded 0 to 4+ based on # of cells per 1x1mm slit beam field
- Flare: based on protein haze/iris detail
- Vitreous haze: useful in intermediate/posterior cases

Topical Therapy

Corticosteroids:

- Prednisolone acetate 1% (+ phenylephrine 0.12%) q1h until 2-step reduction in SUN grading
- Reduce to QID maintenance until no cells/flare
- Taper slowly (1 drop/week) only when AC quiet
- Severe cases: consider loading dose q15–30min for 2h
- Caution in HSV Hx → add acyclovir 400mg 5x/day

Cycloplegia:

- Atropine 1%: 1–4x/day (severe)
- Cyclopentolate 1%: BID–TID (mild-moderate)
- Add phenylephrine 2.5% if synechiae
- Stop cycloplegia once taper starts

Monitoring

- Review at 24h
- Look for 2-step improvement or grade 0 cells/flare
- Maintain dosing until quiet, then taper
- Monitor IOP at every visit (risk of steroid response)

Workup (if needed)

- Recurrent, bilateral, granulomatous, resistant → consider:
- CBC, ESR, CRP, ANA, RF, ACE, PPD, VDRL/FTA-ABS, chest X-ray
- Specific: HLA-B27, toxoplasma/tuberculosis serology, Lyme

Referral

- Posterior/intermediate/panuveitis
- Failure of topical tx
- Suspected masquerade syndromes
- Refer to: ophthalmologist, rheumatologist, infectious disease specialist

Prognosis

Highly dependent on:

- Underlying etiology
- Timeliness of treatment
- Recurrence rate: ~24% per year for anterior uveitis
- Always educate patients to return immediately with recurrence of symptoms

Hyphema (Traumatic)

Features

- Pain, blurred vision
- History of blunt trauma
- Risk of ↑IOP from trabecular blockage
- Check for: cells, flare, measure height, IOP

DO NOT:

Perform gonioscopy in acute phase

Management

- Rest, shield, upright head position
- Cycloplegic (e.g. atropine or cyclopentolate)
- Topical steroid if inflammation (pred acetate 1% QID–hourly)
- Anti-glaucoma drops if IOP high at 24h:
  - Timolol 0.5% BID or brimonidine 0.2% TID
- Avoid NSAIDs & aspirin (paracetamol ok)
- Daily follow-up until resolution

Long-Term

- Risk of: rebleed, PAS, corneal blood stain, angle recession
- Follow-up: gonioscopy, IOP monitoring after resolution

Week 8-10: Refractive Errors + Co-Management

8.1. Co-management & Collaborative Care (Dr. Melinda Toomey)

Overview:

This document details how optometrists collaborate with other healthcare professionals, particularly in co-management settings such as post-operative cataract and refractive surgery care, diabetic screening, and glaucoma management. It emphasizes shared care principles, communication standards, and professional responsibilities.

Definitions:

- Collaborative care: Care provided by two or more health practitioners within their scope of practice in consultation with the patient.
- Co-management: Shared responsibility in managing a patient's care where both parties agree and the patient consents.

Types of Collaborative Relationships:

- Ad hoc (informal arrangements).
- Formal shared-care agreements (e.g., glaucoma shared care).

Benefits:

- Patient: Improved healthcare, timely access, consistent communication.
- Practitioner: Resource optimisation, preventive care, mutual learning, enhanced referral dynamics.

Principles of Effective Care (Kassam, Mitchell):

- Patient-centred, evidence-based.
- Shared goals, clear roles, mutual trust.
- Effective communication, measurable outcomes.

Optometry Board of Australia Recommendations:

Clarity on diagnoses, treatment roles, information formats, timelines, follow-up responsibilities.

Communication – The 7 C's:

- Clear: Purpose evident.
- Concise: Brief and focused.
- Concrete: Specific and tangible.
- Correct: Accurate, appropriate level.
- Coherent: Logical structure.
- Complete: Includes all required info.
- Courteous: Respectful and open.

Co-management Relationships:

- With GPs: Diabetes management, hypertensive retinopathy, red eye management.
- With Ophthalmologists: Cataract/refractive surgery, corneal grafts, glaucoma.

Glaucoma Shared Care:

Follow NHMRC guidelines for target IOP, management planning, and review frequency.

Referral required if:

- Glaucoma is not stable.
- Surgery or laser needed.
- Side effects from initial treatment.
- 4 months post-initial diagnosis without specialist review.

Referral Protocol:

- Triggers: Beyond scope, patient request, second opinion.
- Must include: Reason, test results, history, medications, and expectations.
- Maintain patient contact if referral unfulfilled.

Urgency Guidelines:

- Emergency: Same-day (e.g., disc swelling, sudden vision loss).
- Urgent: 1–30 days.
- Routine: >30 days.

8.2. Cataract and Refractive Surgery Co-Management (Dr. Melinda Toomey)

Preoperative Assessment of Cataracts

- Is the second leading cause of vision impairment in Australia

Minimum testing	Recommended ancillary testing
Visual acuity Pupil reactions and Pupil size Eye dominance Ocular motility Visual field screening Refraction IOP measurement Anterior segment examination Fundus examination (dilated) OCT Imaging (macula)	Contrast sensitivity Glare test Potential acuity testing

Ancillary Test	What?
Contrast sensitivity	Contrast sensitivity: smallest difference in lightness & darkness between an object & its background that can be distinguished • Decreased CS contributes to symptoms of poor vision in cataract px ○ Cataract px can still have good VA despite reporting poor vision since the opacification of the lens can impact CS but not central vision • Good CS is important for VA, VF, motion detection, dark adaptation, pattern recognition
Glare test	Glare: visual perception created when external light becomes scattered within the eye causing a hazy veil • Cataract px can have good VA in a dim room but poor VA in bright illumination due to glare • Glare testing quantifies the effect of light scatter on visual performance
Potential acuity testing	• Is it important to predict post-operative VA with cataract px prior to surgery ○ Poor BCVA & poor potential acuity may reveal that cataract is not the sole reason for a px's decreased vision & symptoms

Lecture Goals:

- Equip optometrists to handle pre- and post-operative care.
- Manage surgical complications.

Cataract Overview:

- Prevalence: 2nd leading cause of vision loss; 1.5M Australians over 55 affected.
- Optometrists' role: Early diagnosis, ongoing management, referrals.

Pre-op Testing:

- Minimum: VA, refraction, IOP, pupil response, anterior and posterior segments, ocular motility, VF.
- Ancillary: CS (Pelli-Robson, SPARCS), glare (BAT, C-Quant), potential acuity testing.

Grading Cataracts:

- LOCS III: Classifies nuclear, cortical, PSC types.

Ocular Comorbidities:

- Posterior: DR, RVO, ERM → Increased CME risk.
- Anterior: EBMD, Fuchs' dystrophy, uveitis, DED → Affects surgical planning and IOL choice.

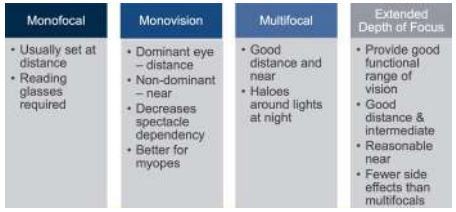
Vision-related Activity Limitation Tools:

- CAT-PROM-5, Catquest-9SF: Measure functional vision impairment for public referrals.

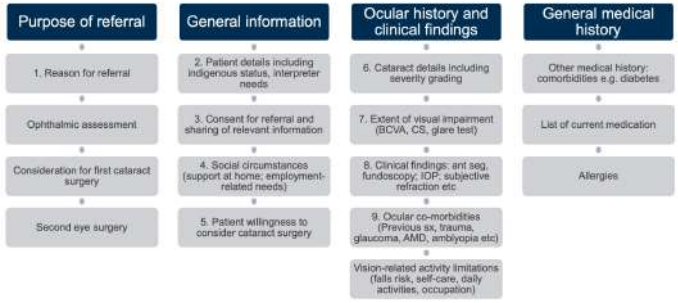
IOL Options:

- Monofocal: Distance or near, reading glasses needed.
- Monovision: One eye near, one distance.
- Multifocal/EDOF: Reduce glasses dependency, possible haloes.

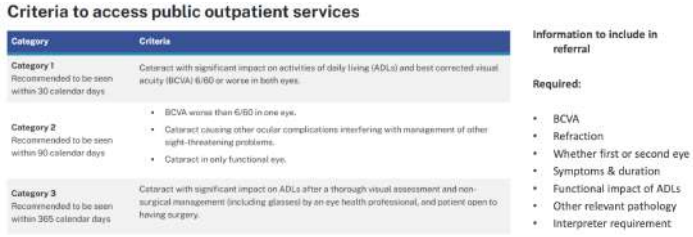
Types of IOLs



Cataract Referral Letter Components



NSW Health Cataract Referral Criteria



AB Prophylaxis:

- Intra-op: Intracameral cefazolin (5× ↓ endophthalmitis).
- Post-op: Chloramphenicol 0.5% qid for 7 days.

Inflammation Control:

- Standard: Steroid + NSAID drops × 1 month.
- Novel: Intracameral dexamethasone, intracanalicular inserts.

Follow-up Timeline:

- Day 1–2: IOP, wound check.
- 1 week: Outcome review, second eye consideration.
- 1 month: Rx finalisation, DFE.

Post-Operative Examinations & Findings

Interval history	• Post-surgery medication use • New symptoms • Self-assessed vision
Visual function	• VA & pinhole VA • Retinoscopy

	• Refraction
IOP	Performed at every visit
Anterior segment	• Examine incision site • Check for any AC reaction
DFE	Routine DFE exam in late post-surgery follow up
Educate	Advise px of post-surgery care, sign/symptoms to expect or look out for

Complications:

- PCO: Most common; treat with Nd:YAG.
- Endophthalmitis: Rare but critical (pain, hypopyon, redness).
- CME: Peaks ~6 weeks.
- High IOP: Early (retained viscoelastic); Late (steroid response).
- Corneal oedema, RD.

Special Populations:

- Dementia: High visual impairment rates; early surgery reduces cognitive decline risk.
- 8x higher hazard of dementia among the visually impaired
    - There are high rates (66.9%) of undiagnosed vision loss in people with dementia
- Cataract extraction is associated with a lower risk of dementia development
    - Best to receive surgery sooner rather than later

Refractive Surgery Pre-Operative Assessment

DISCONTINUE SCL WEAR FOR 1 WEEK PRIOR TO SURGERY
DISCONTINUE RGP WEAR FOR ~3 WEEKS PRIOR TO SURGERY

- Distance & near VA with and with correction
- Manifest & cycloplegic refraction
- Corneal topography/tomography
- Central corneal thickness measurement
- Tear film and ocular surface evaluation
- Anterior segment assessment
- Ocular motility and alignment evaluation
- Fundus examination
- IOP measurement

REFERRALS

EMERGENCY REFERRALS: WITHIN 1-24 HOURS
URGENT REFERRALS: WITHIN 1 WEEK TO 30 DAYS
ROUTINE REFERRALS: >30 DAYS

Emergency Referrals	Urgent Referrals
• Stroke (Immediate referral – call 000) • CRAO (Immediate referral within 1 hour • Trauma to lids, orbit, ocular structures • Penetrating eye injuries • Retained intraocular foreign bodies • Hyphemas – severe (Grade 2/>) • Chemical injuries (Acid – Grade 3/>; Alkali – Grade 2/2) • Acute Angle Closure Glaucoma • IOP ≥ 45mmHg • Orbital Cellulitis • Papilloedema • Retinal detachment, symptomatic retinal tears • Sight-threatening keratitis • Sudden, severe ocular pain • Unexplained pre-retinal haemorrhage • Unexplained sudden vision loss	• Acute dacryoadenitis • Acute dacryocystitis • CMV and candida retinitis • Commotio retinae • CRVO with elevated IOP • Disc haemorrhage • IOP >35mmHg (and ≤45mmHg) • Retrobulbar/optic neuritis • Rubeosis • Scleritis • Squamous cell carcinoma • Sudden onset of diplopia • Wet macular degeneration/choroidal neovascular membrane • Diabetic conditions with sudden or severe loss of vision

8.3. Antimicrobial Resistance (PDF – Prof. Mark Willcox)

Antibiotic: substance that kills or halts the growth of a microorganism
MRSA: Methicillin resistant Staph aureus
Definitions:

- Antibiotic: Drug made by microbes to inhibit other microbes.
- Mechanisms: Inhibit wall synthesis, protein/DNA synthesis.
    1. Structural attack: affects cell walls or cytoplasmic membranes
    2. Metabolic attack: affects cellular metabolism & reproduction by acting on proteins & nucleic acid

Resistance Mechanisms:

1. Mutation of target proteins.
    - Altered bacterial protein (eg. streptomycin)
    - No longer recognises the antibiotic but continues to perform normal function in bacterial growth
2. Enzymatic destruction (e.g., penicillinase).
    - Bacteria acquires a new gene which codes for an enzyme that destroys the antibiotic (eg. penicillinase (enzyme) destroys penicillin)
3. Efflux pumps (e.g., tetracycline).
    - Bacteria acquires a new gene which codes for a protein which pumps the antibiotic back outside the cell (eg. tetracycline resistance)
4. Membrane impermeability.
    - Bacteria alters their cell wall/ membrane so that the antibiotic cannot penetrate the cell (eg. polymyxin B)
    - Beta-lactamase enzymes hydrolyse & destroy beta-lactam penicillin type antibiotics
        - Changes to the proteins in the outer membrane of gram negative bacteria known as porins decreases the ability of antibiotics (eg. beta-lactams) to gain access to cell walls of gram negative bacteria
            - Porins are important for nutrient uptake in the cell
        - eg. penicillinase destroys penicillin by hydrolysing (adding water) to the beta-lactam ring structure
            - Breaks the beta-lactam ring where it is no longer able to inhibit enzymes in the cell wall of bacteria
    - Cells can also use proteins (eg. porins) to quickly remove antibiotics from their insides (eg. fluoroquinolone resistance)
        - Antibiotic concentration in the cell does not reach a level that inhibits cellular function
    - Most common way of drug resistance is target site modification (eg. fluoroquinolone resistance)
        - Target enzymes, DNA gyrase & topoisomerase IV are commonly altered

Multiple Resistance

- Some resistance was already naturally present in bacterial populations
    - Presence of higher concentrations of antibiotics in the environment increased the pressure by natural selection
    - Resistant bacteria passed their resistant genes onto other bacteria

Genetic Acquisition:

- Transformation: Uptake from environment.
- Conjugation: Plasmid transfer.
- Transduction: Bacteriophage-mediated.
- Transposons, Integrons: Mobile resistance elements.

Genetic Basis Of Resistance

Mode	What?	Process
Spontaneous mutations in endogenous genes	• Structural genes: expanded spectrum of enzymatic activity, activity, target-site modification, transport defect • Regulatory genes: increased expression	• A single bacterial cell under ideal conditions can replicate itself within 30 minutes 1. Presence of a mutagenic agent causes DNA mutations in dividing bacteria 2. Growth & division of bacteria produces naturally-occurrin

		g mutants • Mutation can cause the binding proteins to no longer recognise the antibiotic 3. Bacteria continues to multiply, giving rise to different mutants 4. Exposure to antibiotic selective pressure causes mutants with the least fitness cost to proliferate & disseminate 5. Bacteria are killed or prevented from growing except for specific resistant mutants • Allows mutated proteins to grow in place of the dead normal bacteria
Acquisition of exogenous genes	• Involves obtaining resistant genes from other resistant bacteria ○ Usually genes that conde inactivating enzymes or modified targets, regulatory genes • Mechanisms of DNA transfer include: ○ **Conjugation** (cell-cell contact): transfer of DNA from bacterial donor cell to a recipient cell by cell-cell contact ■ Transfer is mediated by a pilus	1. 2 populations of different bacteria (sensitive vs. resistant bacteria) are growing in the same environment 2. Exchange of DNA occurs between bacteria at a low frequency 3. Sensitive bacterial species acquires a resistance gene 4. Exposure to antibiotic selective pressure causes cells with acquired resistance to proliferate & sensitive bacteria to die/ prevented from growing
	• Pilus: tube-like structure formed between cells ○ **Transformation** (uptake of DNA in solution): process of genetic alliteration by pure DNA ■ Recipient cells acquire genes from DNA outside the cell ■ DNA is taken up by the cell & recombinations with genes on a bacterial chromosome ■ May alter phenotype of recipient cells ○ **Transduction**: bacterial DNA is transferred from one bacterial cell to another by a bacteriophage (virus) ■ Transferred DNA can be integrated into the host chromosome by recombination	

Resistance vs Tolerance:

- Resistance: Heritable, high MIC.
- Tolerance: Non-heritable, transient survival (MBC/MIC ≥32).
- Tolerance may lead to resistance with prolonged exposure.

Testing:

- MIC and MBC assays.
- TDtest to measure persistent colonies post-antibiotic.

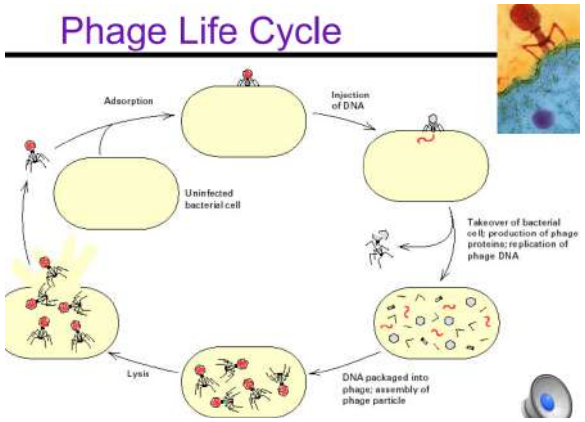
Clinical Evidence:

- Resistance detected in ocular S. aureus and P. aeruginosa (keratitis).
- Ciprofloxacin: Higher MIC → ↑ scar size, ↓ VA, slower healing (SCUT trial).

Global Impacts:

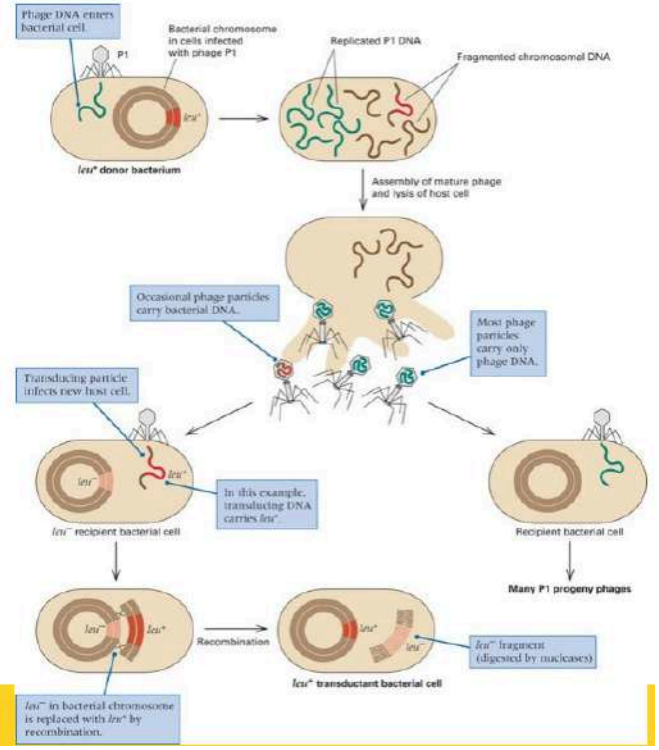
- High resistance in TB, gonorrhoea, malaria.
- Misuse: AB prescribed for viral illness.

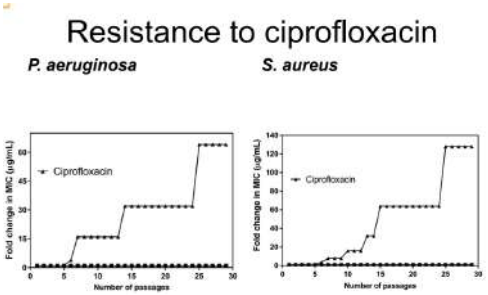
Bacterial Resistance (PPT – Prof. Willcox)\



Phage Life Cycle

1. Virus binds to a bacteria & injects DNA
2. Virus takes over the bacterial cell to produce phage proteins that are involved in replication of phage DNA
3. DNA is package into phage, assembling a phage particle
4. Lysis: disintegration of a cell by rupture of a cell wall membrane





Transposons & Integrons

Transposons ('jumping genes'): DNA sequences that can move from one location in a genome to another

Integrons: genetic elements in bacteria that act as mobile platforms for capturing & expressing new genes, particularly those conferring antibiotic resistance

Resistance & Tolerance

Resistance: inherited ability of microorganisms to grow at high concentrations of an antibiotic, irrespective of the duration of treatment

- Quantified by MIC of the particular antibiotic

Tolerance: ability (inherited or not) of microorganisms to survive transient exposure to high concentrations of an antibiotic without a change in the MIC

- Achieved by slowing down an essential bacterial process
- eg. beta-lactams require active cell wall assembly to kill bacteria so slower growth will result in alonger minimum treatment duration to achieve the same level of killing, regardless of the concentration of the antibiotic
- Can be a precursor to resistance especially if the resistance is due to mutations to existing genes within the bacteria
    - As the bacteria persists during antibiotic exposure, they have a greater chance of accumulating mutations that make them resistant & allow them to begin to grow in the presence of the antibiotic
- To measure tolerance:
    - Perform normal MIC & minimum bactericidal concentration (MBC) tests
    - If MBC/ MIC ≥ 32, bacteria is considered to show tolerance to the antibiotic

Complications of Antibiotic Resistance

1. Increased morbidity & mortality
    - "best-guess" therapy may fail with the px's condition deteriorating before susceptibility results are available
    - no antibiotics left to treat certain infections
2. Greater health care costs
    - More investigations required
    - More expensive, toxic antimicrobials required
    - Expensive barrier nursing, isolation, procedures
    - Therapy priced out of the reach of some third-world countries

8.4. Controlling Myopia Reading (IMI 2019)

Intervention Categories:

Optical

- Spectacles: PALs/bifocals → minor effect.
- Multifocal CLs: Effective, centre-distance designs induce peripheral myopic defocus.
- Orthokeratology: Overnight corneal reshaping, effective but with keratitis risk.

Pharmacological

- Atropine:
    - 1% most effective, but significant side effects.
    - 0.01% → fewer effects, moderate efficacy.
- 7-MX: Promising oral agent, not mainstream.
- Pirenzepine, Timolol: Minimal benefit.

Behavioral

- Outdoor time: ↑ exposure → ↓ onset & progression (dopamine, light intensity effects).
- Near work: ↑ myopia risk; wearables used to track behavior.

Surgical/Experimental

- Posterior scleral reinforcement: Limited clinical use.
- Collagen cross-linking, injectables: Experimental.

Combination Therapies

- Example: Atropine + OK → 50% more effective than OK alone.
- Smart lenses/drug delivery in development.

Conclusion:

- No universal cure.
- Best outcomes may come from personalised, combination-based approaches.
- More longitudinal and mechanistic research needed.

8.5. Clinical Antibiotic Resistance (Hui)

Lecture Objectives

- Discuss antibiotic resistance as encountered in clinical optometry practice.
- Guide how antibiotic-resistant ocular infections should be managed.

Choosing the Right Antibiotic to Prescribe

Key considerations:

- Diagnosis, risk factors, and pathogenesis: Accurate diagnosis impacts treatment decisions.
- Antibiotic spectrum of activity: Whether broad or narrow spectrum.
- Local environment: Awareness of common pathogens and resistance profiles.
- Additional testing: Including cultures and sensitivities.
- Patient factors: Allergies, comorbidities, and age.
- Pharmacology: Includes drug bioavailability and penetration at the site of infection.
- Cost, availability, and route of administration are also relevant.

Empirical vs Directed Therapy

- Empirical Therapy (best guess based on clinical signs/symptoms):
    - Relies on clinician's interpretation of symptoms.
    - Offers timely treatment but carries misdiagnosis risk.
- Directed Therapy (lab-confirmed)
    - Based on culture or organism identification.
    - More accurate but delayed due to test turnaround.

<table>
<tr><th colspan="2">Empirical Therapy</th><th>Directed Therapy</th></tr>
<tr><td colspan="2">• Medical 'best guess' of the cause of signs & symptoms & what treatment could work
• Relies on clinician interpretation of data & experience
• Trades off potential for misdiagnosis with benefit of timely treatment</td><td rowspan="2">• Identifies causative organism for disease and potential effective treatments
• Trades off more certainty in treatment with delay in treatment</td></tr>
<tr><th>Broad Spectrum Antibiotics</th><th>Narrow Spectrum Antibiotics</th></tr>
<tr><td>Advantages:
• Affects multiple species hence we do not need to know the exact cause of infection
• Provides prophylaxis against secondary infection
Disadvantages:
• Could damage the normal flora found on the eye
• Selection of resistance from non-causative organisms</td><td>Advantages:
• Targeted elimination of the causative organism with less effect on other natural flora
• Less chance of multiple organisms developing resistance

Disadvantages:
• May require lab testing to identify organisms which delays treatment
• May not be readily available</td><td></td></tr>
</table>

Spectrum of Antibiotic Activity

Broad Spectrum	Narrow Spectrum
✅ Targets multiple species; helpful when pathogen is unknown.	✅ Specific to the pathogen; minimal impact on normal flora.
❌ Disrupts normal flora; risk of resistance in non-causative organisms.	❌ Requires lab tests and may delay treatment.

Local Environment and Resistance

- Example: Queensland Microbial Keratitis Database (2005–2015) showed resistance trends.
- Increasing fluoroquinolone resistance reported in the USA for Staphylococcus species (Miller et al. 2006).

Clinical Signs of Antibiotic Resistance

Important questions:

- Is the condition improving as expected?
- If not, is the dose correct and is the patient compliant?
- Should you revisit the differential diagnosis?

Culture and Sensitivity Testing

- Used for directed therapy.
- Guides the clinician to choose a more targeted treatment.
- Takes time, but vital if empirical therapy fails.

Summary

- Clinical presentation, local resistance, and patient specifics drive antibiotic choice.
- Lack of improvement should prompt reevaluation of diagnosis, resistance, or compliance.
- Cultures and sensitivity testing, although delayed, are valuable in guiding effective treatment.

8.6. Eye Swabbing – The Why and How for Australian Optometrists – Fiona Moore

Introduction

- Eye swabbing is becoming more essential as most Australian optometrists are therapeutically endorsed.
- 420,000 GP eye-related consultations per year in children; 62% are for conjunctivitis.
- Global conjunctivitis epidemics are often misdiagnosed due to unknown aetiology, leading to treatment errors and economic burden.

Why Perform Eye Swabs?

- Perform only when results impact treatment.
- Most conjunctivitis is self-limiting, but some cases need targeted treatment or referral.
- Distinguishing bacterial, viral, and allergic conjunctivitis is difficult—swabs improve diagnostic accuracy.
- Chlamydia conjunctivitis:
    - Often overlooked.
    - Not notifiable in Australia.
    - Prevalence estimated between 1.8–5.6% of acute cases.

Current Management of Conjunctivitis

- Treatments: Antimicrobials, antihistamines, lubricants.
- Many patients self-medicate with Chlorsig® before consulting.
- Empirical antibiotic treatment may not benefit all and contributes to misuse.

How Eye Swabs Influence Treatment

- NZ studies: Blanket antibiotic use relieved symptoms but wasted resources.
- Targeted approach: Swab + culture = more accurate treatment with fewer unnecessary antibiotics.
- Preliminary culture results (available next day) are highly predictive and useful.

Guiding Principles for Antimicrobial Use (MINDME)

- Microbiology guides therapy
- Indications should be evidence-based
- Narrowest spectrum used
- Dosage site-appropriate
- Minimise treatment duration
- Employ monotherapy where possible

Accessing Eye Swabs

- Partner with local pathology providers (e.g., QML).
- Two options:
    - Swab in-practice and send it.
    - Patient takes referral form to collection centre.
- Provide history, eye sampled, antibiotics used, and requested tests on the pathology form.

Swabbing Procedure

1. Informed Consent: Especially when chlamydia is suspected.
2. Infection Control: Wash hands, gloves, clean surfaces, dispose of PPE.
3. No Anaesthetic or Fluorescein before swabbing.
4. Bacterial swabbing: Gentle sweep of lower fornix from nasal to temporal side.
5. Viral/chlamydia: Vigorous rubbing of tarsal conjunctiva (needs cell bodies).
6. Place swab in culture medium and label properly.
7. Pathology Form: Include eye, laterality, any prior antibiotics, suspected conditions.
8. Notify patient: Turnaround ~24 hrs. Arrange review especially if diagnosis = STI.
9. Arrange pickup of samples (usually via email to pathology).

What Do You Test For?

- HSV
- Varicella
- Chlamydia
- MC&S for bacteria, fungi, gonococcus

Cost to the Patient

- No Medicare rebate for optometrist-initiated swabs.
- Fee = $0–$170, depending on pathology provider.
- Inform patient beforehand.
- Expanding swabbing may help build case for rebate advocacy.

If It's Chlamydia

- Refer patient to GP or Sexual Health Services.
- Anonymous web-based testing available for reluctant patients.

Additional Theory Notes

Resources To Consult

- Product Information (PI) Leaflet – find out more about special populations or drug interactions
    - Scientific, objective account of a medicine's usefulness and limitations
    - Describes both active and inactive ingredients – how the medicine acts, metabolised and eliminated
    - Clinical trials
    - Indications, contraindications, precautions, known adverse reactions
- TGA (Therapeutic Goods Association) – responsible for evaluating, assessing and monitoring products that are defined as therapeutic good
- Consumer Medicine Information (**CMI**) – based on PI, written specifically for the consumers
    - Information and instructions on how to use your medicine safely and effectively
- National Prescribing Services (**NPS**) Medicine Wise – working towards QUM (Quality Use Medicines) in Australia\
    - Provides practical tools and information about medicines, health conditions and medical tests (including access to CMI on many approves drugs)
- Australian Register of Therapeutic Goods (**ARTG**) – a searchable engine that contains information on therapeutic goods that can be supplied in Australia (this includes their schedule)
- Australian Medicines Handbook (**AMH** Online)
    - Detailed profiles of individual drugs (prescribing guidelines, drug interactions, off-label uses and PBS status)
- eTG complete Therapeutic guidelines
    - Condition based treatment algorithms (management of clinical presentations based on best evidence)
    - Therapeutic management – first, second line drugs and when to escalate or refer
    - Non pharmacological treatment guidelines
    - Link to clinical decision tools (i.e. when to refer) and relevant drug protocols (e.g. treating HSV keratitis)

Week 1 Readings:

Optometrical Items: Optometrist PBS prescribing

- Optometrist PBS prescriptions are identifiable by colour and include the words 'PBS/RPBS optometrist'.
- Prescriptions must include the optometrist's PBS prescriber number.
- The same optometrist prescription form is used to prescribe unrestricted, restricted or authority items.
- **Only one item is allowed per form.**
- **Optometrist PBS prescriptions may include repeats**
- **Authority Rx:** for authority required items or for increased quantities or repeats, require prior approval from Services Australia or the Department of Veterans' Affairs (DVA) for each prescription. DVA approval for non-Schedule items is not available for optometrist prescribing.
- **RPBS:** Optometrists approved as PBS prescribers may write prescriptions for supply under the RPBS. The medicines available under the RPBS are the same as those available under the PBS. There are **no optometrist listings in the Repatriation Schedule for prescribing for veterans only.**

Optometry Board of Australia Guidelines for use of scheduled medicines

1	Endorsement for scheduled medicines – optometrists whose registration is endorsed for scheduled medicines must be qualified & competent to obtain, possess, administer, prescribe, supply or use specified scheduled medicines and use those medicines appropriately for the purposes of the practice of optometry
2	Use of scheduled medicines by optometrists
2.1	Quality use of medicines selecting management options wisely, choosing suitable medicines, using medicines safely & effectively to get best possible results
2.2	Maintaining competence optometrists must meet requirements set out in the Optometry Board's registration standard including endorsement for scheduled medicines
2.3	Prescriptions practitioners must ensure that, when authorising the supply of a schedule 4 medicine to a patient, the prescription is handwritten or computer generated when prescribing a schedule 2 or 3 medicine, the Board encourages endorsed optometrists to issue a prescription to help ensure effective communication with the pharmacist *Prescriptions must include date of issue, details of the prescriber & px, medicine including active ingredient, strength, quantity, precise directions, prescriber's signature *advised against self-prescribing schedule 4 medicines
2.4	Practice procedures scheduled medicines must be stored securely & in accordance with the manufacturer's recommendations details of in-practice administration of scheduled medicines are recorded on the px clinical record
2.5	Adverse event reporting TGA collects reports of adverse events associated with medicines & medical devices – allows TGA to investigate & take action on safety issues of meds
3	Supply of scheduled medicines expertise of pharmacist in counselling
	px is important in the follow-up care of the patient
4	Guidelines for the use of topical antimicrobials all optoms treating anterior eye infections with scheduled medicines must have a clear understanding of: • Microbiological and pathological processes relevant to anterior eye infection and their natural histories • Typical presentations of ocular surface infections and the differential diagnoses of these from other anterior eye conditions • Ocular infections that constitute true ocular emergencies and need immediate treatment • Identification of risk factors for ocular surface infections • Indications and mechanisms of referral for microbiological investigation • Interpretation of the results of microbiological investigation and appropriate management stratagems arising from these results
4.1	Antimicrobial resistance selection of antimicrobial should always involve consideration of risk that microbial resistance could develop tx regimens that could result in inappropriate drug selection, insufficient therapeutics, overuse or inappropriate dosage should be avoided *fluoroquinolones should not be used when alternative, equally effective agents could be used instead (to maintain maximal efficacy of fluoroquinolones)
5	Guidelines for use of topical steroidal preparations: Optometrists seeking to treat ocular inflammation with scheduled medicines must have a clear understanding of the: • Immunological processes relevant to inflammatory conditions of the eye and their natural histories • Typical presentations of inflammatory conditions of the eye and the differential diagnoses from other anterior eye conditions • Inflammatory conditions of the eye that constitute true ocular emergencies and need immediate treatment • Identification of risk factors for developing ocular inflammation • Potential side effects of topical steroid preparations, including a propensity for raising intraocular pressure in susceptible • Individuals and the potential development of cataract • Management strategies for steroid-related intraocular pressure rises
6	Working with other practitioners
6.1	Role, responsibilities and communication in collaborative care of patients each party must have clear info of diagnoses, tx & ongoing recommendations to px of other treating practitioners; timeframes in which this information should be provided; preferred format for this info; who is responsible for ongoing patient care & follow up of px who miss scheduled appts; & the roles and responsibilities of each person participating in the shared care
6.2	Patient involvement px must consent to arrangement & be clearly informed about who is responsible for primary care & when they need to attend reviews with each practitioner

AMH Online: Australian Medicines Handbook Online (Section 14 Guides - for Special Populations & Adverse Effects)

- Inappropriate prescribing can lead to ineffective treatment, patient harm, and increased costs
- Always consider non-drug treatments and involve the patient in decision making
- Choose treatment based on Efficacy, safety, suitability and cost
- Avoid new expensive drugs unless they are proven superior
- Overprescribing can cause side effects, addiction, and waste, and underprescribing can lead to ineffective treatment and further complications

Prescribing generic products: same active ingredients as the first marketed brand, bioequivalent indicated by an 'a' or 'b', if narrow therapeutic range then monitor, also may confuse px

Prescribing biosimilar products: complex biological medicines similar to but not identical to originator counterparts. Use the same brand throughout treatment and avoid substitution without prescriber approval

Prescribing for children: have different pharmacokinetics, drug clearance higher in early childhood, dosing based on weight (preferred) or body surface area, Many medications lack paediatric-specific formulations and dosing guidelines. Compliance issues arise from taste, complex schedules, and caregiver attitudes. Once or twice-daily dosing is preferred, Safe storage and child-resistant packaging reduce accidental ingestion. Liquid medications should be carefully formulated to avoid sugars, ethanol, and allergens. Avoid injections unless necessary, and use alternative forms like sprinkles or suppositories when possible.

Prescribing for the elderly: reduced renal function increases drug toxicity risk, adjusted dose. More sensitive to certain drugs eg benzodiazepines, opioids. Polypharmacy: Multiple medications raise the risk of interactions and adverse effects. Noncompliance: Forgetfulness or intentional avoidance can impact medication adherence. Use non-drug treatments when possible, prescribe the lowest effective dose, simplify regimens, provide clear instructions, and regularly review medications.

Prescribing for palliative care: Focus on symptom control and quality of life for patients with incurable diseases.
Rational Prescribing: Use the fewest drugs necessary, balance analgesia with patient preferences, and anticipate opioid-induced constipation.
Ongoing Care: Continue essential treatments while stopping non-essential preventive medications.
Administration Routes: Oral is preferred; SC infusions, transdermal, rectal, or sublingual options are used when needed.
Medication Management: Check drug compatibility in SC infusions, monitor pain and side effects, and adjust treatment accordingly.

Prescribing for pregnant women: Risk Awareness: Drugs can harm the fetus at any stage of pregnancy, causing malformations, growth issues, or complications at birth.
Prescribing Caution: Only prescribe if the benefits to the mother outweigh risks to the fetus.
Avoid Unnecessary Drugs: Counsel pregnant women to limit drug and chemical exposure.
Drug Safety: Few drugs are proven teratogenic, but no drug is completely risk-free in early pregnancy.
Prefer Established Drugs: Use well-studied medications over newer, untested ones.
Seek Expert Advice: Consult pregnancy drug information centers for detailed guidance.

Prescribing for breastfeeding women: Some drugs enter breast milk in significant amounts, potentially harming the infant or affecting lactation.
Prescribe Only When Necessary: Use only well-studied drugs with known safety profiles over newer, untested ones.

Timing Matters: Taking medication after breastfeeding may reduce infant exposure.
Seek Expert Guidance: Consult pregnancy drug information centers or LactMed for safety data.

Prescribing in renal impairment: Risk of Accumulation: Drugs excreted by the kidneys can build up in renal impairment, increasing adverse effects.
Dose Adjustments: Modify dose, reduce dosing frequency, or choose non-renal excreted drugs when needed.
Renal Function Assessment: Use estimated glomerular filtration rate (eGFR) or creatinine clearance (CrCl) before prescribing and adjust doses accordingly. CrCl Calculation: Cockcroft-Gault equation is commonly used, especially for elderly patients and specific drugs like gentamicin.
Monitoring: Regularly check renal function and drug levels to prevent toxicity.
Prescribing in obesity: Obesity affects drug distribution & clearance, potentially altering half-life.
Lean weight estimates are used to adjust dosing, but no universal method exists.
Limited clinical trial data on obese patients make dosing challenging.
Dose modifications may be required; consult specialist sources for guidance.
Adverse effects: Harmful effects from drugs at normal doses, which can be predictable (related to drug's action) or unpredictable (e.g., genetic, immune reactions).Predisposing Factors: Age, gender, dose, polypharmacy, history of adverse reactions, and genetics.
Recognition: Full profiles of adverse effects may take years to emerge, so careful monitoring and reporting are essential. Report all suspected reactions, especially severe ones (death, hospitalization, and birth defects). Use TGA's online portal or other reporting methods. Inform patients about potential adverse effects and provide Consumer Medicine Information (CMI) when available.

Therapeutic Drug Monitoring (TDM)

- Purpose: TDM involves measuring drug concentrations to improve patient outcomes, especially for drugs with narrow therapeutic ranges, significant risks (e.g., seizures, transplant rejection), or non-linear pharmacokinetics.
- When to Use TDM:
  - Narrow therapeutic ratio drugs.
  - Serious consequences of therapeutic failure.
  - Non-compliance, treatment failure, or drug toxicity.
  - Starting or stopping other drugs, or changes in physiology (e.g., renal or hepatic impairment).
  - Assessing overdose.
- Characteristics of Drugs for TDM:
  - Evidence-based therapeutic range.
  - Defined concentration-effect relationship.
  - No appropriate direct effect measure (unlike INR for warfarin).
  - A suitable and accessible lab assay.
- Practical Considerations:
  - Adjust doses based on patient need, not just lab results.
  - Blood samples are typically taken at trough concentration for consistency.
  - Provide drug info (e.g., dose, timing) with samples for accurate interpretation.
  - Consult with clinical pharmacology services for complex cases

Fact sheet - Recommendations for safe use of medicines terminology [^]

- QD = everyday ; QOD = every other day ; TIW = three times a week ; SC or SQ = subcutaneous
- S/L or sl (to mean 'sublingual') ; E (to mean 'eye' or 'ear') ; ug, µg, or mcg (to mean 'microgram')
- d (to mean 'daily') ; QD or qd (to mean 'every day') ; M (to mean 'morning') ; N (to mean 'night')
- 6/24 (to mean 'every six hours') ; X 3d (to mean 'for three days')

Topical ophthalmic medications: What potential for systemic side effects and interactions with other medications? (Goldberg, et al.)

- Formulation, physical, chemical and microbiological stability and utility of the preparation must be considered when compounding drugs
  - also px dosage, potential drug-drug interactions with other medications being taken, contraindication, precautions
- Complex compounding is the extemporaneous preparation of products such as those that must be aseptically prepared or be sterile (eg. ophthalmic use & injections)
- Most commonly recommended over-the-counter eye preparations were for dry eye management ; Most frequently prescribed Schedule 4 medicines were anti-inflammatories
- Some antibiotic combination products and low-strength (0.01%) atropine eye drops may require compounding
- Reasons for Compounding:
  - liquid formulation needed because the patient can-not swallow a tablet or capsule
  - px has allergy or sensitivity to an excipient (ingredient in formulation which is not the active ingredient eg. colouring agent or preservative) in the commercially available product
  - dose or concentration available in commercial products not suitable to administer the required dose to attain the desired effect
  - commercially available product has been discontin-ued, is out of stock or is unavailable
  - product is not very stable and must be prepared freshly to maintain potency and stability
  - specific formulation or presentation is required for a clinical trial

Prescription of compounded ophthalmic medications: a pharmacy perspective: Compounded prescriptions (Weekes, et al.)

- 'Compounding' can be defined as the preparation, mixing, assembling, packaging and labelling of a medicinal product based on prescription
- It is only appropriate to prescribe a compounded medicine when there is no commercially available product or therapeutic alternative (route of administration, potency, formulation)
- Preparing the product: ingredients, formula, preservatives & excipients, expiry date, policy & procedures

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Week 2 Readings:

AMH Online: Australian Medicines Handbook Online (Section 11.1 - Drugs for Eye Infections) [^]

- Aminoglycosides' adverse effects: ocular irritation, superficial punctate keratitis, renal toxicity
  - **Framycetin** indicated for bacterial conjunctivitis
  - **Gentamicin** indicated for bacterial keratitis (in conjunct with cefazolin eye drops)
  - **Tobramycin** indicated for bacterial conjunctivitis (reserved for use by ophthalmologists & serious infections not responding to tx with other topical antibacterials
- Quinolones' adverse effects: mild transient ocular irritation, allergic reactions-hypersensitivity
  - **Ciprofloxacin** indicated for bacterial keratitis or severe bacterial conjunctivitis (other antibacterials preferred for empirical tx of conjunctivitis) // unpleasant taste
  - **Ofloxacin** indicated for bacterial keratitis or severe bacterial conjunctivitis
- Other Antibacterials
  - **Chloramphenicol** indicated for bacterial conjunctivitis or blepharitis, prophylaxis of infection after corneal abrasion & after ocular surgery // stinging/burning after instillation
    - evidence of benefit in corneal abrasion lacking (limit use to only cases with higher risk of infection)
    - broad-spectrum bacteriostatic activity (not pseudomonas & anaerobic bacteria)
- Antivirals
  - **Aciclovir** indicated for herpes simplex epithelial keratitis // adverse effects: transient mild stinging, superficial punctate keratitis, allergic reactions

Australian medicines handbook : (AMH Online). (Section 11.3 - Drugs for allergic & inflammatory eye conditions)

- **Allergic conjunctivitis:** Itchy, red, swollen, and watery eyes may occur with allergic rhinitis. It is seasonal and Perennial. Prevent by identifying and avoiding triggers. Refer severe cases or CL users to a specialist.
- Treatments: aims for symptom control
  - General Measures: cold compress, saline irrigation 0.9% twice daily,, ocular lubricants
  - Topical Antihistamines: Preferred for quick relief (e.g., azelastine, ketotifen, olopatadine).
  - Mast Cell Stabilisers: Preventive use; slow onset (2–4 weeks).
  - NSAIDs (Ketorolac): Used if other treatments fail; may cause irritation.
  - Corticosteroids: Specialist use only due to serious side effects.
  - Vasoconstrictors: Not recommended due to rebound hyperaemia.
- **Practice Tip: Treating allergic rhinitis (e.g., intranasal corticosteroids) may help relieve symptoms.**

Vasoconstrictors (eye): alpha-adrenoreceptor agonists that constrict conjunctival blood vessels, reducing ocular redness and discomfort
Indications: Temporary relief of eye redness due to minor irritation
Precautions: acute angle closure crisis may rarely be precipitated, Pregnancy and Breastfeeding safe to use
Adverse effects: common >1% rebound hyperaemia, stinging on instillation, mild mydriasis, blurred vision

- **Naphazoline** (similar efficacy and adverse reactions to tetrzoline)
  - Dose: Adult and child >12 years, 1 drop every 6-12 hours as required
  - eye drop, 0.01%, 10 mL, Optrex
  - eye drop, 0.012%, 15 mL, Murine Clear eyes
  - eye drop, 0.1%, 0.6 mL (single use), 10, Murine Red eyes
  - eye drop, 0.1%, 15 mL, Albalon, Naphcon Forte, Systane Red Eyes
- Naphazoline with antazoline: indicated for mild allergic conjunctivitis.

    - eye drop, naphazoline 0.05%, antazoline 0.5%, 15 mL, Albalon-A
    - Adult, 1 eye drop every 6-12 hours as required
- **Naphazoline** with **pheniramine**: indicated for mild allergic conjunctivitis.
    - Dose: Adult and child >12 years, 1 drop every 6-12 hours as required
    - eye drop, naphazoline 0.025%, pheniramine 0.3%, 15 mL, Naphcon-A, Visine Allergy
- **Tetryzoline**: indicated for temporary relief or eye redness due to minor irritation
    - Dosage: Adult, child >6 years, 1 eye drop every 6-12 hours as required
    - eye drops, 0.05%, 15 mL, Murine Sore Eyes, Visine Advanced Relief, Visine Clear
    - their benefit is doubtful; **rebound hyperaemia** leads to overuse, although advertised as being useful for relieving eye redness due to minor irritants such as dust, smoke and contact lens wear, a cool compress is as beneficial and is safer; ocular lubricants may also relieve irritation due to dry eyes

Antihistamines (eye)

Types:
- Antihistamines: Levocabastine, Antazoline (with Naphazoline), Pheniramine (with Naphazoline).
- Antihistamine-Mast Cell Stabilisers: Azelastine, Ketotifen, Olopatadine.

Mode of Action: Block (bind to) H1 receptors and stabilise to inactive form to reduce histamine effects; some (Azelastine, Ketotifen, Olopatadine) also stabilise mast cells.
Indication: Allergic conjunctivitis.
Precautions: Pregnancy/Breastfeeding: Safe to use.
Adverse Effects (Common >1%): Stinging on instillation, mild eye irritation, headache.

Comparative Info: All have similar efficacy; mast cell stabilisers may also prevent symptoms. Antazoline and Pheniramine are only available with Naphazoline (vasoconstrictor).

Azelastine: Adults & Children (>4 years): 1 drop twice daily; may increase to 1 drop 4 times daily if needed. Product: Eyezep (0.05% eye drops, 6 mL). Common Adverse Effects (>1%): Stinging on instillation, mild eye irritation, headache.
Infrequent (0.1–1%): Bitter taste.
Rare (<0.1%): Allergy
Ketotifen: Common Adverse Effects (>1%): Stinging on instillation, mild eye irritation, headache.
Infrequent (0.1–1%): Allergy, blurred vision (on instillation), dry eye, dry mouth, sedation.
Dosage: Adults & Children (>3 years): 1 drop twice daily
Products: Zaditen (0.025% eye drops, 0.4 mL (single-use, 20 pack), 5 mL bottle
Levocabastine: Common Adverse Effects (>1%): Drowsiness, stinging on instillation, mild eye irritation, headache.
Dosage: Adults & children (>6 years): 1 drop twice daily, may increase to 3–4 times daily if needed.
Counselling: May cause drowsiness; avoid driving or operating machinery if affected, especially with alcohol.
Products: 0.05% suspension (4 mL) – Livostin, Zyrtec Levocabastine.
Olopatadine: Common Adverse Effects (>1%): Hyperaemia, keratitis, dry eye, stinging on instillation, mild eye irritation, headache.
Dosage: Adults & children (>3 years): 1 drop twice daily.
Products: 0.1% eye drops (5 mL) – Paladopt, Patanol.

Mast cell stabilisers (eye)

Mode of Action: Inhibit the release of inflammatory mediators from mast cells.
Indications: Allergic conjunctivitis, keratoconjunctivitis.
Common Adverse Effects (>1%): Transient stinging and burning on instillation.
Counselling: Full effect may take several weeks.
Practice Points:
- Delayed onset; use for at least 2 weeks before assessing effect.
- Start 2–4 weeks before hay fever season or combine with antihistamines initially.
- Ocular antihistamines with mast cell stabilising properties are generally preferred for faster relief.

Cromoglycate: Indicated for allergic conjunctivitis and vernal keratoconjunctivitis
Safe in pregnancy and breastfeeding
Dosage: adult, child, 1 eye drop 4-6 times daily
Products: eye drop, 2%, 10 mL, Cromo Fresh
Lodoxamide:
No human data but considered safe in pregnancy
No human data in breastfeeding but unlikely to be a concern
Dosage: Adult, child>4 years, 1 eye drop 4 times daily
Products: eye drop: 0.1%, 10ml, Lomide

NSAIDs (eye)

Mode of Action: Inhibits cyclo-oxygenase, reducing prostaglandin-mediated inflammation.
Precautions: Contraindicated in NSAID-induced hypersensitivity; may delay healing when used with corticosteroids; may affect intraocular pressure with prostaglandin analogues; can mask infection symptoms.
Pregnancy/Breastfeeding: Short-term use is safe.
Common Adverse Effects (>1%): Stinging on instillation, ocular irritation, and delayed healing.
Rare but Serious Effects: Corneal issues (keratitis, thinning, ulceration), systemic allergic reactions.
Practice Point: Used as an alternative or adjuvant to ocular corticosteroids.
Ketorolac: indicated for short-term treatment of seasonal allergic conjunctivitis, prevention and reduction of inflammation associated with cataract surgery
Dosage:
- Allergic Conjunctivitis: 1 drop 4 times daily for up to 4 weeks (≥3 years).
- Cataract Surgery: 1 drop 4 times daily, starting 24 hours before surgery; continue for 2–4 weeks if needed.

Practice Point: May help prevent/treat cystoid macular oedema post-cataract surgery.
Product: Acular (0.5% eye drops, 5 mL).
- Nepafenac: Indications: Used in cataract surgery to reduce pain, inflammation, and risk of postoperative macular oedema in non-proliferative diabetic retinopathy.
- Dosage: 1 drop once daily, starting 24 hours before surgery; give an additional drop 30–120 minutes before surgery. Continue for up to 14 days post-surgery (or up to 60 days for diabetic patients).
- Product: Ilevro (0.3% suspension, 3 mL).

Corticosteroids (eye)

Mode of Action: Regulate gene expression, providing anti-inflammatory and immunosuppressive effects.
Precautions: Increased risk of ocular hypertension, infection masking, delayed healing, and cataracts (especially in children and with long-term use). Avoid in most infections, particularly herpes simplex and fungal keratitis.
Adverse Effects: Common: ocular hypertension, infection, delayed healing. Rare: cataracts, adrenal suppression.
Monitoring: Requires close ophthalmologist supervision, including slit lamp exams and intraocular pressure checks if used for >10 days.
Comparative Efficacy: Prednisolone acetate (1%) is the most potent, followed by dexamethasone (0.1%) and fluorometholone (0.1%).
Practice Points: Gradual tapering may be required to prevent rebound inflammation. Some corticosteroids (e.g., loteprednol, triamcinolone) are available through special access schemes.
Dexamethasone
Indications for Dexamethasone
- Allergic and inflammatory conditions of the lids, conjunctiva, cornea, iris, and ciliary body, including postoperative inflammation.
- Intravitreal implant (specialist advice needed) for macular oedema due to:
    - Diabetes
    - Central retinal vein occlusion
    - Branch retinal vein occlusion
    - Non-infectious posterior uveitis

Dosage for Dexamethasone: Allergic and inflammatory conditions: 1 eye drop 2–6 times daily, adjusted based on inflammation severity. Severe inflammation: Initially 1 eye drop every hour, then reduce frequency gradually.
Products: Eye drop (0.1% suspension), 5 mL: Maxidex, Implant (700 mcg): Ozurdex (for macular oedema, non-infectious posterior segment uveitis)
Fluorometholone
Indications: Selected allergic and inflammatory conditions of the lids, conjunctiva, and cornea.
Dosage: Adults and children >2 years: 1 eye drop 2–4 times daily, adjusted based on the severity of inflammation.
Products: Eye drop (0.1% suspension), 5 mL: FML.
Hydrocortisone
Indications: Inflammatory conditions of the eyelids (e.g., eyelid dermatitis) Adjunct to corticosteroid eye drops for anterior uveitis
Dosage: Eyelid inflammation: Apply eye ointment 2–4 times daily. Adjunct to corticosteroid eye drops: Apply eye ointment at night.
Products: Eye ointment (1%, 5 g): Siguent Hycor.
Prednisolone (eye)
Indications: Selected allergic and inflammatory conditions of the conjunctiva and cornea
Dosage: Usual range: 1 eye drop 2–4 times daily. Severe inflammation: Initially, 1 eye drop every hour, then gradually reduce frequency.
Products: Eye drop (Prednisolone sodium phosphate 0.5%, 0.5 mL [single use], 20): Minims Prednisolone, Prednisolone combinations: Prednisolone with phenylephrine.

eTG complete Therapeutic guidelines.

Principles of antimicrobial use
- Appropriate antimicrobial prescribing (for prophylactic, empirical and directed therapy)
    - Decide if antimicrobial therapy is indicated
    - Microbiological testing indicated? Collect samples for testing BEFORE the first dose of antimicrobial
    - Select a drug that is consistent with the clinical guidelines recommendations or advice from someone reliable. Consider the required spectrum of activity (use the narrowest spectrum therapy required), potential SEs, drug interactions, and costs, and factor in Px Hx of hypersensitivity, pregnancy, breastfeeding, etc.
    - Select appropriate dose, frequency, and route (use oral when clinically appropriate)

- Prescribe the shortest possible duration of therapy
- Document in records/ charts, including the indication and intended duration of therapy before further review or cessation

- Prophylactic, empirical and directed therapy
  - Prophylactic → aims to prevent infection when there is significant clinical risk of infection developing
  - Empirical → used to treat an established infection when the pathogen has not been identified, base it on most likely or important pathogen
    - This therapy is reasonable in the following circumstances:
      - When Tx MUST be started before the results of a culture
      - If a simple culture cannot be obtained
    - Review therapy ASAP
  - Directed → treat an established infection when pathogen has been identified
- Optimal route of administration
  - Oral and enteral: oral usually assoc. w/ less severe ARs than parenteral therapy. Lower drug and administration costs, good oral bioavailability, adequate tissue penetration, tolerable
  - Parenteral: (intravenous, intramuscular) used when oral is not possible, GI absorption is significantly reduced, and when urgent Tx is required fr severe and rapidly progressing infection
- Tips for optimising antimicrobial dosage
  - Take into account Px factors, the pharmacokinetic and pharmacodynamic properties of the drug and potential drug interactions, and patients with altered pharmacokinetics (septic shock, severe burns, cystic fibrosis, preggo women, obese patients)
- Duration of antimicrobial therapy
  - Use the shortest possible duration (<7 days), consistent with the condition being treated and Px clinical response
  - Prolonged duration of antimicrobial therapy is associated. w/ increased risk of ARs
- Types of ARs
  - diarrhoea, rash or more serious hypersensitivity reactions.
  - mild yeast infections (eg thrush) through to more serious infections (eg Clostridioides difficile).
  - Multidrug-resistant bacteria (known as 'superbugs') can spread between people, affecting other family members and the community.
  - (eg anaphylaxis) or delayed (eg drug rash with eosinophilia and systemic symptoms [DRESS], Stevens–Johnson syndrome / toxic epidermal necrolysis [SJS/TEN]), and subsequent exposure to the drug could be fatal

Optometry Australia's Infection Control Guidelines 2020 (Hart, et al.)

Main Modes of Transmission in a Healthcare Setting

- Contact Transmission: direct - pathogen transferred from one person to another via contact with mucous membrane or breaks in skin

vs

- Indirect - pathogen transferred through contact with contaminated intermediate objects
- Droplet Transmission: when an infectious particle greater than 5 microns is spread via coughing, sneezing or talking, or during certain procedures
- Airborne Transmission: when droplets smaller than 5 microns (aerosols) containing an infectious agent remain suspended in the air over long distances & time (via breathing, talking, coughing, sneezing)

Standard Precautions Implemented in Optometric Practices

(1) hand hygiene

Washing hands with hand wash (water and soap or a soap solution), either non-antimicrobial or antimicrobial, when hands are visibly soiled, or Applying an alcohol-based hand rub to the surface of hands (e.g. liquids, gels, foams), where hands are visibly clean

→ **Alcohol-based antiseptics (used in hand rub):** Suitable products should contain 60-80% v/v ethanol; effective against Gram-positive and Gram-negative bacteria (including multidrug-resistant pathogens like methicillin-resistant staphylococcus aureus (MRSA) and vancomycin-resistant enterococci (VRE), mycobacteria, fungi and some enveloped viruses (e.g. herpes simplex virus, HIV, influenza virus, SARS-CoV -2); very poor activity against bacterial spores, protozoan cysts (e.g. Acanthamoeba), tropical parasites and some non-enveloped viruses (e.g. norovirus, rotovirus, hepatitis A)

→ **Chlorhexidine** (used in hand rub and hand wash): prevents rapid regrowth of skin organisms and increases the duration of skin antisepsis; Preparations containing 2% chlorhexidine are slightly less effective than those using 4% chlorhexidine; Addition of low concentrations (0.5-1.0%) of chlorhexidine to alcohol-based formulations results in greater residual activity than alcohol alone; good activity against Gram-positive bacteria, but less against Gram-negative bacteria and fungi and shows minimal effect against mycobacteria; in vitro activity against enveloped viruses (e.g. herpes simplex, HIV, cytomegalovirus, influenza, respiratory syncytial virus), but significantly less activity against non-enveloped viruses (e.g. adenovirus, rotavirus, enteroviruses) ; may cause conjunctivitis or corneal damage

→ **Iodine and iodophors (used in hand wash):** effective against Gram-positive, Gram-negative and some spore-forming bacteria (e.g. clostrida, Bacillus spp), mycobacteria, viruses and fungi. Most iodophor formulations used for hand hygiene contain 7.5-10% povidone-iodine. Iodophors cause less skin irritation and fewer allergic reactions than iodine, but more irritant contact dermatitis than other antiseptics used for hand hygiene

→ **Triclosan (used in hand wash)** – Not recommended: banned by the FDA, still be found in hand sanitiser, toothpaste and mouthwash, as well as in healthcare settings, Concerns have been raised about its effect on human endocrine and reproductive systems, as well as associations with allergies, asthma and cancer

(2) personal protective equipment (PPE)

→ Gloves, Masks, Protective Eyewear, Protective Footwear

(3) safe handling and disposal of sharps

After they are used, single-use sharps should be disposed of immediately in a clearly labelled, puncture- and leak-proof container

(4) routine environmental cleaning

Neutral pH detergents (for example, dish washing liquid) are recommended as they are less likely than acid or alkali detergents to damage metals such as stainless steel or to cause skin irritation

Clinell Universal Wipes can be used as a single-step detergent and disinfectant for all surfaces

Tristel Jet is recommended as a surface cleaner and disinfectant, given its sporicidal, virucidal, mycobactericidal, fungicidal and bactericidal properties. (TGA approved and registered as a hospital-grade disinfectant)

A bleach solution diluted to 1000 ppm is used to disinfect hard, non-porous surfaces. Household bleach is typically 5-6% sodium hypochlorite, so a 1:50 dilution of bleach (1 part bleach, 49 parts water) equals 1000ppm

An alcohol solution may be used, with the recommended concentration being 60-90%

Table 3. Examples of recommended environmental cleaning in optometric practices.

Item	Frequency	Method/Agent
Waiting room chair	Weekly and when visibly soiled	Detergent
Consulting room chair	Daily and when visibly soiled	Detergent
Computer, keyboard and mouse (in consulting room)	Weekly and when visibly soiled	Detergent Follow manufacturer's recommendations Instal keyboard covers or washable keyboards where possible
Surfaces in consulting room (e.g. bench tops)	Weekly and when visibly soiled	Detergent and disinfectant
Hand washing sink, door knob of consulting room	Daily	Detergent
Light switches	Weekly	Detergent
Waste receptacle	Weekly and spot clean when visibly soiled	Detergent

(5) reprocessing of reusable equipment, instruments and contact lenses

Cleaning: removal of foreign material (e.g. organic material) from objects and is normally carried out using detergent, water and physical action. An object cannot be disinfected or sterilised unless it has been cleaned.

Disinfection: process that inactivates non-sporing infectious agents, using either thermal (moist or dry heat) or chemical means. Disinfection can be either low-, intermediate- or high-level

Sterilisation: destroys all microorganisms on the surface of an instrument or device. Sterilisation is usually achieved through reprocessing of heat-resistant items using steam, although low-temperature sterilisation technology (chemical) is available for reprocessing heat and moisture-sensitive items

Tonometry 1:10 dilute bleach is a high-level disinfectant with broad efficacy against common infectious agents (70% isopropyl alcohol & 3% hydrogen peroxide can damage tonometer tips)

Options for disinfection of reusable contact instruments would involve either sodium hypochlorite or Tristel Duo OPH:

Sodium Hypochlorite Note: The appropriate concentration of sodium hypochlorite is 5000ppm, approximately 0.5%.

Household bleach is typically 5-6% sodium hypochlorite, so a 1:10 dilution of bleach (1 part bleach, 9 parts water) equals 5000ppm

Method: (1) Clean with mild pH neutral detergent registered by the TGA as a Class I medical device cleaning product (2) Rinse with sterile water/saline before disinfecting (3) Soak in sodium hypochlorite (5000ppm) for 5-10 minutes (4) Rinse thoroughly with a steady stream of sterile water/ saline (5) Air dry or dry with sterile, soft disposable cloth.

Tristel Duo OPH

Method: (1) Clean with mild pH neutral detergent registered by the TGA as a Class I medical device cleaning product (2) Rinse with sterile water/saline before disinfecting (3) Dispense 2 aliquots of Tristel Duo onto a Tristel Dry Wipe or directly onto the instrument (4) Spread the foam over the surface of the instrument (5) Wait 2 minutes (6) Rinse with sterile water/saline (7) Air dry or dry with sterile, soft disposable cloth

Table 4. Levels of reprocessing of medical devices.[4,7,12,24]

Level of risk	Application	Process	Example(s)
Critical: High risk for infection if contaminated with any microorganism	Entry/penetration into sterile tissue, cavity or bloodstream	All items must be sterile (e.g. steam under pressure, chemical sterilants)	Needles, lacrimal cannulae. Note: needles used in optometry practice for foreign body removal MUST be single use
Semi-critical	Contact with mucous membranes or non-intact skin	Items should be sterile or undergo high-level disinfection as a minimum (e.g. steam sterilisation, thermal disinfection or high-level chemical disinfection)	Tonometer probes, gonioscopy lenses, ophthalmic ultrasound probes (including pachymeters), epilation forceps, golf club spud, trial contact lenses
Non-critical	Contact with intact skin	Items must be clean or undergo low/intermediate-level disinfection	Phoropters, head and chin rests, trial frames

Note: Levels of disinfection are defined in Appendix Table A1.

Table 5. Disinfection of RGP, PMMA, soft and hybrid lenses.

Step	RGP and PMMA lenses	Soft and hybrid lenses
Cleaning	Rub thoroughly with a daily surfactant cleaner	Rub thoroughly with a daily multipurpose solution (MPS)
Rinsing	Rinse with sterile saline or MPS for at least 30 seconds	
Inspection	Inspect lens for damage/defect and dispose of appropriately if necessary	
Preparation	Fill a non-neutralising CL case with 3% hydrogen peroxide	
Disinfection	Soak lens for at least 3 hours without neutralising	
Neutralisation	Not required	Refill a neutralising case with fresh hydrogen peroxide and add neutralising tablet/disk. Soak for recommended neutralisation time (e.g. 6 hours)
Rinsing	Rinse with sterile saline (or MPS)	
Storing	Store RGP lens dry after wiping with clean, lint-free tissue.	Use disinfected tweezers to store lens in MPS. It is recommended to repeat the full disinfection process every 28 days.

(6) respiratory hygiene and cough etiquette
(1) Covering the mouth/nose with a disposable tissue during coughing, sneezing, wiping or blowing the nose. A flexed elbow can be used as an alternative for coughing/sneezing if tissues are unavailable
(2) Immediate disposal of tissues into a hands-free waste receptacle
(3) Keep contaminated hands away from the face, particularly the mucous membranes of the mouth, eyes and nose
(4) Hands must be washed with soap and water after coughing, sneezing, using tissues, or contact with respiratory secretions or objects contaminated by these secretions
(7) Aseptic and clean technique to reduce the microorganisms on surfaces or equipment
➔ Aseptic: Asepsis or aseptic means to be free from pathogenic organisms. Thus, an aseptic technique aims to prevent the transfer of pathogenic organisms, in sufficient numbers to cause infection, to
susceptible sites. Sterile equipment is used (eg. FB removal, IV injection)
➔ Clean: A clean technique aims to reduce the overall number of microorganisms and lower the risk of transmission. A clean technique may involve the use of clean, disinfected or sterilised equipment
(eg. instilling multiuse drops into the eye, tonometry)
(8) Waste management Material is to be placed in yellow containers with a black biological hazard symbol; consider using a sharps container for this purpose. Waste should be disposed of according to local state-based regulations

Antibiotics versus placebo for acute bacterial conjunctivitis (Chen, et al)

- Should we prescribe antibiotics to someone with bacterial conjunctivitis?
- Does prescribing vs not prescribing make a difference to remission

Objective: To assess the benefits and side effects of antibiotics in treating acute bacterial conjunctivitis.
Main Finding: Antibiotics likely improve clinical and microbiological cure compared to placebo, with antibiotics also reducing the likelihood of treatment incompletion and persistent infection.
Effectiveness:

- Antibiotics led to a 26% improvement in clinical cure compared to placebo.
- Antibiotics also increased microbiological cure by 53% compared to placebo.

Side Effects:

- There were no serious systemic side effects in either group.
- Fluoroquinolones (FQs) had fewer ocular side effects than non-FQs, though the evidence was of very low certainty.
- No evidence of systemic side effects like headache or altered sense of smell.

Conclusion: Topical antibiotics modestly improve outcomes in acute bacterial conjunctivitis compared to placebo, with no serious side effects reported. However, non-FQs may cause more unwanted ocular effects than FQs.
Limitations: The Evidence does not address neonatal conjunctivitis or compare short vs. long treatment durations.

AMH – drugs for examinations and procedures
Mydriasis and Cycloplegia
Key Concepts

- **Mydriasis**: Pupil dilation (iris sphincter relaxation)
- Cycloplegia: Paralysis of accommodation (ciliary muscle relaxation)
- Agents:
  - **Anticholinergics** (Atropine, Cyclopentolate, Tropicamide)
  - **Sympathomimetic** adjunct (Phenylephrine)

Relevance to OPTM6413

- Used for fundus examination, cycloplegic refraction, and management of uveitis
- Critical knowledge for drug selection based on age, iris pigmentation, and systemic risk
- Forms the basis of prescribing, co-management, and pharmacological safety education

Ocular Anticholinergics
Mechanism of Action

- Block muscarinic **ACh** receptors, causing:
  - Iris sphincter relaxation ➔ Mydriasis
  - Ciliary muscle relaxation ➔ Cycloplegia

Indications

- Diagnostic mydriasis/cycloplegia
- Pain relief in uveitis
- Amblyopia penalisation (Atropine)
- Myopia control (low-dose Atropine)

Precautions

- Head injury: Use short-acting agents, consult neurosurgeon
- **Angle-closure glaucoma:** Risk of precipitating acute crisis
- Children <1 year: Higher risk of systemic toxicity
- Diabetes/dark irides: Reduced/slow response

Ocular Anticholinergics

Drug/Use	Peak Effect	Duration	Comments
Atropine 1%	30–40 mins (myd)	7–10 days (myd)	Rarely used diagnostically; long-acting. Used in myopia control, amblyopia.
	1–3 hrs (cyclo)	7–14 days (cyclo)	
Cyclopentolate 0.5–1%	30–60 mins (myd)	1 day (myd)	Preferred for cycloplegic refraction; some CNS side effects in children.
	25–75 mins (cyclo)	6–24 hrs (cyclo)	
Tropicamide 0.5–1%	20–40 mins (myd)	6 hrs (myd)	Preferred for mydriasis; less reliable cycloplegia
	30–40 mins (incomp.)	2–6 hrs (cyclo)	May combine with phenylephrine for stronger dilation

Counselling Points (All Anticholinergics)

- Blurred vision, photophobia ➔ use sunglasses
- Avoid driving/operating machinery
- Warn of prolonged duration for Atropine (up to 10 days!)
- Children: Impaired task performance post-dilation
- Seek help if eye becomes painful/red or vision worsens

Drug-Specific Notes
Atropine

- Very long-acting
- Used in:
  - Amblyopia (1% weekend penalisation)
  - Myopia control (0.01% daily)
  - Uveitis (1% OD to QID)
- Practice Point: Rarely used for routine cycloplegia in school-aged children due to duration.

Cyclopentolate

- Preferred agent for diagnostic cycloplegia
- Safe in children >1 month with observation post-instillation
- CNS effects (e.g. drowsiness, hyperactivity) may occur

Tropicamide

- Preferred for mydriasis in clinic
- Short duration makes it ideal for routine examination
- Less effective cycloplegia; often used with phenylephrine

Phenylephrine (Sympathomimetic Adjunct) ➔ Mydriasis
Mechanism of Action

- α1 agonist ➔ Contracts iris dilator muscle ➔ mydriasis (no cycloplegia)

Indications

- Adjunct for mydriasis in dark irides, diabetics
- Used in uveitis to break posterior synechiae

Precautions

- Avoid 10% drops in elderly, infants, or those with:
  - Recent MI, unstable angina, insulin-dependent diabetes
  - MAOI treatment (↑ risk of hypertensive crisis)

Counselling

- Same as anticholinergics; note that phenylephrine does not impair accommodation

Ocular Local Anaesthetics
Mechanism

- Reversibly block sodium channels in corneal nerves → temporary anaesthesia

Indications
- Used for short ophthalmic procedures (e.g. tonometry, FB removal)

Practice Points
- NEVER prescribe for home use
- Use only for short periods (<20 mins)
- Prolonged use → epithelial toxicity, keratopathy
- Increases penetration of other drugs—instil first

Ocular Local Anaesthetics

Drug	Type	Onset	Duration	Initial Sting	Epithelial Damage
Lidocaine[1]	Amide	2–5 min	20–30 min	30 sec	+
Oxybuprocaine	Ester	20 sec	10–20 min	10 sec	++
Proxymetacaine	Ester	20 sec	10–20 min	10 sec	++
Tetracaine	Ester	20 sec	~20 min	30 sec	+++

[1] Lidocaine not available alone (usually combined with fluorescein)

Relevance to OPTM6413 – Therapeutic Optometry

Topic	Relevance
Optometric Prescribing	Detailed dosage regimens, indications, safety considerations and practice counselling forms basis for real-life prescribing questions.
Ophthalmic Medications	Comprehensive understanding of anticholinergics, sympathomimetics, and local anaesthetics is essential. Includes MOA, ADRs, and patient-specific choices.
Refractive Errors & Cycloplegic Refraction	Guides when to choose Cyclopentolate vs Atropine vs Tropicamide depending on age, refractive error, and ocular pigmentation.
Inflammatory Eye Disease	Understanding the role of mydriatics in uveitis management (pain relief, posterior synechiae prevention) is exam-critical.
Co-management & Referrals	Knowledge of risks (e.g., angle-closure) is essential for safe patient triage and referral decisions.
Script Writing	Accurate PBS product references (e.g. Eikance 0.01%, Minims) and dosing instructions directly support prescription-writing tasks.

Week 3 Readings:

Topical Corticosteroids as Adjunctive Therapy for Bacterial Keratitis (Samantha Herretes, et al.)
- To control inflammation using the smallest amount of the drug
- Requires optimal timing, concomitant antibiotics and careful follow-up
- Randomised controlled trials
- Visual acuity - no important difference between the corticosteroid group and the control group.
- No important differences in time to re-epithelialization/ re-epithelialization in the steroid group was slower than the placebo group/
- Inadequate evidence of the effectiveness and safety of adjunctive topical corticosteroids compared to no topical corticosteroids

Condition: Pain Management Oral Medicine Guidelines (Optometrists and Dispensing Opticians Board)

OTC pain medication or topical ophthalmic drugs (eg, cyclopentolate) sufficient to ease discomfort

Non-narcotic oral pain relief (aspirin, nefopam hydrochloride, paracetamol) or NSAIDs may be used to reduce ocular pain
- Before initiating drug therapy, it is essential to assess Medical hx, Drug hx, allergy hx, drug reactions
- Indications for temporary pain management: corneal abrasion, FB removal, trauma, co-management with ophthalmologist after ocular surgery
- Nature of pain, including severity & location, should be assessed before initiating tx

Topical Tx
- NSAIDs reduce inflammation, maintain pupil dilation, & induce analgesic effects without sight-threatening effects of topical steroids
- No epithelial damage // adverse effects: punctate epitheliopathy, immune rings, persistent epithelial defects, corneal melting
- Cycloplegics temporarily induce pupillary dilation & paralyse ciliary muscle // can significantly impair visual function
- – Topical Corticosteroids indicated only for cessation of ocular inflammation
- Bandage CLs indicated in cases of large corneal abrasion or severe recurrent corneal erosion syndrome; often used in combination with prophylactic, preservative-free antibiotic cover

Oral Tx
- **Aspirin** has anti-inflammatory, analgesic, and antipyretic effect
- Paracetamol narrow therapeutic range // caution in px with alcoholism, anorexia, malnutrition & dehydration
- **Nefopam** Hydrochloride non-narcotic analgesic; no anti-inflam properties // contraindicated in px with convulsive disorders & those taking MAOIs due to blood pressure rise
- **Ibuprofen** analgesic & antipyretic properties; indicated for relief of chronic & acute pain with inflammatory component // caution in px with CVS, hx of GI bleeding or perforation, peptic ulcers
- **Diclofenac** (Voltaren) analgesic effect with rapid onset of action // increased risk of cardiothrombotic events, onset of hypertension, GIT events, severe skin reactions

Contact Lenses (Phillips, Anthony J) - Chapter 17 Section 5

A full patient history, including allergies and medication reactions, is essential for proper diagnosis and management. Antibiotic use for contact lens-related complications should be judicious, focusing on treating or preventing sight-threatening infections. The concurrent use of soft contact lenses and topical ophthalmic medications is generally discouraged due to toxicity concerns and preservative interactions.

Antimicrobial Agents Antibacterials: Used to manage bacterial infections in contact lens wearers. Commonly prescribed fluoroquinolones include ciprofloxacin, levofloxacin, moxifloxacin, and gatifloxacin. Resistance is a growing concern. Antifungals: Natamycin is the most common, but voriconazole and amphotericin B are also used. Antiamoebic Drugs: Used for Acanthamoeba keratitis (AK), including biguanides (PHMB, chlorhexidine) and diamidines (propamidine, hexamidine).
Anti-Inflammatory Agents Corticosteroids: Strongest anti-inflammatory agents but should be used cautiously due to risks like increased intraocular pressure (IOP) and cataract formation. NSAIDs provide pain relief and are less potent than corticosteroids but have fewer long-term side effects. Immunomodulators: Cyclosporine A is commonly used for chronic inflammation. Histamine Anti-Allergy Agents: Used for allergy-related contact lens complications.
Contact Lens-Induced Pathology Microbial Keratitis (MK) Bacterial Keratitis: Caused by bacteria like Pseudomonas aeruginosa and Staphylococcus aureus, diagnosed through corneal cultures and PCR. Acanthamoeba Keratitis (AK): Caused by a free-living amoeba, leading to severe pain and corneal infiltrates. Fungal Keratitis: Common in warm climates, presenting with serrated ulcer margins and raised slough. Adjunct Topical Corticosteroids for Corneal Infections Used in bacterial keratitis after initial antibiotic therapy. Not recommended for fungal keratitis as it can exacerbate the infection.
Contact Lens-Associated Inflammatory Diseases
Contact Lens-Associated Red Eye (CLARE): Caused by Gram-negative bacteria, particularly Haemophilus species.
Symptoms: Redness, photophobia, watery discharge.
Treatment: Discontinuation of lens wear, mild steroids, and prophylactic antibiotics.
Contact Lens Peripheral Ulcers (CLPU) Small focal corneal ulcers are often linked to Gram-negative bacterial colonization. Treatment: Mild steroids, prophylactic antibiotics, and close monitoring.
Other Sterile Infiltrates Can result from allergy, trauma, or toxic reactions.
Treatment: Discontinuing lens wear, mild steroids, and saline rinses.
Limbal Stem Cell Deficiency (LSCD) Rare, but can result from chronic lens wear.
Symptoms: Whorl-like epitheliopathy, impaired corneal healing. Treatment: Increased lubrication, discontinuation of lens wear, and in severe cases, mild steroids or cyclosporine. Corneal Erosions Caused by mechanical trauma from lens wear. Treatment: Discontinuation of lenses, prophylactic antibiotics, and pain management.
Giant Papillary Conjunctivitis (GPC/CLPC) is caused by an allergic reaction to lens deposits or mechanical irritation.
Treatment: Mast cell stabilizers, antihistamines, and, in severe cases, corticosteroids.
Ocular Comorbidities Allergic Conjunctivitis: Common in contact lens wearers and can be seasonal or perennial. Treatment: Cold compresses, allergen avoidance, lubricating drops, antihistamines, and mast cell stabilizers. Other Comorbidities:

Include meibomian gland dysfunction, blepharitis, and dry eye, which can be exacerbated by contact lens wear.

Conclusion The management of contact lens-related complications requires careful diagnosis, appropriate use of antimicrobial and anti-inflammatory agents, and modifications in lens wear habits. Close follow-ups and patient education on hygiene and appropriate lens use are critical to preventing complications.

The diagnosis and management of contact lenses: related microbial keratitis (Carnt, Nicole et al.)

The distinction between sterile inflammation and microbial infection is often blurred

MK is a serious infection of the cornea assoc. w/ CLs

Epidemiology:

- The incidence of CL-related MK is 2–4 per 10,000 wearers per year for daily soft lens users and 20 per 10,000 for overnight wearers.
- Daily disposable lenses result in less severe infections, while rigid lenses have the lowest risk
- Orthokeratology and cosmetic lenses pose higher risks, especially when improperly used

Risk factors

- Overnight wear and poor hygiene are the biggest contributors.
- Swimming with lenses and traveling increase the risk due to exposure to bacteria.
- Males are more prone, likely due to non-compliance and delayed care-seeking
- Pathogen
- 90% of CL-related MK cases are bacterial, with Pseudomonas aeruginosa being the most common pathogen.
- Acanthamoeba and fungal keratitis are less frequent but more severe.
- CL storage cases and environmental contamination (e.g., water, hands) contribute to bacterial transmission.

Diagnosis

- Bacterial Keratitis: Rapid onset, corneal ulcer, stromal infiltration, and purulent discharge
- Acanthamoeba Keratitis: Early signs are subtle but may involve perineural infiltrates (pathognomonic).
- Fungal Keratitis: Slow progression with feathery-edged infiltrates and satellite lesions.
- Differentiating sterile inflammation (e.g., contact lens peripheral ulcers) from MK is clinically challenging.

Management

- Bacterial MK: Treated with topical fluoroquinolone antibiotics (e.g., moxifloxacin, ciprofloxacin). Severe cases require fortified antibiotics.
- Acanthamoeba Keratitis is managed with biguanides (PHMB, chlorhexidine) and diamidines (propamidine, hexamidine).
- Fungal Keratitis: Natamycin is the first-line treatment, with oral voriconazole for deeper infections.
- Corticosteroids are controversial but may help in bacterial keratitis if used cautiously

Future directions

- New treatments: Corneal crosslinking, nanoparticle drug delivery, and bioengineered therapies.

Conclusion:

- CL-related MK is preventable through proper hygiene, avoiding overnight wear, and regular lens case cleaning
- Early diagnosis and appropriate treatment are crucial to prevent vision loss.

Week 4 Readings:

Dry Eye Resource Guide 2024

AHM Online: Australian Medicines Handbook Online (Section 11.4: Drugs for Dry Eye)

Dry eye syndrome: common chronic condition, prevalence increases with age, due to evaporation of tears and/or decreased tear production

Symptoms: pain, blurred vision, itching and gritty or FB sensation

Before starting treatment: consider 2ndary causes eg MGD, Sjorgen's syndrome, rheumatoid arthritis, contact lenses, adverse drug effects (eg anticholinergics, estrogens, isotretinoin)

Drug choice

- Ocular Lubricants(artificial tears): relieve symptoms, 1st line, no evidence one is more superior, px preference after empirical trial determines choice. Specialist if inadequate response after 2-4 weeks
- Eye drops and gels: Drops are for frequent administration, and gels provide longer relief than drops but blur vision (less than ointments). Risk of preservative toxicity due to reduced tear secretion. Preservative free for px using >4 times daily, more important than choice of lubricant
- Eye ointments blur vision, overnight for symptoms that interrupt sleep/occur on awakening. do not contain preservatives, but lanolin may irritate
- Eye sprays: Phospholipid liposomes (soy lecithin), stabilise lipid layer of tear film to reduce tear evaporation. Spray 10 cm away from eyelid, close eye briefly afterwards.

Other drug treatments

- Corticosteroid eye drops: short periods in severe cases to reduce ocular surface inflammation, need specialist supervision and monitor adverse effects
- Ciclosporin/lifitegrast eye drops: anti-inflammatory effects in moderate-severe dry eye where ocular lubricants are insufficient. Ciclosporin appears to have a slower onset of action
- Actylcysteine

TFOS DEWS II Management and Therapy Report (Jones, et al.)

https://www.centreforeyehealth.com.au/wp-content/uploads/2018/12/Dry-Eye-1.pdf

Diagnosis & Classification

- DED is multifactorial, with evaporative and aqueous-deficient forms often overlapping.
- Accurate diagnosis is crucial before selecting treatment.

Management Strategies

- Artificial tears: Mainstay of therapy but only provide temporary relief.
- Lipid-based drops: Beneficial for evaporative dry eye (e.g., meibomian gland dysfunction).
- Anti-inflammatory therapies: Includes corticosteroids, cyclosporine, and lifitegrast for reducing inflammation.
- Punctal occlusion: This is used for tear conservation but should be done cautiously in inflamed eyes.
- Moisture chambers and environmental modifications: Help minimize tear evaporation.
- Tear stimulation therapies: Secretagogues (like diquafosol) and neurostimulation approaches.
- Biologic therapies: Autologous serum and platelet-rich plasma for severe cases.
- Nutritional supplements: Omega-3 fatty acids show potential benefits.
- Surgical interventions: Reserved for severe, refractory cases.

Staged Treatment Approach

Mild Dry Eye

- Lifestyle modifications: Avoiding air conditioning, improving hydration
- Artificial tears: Non-preserved, lipid-containing drops work best.
- Lid hygiene: Warm compresses, lid scrubs (especially for meibomian gland dysfunction).

Moderate Dry Eye

- Anti-inflammatory treatments:
- Cyclosporine (Restasis), Lifitegrast (Xiidra) – reduce inflammation.
- Short-term corticosteroids are effective but have side effects.
- Punctal occlusion: Blocks tear drainage to keep more moisture on the eye.

Severe Dry Eye

- Autologous serum eye drops: contain growth factors for healing.
- Platelet-rich plasma (PRP): Similar benefits to autologous serum
- Surgical interventions: Includes tarsorrhaphy (partial eyelid closure) for extreme cases.

Challenges & Future Directions

- More randomized controlled trials (RCTs) are needed to evaluate newer therapies.
- There is a need for personalized treatment approaches based on individual tear film characteristics.

Conclusion:

- The report emphasizes a stepwise, evidence-based approach to managing DED, incorporating lifestyle modifications, artificial tears, anti-inflammatory treatments, and advanced therapies based on disease severity. However, more research is required to optimize treatment selection.

Over-the-Counter (OTC) Artificial Tear Drops for Dry Eye Syndrome (Pucker, et al)

➔ Environmental interventions aimed to increase air moisture & reduce particles in the air

➔ Non-invasive therapies: nutritional supplements with essential fatty acids

➔ Newer therapies target inflammatory pathways associated with dry eye and include several anti-inflamm agents such as corticosteroids, cyclosporine A, NSAIDs & tetracyclines

➔ Many OTC products include chemical formulations, e.g. cellulose ethers, carbomers, polyvinyl alcohol, and lipid-based formulations, which provide additional viscosity and adhesion and allow for even distribution over the ocular surface

➔ 1st line pharmacotherapy for treating dry eye consists of over-the-counter (OTC) artificial tear drops, gels, ointments, or lubricants with products that appear to mimic the different layers of the tear film in order to maintain ocular hydration

artificial tears are typically preserved with chemicals (e.g. benzalkonium chloride (BAK), ethylenediaminetetraacetic acid (EDTA), purite – associated with ocular allergies and toxicities

Chemical Preservatives Comparison:

– Carbomer:

- Five participants in the carbomer group experienced treatment-related ocular adverse events: local allergic reaction, mild hyperemia, and mild foreign body sensation
- Carbomer gel was found to be more effective at treating each symptom than the CMC artificial tear
- Change in Schirmer's test values from baseline were significantly greater in the carbomer-based gel group than in the HP-guar gel group at weeks two and four

– PEG:

- Participants using PEG reported significantly less dryness, grittiness, burning, and lash crusting symptoms than the CMC group at week four

- Significantly less overall dryness, scratchiness, soreness, and burning symptoms were reported in the 0.25% PEG 400 group compared with the 1% CMC group
- Significantly lower frequency of foreign body sensation in the PEG 400/PG/HP-guar group compared with the 0.5% CMC group
- PEG-containing drops were more effective than the CMC-containing drops at improving corneal staining scores

– CMC:

- Three participants in the CMC group reported keratitis, conjunctival hemorrhage, hypersensitivity, and upper limb fracture

– Sodium Hyaluronate:

- Six participants in the hyaluronic acid group reported eye disorders, including keratitis, conjunctival hyperemia, viral conjunctivitis, instillation site pain, amyotrophic lateral sclerosis, and erythema

— HPMC

- Showed a non-significant improvement in dryness, itching, and foreign body sensation symptoms at 12 weeks

Interventions for Dry Eye: An Overview for Systematic Reviews (McCann et al)

Purpose: To evaluate the reliability and effectiveness of systematic reviews on dry eye treatments and identify evidence gaps.

Key Findings

- No conclusive evidence for any dry eye treatment; most findings are inconclusive due to low-certainty evidence.
- Some evidence suggests potential benefits for:
    - Environmental & Dietary Modifications: Indoor humidifiers (low certainty), omega-3 supplements (moderate certainty).
    - Artificial Tears & Lubricants: May improve symptoms, but no clear superiority between formulations.
    - In-Office Procedures:
        - Punctal Plugs: May improve symptoms but have side effects (e.g., excessive tearing).
        - IPL Therapy & LipiFlow: Possible short-term benefits but low-certainty evidence.
    - Prescription Therapies:
        - Topical corticosteroids, cyclosporine A, secretagogues: May help but have risks (e.g., stinging, increased eye pressure).
    - Autologous Serum: Some symptom improvement but concerns over contamination.

Gaps in Evidence

- No reliable reviews for lifitegrast, oral antibiotics, or moisture chamber devices, despite existing RCTs.
- Many systematic reviews were methodologically weak, leading to fragmented and biased evidence.

Conclusion & Future Research Needs

- Higher-quality systematic reviews needed for key treatments.
- More research required on long-term effectiveness, dosing, and formulation comparisons.
- Despite some promising interventions, no treatment has definitive proof of effectiveness.

Australian Medicines Handbook, 2025 – Dry Eye Syndrome Section)

Overview of Dry Eye Syndrome (DED)

- **Definition**: A chronic condition due to increased tear evaporation and/or reduced tear production.
- **Symptoms**: Irritation, pain, blurred vision, itching, foreign body sensation, and grittiness.
- **Secondary Causes**:
    - Meibomian gland dysfunction (MGD)
    - Autoimmune disorders (e.g. Sjögren's syndrome, RA)
    - Drug-induced (e.g. isotretinoin, anticholinergics)
    - Contact lens use

Treatment Pathway

1. **Step 1 – Ocular Lubricants**
    - First-line therapy.
    - Choice depends on *patient preference*; no single formulation shown to be superior.
    - **Drops**: Up to hourly in severe cases.
    - **Gels**: Longer-lasting, may blur vision.
    - **Ointments**: Preferred at night due to vision blurring.
    - **Sprays**: Soy lecithin liposomes to stabilise tear film lipid layer.
    - **Preservative-Free** recommended for:
        - Frequent users (>4x/day)
        - Multiple concurrent drops
2. **Step 2 – Anti-Inflammatory Therapy**
    - **Ciclosporin** (Cequa®, Ikervis®)
        - Increases tear production; anti-inflammatory.
        - Onset: Delayed (1 month+)
        - Dosing:
            - *Cequa*: BD
            - *Ikervis*: QHS
    - **Lifitegrast** (Xiidra®)
        - LFA-1 antagonist; newer drug.
        - BID dosing.
        - Onset: Improvement by 2 weeks, reassess at 2–3 months.
3. **Step 3 – Advanced/Adjunctive Options**
    - Short-term corticosteroids (*under supervision only*)
    - Punctal plugs
    - Autologous serum drops
    - Acetylcysteine (for filamentary keratitis)

Practice Points

- Treat MGD/blepharitis first (e.g. lid scrubs, warm compresses).
- Minimise tear evaporation (humidifiers, wrap-around glasses).
- Monitor screen time → reduced blinking exacerbates symptoms.
- Always **counsel** patient on expected response time and how to use the medication (e.g. 15-minute gap between lubricants and ciclosporin/lifitegrast).
- Evaluate treatment success at:
    - **2–4 weeks** for lubricants
    - **1–3 months** for immunomodulators

Table – Comparison of ocular lubricants

Brand®	Preservative	Key ingredients	Pack size, PBS status
Drops (multidose) – with preservative			
Blink Intensive Tears	oxychloro complex[1]	macrogol 400, sodium hyaluronate	15 mL
GenTeal, In a Wink Moisturising	sodium perborate	hypromellose	10 mL, PBS-R2
Liquifilm Tears, PVA Tears	benzalkonium chloride[3]	polyvinyl alcohol	15 mL
Methopt	benzalkonium chloride[3]	hypromellose	15 mL, PBS-R2
Murine Tears	benzalkonium chloride[3]	polyvinyl alcohol, povidone	15 mL
Optive	oxychloro complex[1]	carmellose, glycerin	15 mL, PBS-R2
Optive Advanced	oxychloro complex[1]	carmellose, glycerin, polysorbate 80	15 mL
Optive Fusion	oxychloro complex[1]	carmellose, glycerin, sodium hyaluronate	10 mL
Optive Gel Drops	oxychloro complex[1]	carmellose, glycerin	10 mL
Optix	oxychloro complex[1]	macrogol 400, propylene glycol	15 mL, PBS-R2
Poly-Tears, Tears Naturale	polyquaterniu m	hypromellose, dextran 70	15 mL, PBS-R2
Refresh Liquigel, Refresh Tears Plus	oxychloro complex[1]	carmellose	15 mL, PBS-R2

Revive Tears	sorbic acid	hypromellose	10 mL, PBS-R2
Systane	polyquaternium	hydroxypropyl guar, macrogol 400, propylene glycol	15 mL, PBS-R2
Systane Balance, Systane Complete	polyquaternium	hydroxypropyl guar, mineral oil, propylene glycol, sorbitol	10 mL
Systane Ultra	polyquaternium	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	10 mL

Preservative-free (multidose)			
Cationorm	none	heavy mineral oil, light mineral oil, glycerin	10 mL, PBS-A2
Evolve Carmellose	none	carmellose	10 mL, PBS-A2
Evolve Hypromellose	none	hypromellose	10 mL, PBS-A2
Hylo-Forte, Hylo-Fresh	none	sodium hyaluronate	10 mL, PBS-A2
Murine Dry Eyes	none	sodium hyaluronate	10 mL
NovaTears	none	perfluorohexyloctane	3 mL, PBS-A2
Systane Complete Preservative-free	none	hydroxypropyl guar, mineral oil, propylene glycol, sorbitol	10 mL
Systane Ultra Preservative-free	none	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	10 mL

Drops (single use)			
Blink Intensive Tears	none	macrogol 400, sodium hyaluronate	20
Cationorm	none	heavy mineral oil, light mineral oil, glycerin	30
Cellufresh, Celluvisc	none	carmellose	30, PBS-A[90]2
Optifresh Plus, Optifresh Tears, Refresh Plus	none	carmellose	30
Optive Advanced	none	carmellose, glycerin, polysorbate 80	30
Optive Fusion	none	carmellose, glycerin, sodium hyaluronate	30
Optive Sensitive	none	carmellose, glycerin	30
Refresh	none	polyvinyl alcohol, povidone	10, 30
Systane	none	hydroxypropyl guar, macrogol 400, propylene glycol	30, PBS-A[60]2
Systane Hydration	none	hydroxypropyl guar, macrogol 400, propylene glycol, sodium hyaluronate	30
Systane Ultra	none	hydroxypropyl guar, macrogol 400, propylene glycol, sorbitol	24
TheraTears	none	carmellose	24
Eye ointment			
Refresh Night Time	none	paraffin	3.5 g, PBS[2]/PBS-R[2]^2
Poly Visc	none	paraffin, lanolin	3.5 g, PBS[2]/PBS-R[2]^2
VitA-POS	none	retinol palmitate, paraffin, lanolin	5 g
Eye spray			
Murine Eye Mist	none	phospholipid liposomes	15 mL
Optrex Actimist	phenoxyethanol		10 mL
Tearsagain	phenoxyethanol		10 mL, PBS-A[2]^2

Topical Antihistamines and Mast Cell Stabilisers for Treating Seasonal and Perennial Allergic Conjunctivitis (Castillo, et al.)

- seasonal/ perennial allergic conj. Is the most common allergic conj.
- Acute manifestations when exposed to allergens and with typical signs and symptoms, including itching, redness and tearing
- Clinical signs and symptoms are mediated by the release of histamine by mast cells
- Antihistamines inhibit the action of histamine by blocking HI receptors, antagonising the vasoconstrictor and vasodilator effects of histamine
- Mast cell stabilisers inhibit degranulation of mast cells and consequently the release of histamine by interrupting the normal chain of intracellular signals
- Topical Tx: antihistamines, mast cell stabilisers, NSAIDs, combinations, corticosteroids
- Study: Randomised controlled trials (RCTs) comparing topical antihistamines and mast cell stabilisers, alone or in combination, with placebo, no treatment or to any other antihistamine or mast cell stabiliser, or both, that examined people with seasonal or perennial allergic conjunctivitis, or both.
- Results: Trials evaluated only short-term effects, with a range of treatment of one to eight weeks. There was some evidence to support that topical antihistamines and mast cell stabilisers reduce symptoms and signs of seasonal allergic conjunctivitis when compared with placebo. There were no reported serious adverse events related to the use of topical antihistamine and mast cell stabilisers treatment.
- Conclusions: all reported topical antihistamines and mast cell stavilisers reduce symptoms and signs of seasonal allergic conjunctivitis when compared with placebo in SHORT term. No long-term data on their efficacy. Overall, topical antihistamines and mast cell stabilisers appear to be safe and well-tolerated.

Interventions for Blepharitis (Lindsey et al.)

➔ **Anterior Blepharitis:** affects eyelid skin, base of eyelashes & lash follicles // can be staphylococcal, seborrheic, demodectic or mixed

*demodex infestation increases with age (84% of population aged 60, 100% aged over 70 years)

➔ **Posterior Blepharitis:** caused by MGD – characterised by chronic, diffuse abnormality of meibomian glands resulting in duct obstruction & changes in glandular secretion

Sequence of testing recommended for ocular surface disease evaluation:

1. Administration of a validated dry eye questionnaire
2. Calculation of blink rate and blink interval
3. Measurement of tear meniscus height
4. Non-invasive break-up time (NIBUT)
5. Measurement of tear osmolarity
6. Tear break-up time (TBUT) with minimal fluorescein (only if NIBUT not possible)
7. Grading of conjunctival and corneal staining with fluorescein and/or lissamine green, including staining of the lid wiper area
8. Schirmer or Phenol Red Thread test
9. Eyelid and MGD assessment – quantification of morphological lid and lash features, meibomian gland expressibility and quality of meibum, and meibography to grade or quantify meibomian gland drop out

TFOS DEWS II diagnostic criteria for positive diagnosis for dry eye disease:

➔ OSDI symptom score of ≥ 13 or a DEQ-5 score of ≥ 6 plus one or more positive sign of loss of tear film homeostasis indicated by:

– a non-invasive tear break up time < 10 seconds

– osmolarity ≥ 308 mOsm/L in either eye or an inter-ocular difference of > 8 mOSm/L

– corneal (> 5 spots)

– conjunctival staining (> 9 spots)

– lid wiper epitheliopathy (> 2 mm in length and > 25% of lid margin width)
➔ Management Options (for DED)
– OTC Management
➔ Lid cleansing, Warm compresses, Diet, Environment control, Tear supplementation, Manuka honey drops / gel
– In-Office Management
➔ Intense pulsed light therapy, Tea Tree Oil, Therapeutic gland expression, Thermal pulsation, Intraductal meibomian gland probing, Amniotic membrane
– Topical Pharmacological Treatment
➔ Topical antibiotics, Topical corticosteroids, Topical cyclosporine, Tacrolimus ointment
– Oral Pharmacological Treatment
➔ Tetracyclines, Macrolides, Ivermectin

Week 5 Readings

Oral Acyclovir for Herpes Simplex Virus Eye Disease: Effect on Prevention of Epithelial Keratitis and Stromal Keratitis (Beck et al)
Objective: to investigate the effect of oral acyclovir therapy for recurrences of HSV epithelial keratitis and stromal keratitis and to determine if certain patients derive differential benefit
Design: randomised, double masked clinical trial
Results: cumulative probability of a recurrence of any type of ocular HSV during the one year treatment period was 19% in the acyclovir group vs 32% in the placebo group. A benefit was seen for preventing both epithelial keratitis and stromal keratitis. The magnitude of benefit was highest among patients with the highest number of previous ocular HSV disease episodes. The benefit in preventing stromal keratitis was seen solely among patients with a history of stromal keratitis.
Conclusions: Long-term suppressive oral acyclovir therapy reduces the rate of recurrent HSV epithelial keratitis and stromal keratitis. Acyclovir's benefit is greatest for patients who have experienced prior HSV stromal keratitis
Evidence Based Treatment of Herpes Simplex Virus Keratitis: A systematic review **(Guess et al)**
ABSTRACT: This article systematically reviews randomized clinical trials (RCTs) on herpes simplex virus (HSV) epithelial and stromal keratitis to guide evidence-based treatment. The best evidence supports treating HSV epithelial keratitis with topical trifluridine or acyclovir (topical or oral), with possible benefits from topical interferon. For stromal keratitis, RCTs support using topical corticosteroids combined with antiviral therapy to shorten active disease and long-term oral acyclovir to reduce recurrences.
Introduction: Ocular manifestations of HSV include: blepharoconjunctivitis, epithelial and stromal keratitis, uveitis and retinitis, last 3 can cause blindness.

A. Primary Ocular HSV Infection

➔ acquired by direct contact with infected lesions/secretions OR commonly by exposure to virus shed asymptomatically at mucosal sites
➔ can be transmitted to neonates as they pass through mother's birth canal with genital infection ➔ can be threatening to ocular surface and life
➔ POI has incubation of 1-28 days
➔ typically manifests as **unilateral blepharoconjunctivitis** with grouped vesicular eruptions on the eyelids and adjacent skins accompanied by a follicular conjunctival inflammatory response and preauricular lymphadenopathy
➔ associated with upper respiratory tract infection in about ⅓ of patients.

B. Recurrent Ocular HSV Infection

➔ following primary infection of any skin or mucosal epithelium ➔ HSV spreads via retrograde axonal transport to establish latent infection in associated sensory nerve ganglia, including the trigeminal ganglion.
➔ recurrent epithelial infection recurs as dendritic epithelial keratitis ➔ may occur any time after latency is established

C. Clinical Presentations of HSV Keratitis

➔ HSV epithelial keratitis presents as punctate, linear, dendritic or geographic epithelial ulceration with viral cytopathic effects in corneal epithelial cells at the base and edge of the epithelial ulceration
➔ classic Classic dendritic epithelial keratitis begins as a punctate epithelial keratitis, which coalesces to a linear and, finally, a branching epithelial ulceration with terminal bulbs
➔ corticosteroids contraindicated ➔ can cause dendritic EK to progress to larger geographic epithelial ulceration
➔ corneal hypoesthesia (numbness) can develop after repeated attacks of DEK or one attack of GEK

Stromal HSV

1. Herpetic Interstitial Keratitis ➔ presents as unifocal or multifocal interstitial haze or whitening of stroma in the **absence** of epithelial ulceration ➔ longstanding = corneal vascularisation possible.
2. Herpetic necrotising keratitis ➔ presents as suppurative corneal inflammation, may be severe or progress rapidly ➔ looks similar to fungal keratitis or fulminant bacterial keratitis ➔ can involve neovascularisation
3. Herpetic Disciform Keratitis ➔ corneal stromal and epithelial edema, typically in a round or oval distribution, associated with keratic precipitates underlying the zone of edema

D. Clinical Presentations of HSV Keratitis

➔ lab test all neonatal suspected herpes. HSV cultures in px with stromal keratitis are non-revealing. Most adults will have the antibodys and therefore results will be positive. It is more useful when it is a negative result, as it can rule out HSV.

Summary of RCTS in HSV Epithelial Keratitis
➔ Overall, trifluridine exhibited greater (or no less) efficacy than other topical agents tested
➔ In general, 3% acyclovir ointment exhibited greater (or no less) efficacy than other topicaL therapies tested
➔ **acyclovir alone led to resolution in an average time of 7 days**
➔ Penetrating keratoplasty for stromal scarring caused by HSV historically resulted in poor outcomes due to recurrent keratitis and graft rejection
Summary of RCTS in HSV Stromal Keratitis
➔ In 30 patients with disciform keratitis, disciform keratitis treated with topical betamethasone (0.1%) (CORTICOSTEROID - in Australia use **DEXAMETHASONE** 0.01%)) and **acyclovir** ointment resolved in 21.8 days on average, while the keratitis treated with topical placebo and acyclovir ointment resolved in 34.5 days on average.

Conclusions of the Article
➔ HSV EK generally self-limiting ➔ acyclovir ointment 3% is good applied times a day
➔ Visually significant herpetic **interstitial** keratitis can be treated initially with 1% prednisolone drops every 2 hours accompanied by a prophylactic antiviral drug, oral agent such as acyclovir (400 mg bid) . The prednisolone drops are tapered every 1 to 2 weeks, depending on the degree of clinical improvement. The antiviral is administered to prevent severe epithelial keratitis should the patient shed HSV while on corticosteroid drops, and it is continued until the patient is completely off corticosteroids or is receiving less than one drop of 1% prednisolone per day.

Patching for Corneal Abrasion (Lim et al)
Objective: Assess the effects of patching for corneal abrasion on healing and pain relief
Corneal Abrasions are superficial defects involving the corneal epithelium that mostly arise from mechanical injuries.
Results
➔ Participants receiving a patch may be less likely to have complete healing after 24 hours than participants receiving no patch
➔ At cases which had gone to 48 hours: suggested a negligible difference between the patch and no-patch groups
➔ 72 hours: the abrasion would have healed regardless
therefore reasonable to conclude that patching of the eye is **not useful** for the treatment of simple, traumatic corneal abrasions.
Interventions for Recurrent Corneal Erosions (Stephanie Watson et al)
RCE: predisposes the cornea to infection, may follow corneal trauma
Objective: To assess the effectiveness and adverse effects of regimens for the prophylaxis of further recurrent corneal erosion episodes, the treatment of recurrent corneal erosion and prophylaxis of the development of recurrent corneal erosion following trauma.
Results:
➔ therapeutic contact lens vs topical lubrication ➔ no important difference in pain score. Note this therapeutic CL is no longer available
➔ low-certainty evidence of an increased risk of recurrence with phototherapeutic keratectomy compared with alcohol delamination
➔ similar adverse effects between diamond burr superficial keratectomy after epithelial debridement compared with sham diamond burr treatment after epithelial debridement
➔ using subepithelial laser ablation ➔ small increased risk of recurrence of CE at one year follow up in people with transepithelial compared with subepithelial technique

Topical Non-Steroidal Anti-Inflammatory Drugs for Analgesia in Traumatic Corneal Abrasions (Wakai et al)
Objective: Identify and evaluate RCTS comparing the use of topical NSAIDs with placebo or any alternative analgesic interventions in adults with traumatic corneal abrasions to reduce pain and its effects on healing time
Results:
➔ 93% of abrasions were healed within 24 hours and the remainder within 48 hours
➔ although Prophylactic antibiotics with or without cycloplegia are typically used, there was little evidence to support the use of topical NSAIDs in traumatic corneal abrasions.

Antiviral Treatment and Other Therapeutic Interventions for HSVEK (Kirk Et al)
Objective: to compare the relative effectiveness of antiviral agents, interferon and corneal debridement in the treatment of HSV epithelial keratitis
Results:
➔ No significant differences in healing emerged among trifluridine, acyclovir, brivudine, and foscarnet although few studies compared brivudine or foscarnet with other antivirals.
➔ The first antivirals, idoxuridine and vidarabine, seem better than no treatment in healing HSV dendritic keratitis within two weeks
➔ The evidence is conflicting whether ganciclovir is as good as or better than acyclovir.
➔ IInterferon, a natural part of the immune system that can be given as an eye drop, is active against HSV infection of the cornea.
➔ The integrated use of interferon and an antiviral drug might be slightly better than an antiviral drug by itself.
➔ debridement is not consistently better than just an antiviral medication

Week 7 Readings

Standardisation of Uveitis Nomenclature (SUN) for Reporting Clinical Data (Jabs et al)

Type	Primary Site of Inflammation	Includes
Anterior Uveitis	Anterior Chamber	Iritis, Iridocyclitis, anterior cyclitis
Intermediate uveitis	Vitreous	Pars planitis, Posterior cyclitis, Hyalitis
Posterior uveitis	Retina or Choroid	focal/multiple/diffuse choroiditis, chorioretinitis, retinochoroiditis, retinitis, neuroretinitis
Panuveitis	Anterior chamber, vitreous and retina or choroid	

Category	Descriptor
Onset	Sudden, Insidious
Duration	Limited ➙ less than 3 months duration Persistent ➙ over 3 months
Course	Acute ➙ episode characterised by sudden onset and limited duration Recurrent ➙ repeated episodes separated by periods of inactivity without treatment, greater than 3 months in duration Chronic ➙ persistent uveitis with relapse in less than 3 months post discontinuing treatment

Anterior Chamber Cells

Grade	Cells in Field: Field size is a 1mm by 1mm slit beam
0	<1
0.5+	1-5
1+	6-15
2+	16-25
3+	26-50
4+	>50

Anterior Chamber Flare

Grade	Description
0	None
1+	Faint
2+	Moderate (iris and lens details clear)
3+	Marked (iris and lens details hazy)
4+	Intense (fibrin or plastic aqueous

Terms to Describe Disease State

Inactive	Grade 0 cells
Worsening activity	Two step increase in level of inflammation (e.g. anterior chamber cells, vitreous haze, or increase from grade 3+ to 4+)
Improved activity	Two step decrease in level of inflammation (e.g. anterior chamber cells, vitreous haze or decrease to grade 0)
Remission	Inactive disease for over 3 months after discontinuing all treatments

Medical Interventions for Traumatic Hyphema (Gharaibeh et al)

Interventions included: antifibrinolytic agents, corticosteroids, cycloplegics, miotics, aspirin, conjugated estrogens, traditional chinese medicine, monocular vs bilateral patching, elevation of head and bed rest.

Results: No intervention had a significant effect on visual acuity whether measured at two weeks or less after the trauma or at longer time periods.

➙ Antifibrinolytic drugs are often used to treat traumatic hyphema and are thought to be effective, because they delay absorption of blood clots until complete healing of the damaged blood vessels takes place. This review found that antifibrinolytics did not affect final vision, but did appear to reduce the risk of secondary bleeding.

➙ recurrent hyphema increases time to visual recovery and is associated with poorer visual outcomes

➙ Corticosteroids, reduce inflammation, stabilise the blood-ocular barrier or direct inhibition of fibrinolysis thus preventing secondary rebleeds ➙ **prednisolone**

Episcleritis and Scleritis ➙ clinical features and treatment results (Jabs et al)

➙ no patient with episcleritis had decreased VA, but 16% of scleritis cases did

➙ for most episcleritis cases, topical corticosteroids were sufficient

➙ 30% of scleritis cases needed NSAIDs, oral prednisolone or systemic immunosuppressive drugs

➙ px with necrotising scleritis or posterior scleritis were more likely to be treated with oral corticosteroids or immunosuppressive drugs than patients with diffuse anterior or nodular anterior scleritis

➙ about a 20% risk of scleritis developing in the other eye after a year

➙ complications: necrotising scleritis could lead to anterior uveitis, elevated IOP in scleritis (low chance). Posterior scleritis involved risks of vitritis, cystoid macular edema and exudative retinal detachment

➙ Nodular anterior scleritis and diffuse anterior scleritis were more likely to respond to oral nonsteroidal anti-inflammatory drugs.

➙ Nodular anterior scleritis most often responded to oral nonsteroidal anti-inflammatory drugs (57.1%) and least often required immunosuppressive drugs (7.1%).

➙ diffuse anterior scleritis mostly responded to oral NSAIDs

➙ necrotising scleritis required systemic immunosuppressive agents

➙ posterior scleritis was most often treated with oral corticosteroids and sometimes immunosuppressive drugs

Oral Medication Guidelines - Episcleritis

➙ although self-limiting, patients with nodular episcleritis may have longer attacks which are more painful

➙ px may feel relief with no pharmacological treatment ➙ cool compresses or lubricant drops

➙ fluorometholone 0.1% may be used four times a day for 1-2 weeks. OR prednisolone acetate 1% up to four times a day

➙ topical NSAIDs do NOT provide additional benefit over artificial tears and therefore should not be used e.g. keterolac

➙ when topical corticosteroids are not sufficient, oral NSAIDs may be used

➙ in cases of episcleritis where only lubricant eye drops are used, no follow up needed unless the patient does not feel resolved

➙ of patients using topical CS ➙ follow up based on potential risk of adverse events: prednisolone acetate = 1-2 weeks (monitor changes of IOP)

Topical Flurbiprofen: An Effective Treatment for Episcleritis? (Lyons et al)

- **Flurbiprofen** is one of the few non-steroidal anti-inflammatory drugs which has been found to be effective in the management of scleral inflammatory disease
- Recently available in topical form ➙ marketed for use in prevention of surgically induced miosis
- Possesses anti-inflammatory, analgesic and antipyretic activity ➙ acts by inhibition of prostaglandin synthetase ➙ does **not** appear to be a useful treatment in episcleritis when used topically.
- **Treatment with topical prednisolone is superio**r but its use should be restricted to inflammation which has failed to improve spontaneously after one week.

Week 8 Readings

Optical and Pharmacological Strategies of Myopia Control (Kang et al)

- Refractive myopia ➙ the eye's refractive elements have too high a power relative to the eye's focal length
- Axial myopia ➙ the eye is too long and therefore the focal point falls within the eye
- Excessive axial elongation is usually due to increase in vitreous chamber depth
- Animal studies have revealed that eye growth involves constant visual feedback and that manipulating the visual experience has a profound impact on eye growth

Single Vision Spectacles and Contact Lenses

- Minimal side effects while correcting myopia = first line treatment, actually not that effective.

Under Correction

- Deliberate under correcting of distance refractive error to induce myopic retinal defocus during distance viewing and/or reduce accommodative demand and response during near viewing
- Undercorrection by 0.50 to 0.75D
- Recent investigations say this may actually quicken the rate of progression in children.

Multifocal Spectacle Lenses

- Multifocal specs = bifocal + progressive addition lenses (PALS) with near additions varying between +1.00 to +2.00
- Under the premise that near addition correction reduces accommodative demand and error during near viewing or induces changes in peripheral retinal blur which may remove the stimulus for myopic progression

- Accommodative error = lead or lag. Myopic children typically have a lag even before developing myopia. The resultant larger retinal hyperopic defocus has been hypothesised to encourage axial elongation
- Children who read for more than 30 minutes continuously and who hold their reading closer are at higher risk of being myopic = association between myopia and near work.
- However, studies proved progressives and bifocals have minimal impact/benefit over single vision correction for myopia control.

Novel Multifocal Spectacles and Contact Lenses

There are two main types of lens designs:

1. Concentric ring (bifocal) designs – alternating zones for distance correction and plus power to simultaneously manage central vision and impose peripheral myopic defocus.
2. Progressive power (peripheral add) designs – a gradual shift from central correction to increased plus power in the periphery.

Numerous studies have tested these lenses, with modest success:

1. Concentric lenses typically show greater control (≈31 D/year refractive error reduction, −0.12 mm/year axial length) than progressive designs (≈22 D/year, −0.10 mm/year). Studies are limited by short duration and high dropout rates.
2. Some commercial multifocal lenses, originally intended for presbyopia, have shown off-label promise in reducing myopia progression by up to 50% in children.
3. The CONTROL study reported the most significant myopia control (72% refractive error and 80% axial length reduction) using individualised concentric lens prescriptions for children with specific visual disparities.

Ortho-Keratology

Orthokeratology (OK) involves overnight wear of rigid contact lenses that temporarily reshape the cornea to correct myopia, providing clear daytime vision without glasses or lenses.

Beyond refractive correction, OK has shown significant myopia control effects, largely attributed to induced peripheral myopic defocus along horizontal and vertical meridians.

Effectiveness:

- Multiple studies (including a landmark two-year study by Cho et al.) have demonstrated around 50% reduction in axial elongation compared to spectacle wear.
- Meta-analyses report a 41–45% average reduction in myopia progression with OK.
- OK is effective even in children with astigmatism and high myopia (when partially corrected with lenses and residual correction via spectacles), though such studies often suffer from high dropout rates.

Long-Term Effects:

- A 5-year study (Hiraoka et al.) showed significant effects in the first two years, but diminishing returns beyond that, likely due to natural slowing of eye growth with age.
- A clinical retrospective study reported apparent stabilisation in 64% of eyes after ~5 years of OK wear, though based on refractive error rather than axial length.

Discontinuation Effects (DOEE Study):

- Discontinuing OK leads to rapid axial elongation, suggesting ongoing lens wear is needed for sustained effect.
- After resuming OK, progression slows again, showing that re-initiation is beneficial.
- Data suggest that the control effect attenuates after ~2 years, but continued wear is still advisable for children under 14.

Mechanisms of Myopia Control:

Peripheral Myopic Defocus Theory: Reshaped cornea causes peripheral image focus in front of the retina, slowing eye growth.

Binocular Vision Theory: OK may improve accommodative-vergence function (e.g., reduced accommodative lag), thereby reducing hyperopic defocus and contributing to control.

Pharmacological Strategies

1. Atropine → results in cycloplegia
- Initially thought to work by affecting accommodation, studies (especially in animal models like chicks) suggest atropine's myopia control effects are non-accommodative.
- Muscarinic receptors in the retina, choroid, and sclera are potential sites of action, but the exact mechanism remains unclear.
- Higher concentrations (e.g., 1%) show significant rebound myopia progression after treatment stops.
- 0.01% atropine had minimal rebound, suggesting better long-term stability.

Pirenzepine

- Pirenzepine is a selective antimuscarinic agent with high affinity for M1 and M4 receptors.
- Unlike atropine, it causes minimal cycloplegia and mydriasis, making it initially promising for myopia control.
- Animal studies showed that pirenzepine reduced form deprivation myopia and eye elongation.
- Clinical trials in children (8–12 years) using 2% pirenzepine gel twice daily for 2 years showed reduced myopia progression compared to placebo.

IMI: Clinical Management Guidelines Report (Gifford et al)

Risk Factors

Refractive Error and Eye Growth	- Normal eye progresses from hyperopia to emmetropia. - The process of emmetropization is designed to match the increasing axial length of the eye with the focal lengths (reducing power) of the cornea and crystalline lens. - Axial elongation is faster in younger than older children - In a myopic eye, the fastest growth in axial length appears to be the year before onset, - Suspect abnormal: less than +0.75 by age 6, less than +0.50 around age 7-8, less than +0.25 age 9-10, and emmetropia at age 11 -
Age	The younger the age of onset, the faster the progression/the higher the extent of myopia
Family history and ethnicity	- Risk of developing myopia increased threefold or more among children with two myopic parents compared with children with no myopic parents. - East asian more likely, europeans least likely
Visual Environment	- Spending less time outdoors - Reading up close (less than 20cm) for continuous periods (over 45min) - Asian countries: harder curriculum demanding more near work can induce myopia - Higher levels of esophoria and accommodative lag at near and high AC/A

Key Elements of the Baseline Exam for Myopia Control

1. History taking: age, sex, ocular history, general health, ocular surgery, what age of onset for parents, previous myopic controls
2. Refraction: noncyclo or cyclo as indicated
3. BCVA
4. Binocular vision and accommodative tests
5. Anterior eye health exam using slit lamp + IOP measurement
6. Corneal topography if needed to fit CL
7. Axial length measurement
8. Fundus exam
9. Dry eye evaluation
    a. DED can surface or be exacerbated in response to myopia control treatments or exposure to environmental risk factors

Selecting a Treatment Strategy

1. Predicting Progression Rate: based on the meta-analysis, annualized progression rates for myopic children of Asian and Caucasian ethnicities ranged from about 0.50 to 1.00 D and varied by age and sex.
2. Selecting a Treatment: consider:
    a. Basic Rx: Earlier onset of myopia often results in higher degrees of myopic refractive error
    b. BV abnormalities → Greater myopia control effects with progressive addition (spectacle) lenses (PALs) were reported in children with larger lags of accommodation and near esophoria.
    c. Compliance? Clinicians must be aware of contraindications to atropine eye drop use so that it can be safely administered. If Ortho-K, can they demonstrate that they can put the lenses in and out safely?
    d. Cost?

Guidelines for Advice and Clinical Care

- Children should be encouraged to wear their myopic correction full time, as undercorrection of myopia has been shown in some studies to increase myopia progression
- Younger myopic patient = beware of risk of amblyopia
- Encourage OK wear to be at least 8 hours each night.
- Educate patients on CL risks and Atropine risks
- Reviews: generally should be at least every 6 months to monitor safety and efficacy of treatment
    - Atropine: 4-7 days, then one month, 3 months, 6 months
    - OK: 1 day, 4-7 days, 1 month, 3 month, 6 monthly
    - Multifocal SCLs: 4-7 days, 1 month, 6 monthly
    - PAL/bifocal specs: 1 month, 6 monthly
- Atropine: **consider that this needs to be compounded** in order to achieve the low dose needed for myopia control. Patients undergoing atropine therapy → require distance correction. May require near addition (not SV) due to cycloplegic effect → expensive. Long term use of atropine is not well understood!

- Ortho-K/myopia control using only OK should not be discontinued before age 14

IMI - Interventions for Controlling Myopia Onset and Progression Report (Wildsoet et al)

Optical Interventions for Myopia Management

Spectacles	• Easy to fit, well tolerated, affordable, minimally invasive • SV, BF and PAL lenses
Undercorrection with Spectacles	• Originally considered to slow the progression of myopia by reducing the accommodative demand during near tasks. • Undercorrection by 0.5-0.75D • Possibility that undercorrection for distance leads to child not wanting to go outside which means more myopia
SV Peripheral Defocus-Correcting Lenses	• Thus, it has been hypothesized that the hyperopic defocus experienced by the retinal periphery may drive further axial elongation.
Bifocal Spectacles	• Targeted at reducing lags of accommodation ➔ being a potential source of hyperopic defocus
PALS	• Reduce the accommodative demand and/or reduce accommodative lag during near tasks.
Contact Lenses	• However, Atchison37 has shown with optical modeling that spherical contact lenses will produce more peripheral myopic shift than spherically surfaced spectacle lenses.
Gas Permeable CL	• More recent, well-conducted studies showed that the use of these lenses did not impact axial elongation and that the apparent control of myopia progression observed with GPs was most likely induced by corneal flattening.44,45
Soft Multifocal Contact Lenses	• In these designs, the peripheral region of the lens has relatively more positive (plus) power, incorporated as a gradual increase toward the periphery (progressive design) or presented in distinct zones (concentric ring design).

The article *IMI – Interventions for Controlling Myopia Onset and Progression Report* offers a comprehensive synthesis of current strategies to combat the rising global prevalence of myopia, **which is projected to affect up to 50% of the population by 2050.** It categorizes interventions into four main groups: optical (e.g. multifocal lenses, orthokeratology), **pharmacological** (e.g. low-dose atropine, 7-methylxanthine), behavioral (e.g. increasing outdoor time), and surgical (e.g. posterior scleral reinforcement). Although various treatments have demonstrated efficacy—particularly in randomized controlled trials—their individual effectiveness remains highly variable, with no single therapy universally successful. Of special relevance to the OPTM6413 curriculum, this report reinforces the therapeutic importance of axial length monitoring, evidence-based prescribing, and emerging trends like **combination therapies (e.g. atropine with orthokeratology or contact lenses with embedded drug delivery)**. The variability in patient response underscores the clinical need for personalized treatment planning, which aligns with the course's emphasis on co-management, prescribing in special populations, and myopia control. Additionally, the paper highlights the need for continued research into the underlying mechanisms of myopia progression, offering optometrists both a challenge and opportunity to integrate evolving therapeutic approaches into everyday practice .

IMI Risk Factors for Myopia (Morgan et al)

In-Depth Summary of the Article

Overview and Purpose

This article, published as part of the International Myopia Institute (IMI) special issue, provides a comprehensive review of the **epidemiological and biological risk factors** for the onset and progression of myopia. It aims to distinguish between **causal and associative** factors, evaluates confounders, and highlights the **need for improved methodologies** such as objective exposure measurement and randomized controlled trials (RCTs).

Key Risk Factors for Myopia

1. Education (Causal)

- There is **strong and consistent evidence** that more years of education are causally associated with increased prevalence of myopia, confirmed through **Mendelian randomization** studies.
- Mechanism remains unclear but is likely linked to increased **nearwork**, reduced **outdoor time**, and **light exposure**.

2. Time Outdoors (Causal and Protective)

- Robust evidence from **RCTs** supports that **more time outdoors** significantly reduces incident myopia in children.
- Hypothesized mechanisms include:
  - **Dopamine release** stimulated by **bright light**.
  - Possible effects on **circadian rhythms** and **diurnal regulation** of axial growth.
- School-based interventions increasing outdoor time by 40–80 mins/day have been successful in **preventing onset**.

3. Nearwork and Screen Time (Equivocal)

- Evidence linking **nearwork (including screen time)** to myopia is **weak and inconsistent**, though meta-analyses suggest a modest association.
- It is unclear if screen time adds a unique risk beyond traditional near tasks like reading.
- Historical data shows myopia epidemics predate digital devices, suggesting **reading and educational demands** are bigger drivers.

Other Factors (Mostly Non-Causal or Inconsistent)

Category	Risk Factor	Strength of Evidence
Basic Birth Factors	Ethnicity, birth order, sex	Weak/Inconsistent
Personal	Height, IQ, physical activity	Mostly Weak/Confounded
Family	Parental myopia, SES	Strong (but genetic/environmental confounding)
Environment	Urbanization, light pollution	Moderate
Cultural Beliefs	Reading posture, font size	Weak, anecdotal

- Several commonly held beliefs (e.g., posture, reading in dim light) **lack scientific evidence** and are culturally rooted.
- Socioeconomic status often acts as a **proxy** for other factors like education and access to outdoor environments.

Research and Methodological Recommendations

- **Improve exposure measurement** using wearable sensors (e.g., Clouclip, FitSight).
- Employ **cycloplegic refraction** for accurate diagnosis.
- Use **mediation analysis**, **regression discontinuity**, and **Mendelian randomization** to distinguish causality.
- Validate interventions in **RCTs**, not just observational studies.

Conclusions

- **Two primary modifiable risk factors**: education (causal and adverse) and outdoor time (causal and protective).
- Despite the large list of proposed risk factors, only a few have demonstrated causal associations.
- **Only increased outdoor time** has been successfully translated into a **validated preventive intervention**.

Relevance to OPTM6413

1. **Myopia Prevention Strategies**:
   - The article strengthens your understanding of **evidence-based preventive interventions** such as **increasing outdoor time**, which aligns with OPTM6413's

focus on myopia management in paediatric patients and public health contexts.

2. **Critical Evaluation of Risk Factors**:
    - Offers insight into **confounding vs causality**, vital when recommending interventions or counseling parents on modifiable behaviours (e.g., outdoor play vs limiting screen time).
3. **Pharmacological and Behavioral Comanagement**:
    - Although this paper is not pharmacological, it complements discussions on non-pharmacological strategies like **outdoor time**, which can be **combined with low-dose atropine or optical interventions**, a key concept in OPTM6413 combination therapy frameworks .
4. **Clinical Relevance for Practice**:
    - As future prescribers, understanding **why and how interventions work** allows you to justify treatment plans with scientific evidence, a key skill assessed in script writing and clinical reasoning components of the course.
5. **Informing Public Health and Policy**:
    - This article's findings are the foundation for **school-based public health strategies** adopted in places like Taiwan and China. Knowledge of such programs is vital for OPTM6413's emphasis on **system-level interventions and referrals**.

Advances in Amblyopia: What Have We Learned From PEDIG Trials? (Kammi, et al.)

Background

- **Amblyopia** is the most common cause of monocular vision loss in children (1%–3.5% prevalence).
- It arises from *neurological inhibition* of the amblyopic eye during development, often due to **anisometropia**, **strabismus**, or **visual deprivation**.
- The **critical window** for treatment is birth to ~6–8 years, where neuroplasticity allows for successful intervention.

Study Methods; PEDIG conducted numerous multicenter RCTs examining:

- Spectacle-only correction
- Patching regimens (2h, 6h, full-time)
- Atropine penalization (daily vs. weekend only)
- Combined strategies
- Compliance, psychosocial impact, and long-term stability

Key Findings

1. Refractive Correction Alone

- For anisometropic and strabismic amblyopia, **glasses alone** improved VA ≥2 lines in **77%** of children (average gain: **2.9 lines**).
- ~27% had **complete resolution** with just spectacles.
- Significant visual improvement occurred within **15 weeks**.

2. Patching Therapy

Severity	Patching Duration	Vision ≥20/30	Visual Gain
Moderate	2h/day	62%	2.4–3.2 lines
Moderate	6h/day	62%–79%	3.16 lines
Severe	6h/day	25%	4.8 lines
Severe	Full-time	25%	4.7 lines
Severe	2h/day	–	3.7 lines

- **No significant difference** in final outcome between 2h and 6h for moderate amblyopia.
- Full-time patching did **not outperform** 6h in severe amblyopia. **Compliance** was critical; objective monitoring revealed over-reporting of patch hours.

3. Atropine Penalization

Regimen	Population	Vision ≥20/40	VA Gain
Daily drops	Moderate amblyopia	74%–75%	~2.8–3.2 lines
Weekend only	Moderate	53%	2.3 lines
Weekend only	Severe	21%	4.5 lines

- **Equivalent efficacy** to patching in moderate cases.
- **Better tolerated** psychosocially—less stigma, easier compliance.
- Daily and weekend atropine had **no significant difference** in visual outcomes.

4. Functional Outcomes After Treatment

- **Contrast Sensitivity:** Slight persistent deficits in low contrast, fine detail.
- **Stereoacuity:** Generally remains **reduced** even after treatment; better outcomes if baseline stereoacuity was present.
- **Reading Fluency:** Slight monocular delay; comprehension unaffected.
- **Binocular Reading:** Results mixed—may or may not be impaired.

5. Treatment Cessation & Recurrence

- **Recurrence rate:** 21%–25% in 1 year if therapy stopped abruptly.
- **Tapering** (e.g., 6h to 2h) significantly **reduced recurrence**.
- Factors increasing recurrence:
    - More VA gain during treatment
    - High final VA
    - Previous relapse

6. Older Children (7–17 years)

- Still benefit from treatment:
    - **53%** of 7–12 y/o showed ≥10 letters improvement
    - **25%** of 13–17 y/o improved, but **47%** improved if untreated previously
- Gains were **sustained**, making treatment worthwhile even later.

Mechanism Notes

- **Patching:** Forces use of amblyopic eye by occluding the stronger eye.
- **Atropine:** Paralyzes accommodation in the fellow eye → penalizes near vision → encourages amblyopic eye use.
- **Under-correction** of the fellow eye with atropine was **not more effective**.

Relevance to OPTM6413

Optometric Prescribing

- Understanding of **when to initiate therapy** (e.g., trial of glasses alone first).
- Justifies prescribing spectacles as **standalone initial treatment**.
- Supports **adjusted prescribing duration** (2h patching may be enough).
- Knowledge of **psychosocial impact** helps improve **patient compliance and education**.

Refractive Error & Amblyopia Management

- Directly applicable to **co-management** of amblyopia.
- Confirms **anisometropic and strabismic amblyopia** benefit from targeted approaches.
- Emphasizes **early correction** to maximize neuroplasticity.

Co-management & Referral

- Optometrists can:
    - Monitor progress under glasses-only phase
    - Educate parents on **treatment expectations**
    - Identify and refer cases with **risk of non-compliance** or **relapse**

Therapeutic Decision-Making

- Reinforces **customizing treatment**: atropine vs patching based on lifestyle and preference.
- Monitoring for **recurrence post-therapy** is essential—suggests **1-year follow-up** after cessation.

Final Takeaways

Treatment Aspect	Optimal Choice or Finding
Initial management	Glasses alone (especially in anisometropia)
Moderate amblyopia	Patching 2h/day = 6h/day; atropine equally effective
Severe amblyopia	6h or weekend atropine works; 2h/day may suffice
Compliance	Critical—psychosocial support improves it
Age consideration	Treatment effective even up to 17 y/o
Recurrence prevention	Taper treatment + monitor 12 months post-therapy

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Week 9

Evidence Based Guidelines For Cataract Surgery (Lundstrom, et al.)

1. Postoperative Complications and Their Management Highly relevant to optometrists co-managing cataract patients post-surgery.

- **Endophthalmitis** (incidence 0.036%):
  - *Clinical Relevance*: Know red flags (pain, vision loss, hypopyon) and refer immediately.
  - *Prescribing Relevance*: Optometrists should understand antibiotic prophylaxis (e.g., intracameral cefuroxime) and systemic signs requiring escalation.

- **Persistent Corneal Edema** (0.15%):
  - *Clinical Relevance*: Can present as blurred vision; rule out corneal decompensation.
    Management: Lubricants, steroids, or referral if bullous keratopathy suspected.

- **Posterior Capsule Opacification (PCO)** (0.21%):
  - *Clinical Relevance*: Common cause of vision loss post-op. Presents like cataract return.
  - *Optometric Role*: Diagnose using slit lamp retroillumination; refer for Nd:YAG capsulotomy.
- **Uveitis** (0.35%):
  - *Prescribing Relevance*: Familiarity with **topical corticosteroid** prescribing (e.g., prednisolone acetate 1%) and tapering regimens.
  - *OPTM6413 Link*: Covered in inflammation and steroid prescribing modules.

- **Uncontrolled IOP** (0.06%):
  - *Clinical Relevance*: IOP spikes post-surgery can occur (especially in steroid responders).
  - *Optometric Role*: Measure IOP post-op; initiate or adjust IOP-lowering therapy within scope.

2. Preoperative and Visual Outcomes

Important for understanding the **rationale for surgery** and **post-op expectations**.

- **Indications for Surgery**: No fixed acuity cut-off; decisions are symptom-based (e.g., glare, daily activity difficulty).
  - *Optometric Role*: Contribute to surgical referral decisions and document pre-op acuity.
- **Visual Outcome**: 94.4% reach 6/12 or better post-op; 97.2% if no ocular comorbidity.
  - *Patient Counselling*: Useful when managing expectations, especially if AMD or glaucoma present.

3. Refractive Outcomes and IOL Selection

Essential for patient counselling and **refractive planning** post-op.

- **Biometry Error**: Mean error 0.55D; 91.5% within ±1.0D.
  - *Optometric Role*: Evaluate final refraction and prescribe accurately.
- **Premium IOLs (Multifocal/Toric)**: Rarely used, require specific counselling.
  - *Counselling Role*: Understand pros/cons for patient discussions.

4. Anaesthesia and Surgical Techniques

Relevant for patient understanding and informed consent.

- **Anaesthesia Types**: Topical (most common), Sub-Tenon, Retrobulbar, General.
  - *Clinical Relevance*: Know when a patient might be unsuitable for topical anaesthesia alone.
- **Phacoemulsification**: Standard technique. ECCE used selectively.
  - *Knowledge Expectation*: Be able to explain modern cataract surgery simply to patients.

5. Comorbidities and Risk Stratification

Supports co-management and referral decisions.

- **Common Ocular Comorbidities**: AMD (12.4%), Glaucoma (7.7%).
  - *Clinical Relevance*: These reduce the likelihood of perfect outcomes.
  - *OPTM6413 Link*: Differential diagnosis and shared care with ophthalmology.
- **Difficult Surgeries**: Predictable risks like small pupils, corneal opacities, etc.
  - *Referral Notes*: Important for pre-referral notes and post-op monitoring plans.

Topic	Clinical/Prescribing Link
Post-op complications	Co-management, corticosteroid use, emergency recognition
IOP control	Glaucoma co-management, IOP-lowering drugs
Uveitis	Corticosteroid prescribing, inflammation recognition
Visual and refractive outcomes	Prescription adjustment, counselling
PCO	Diagnosis, referral for Nd:YAG
IOL types	Counselling, refractive outcomes
Referral criteria	Recognising surgical need based on symptoms

Non-steroidal anti-inflammatory drugs versus corticosteroids for controlling inflammation after uncomplicated cataract surgery (Juthani, et al.)

Article Overview

Objectives

To determine whether **topical NSAIDs (alone or in combination)** are more effective than **topical corticosteroids alone** in:

- Controlling **intraocular inflammation**
- Improving **best-corrected visual acuity (BCVA)**

- Reducing postoperative complications like **cystoid macular oedema (CME)**
- Assessing side effects such as **elevation in IOP**
- Evaluating **patient comfort and cost-effectiveness**

Study Design & Methods

- **Study Type:** Systematic review of **48 randomized controlled trials (RCTs)**

- **Inclusion Criteria:** Participants undergoing **uncomplicated phacoemulsification**
- **Interventions compared:**
    - NSAIDs vs corticosteroids
    - NSAIDs + corticosteroids vs corticosteroids alone
- **Follow-up period:** Mostly 1 week to 3 months
- **Primary Outcomes:**
    - **Anterior chamber inflammation** (via flare/cell count)
    - **BCVA**
- **Secondary Outcomes:**
    - **Incidence of CME**
    - **Intraocular pressure (IOP) changes**
    - **Adverse effects**

Key Results

1. Inflammation Control

- NSAIDs may slightly reduce **anterior chamber flare** at 1 week post-op, but **evidence was weak** and inconsistent.
- **No conclusive superiority** of NSAIDs over corticosteroids in reducing inflammation.

2. Cystoid Macular Oedema (CME)

- NSAIDs may **reduce the risk of CME**, either alone or in combination with corticosteroids:
    - **NSAIDs alone vs corticosteroids**: Slightly lower CME rates (low-certainty evidence)
    - **NSAIDs + corticosteroids vs corticosteroids**: Some studies showed a **benefit**, especially in diabetic patients.

3. Visual Acuity (BCVA)

- No consistent evidence of **significant BCVA improvement** with NSAIDs over corticosteroids after 3 months.
- Short-term recovery may be better with NSAIDs in **diabetic patients**, but not in the general population.

4. Adverse Effects

- Corticosteroids linked to **IOP elevation**
- NSAIDs associated with **rare but serious complications**:
    - Corneal melt
    - Keratitis or ulceration
- **No major difference in overall adverse events**, but NSAIDs require careful use especially in those with **ocular surface disease**.

5. Cost-Effectiveness

- Not assessed in the review but mentioned as an area of interest.
- NSAIDs are generally **more expensive** than corticosteroids.

Authors' Conclusions

- **Evidence insufficient** to declare NSAIDs (alone or combined) superior to corticosteroids for post-cataract inflammation.
- NSAIDs may help **reduce CME** incidence but the benefit is **limited and uncertain**.
- **Future RCTs** are needed, with:
    - Consistent outcome definitions (e.g., SUN criteria)
    - Standardized drug types and dosing
    - Reporting in dichotomous outcomes (yes/no inflammation) rather than means

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Relevance to OPTM6413

Optometric Prescribing

- Supports a **conservative, evidence-based approach** to postoperative inflammation management.
- Highlights risks of NSAIDs in **ocular surface disease**, a common presentation in optometry.
- Emphasizes importance of **monitoring IOP** with corticosteroids — critical for **glaucoma suspects** and routine follow-up.

Therapeutics Knowledge

- Reinforces that NSAIDs and corticosteroids act via **distinct anti-inflammatory pathways**:
    - NSAIDs inhibit COX enzymes → ↓ prostaglandins
    - Corticosteroids suppress phospholipase A2 → broader immunosuppression

Ocular Surface Disease

- NSAID use is contraindicated or requires **caution** in dry eye and epithelial compromise (e.g., post-PRK, corneal melts).

- Highlights need for **careful therapeutic judgment**, especially in elderly or diabetic patients undergoing cataract surgery.

Inflammatory Disease

- Insight into the **management of postoperative uveitis** or inflammation
- Combination therapy may be necessary in certain populations but should be **individually tailored**.

Co-management & Referrals

- While optometrists may initiate therapy, this review suggests:
    - Referral is warranted for **persistent inflammation**
    - Collaborative care with ophthalmology for **CME detection**, particularly in diabetics
- Emphasizes **optometrist role** in early detection and referral rather than definitive CME treatment

Summary Table

Feature	NSAIDs	Corticosteroids
Anti-inflammatory effect	Moderate	Strong
Mechanism	COX inhibition	Phospholipase A2 inhibition
IOP elevation	Rare	Common
Risk of CME	Lower? (low certainty)	Higher
Adverse events	Corneal melt, keratitis	IOP, delayed wound healing
Use in optometry	Prescribable (under OBA list)	Also widely used
Role in Ocular Surface Disease	Caution	More tolerable but monitor IOP

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Chair-side Reference: Dry Eye – TFOS DEWS II

UNSW School of Optometry & Vision Science

Optometry NEW SOUTH WALES AUSTRALIAN CAPITAL TERRITORY

Dry Eye Definition: TFOS DEWS II

Dry eye is a multifactorial disease of the ocular surface characterised by a loss of homeostasis of the tear film and accompanied by ocular symptoms in which tear film instability and hyperosmolarity, ocular surface inflammation and damage, and neurosensory abnormalities play aetiological roles.

Examination and differential diagnosis workflow

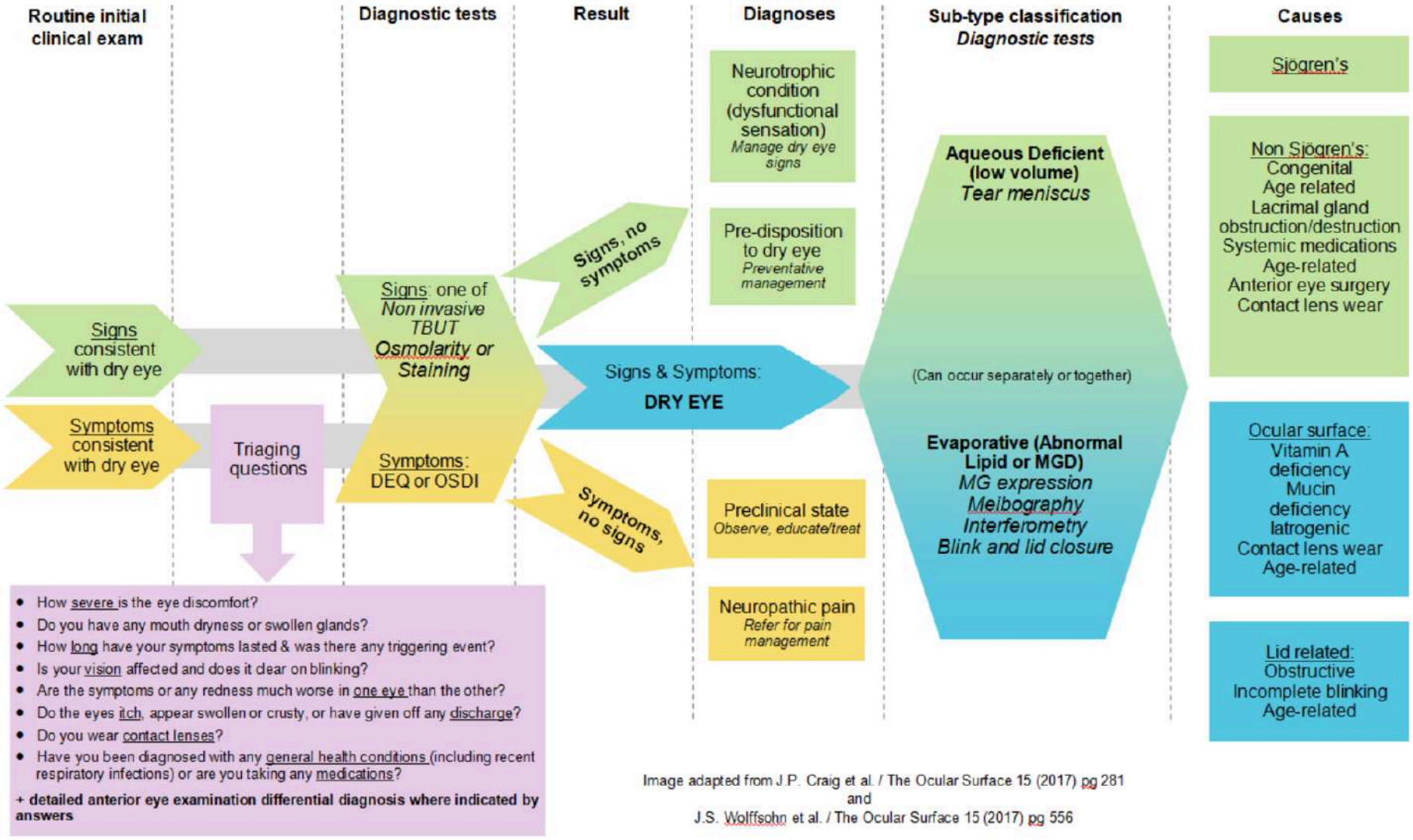


Image adapted from J.P. Craig et al. / The Ocular Surface 15 (2017) pg 281
and
J.S. Wolffsohn et al. / The Ocular Surface 15 (2017) pg 556

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Chair-side Reference: Dry Eye – TFOS DEWS II

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Risk Factors: Modifiable

Consistently shown	Probable	Inconclusive
Androgen deficiency Computer use CL wear Hormone replacement therapy Haematopoeitic stem cell transplantation Environment (pollution, low humidity)	Low intake of FFA Refractive Surgery Allergic Conjunctivitis	Smoking Alcohol Pregnancy Demodex Botox injection
Medications		
Antihistamines, Antidepressants, Anxiolytics, Isotretinoin	Anticholinergics, Diuretics, Beta-blockers	Multivitamins / Oral contraceptive

Risk Factors: Non-Modifiable

Consistently shown	Probable	Inconclusive
Age Female Asian MGD Connective tissue disorder Sjögren's syndrome	Diabetes Rosacea Viral infection Thyroid disease Psychiatric conditions Pterygium	Hispanic Menopause Acne Sarcoidosis

TFOS DEWSII staged management and treatment recommendations

Notes: 1. These steps are not designed to be a rigid, fixed process, but more an organisational tool
2. Options within a step are not ranked by importance
3. It is possible and likely that one or more option within each step will be needed/utilised
4. Earlier steps should be continued if moving to higher steps

STEP 1

Education
Modification of local environment
Dietary modifications including oral essential fatty acid supplementation

Identify and potentially modify offending systemic and topical medications
Ocular lubricants of various types: if MGD is present, then consider lipid containing supplements
Lid hygiene and warm compresses of various types

STEP 2

Non-preserved ocular lubricants
Tea tree oil treatment for Demodex (if present)
Tear conservation (Punctal occlusion / Moisture chamber goggles)
Overnight treatments (ointment or moisture chamber devices)
In-office, physical heating and expression of the meibomian glands (including device-assisted therapies)
In-office intense pulsed light therapy for MGD

Prescription drugs to manage dry eye disease
Topical antibiotic or antibiotic/steroid combination applied to the lid margins for anterior blepharitis (if present)
Topical corticosteroid (limited-duration)
Topical secretagogues
Topical non-glucocorticoid drugs (such as cyclosporine)
Topical LFA-1 antagonist drugs (such as Lifitegrast)
Oral macrolide or tetracycline antibiotics

STEP 3

Oral secretagogues
Autologous/allogeneic serum eye drops
Therapeutic contact lens options
Soft bandage lenses
Rigid scleral lenses

STEP 4

Topical corticosteroid for longer duration
Amniotic membrane grafts
Surgical punctal occlusion
Other surgical approaches: tarsorrhaphy, salivary gland transplantation